ELMFIRE Verification Guide

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ELMFIRE version: not declared
Compiled: October 22, 2026

About This Guide

This guide examines whether ELMFIRE reproduces prescribed physical and numerical responses under controlled conditions. Each case develops an expected response, defines quantitative comparisons and acceptance criteria, and interprets the available simulation results. The opening summary presents the calculated decisions and the computational information available during report preparation.

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1 How to Read This Guide

Start with the verification summary for the recorded computational environment, declared parallel allocation, and scientific decision table. The case identifiers in the summary link to the corresponding case-report entries. The complete case reports follow in numerical order; each presents its own scientific context, equations, metrics, citations, acceptance criteria, and limitations.

Interpret PASS and FAIL against each case's stated acceptance criteria. Missing results, incomplete simulations, or an unavailable model capability are shown as NOT EVALUABLE. A completed simulation or a complete narrative alone does not establish agreement with the expected response.

The computational environment recorded during report preparation is not necessarily the environment used for every simulation. Within each case, read the formulation and derived expectation before interpreting the measured response. The scope and limitations distinguish the behavior demonstrated by the prescribed conditions from behavior that remains unverified.

2 Verification summary

2.1 Computational environment recorded during report preparation

The following information describes the environment available during preparation of this guide. It does not establish the environment used for earlier simulations. An available compiler is not necessarily the compiler used to build the model. ELMFIRE was not executed to obtain these observations; an undeclared model release remains unknown.

Configuration	Recorded value
Recorded at (Coordinated Universal Time)	2026-10-22T05:27:03+00:00
Declared ELMFIRE release	not declared
Operating system	macOS-15.7.7-arm64-arm-64bit
Processor architecture	arm64
Message Passing Interface (MPI) implementation	not available
Available Fortran compiler	GNU Fortran (Homebrew GCC 14.2.0_1) 14.2.0
Python version	CPython 3.12.9
Scientific analysis libraries	NumPy 1.26.4; Matplotlib 3.9.2; PyYAML 6.0.1; Rasterio 1.5.0; GDAL not installed; Pandas 2.2.2; GeoPandas not installed; Shapely not installed

2.2 Comparison type and declared parallel allocation

The comparison type distinguishes isolated and coupled processes or the spatial scale of validation. The process count is the allocation declared for each case, not a measurement of resources used in an earlier simulation. Missing declarations are not inferred.

Case	Comparison type	MPI processes
CASE01_BET	Coupled-process verification	not declared
CASE02_FGR	Coupled-process verification	not declared
CASE03_ETC	Coupled-process verification	not declared
CASE04_STI	Coupled-process verification	not declared
CASE05_TDW	Coupled-process verification	not declared
CASE06_F2D	Coupled-process verification	not declared
CASE07_IPC	Coupled-process verification	not declared
CASE08_FBC	Coupled-process verification	not declared
CASE09_MMD	Coupled-process verification	not declared
CASE10_WSR	Coupled-process verification	not declared
CASE11_WSC	Coupled-process verification	not declared
CASE12_TRS	Coupled-process verification	not declared
CASE13_PRM	Coupled-process verification	not declared
CASE14_SRS	Coupled-process verification	not declared
CASE15_CRO	Isolated-process verification	not declared
CASE16_NSP	Isolated-process verification	not declared
CASE17_SCV	Isolated-process verification	not declared
CASE18_TCV	Isolated-process verification	not declared
CASE19_WTH	Coupled-process verification	not declared
CASE20_PIG	Coupled-process verification	not declared
CASE21_WDE	Coupled-process verification	not declared
CASE22_VSL	Coupled-process verification	not declared

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Case	Comparison type	MPI processes
CASE23_FQD	Coupled-process verification	not declared
CASE24_CXI	Coupled-process verification	not declared
CASE25_MQD	Coupled-process verification	not declared
CASE26_CNF	Coupled-process verification	not declared
CASE27_FBT	Coupled-process verification	not declared
CASE28_OVA	Coupled-process verification	not declared
CASE29_SUP	Coupled-process verification	not declared
CASE30_MIT	Coupled-process verification	not declared
CASE31_PFT	Coupled-process verification	not declared
CASE32_RCV	Coupled-process verification	not declared
CASE33_SDO	Coupled-process verification	not declared
CASE34_FMS	Isolated-process verification	not declared
CASE35_WSS	Isolated-process verification	not declared
CASE36_SLS	Isolated-process verification	not declared
CASE37_DMS	Isolated-process verification	not declared
CASE38_LMS	Isolated-process verification	not declared
CASE39_DHC	Isolated-process verification	not declared
CASE40_WSD	Coupled-process verification	not declared
CASE41_CFP	Isolated-process verification	not declared
CASE42_WAF	Coupled-process verification	not declared
CASE43_WER	Coupled-process verification	not declared
CASE44_WHP	Coupled-process verification	not declared
CASE45_HRS	Coupled-process verification	not declared
CASE46_WUT	Coupled-process verification	not declared
CASE47_UWT	Coupled-process verification	not declared
CASE48_WGC	Coupled-process verification	not declared
CASE49_WTC	Coupled-process verification	not declared

2.3 Scientific decision summary

The decisions below follow the established calculated metrics and case-specific acceptance criteria. Completing a simulation or preparing a report does not establish PASS. Missing or insufficient scientific evidence is reported as NOT EVALUABLE.

Total cases: 49 **PASS:** 31 **FAIL:** 11 **CHARACTERIZED:** 0 **NOT EVALUABLE:** 7

Case	Category	Title	Decision
CASE01_BET	Coupled-process verification	Biomass Emission-Time Resolution Independence	**PASS**
CASE02_FGR	Coupled-process verification	Finite-Duration Firebrand Generation	**PASS**
CASE03_ETC	Coupled-process verification	Eulerian Firebrand Transport Convergence	**FAIL**
CASE04_STI	Coupled-process verification	Steady Transport and Immediate Ignition	**PASS**
CASE05_TDW	Coupled-process verification	Firebrand Transport in Time-Dependent Wind	**PASS**
CASE06_F2D	Coupled-process verification	Two-Dimensional Firebrand-Driven Spread	**PASS**
CASE07_IPC	Coupled-process verification	Ignition Probability and SFT Convergence	**FAIL**

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Case	Category	Title	Decision
CASE08_FBC	Coupled-process verification	Deposited Firebrand Consumption	**FAIL**
CASE09_MMD	Coupled-process verification	Mixed-Mode Ignition-Delay Effects	**PASS**
CASE10_WSR	Coupled-process verification	WUI Structural Resolution Sensitivity	**NOT EVALUABLE**
CASE11_WSC	Coupled-process verification	WUI Surface-Coupling Synchronization	**NOT EVALUABLE**
CASE12_TRS	Coupled-process verification	WUI Temporal-Resolution Sensitivity	**FAIL**
CASE13_PRM	Coupled-process verification	WUI Parametric Response Maps	**NOT EVALUABLE**
CASE14_SRS	Coupled-process verification	Two-Dimensional Spatial-Resolution Sensitivity	**NOT EVALUABLE**
CASE15_CRO	Isolated-process verification	Case 15: Constant rate of spread	**PASS**
CASE16_NSP	Isolated-process verification	Case 16: No-spread limiting behavior	**PASS**
CASE17_SCV	Isolated-process verification	Case 17: Spatial convergence	**FAIL**
CASE18_TCV	Isolated-process verification	Case 18: Temporal convergence	**FAIL**
CASE19_WTH	Coupled-process verification	Case 19: Transient WUI heat flux	**NOT EVALUABLE**
CASE20_PIG	Coupled-process verification	Case 20: Point ignition and isotropic spread	**PASS**
CASE21_WDE	Coupled-process verification	Case 21: Wind-driven elliptical spread	**PASS**
CASE22_VSL	Coupled-process verification	Case 22: Spread across opposing valley slopes	**PASS**
CASE23_FQD	Coupled-process verification	Case 23: Spatially varying fuel quadrants	**PASS**
CASE24_CXI	Coupled-process verification	Case 24: Combined fuel, slope, and wind interactions	**PASS**
CASE25_MQD	Coupled-process verification	Case 25: Spatially varying moisture quadrants	**PASS**
CASE26_CNF	Coupled-process verification	Case 26: Surface, passive-crown, and active-crown fire	**PASS**
CASE27_FBT	Coupled-process verification	Case 27: Lagrangian firebrand generation and transport	**FAIL**
CASE28_OVA	Coupled-process verification	Case 28: Overnight spread-rate adjustment	**PASS**
CASE29_SUP	Coupled-process verification	Case 29: Initial- and extended-attack suppression	**FAIL**
CASE30_MIT	Coupled-process verification	Case 30: Multiple-ignition topology	**PASS**
CASE31_PFT	Coupled-process verification	Case 31: Planar-front transport	**PASS**
CASE32_RCV	Coupled-process verification	Case 32: Raster-orientation covariance	**PASS**
CASE33_SDO	Coupled-process verification	Case 33: Slope-driven directional spread	**PASS**

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Case	Category	Title	Decision
CASE34_FMS	Isolated-process verification	Case 34: Standard fuel-model Rothermel sweep	**PASS**
CASE35_WSS	Isolated-process verification	Case 35: Midflame wind-speed Rothermel sweep	**PASS**
CASE36_SLS	Isolated-process verification	Case 36: Slope-factor Rothermel sweep	**PASS**
CASE37_DMS	Isolated-process verification	Case 37: Dead-fuel moisture and extinction sweep	**PASS**
CASE38_LMS	Isolated-process verification	Case 38: Static live-fuel moisture sweep	**PASS**
CASE39_DHC	Isolated-process verification	Case 39: Dynamic herbaceous curing sweep	**PASS**
CASE40_WSD	Coupled-process verification	Case 40: Wind-slope vector and direction sweep	**PASS**
CASE41_CFP	Isolated-process verification	Case 41: Custom fuel-parameter Rothermel sweep	**PASS**
CASE42_WAF	Coupled-process verification	Case 42: Wind-adjustment-factor coupling sweep	**PASS**
CASE43_WER	Coupled-process verification	Case 43: WU-E ellipse response	**PASS**
CASE44_WHP	Coupled-process verification	Case 44: WU-E heat-parameter response	**FAIL**
CASE45_HRS	Coupled-process verification	Case 45: WU-E heat-to-urban rate-of-spread response	**PASS**
CASE46_WUT	Coupled-process verification	Case 46: Wildland-to-urban WU-E transition and isolated receivers	**FAIL**
CASE47_UWT	Coupled-process verification	Case 47: Urban-to-wildland WU-E transition and isolated wildland receivers	**NOT EVALUABLE**
CASE48_WGC	Coupled-process verification	Case 48: WU-E uniform-community spatial convergence	**NOT EVALUABLE**
CASE49_WTC	Coupled-process verification	Case 49: WU-E uniform-community temporal convergence	**FAIL**

Decision vocabulary. PASS and FAIL are case-defined scientific decisions. CHARACTERIZED denotes descriptive validation metrics without a justified acceptance threshold. NOT EVALUABLE denotes missing, unrun, incomplete, unsupported, or otherwise non-decisional evidence.

Case 01: Biomass Emission-Time Resolution Independence

1 Verification purpose

This case verifies that biomass-emission duration is represented as a physical residence time rather than a grid-dependent number of solver steps. Changing cell size must preserve accumulated firebrand density for the same unit-width source and forcing.

The test exposes incorrect conversion between per-cell and unit-width generation rates, timestep-dependent source integration, and unintended cell-width scaling. Success requires overlapping downwind profiles and mesh-independent far-field accumulation within tolerance.

2 Mathematical formulation and numerical implementation

A surface-fire cell emits firebrands while the flame front crosses that cell. For a square cell of size $\Delta x = \Delta y$ and surface-fire spread rate V_s, the emission duration is

$$t_{\text{emission}} = \frac{\Delta x}{V_s}.$$

Without this duration scaling, changing the resolution would change the time for which a burning cell acts as a firebrand source and would therefore change the accumulated deposition solely through a discretization effect.

This case represents a one-dimensional firebrand-transport problem with a 1 m-wide strip perpendicular to the transport direction. The unit-width firebrand generation rate is

$$\text{GR}_{\text{1m}} = 33.3 \text{ pcs s}^{-1} \text{ MW}^{-1}.$$

ELMFIRE applies the vegetation firebrand production coefficient per unit fire power to the total fire power of a computational pixel. A pixel with fireline intensity I_B and cross-stream width Δy has fire power $I_B \Delta y$. Its firebrand generation rate is therefore

$$\dot{N}_{\text{pixel}} = I_B \Delta y \, \text{GR}_{\text{ELMFIRE}}.$$

To keep the represented source equal to the 1 m-wide source at every grid resolution, preprocessing sets

$$\text{GR}_{\text{ELMFIRE}} = \frac{\text{GR}_{\text{1m}}}{\Delta y} = \frac{33.3}{\Delta x} \quad [\text{pcs s}^{-1} \text{ MW}^{-1}].$$

The configured production coefficients are consequently 6.66, 3.33, and 1.11 pcs/s/MW for the 5, 10, and 30 m simulation conditions. In all three simulation conditions, $\text{GR}_{\text{ELMFIRE}} \Delta y = \text{GR}_{\text{1m}}$, so the unit-width source strength is independent of cell size.

2.1 Physical and numerical processes

The documented residence-time formulation requires a burning cell to emit only while the flame resides in that cell [1]. ELMFIRE applies the prescribed fire-power-proportional source parameters to each burning cell, converting its fire power to firebrand production. The emission duration is determined from the flame residence time. Transport and deposition then distribute the emitted firebrands over the downwind cells. The comparison checks both parts of the discretization: source preparation preserves the intensive one-metre source strength by scaling the configured rate with cross-stream cell

size, and analysis derives the expected density from simulated fireline intensity and spread rate at the same physical state. A source-duration or cell-power scaling error would create a systematic resolution trend. This controlled numerical verification does not validate the empirical generation coefficient or transport distribution against experiments.

3 Derivation of expected behavior

Combining the pixel generation rate with the emission duration gives the number emitted while the front crosses one cell. Dividing that number by the represented bin area eliminates Δx and gives the expected accumulated density:

$$N''_{\text{ref}} = \frac{I_B \Delta y \, \text{GR}_{\text{ELMFIRE}}(\Delta x / V_s)}{\Delta x (1\ \text{m})} = \text{GR}_{1\text{m}} \frac{I_B}{V_s} \quad [\text{pcs m}^{-2}].$$

Thus, the reference is calculated separately from the fire behavior actually produced by each simulation; it is not a fixed nominal number. The postprocessor finds the maximum positive fireline intensity in the usable interior of the latest fireline-intensity field and reads the spread rate from the corresponding cell of the spread-rate field. ELMFIRE writes these fields in kW/m and ft/min for this configuration, so the implemented conversion is

$$N''_{\text{ref}} = 33.3 \, \frac{I_{B,\text{sim}}/1000}{V_{s,\text{sim}}(0.3048/60)} \quad [\text{pcs m}^{-2}].$$

For the current outputs, the mean values across the simulation conditions are $I_{B,\text{sim}} = 2.200042$ MW/m and $V_{s,\text{sim}} = 0.577015$ m/s. These values give $N''_{\text{ref}} = 126.966$ pcs/m^2. The per-simulation condition values and selected raster cells are retained in calculated verification and validation metrics.

The generated numerical substitutions make the cancellation visible. With $CFL_w = 0.5$ and $U = 6.71\ \text{m s}^{-1}$,

Δx (m)	5	10	30
$\Delta t_{\text{generated}}$ (s)	0.372578	0.745156	2.000000
$GR_{\text{ELM}} = 33.3/\Delta x$	6.66	3.33	1.11
$GR_{\text{ELM}} \Delta y$	33.3	33.3	33.3

in pcs s^{-1} MW^{-1} for the last two rows. The 5- and 10-m Courant–Friedrichs–Lewy (CFL) steps divide the requested 500-s duration exactly. At 30 m, the nominal CFL step is 2.235469 s; preprocessing selects the conservative whole-second divisor 2 s, giving 250 steps and preserving the 500-s stop while keeping $CFL_w = 0.4473 < 0.5$. Thus the source-scaling reference is exactly 33.3 in all three simulation conditions, and its relative error is

$$e_{GR} = \max_i \frac{|GR_i \Delta y_i - 33.3|}{33.3}, \qquad e_{GR,\text{ref}} = 0.$$

The run-derived FLIN and VS values enter only as controlled fire-behavior normalizers; the ember-flux field being tested does not determine them. With $0.3048/60 = 0.00508$,

$$N''_{\text{ref}} = 33.3 \frac{FLIN/1000}{VS(0.00508)} = 6.555118 \, \frac{FLIN\ [\text{kW m}^{-1}]}{VS\ [\text{ft min}^{-1}]} \quad \text{pcs m}^{-2}.$$

For each generated mesh the expected far-field error is therefore zero, and the substituted decision requires

$$\text{mean}_{150<x\leq 200} \, |N'' - N''_{\text{ref}}| / N''_{\text{ref}} \leq 0.05.$$

4 Verification metrics and acceptance criteria

For each resolution, the centre-row accumulated count N_i is converted to an areal density for the represented one-metre strip,

$$n_i'' = \frac{N_i}{\Delta x\,(1\ \mathrm{m})}.$$

The independent reference uses the maximum-FLIN valid interior cell from the same run,

$$n_{\mathrm{ref}}'' = GR_{1m}\,\frac{FLIN/1000}{VS\,(0.3048/60)},$$

where FLIN is read in kW m^{-1}, VS in ft min^{-1}, and $G_{R,1m} = 33.3$ pcs s^{-1} MW^{-1}. The comparison mask contains finite physical cells with $150 < x \leq 200$ m; the two-cell halo is excluded.

4.1 Justification of metric quantification

The source-scaling check is an algebraic invariant rather than an empirical model tolerance. Preprocessing computes $G_{R,\Delta x}\Delta x$ from the same double-precision inputs used to write the namelist, so a relative allowance of 10^{-12} admits only floating-point roundoff and still detects any omitted or duplicated width factor. The 5% density-error limit is applied to a far-field mean because the derived expectation is a developed plateau, not a pointwise subcell solution. It permits the cell integration, discrete selection of the run-derived FLIN/VS reference cell, and finite-window averaging to differ slightly while remaining far below the order-one, resolution-proportional error caused by incorrect extensive/intensive scaling. Requiring the limit at every resolution makes the criterion a grid-consistency test rather than an agreement obtained by cancellation in one simulation condition. Completeness and the conjunctive decision are logical requirements: without every source and reference raster, neither the normalization nor the claimed resolution independence is testable.

4.2 Predeclared acceptance criteria

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Output completeness	Final accumulated firebrand deposition, fireline intensity, and spread rate rasters for each prescribed simulation condition; shape, cell size, nodata, and halo are validated.	3 of 3 simulation conditions usable
Unit-width source scaling	Configured rate multiplied by cross-stream cell width, $G_{R,\Delta x}\Delta x$.	33.3 pcs s^{-1} MW^{-1} to 10^{-12} relative tolerance
Far-field mean relative error	$E_r = \mathrm{mean}_{150<x\leq200}\,\lvert n_i'' - n_{\mathrm{ref}}''\rvert / n_{\mathrm{ref}}''$, evaluated separately for every resolution.	$E_r \leq 0.05$ for every simulation condition
Overall decision	Conjunction of completeness, source scaling, and all per-resolution error decisions.	PASS only if every component passes

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for a 10 m grid. These panels show the prepared spatial fields, remove the two-cell numerical buffer, and retain map coordinates in metres. The left panel shows

the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

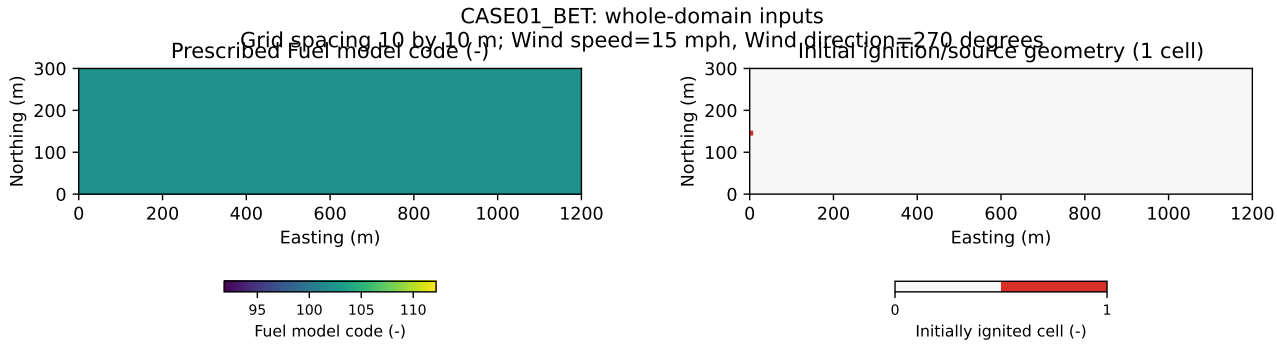


Figure 1: Whole-domain prepared input configuration for the representative simulation condition. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

The preprocessor creates flat, uniform fire behavior fuel model (FBFM) 102 domains at $\Delta x = \Delta y = 5$, 10, and 30 m. The physical simulation domain is the same for every resolution. Each raster also includes a two-cell numerical buffer outside each side of that physical domain. The ignition is placed in the first usable column rather than in the buffer. Crosswind transport and secondary firebrand ignition are disabled so the accumulated deposition can be evaluated as a one-dimensional profile.

ELMFIRE writes the accumulated number of deposited firebrands in each target pixel. The area represented by a streamwise bin in the unit-width problem is $\Delta x \times 1$ m, so the postprocessor converts the pixel count N_i to density using

$$N''(x_i) = \frac{N_i}{\Delta x (1\ \mathrm{m})} = \frac{N_i}{\Delta x} \quad [\mathrm{pcs\,m^{-2}}].$$

The two buffer columns on each end are excluded from this profile.

For each completed simulation condition, the mean accumulated density over $150 < x \leq 200$ m is compared with that simulation condition's FLIN/VS-derived reference. The mean relative error must not exceed 5%. The case passes only when all three simulation conditions have valid outputs and pass independently. Agreement across the three resolutions demonstrates that both the emission duration and the unit-width generation-rate scaling are resolution independent.

6 Actual simulation results

Figure 2 shows the representative accumulated firebrand density from a 10 m grid. The displayed field is taken from the simulated results, masks missing values, and excludes the two-cell numerical buffer. Firebrand counts are normalized by the cell length and the represented 1-m cross-stream strip, giving units of pcs m^{-2}.

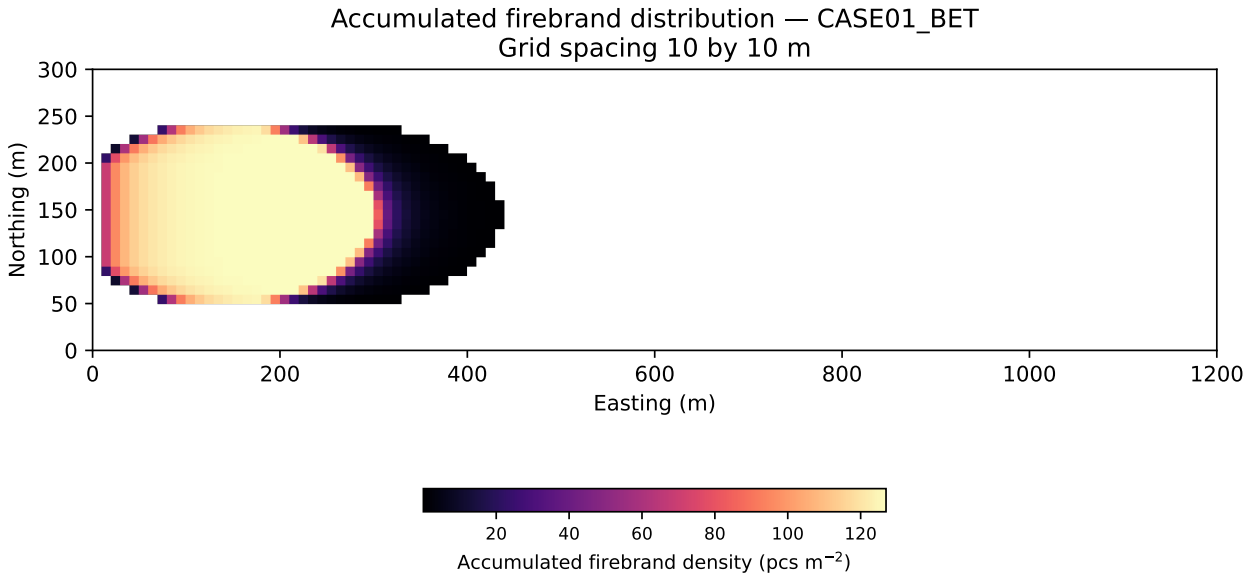


Figure 2: Whole-domain accumulated firebrand density from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

Figure 3 first presents the physical observable in the same three-resolution arrangement used to define the test. All bars are calculated from final ELMFIRE ember-flux rasters, and each dashed line uses the FLIN and spread rate extracted from that same simulation condition.

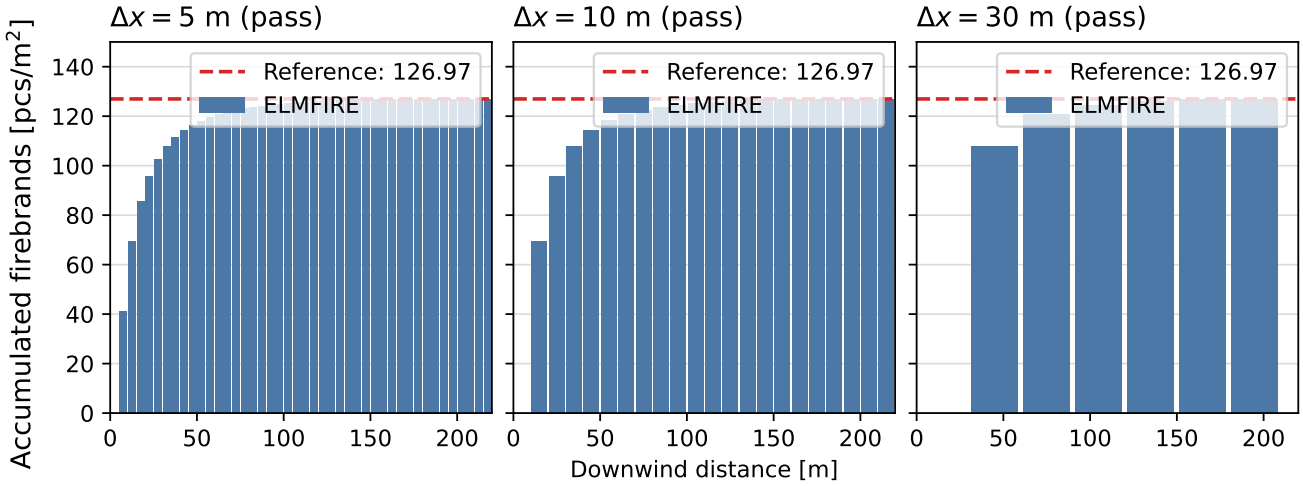


Figure 3: Accumulated firebrand density for the 5, 10, and 30 m grids. The dashed red line in each panel is the run-derived reference. The two-cell numerical halo is excluded from every profile.

7 Calculated metrics and verification decision

The overall decision from the calculated metrics is **PASS**.

Quantity	Value
Overall status	PASS
Mean reference fireline intensity	2.200042 MW/m
Mean reference spread rate	0.577015 m/s
Calculated accumulation reference	126.966 pcs/m^2
Simulation conditions configured	3
Simulation conditions computed	3

Table 2: Biomass emission-time verification summary.

8 Technical interpretation

Agreement between the accumulated deposition density and the independently calculated FLIN/VS reference verifies the complete chain from biomass consumption through emission duration, per-unit-area source scaling, transport, and raster accumulation. Consistent agreement across the three grids further indicates that the implemented source normalization is not tied to cell area.

9 Scope and limitations

This case isolates a uniform, one-dimensional transport problem with crosswind dispersion, delayed ignition, and secondary spotting disabled. It verifies emission-time scaling and accumulated deposition, but does not by itself validate stochastic trajectories or ignition physics. The preprocessing parameters, simulation specification, calculated raster samples, and metric values are retained with the case so that the reported decision can be reproduced.

References

[1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

Case 02: Finite-Duration Firebrand Generation

1 Verification purpose

This case verifies finite-duration firebrand generation when the source lifetime is not commensurate with the solver timestep. It compares timesteps shorter and longer than the residence time while holding the source, grid, wind, and transport distribution fixed.

The test exposes whole-step over-generation, truncation of the last active interval, and timestep-dependent deposition. Success requires both simulation conditions to recover the same integrated source count and downwind profile within tolerance.

2 Mathematical formulation and numerical implementation

Let a burning computational cell generate firebrands at the constant rate GR from ignition time t_{ign} until $t_{\text{ign}} + t_{\text{res}}$. The number generated during an ELMFIRE step $[t, t + \Delta t]$ must use the overlap between that step and the active source interval:

$$\Delta t_{\text{active}} = \max(0, \min(t + \Delta t, t_{\text{ign}} + t_{\text{res}}) - \max(t, t_{\text{ign}})) ,$$

$$\Delta N = \text{GR}\, \Delta t_{\text{active}}.$$

This formulation handles the first and final partial steps and also handles a single timestep that spans the entire residence time. Summing over all steps must give

$$N_{\text{total}} = \text{GR}\, t_{\text{res}}.$$

The case sets GR $= 10.0$ pcs/s and $t_{\text{res}} = 10.0$ s, so the expected total is 100.0 firebrands.

2.1 Physical and numerical processes and verification rationale

The documented finite-duration source formulation identifies a timestep-integration error when the solver step is comparable to or longer than the fireline residence time [1]. ELMFIRE uses an area-proportional source and computes emission from each active cell only during its prescribed emission interval. The Eulerian transport calculation moves the emitted firebrands downwind. Comparing a fine timestep with a timestep exceeding the source duration tests partial-step source integration as well as deposition. The emitted amount must follow the overlap with the active source interval, not the nominal solver step. The total-count metric detects temporal over- or under-integration, while the spatial profile detects correct totals deposited in incorrect cells. Surface spread, ignition, consumption, and crosswind dispersion are disabled so they cannot compensate for source-timing errors. The case verifies numerical accounting and the prescribed transport distribution, not the physical realism of the lognormal distribution.

3 Derivation of expected behavior

Generated firebrands are distributed with a lognormal distance model using $\mu = 2.18$ and $\sigma = 1.23$. The distribution is truncated at its 99th percentile. ELMFIRE rounds that distance to an integer number K of cells and normalizes the retained probability:

$$P_i = \frac{F(i\Delta x) - F((i-1)\Delta x)}{F(K\Delta x) - F(10^{-6})}, \qquad i = 1, \ldots, K,$$

where F is the lognormal cumulative distribution. The analytical count in cell i is $N_i = 100P_i$. The postprocessor reproduces this cell-integrated reference and compares it directly with the final accumulated firebrand deposition raster.

The two time integrations can be reproduced explicitly. For $\Delta t = 0.1875$ s, 53 complete steps cover $53(0.1875) = 9.9375$ s and the final overlap is $10 - 9.9375 = 0.0625$ s, so

$$N_{\text{fine}} = 10[53(0.1875) + 0.0625] = 100.$$

For the $\Delta t = 12.0$ s member, the first step spans the complete source interval and

$$\Delta t_{\text{active}} = \min(12.0, 10) - \max(0, 0) = 10 \text{ s}, \qquad N_{\text{coarse}} = 10(10) = 100.$$

For the spatial reference, $z_{0.99} = 2.3263479$ gives

$$x_{0.99} = \exp[2.18 + 1.23(2.3263479)] = 154.6876 \text{ m}, \qquad K = \text{NINT}(154.6876/10) = 15.$$

The retained cumulative distribution function (CDF) mass is $F(150) - F(10^{-6}) = 0.9893135$. Substitution in the cell formula gives $N_1 = 54.5524$, $N_2 = 20.8937$, $N_3 = 9.4216$ particles, with all 15 bins summing to 100 after normalization. Hence the exact metric references are $E_N = 0$ and $E_{L_1} = 0$; the numerical tests substitute $E_N \leq 0.005$ and $E_{L_1} \leq 0.01$ for both timesteps.

4 Verification metrics and acceptance criteria

The integrated reference count is $N_{\text{ref}} = G_R t_{\text{res}} = 10 \times 10 = 100$ particles. The analytical spatial reference integrates the configured lognormal landing distribution over each downwind cell, truncates it at the 99th percentile, and renormalizes the cell probabilities to N_{ref}. The ignition row and physical columns are read from each simulation specification; all halo cells and nodata values are excluded.

4.1 Justification of metric quantification

Both limits have a direct particle-count interpretation. Since $N_{\text{ref}} = 100$, the integrated-error bound $E_N \leq 0.005$ permits an absolute imbalance of only 0.5 particle. This is large enough for raster roundoff and percentile-edge binning, but too small to hide a one-particle source-duration error. The normalized L_1 bound of 0.01 permits at most one particle's worth of total spatial redistribution after the analytical lognormal probabilities have been integrated over the same cells. Unlike a signed error, this norm cannot conceal compensating upstream and downstream errors. Applying both conditions to the short- and long-timestep simulation conditions separates conservation of the total from correct overlap-limited placement. The exact simulation condition count is required because the intended assertion concerns both sides of the residence-time boundary.

4.2 Predeclared acceptance criteria

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Output completeness	One final accumulated firebrand deposition raster with current simulation specification geometry for each timestep.	2 of 2 simulation conditions usable
Integrated-count error	$E_N = \lvert \sum_{ij} N_{ij} - N_{\text{ref}} \rvert / N_{\text{ref}}$, using the full valid output raster.	$E_N \leq 0.005$ for each timestep
Profile normalized L_1 error	$E_{L_1} = \sum_k \lvert N_k - N_{k,\text{ref}} \rvert / N_{\text{ref}}$ on matched physical downwind cells.	$E_{L_1} \leq 0.01$ for each timestep
Overall decision	Conjunction of output completeness and both errors for both timestep simulation conditions.	PASS only if every component passes

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for a timestep of 0.1875 s. These panels show the prepared spatial fields, remove the two-cell numerical buffer, and retain map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

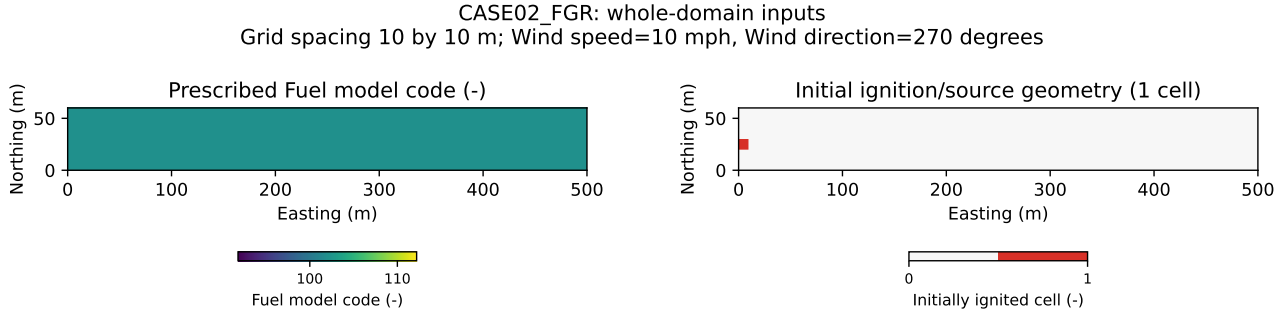


Figure 1: Whole-domain prepared input configuration for the representative simulation condition. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

The preprocessor creates two simulation conditions with $\Delta t = 0.1875$ s and $\Delta t = 12.0$ s. The fine step is exactly representable in binary and leaves a partial final emission step; the coarse step spans the full source lifetime. Both timesteps divide the 120-s stop exactly (640 and 10 steps), so the final dump occurs at 120 s. The verified 10-s source duration is unchanged. Both use a 10 m grid and the same 500 m by 60 m physical domain. This gives 50 by 6 usable cells and a 54 by 10 raster after two numerical buffer cells are added outside all four sides. A single ignition is placed in the first usable column and the middle usable row.

The source is isolated with disabled vegetative surface spread, so it does not create additional burning cells. The current area-proportional generation model calculates

$$\text{GR}_{\text{pixel}} = \text{EMBER_GR}\,\Delta x \Delta y.$$

Using an areal production rate of 0.1 pcs/s/m^2 and a 100 m^2 cell gives the required 10 pcs/s source. A fixed emission duration of 10 s is prescribed rather than calculated from flame residence. Firebrand ig-

nition, consumption, and crosswind dispersion are disabled so that they cannot change the accumulated count.

Each simulation condition must satisfy both conditions:

- total-count relative error no greater than 0.5%;
- profile L_1 relative error no greater than 1%.

The overall case passes only if outputs exist and both timestep simulation conditions pass. The two-cell buffers are excluded from profile extraction, while the total count is checked over the complete raster to detect misplaced firebrands.

6 Actual simulation results

Figure 2 shows the representative accumulated firebrand density from a timestep of 0.1875 s. The displayed field is taken from the simulated results, masks missing values, and excludes the two-cell numerical buffer. Firebrand counts are normalized by the cell length and the represented 1-m cross-stream strip, giving units of pcs m^{-2}.

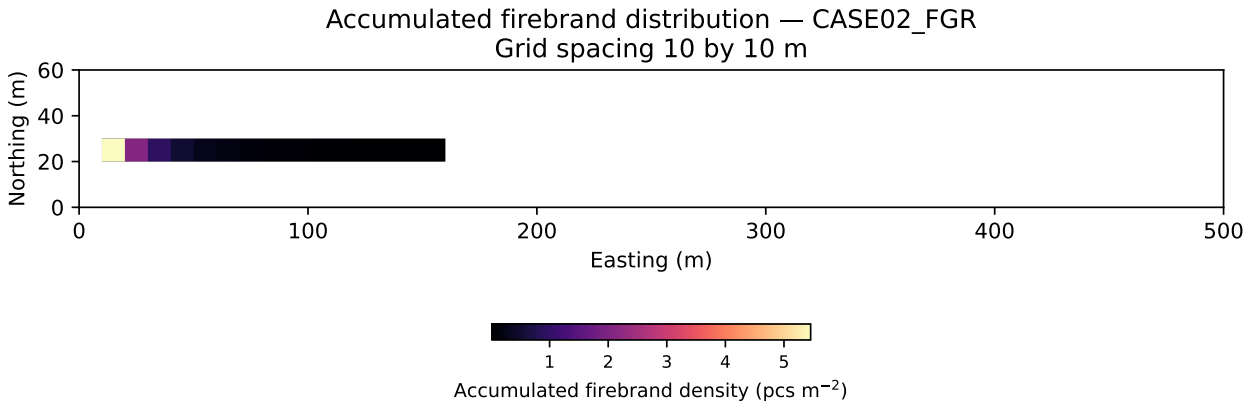


Figure 2: Whole-domain accumulated firebrand density from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

Overall status: **PASS**. Figure 3 first compares each actual ELMFIRE profile with the cell-integrated analytical distribution. Agreement for the 12.0 s simulation condition is the critical result because that timestep exceeds the complete 10 s source duration.

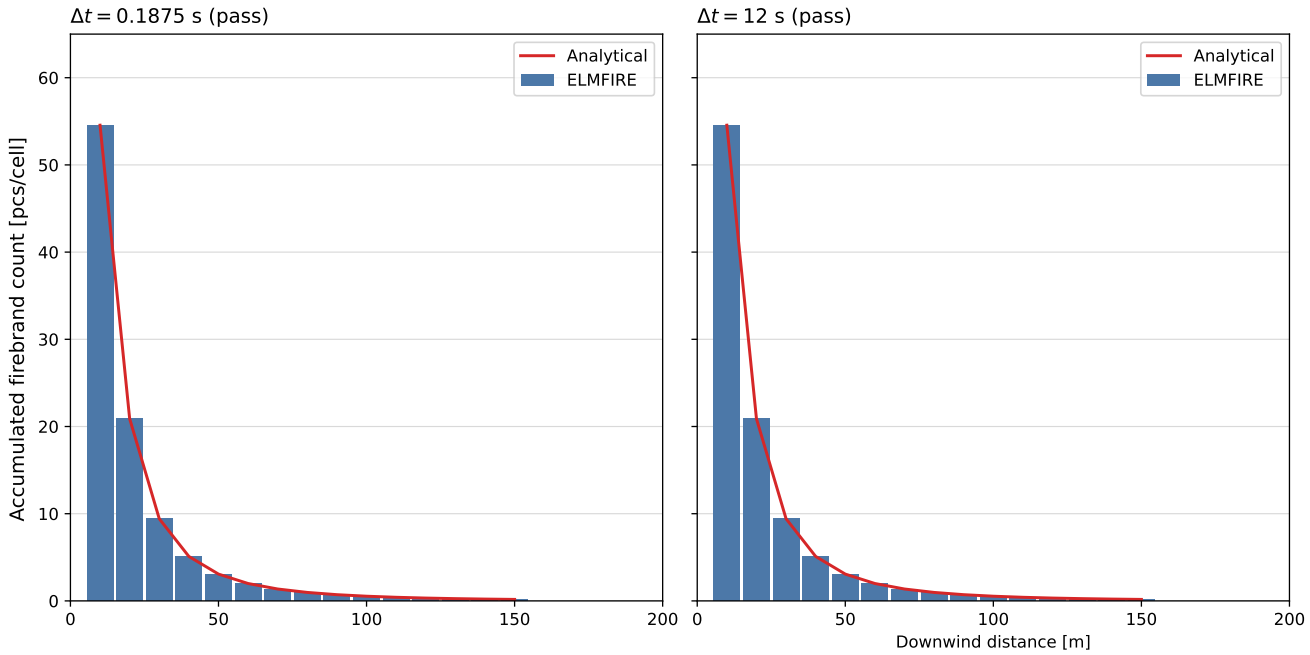


Figure 3: Accumulated firebrands per downwind cell for the 0.1875 and 12.0 s timestep simulation conditions. Blue bars are ELMFIRE output and the red curve is the cell-integrated, truncated-lognormal reference for 100 generated firebrands.

7 Calculated metrics and verification decision

Timestep	Observed total	Total relative error	Profile L_1 error
0.1875 s	100	1.669e-07	6.496e-07
12.0 s	100	2.399e-08	3.449e-07

Table 2: Firebrand-count and spatial-distribution errors for the two timestep simulation conditions.

The overall decision is **PASS** and is taken directly from postprocessing.

8 Technical interpretation

Agreement of both totals and profiles across a timestep longer than the complete source duration demonstrates correct partial-step accounting; a discrepancy isolates temporal source integration or deposition indexing.

9 Scope and limitations

This case isolates emission from one prescribed source over a finite duration. The spatial dimensions, numerical buffer, source location, production rate, and duration stated above define the reproducible experiment. Agreement is assessed from both the emitted total and the spatial deposition distribution; it does not validate the source parameters against physical measurements.

References

[1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

Case 03: Eulerian Firebrand Transport Convergence

1 Verification purpose

This case verifies convergence of Eulerian firebrand transport in a uniform one-dimensional wind. Surface propagation is suppressed and deposited firebrands ignite immediately, so the leading edge isolates generation, transport, deposition, and level-set ignition updates.

The test exposes grid- or Courant–Friedrichs–Lewy (CFL)-dependent advection speed, lost transient output, and inconsistent front extraction. Success requires the measured leading-edge rate to approach the prescribed wind speed over the accepted grid and CFL range.

2 Mathematical formulation and numerical implementation

This sweep reconstructs the wind-Courant-number convergence experiment [1]. Area-proportional firebrand production, Eulerian transport and accumulation, and immediate ignition couple the arriving firebrands to the fire arrival time and level-set field. Because surface spread is suppressed, sustained propagation requires all these processes. The sweep changes only wind displacement per step relative to cell size, with fixed physical domain and areal production. Errors in transport, timestep scaling, immediate ignition, or arrival-time updates can therefore appear as grid- or Courant-number-dependent leading-edge speed. The wind-speed reference is an idealized numerical expectation, not a claim that real firebrands travel deterministically at the ambient wind speed.

3 Derivation of expected behavior

With surface spread disabled and direct ignition enabled, the transported ember field is the sole propagation mechanism. Uniform advection therefore gives the analytical leading-edge trajectory $x_{LE}(t) = u_{wind}t$ and the normalized propagation rate $ROS/u_{wind} = 1$. Grid and wind-CFL refinement should reduce numerical transport error without changing this limiting value.

The generated timestep is not an independent free parameter:

$$\Delta t = CFL_w \frac{\Delta x}{6.71}.$$

For the five required CFL values $\{0.2, 0.4, 0.5, 0.6, 0.8\}$, the generated timesteps are

Δx (m)$\backslash CFL_w$	0.2	0.4	0.5	0.6	0.8
5	0.149031	0.298063	0.372578	0.447094	0.596125
10	0.298063	0.596125	0.745156	0.894188	1.192250
20	0.596125	1.192250	1.490313	1.788376	2.384501
30	0.894188	1.788376	2.235469	2.682563	3.576751

s.

These $4 \times 5 = 20$ members form the acceptance set. Uniform advection over the 300–1500 m comparison interval predicts

$$\Delta t_{\text{ref}} = \frac{1500 - 300}{6.71} = 178.8376 \text{ s}, \qquad \overline{ROS}_{\text{ref}} = \frac{1200}{178.8376} = 6.71 \text{ m s}^{-1}.$$

At the 260-s stop the characteristic displacement is $6.71(260) = 1744.6$ m, plus the generated half-cell initial coordinate where absolute position is needed. Thus every required simulation condition has the exact normalized reference $\overline{ROS}/U = 1$ and $E_U = |1 - 1| = 0$; the implemented allowance is $E_U \leq 0.10$.

4 Verification metrics and acceptance criteria

ELMFIRE writes the level-set field ϕ after every solver time step. For every dump, the postprocessor reads the physical centerline, locates all sign changes, and linearly interpolates between adjacent cell centers to find the subcell locations where $\phi = 0$. The most downwind zero crossing is the leading edge $x_{LE}(t)$. Consecutive dumps give the stepwise rate

$$ROS_i = \frac{x_{LE}(t_{i+1}) - x_{LE}(t_i)}{t_{i+1} - t_i}. \tag{1}$$

The reported average is the time-weighted mean of these stepwise rates while both front observations lie between 300 m and 1500 m. Thus every solver step in the comparison interval contributes, and the result equals the total leading-edge displacement divided by elapsed time. This interval excludes startup behavior, the maximum downwind flight scale near the source, and the downstream boundary. Raster shape, cell size, buffer exclusion, dump-to-time matching, front-observation count, rate-interval count, and a diagnostic least-squares coefficient of determination are recorded for every result. A CSV trajectory preserves each dump time, selected ϕ raster, leading-edge location, and stepwise rate of spread (ROS).

Every simulation condition with $CFL_{wind} \leq 0.8$ passes when

$$|ROS/u_{wind} - 1| \leq 0.10. \tag{2}$$

The higher-CFL points are retained to expose the onset of temporal transport error but do not determine overall pass/fail status.

4.1 Justification of metric quantification

The required range $CFL_{wind} \leq 0.8$ stays below the unit-CFL transport limit and leaves a 20% margin for the coupled source, transport, and direct- ignition updates. Thus the acceptance set probes the intended stable regime; larger CFL values diagnose degradation without being used to redefine that regime after results are known. The 10% speed band is deliberately wider than the largest single-cell position uncertainty divided by the 1200 m comparison length, but much narrower than the order-one error produced by a stalled front or by advancing trackers for only a fraction of a solver step. Time-weighting the rates over 300 m–1500 m further prevents an isolated grid crossing from controlling the decision. Requiring all 20 stable-regime simulation conditions guards against compensating resolution and timestep errors; incomplete trajectories cannot establish convergence and are therefore non-passing.

4.2 Predeclared acceptance criteria

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Required-output completeness	Transient level-set field rasters, dump times, and at least one valid front-rate interval between 300 and 1500 m.	All 20 simulation conditions with $CFL_{wind} \leq 0.8$
Normalized mean-ROS error	$E_U = \lvert\overline{ROS}/u_{wind} - 1\rvert$, where $\overline{ROS}$ is displacement divided by elapsed time over the comparison interval.	$E_U \leq 0.10$ for every required simulation condition
Overall decision	All required simulation conditions must be complete and satisfy the normalized-speed limit; higher-CFL simulation conditions are diagnostic only.	PASS iff all required components pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for a 10 m grid and wind Courant number 0.2. These panels show the prepared spatial fields, remove the two-cell numerical buffer, and retain map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

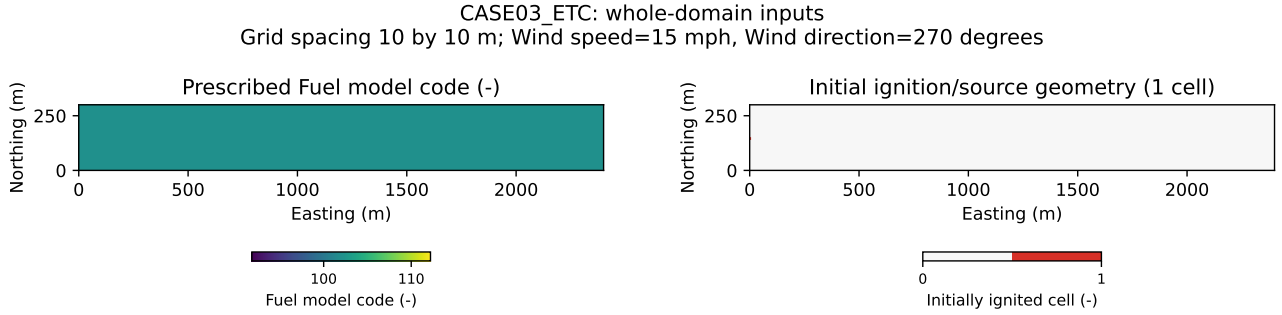


Figure 1: Whole-domain prepared input configuration for the representative simulation condition. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

The active physical domain is 2400 m long by 300 m wide. Two additional numerical cells are placed outside every physical boundary for each resolution. The initial burning cell is the first usable downwind cell and the center row; it is therefore never in the buffer. Fuel model 102, flat terrain, a uniform 15 mph wind from west to east, and deterministic input rasters are used. Crosswind dispersion and ember consumption are disabled.

The current spotting interface uses the following explicit model selections:

- generation: area-proportional;
- spotting distance: empirical;
- accumulation: Eulerian;
- ignition: immediate.

The customized downwind distribution has $\mu = 2.18$, $\sigma = 1.23$, and $P_\varepsilon = 0.01$. Every ignited cell emits for 10 s. The configured areal generation rate is resolution-independent:

$$GR_A = 10\,[\text{pcs m}^{-2}\,\text{s}^{-1}]. \tag{3}$$

ELMFIRE therefore emits $N = GR_A\,\Delta x^2\,\Delta t$ firebrands from one burning square cell during one solver step. This is the appropriate source definition because surface-fire spread and its fireline-intensity-dependent power are disabled.

5.1 Resolution sweep and predicted trend

Four grid spacings, $\Delta x = 5$, 10, 20, and 30 m, are crossed with wind-based CFL values 0.2, 0.4, 0.5, 0.6, 0.8, 1.0, and 1.2, for 28 simulation conditions. Each time step is prescribed by

$$\Delta t = CFL_{wind}\frac{\Delta x}{u_{wind}}. \tag{4}$$

Because timestep variation is the verification variable, these fractional values are not rounded. For each member preprocessing uses $N = \lceil 260/\Delta t \rceil$ and writes $t_{stop} = N\Delta t$, recording the requested 260 s, aligned stop, and integer step count in the simulation specification. All simulation conditions retain the same physical domain; only the number of active and buffer cells changes.

6 Actual simulation results

Figure 2 shows the representative transported level-set field from a 10 m grid and wind Courant number 0.2. The displayed field is taken from the simulated results, masks missing values, and excludes the two-cell numerical buffer. The zero contour identifies the transported front, while the signed field exposes domain-wide numerical structure away from the extracted centerline.

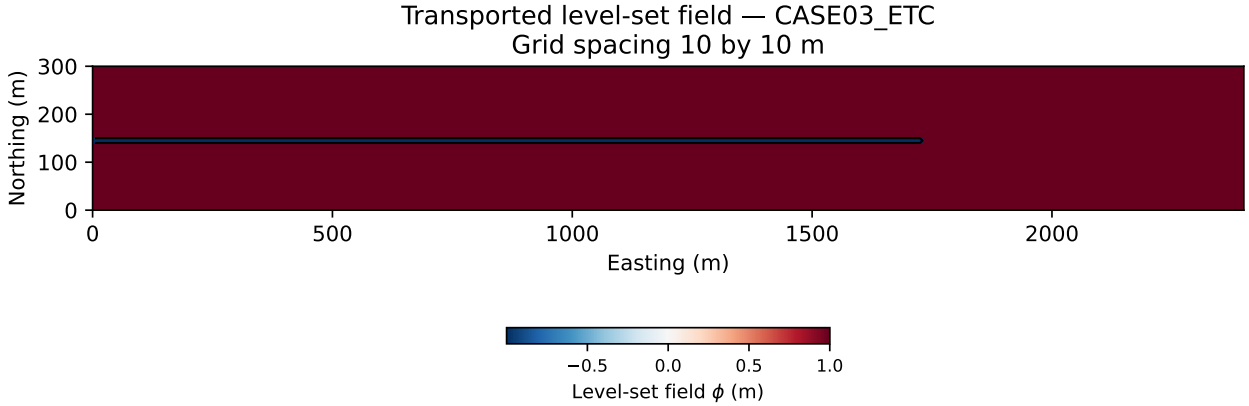


Figure 2: Whole-domain transported level-set field from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

Figure 3 first presents the complete wind-CFL and grid-resolution sweep in the reference coordinate system. Each point is the normalized speed calculated from a complete transient level-set history.

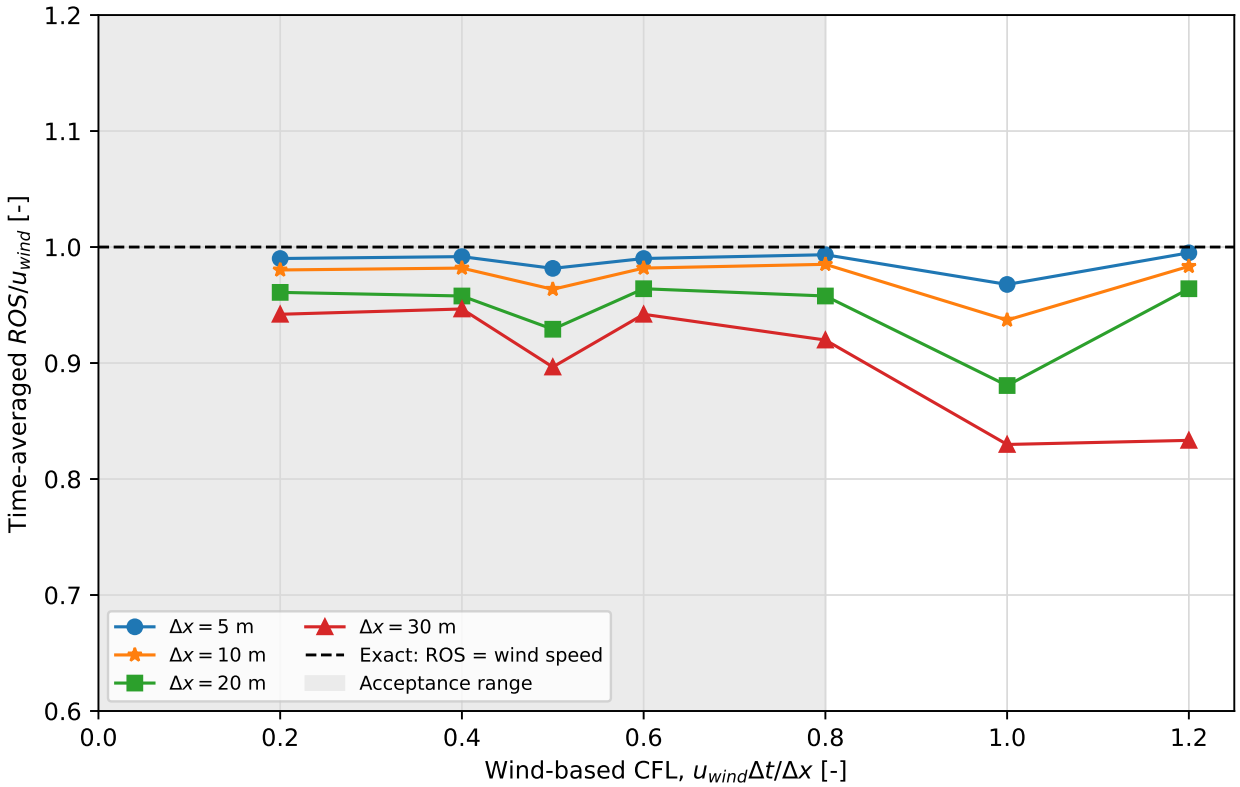


Figure 3: Time-averaged Eulerian firebrand-driven spread speed normalized by the uniform wind speed as a function of wind-based CFL number. The shaded region marks the CFL range subject to the 10% acceptance criterion.

Figure 4 is the supplemental diagnostic that traces one representative point from the level-set contour through the reported mean. It expands one member of the sweep, `dx10_cfl0p5`, to show the complete path from the level-set field to the scalar convergence metric. Panel (a) shows one actual centerline profile, zoomed around the extracted front. In this direct-ignition, zero-surface-spread test, ELMFIRE's level set is nearly binary away from the interface: burned and unburned cell centers remain close to -1 and $+1$. This is expected and is not a corrupted field. Adjacent cell-center values ϕ_j and ϕ_{j+1} with opposite signs bracket the front. Linear interpolation gives

$$x_{LE} = x_j - \phi_j \frac{x_{j+1} - x_j}{\phi_{j+1} - \phi_j}. \tag{5}$$

All centerline sign changes are evaluated, and the most downwind crossing is selected. This matters after spotting because a line can contain several separate burned regions and therefore several fire fronts.

Panel (b) assembles those independently extracted locations into $x_{LE}(t)$. The gray interval is the prescribed 300 m to 1500 m comparison window expressed in time. The dashed least-squares line is a trajectory-linearity diagnostic; its slope is 6.4957 m s^{-1}, but it is not the reported acceptance metric.

Panels (c) and (d) show how that metric is calculated. Panel (c) displays a 20-second detail from the start of the averaging interval so the alternating peak and trough values are individually visible. Each pair of consecutive dumps within the comparison window supplies $ROS_i = \Delta x_{LE,i}/\Delta t_i$. Because the final solver step can have a different duration, the average is explicitly time weighted:

$$\overline{ROS} = \frac{\sum_i ROS_i \Delta t_i}{\sum_i \Delta t_i} = \frac{x_{LE}(t_N) - x_{LE}(t_0)}{t_N - t_0}. \tag{6}$$

The representative result is $\overline{ROS} = 6.4655\,\mathrm{m\,s^{-1}}$, or $\overline{ROS}/u_{wind} = 0.96356$. Thus the oscillatory instantaneous values caused by cell-by-cell front motion are not mistaken for the effective transport speed.

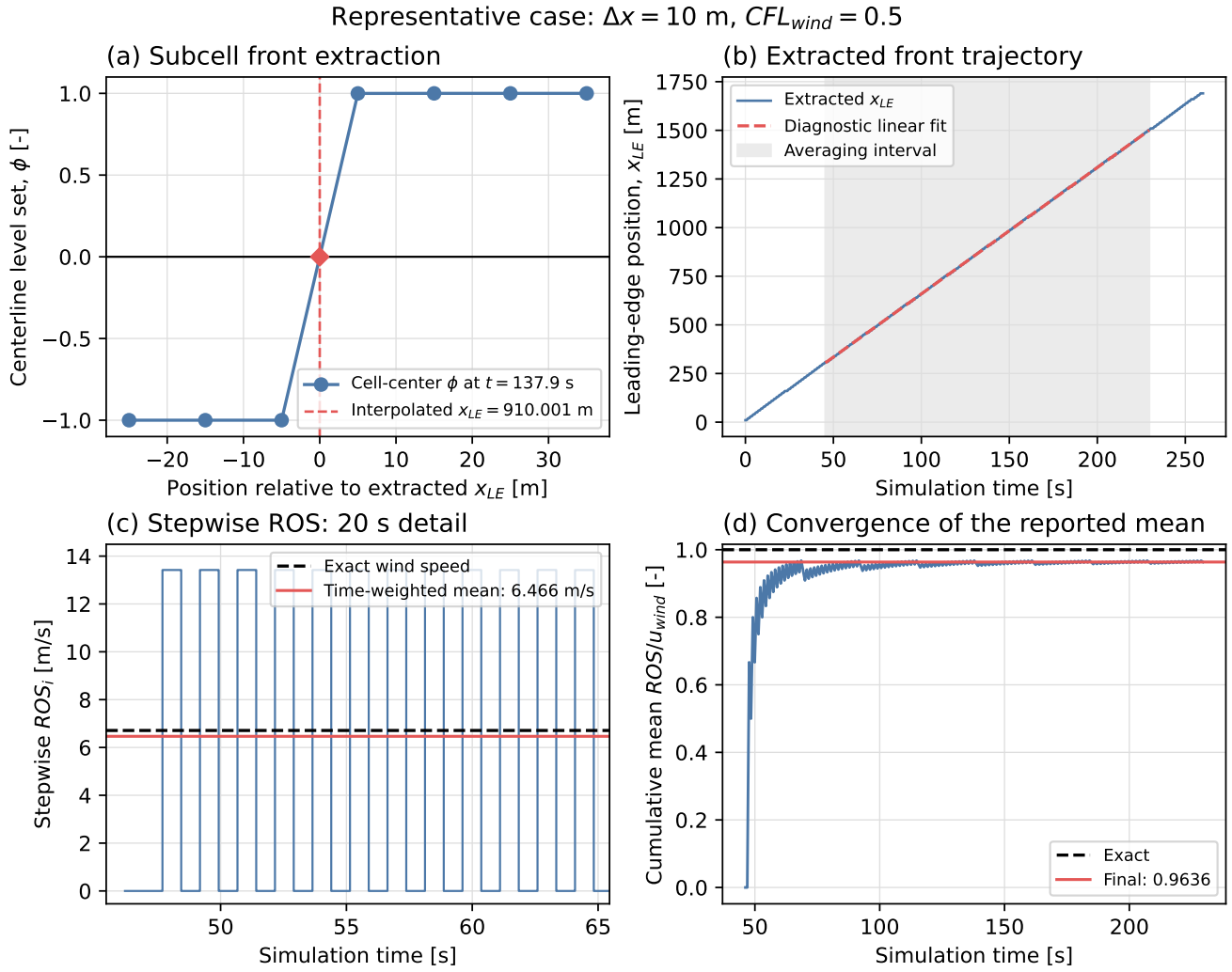


Figure 4: Detailed extraction for the representative $\Delta x = 10\,\mathrm{m}$, $CFL_{wind} = 0.5$ simulation. (a) Linear interpolation of centerline $\phi = 0$ crossings; (b) resulting leading-edge trajectory and diagnostic fit; (c) a 20-second zoom of consecutive-dump finite-difference ROS and its time-weighted mean; (d) convergence of the cumulative normalized mean to its reported value.

7 Calculated metrics and verification decision

8 Technical interpretation

The expanded sweep fails its aggregate criterion because the $\Delta x = 30\,\mathrm{m}$, $CFL_{wind} = 0.5$ member gives

$$ROS/u_{wind} = 0.89655$$

corresponding to a relative error of 0.10345. This is slightly larger than the allowed 0.10. The $\Delta x = 10\,\mathrm{m}$ representative case remains within tolerance and is used above to explain the extraction procedure.

Quantity	Value
Overall status	FAIL
Prepared simulation conditions	28
Simulation conditions with simulation results	28
Required simulation conditions	20
Required simulation conditions evaluated	20
Reference wind speed [m/s]	6.71
Maximum relative error	0.10
Acceptance CFL limit	0.8

Table 2: Verification status after processing all transient level-set outputs.

All 28 input-matched simulation conditions completed and the generated controls match the declared wind, grid, and CFL sweep. The isolated coarse-grid excess is therefore a numerical transport/level-set discrepancy in the current source behavior, not an incomplete run or an admissible configuration correction.

9 Scope and limitations

Each prescribed condition is simulated and analyzed independently. The comparison depends on complete results for the stated grid and timestep sweep, not on whether a simulation or report preparation procedure finishes.

References

[1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

Case 04: Steady Transport and Immediate Ignition

1 Verification purpose

This case verifies the limiting mechanism shared by steady Eulerian transport and zero-delay wildland ignition. The isolated transport simulation condition disables surface spread and uses an areal source; wildland ignition without delay uses a fire-power-based source and additionally checks accumulated population.

The paired design exposes failures in generation, transport, deposition, direct ignition, and time-of-arrival (TOA) updates. Success requires both simulation conditions to recover wind-speed arrival behavior and the wildland simulation condition to recover its one-metre-strip accumulation reference.

2 Mathematical formulation and numerical implementation

For constant wind speed $U = 6.71$ m/s and zero ignition delay, an ember-deposited cell becomes an active source immediately. Translation invariance then gives

$$t_{\text{ref}}(x) = \frac{x - x_0}{U}, \qquad x_{\text{LE,ref}}(t) = x_{\text{LE},0} + U(t - t_0), \qquad \overline{ROS}_{\text{ref}} = U, \tag{1}$$

where x_0 is the first usable cell centre. The final arrival-time raster stores the complete cell-crossing history used for the time of arrival (TOA) profile metric. Leading-edge history is evaluated independently from every dumped instantaneous level-set field. At the physical centre row, the postprocessor linearly interpolates every adjacent pair of cell-centred ϕ values that changes sign and retains the rightmost $\phi = 0$ crossing. Here $x_{\text{LE},0}$ is the first extracted crossing at dump time t_0. Once the zero contour has left the physical domain, that dump has no observable in-domain leading edge and is excluded from the trajectory fit rather than being replaced by a stationary domain-edge value. Mean rate of spread (ROS) is the least-squares slope through all observable dump samples.

For immediate-ignition simulation condition, a burning cell emits during its surface-fire residence time. Using the FLIN and spread rate written by the same executed cell gives

$$t_{\text{emission}} = \frac{\Delta x}{VS}, \qquad N_{\text{ref}} = GR_{1m}\,\frac{FLIN}{1000}\,\frac{\Delta x}{VS}. \tag{2}$$

Here FLIN is in kW m^{-1} and VS is converted from ft min^{-1} to m s^{-1}. The documented nominal FLIN and VS give 1497.2 particles for this illustrative state [1]; the metric uses run-derived values so it remains consistent with the active fuel calculation. ELMFIRE forms pixel fire power from fireline intensity times cross-stream width Δy. The configured coefficient $GR_{\text{ELM}} = 33.3/\Delta y$ pcs/s/MW therefore makes the raw pixel count represent a one-metre strip. For the one-dimensional unit-width profile, each centre-row pixel count is converted to the density of the represented one-metre strip,

$$n_i = \frac{N_i}{\Delta x(1\text{ m})} \quad [\text{pcs m}^{-2}]. \tag{3}$$

The two-cell numerical halo is excluded. The complete physical raster is still inspected, and its all-row deposited total is retained as a conservation diagnostic, but it is not mixed into the one-dimensional plotted observable. For the PER-AREA simulation condition, the configured source gives $n_\infty = GR\Delta x\tau_{\text{ember}}/(1\text{ m})$. For the PER-MW simulation condition, Equation 2 divided by $\Delta x(1\text{ m})$ gives the run-derived density reference. A zero or nodata final burned-cell FLIN or VS value does not define an independent residence-time reference. The evaluator retains only finite cells with $FLIN > 0$ and

$VS > 0$, requires at least two such cells in the declared plateau, and otherwise reports the accumulation component as **NOT EVALUABLE** rather than forming a relative error against zero.

The discrete algorithm is: generate embers in every active source cell; distribute them among downwind bins; accumulate them in landing cells; apply DIRECT ignition with $P_{ign} = 1$ and zero local/cell delay; update ember-ignition, level-set, and TOA states together; then advance the level-set clock. The two simulation conditions differ only where required by their source experiment.

3 Derivation of expected behavior

A deposition distribution with finite support changes which upstream cells contribute near x_0, but not the propagation characteristic in Equation 1. Consequently TOA should be linear, and the leading edge extracted from ϕ should oscillate by at most one cell around $x_{\mathrm{LE},0} + U(t - t_0)$. Its fitted slope should approach U. Every cell marked by the ember-ignition raster must have finite nonnegative TOA.

In both simulation conditions, locations farther than the maximum travel distance receive contributions from a complete set of upstream emitters. Conservation then produces the configuration-consistent plateaus. The PER-AREA source uses $GR = 10$ pcs m^{-2} s^{-1}, $\Delta x = 10$ m, and $\tau_{\text{ember}} = 6$ s, giving 600 pcs m^{-2} under the represented-strip normalization. The PER-MW reference is calculated from co-located final FLIN and VS. The 155–195 m interval contains cells with complete upstream support.

The wind Courant–Friedrichs–Lewy (CFL) upper-bound step is

$$\Delta t_{\text{max}} = 0.5(10)/6.71 = 0.745156 \text{ s}.$$

To retain the requested 100-s stop as an exact multiple, preprocessing selects $N = \lceil 100/0.745156 \rceil = 135$ and $\Delta t = 100/135 = 0.740741$ s. The resulting wind CFL is 0.4970, below the declared 0.5 bound. With $x_0 = 5$ m, representative exact TOA substitutions in the declared plateau interval are

$$t_{\text{ref}}(155) = \frac{155 - 5}{6.71} = 22.3547 \text{ s}, \qquad t_{\text{ref}}(195) = \frac{195 - 5}{6.71} = 28.3159 \text{ s}.$$

Likewise $x_{\mathrm{LE,ref}}(50) = 5 + 6.71(50) = 340.5$ m, and the front reaches the 500-m boundary after $(500 - 5)/6.71 = 73.7705$ s. These substitutions give exact reference values of zero for TOA relative error, leading-edge NRMSE, and ROS relative error before raster quantization.

For the PER-AREA member, the complete-support density is

$$n''_{\infty} = GR\,\Delta x\,\tau_{\text{ember}}/(1 \text{ m}) = 10(10)(6) = 600 \text{ pcs m}^{-2}.$$

For the PER-MW member, the reference evaluated at each co-located FLIN/VS cell is

$$n''_{\text{ref}} = 33.3\frac{FLIN/1000}{VS(0.3048/60)} = 6.555118\frac{FLIN}{VS} \text{ pcs m}^{-2},$$

with FLIN in kW m^{-1} and VS in ft min^{-1}. The historical nominal substitution gives 1497.2 particles; current acceptance uses the generated co-located fields rather than that rounded historical value. Both plateau references therefore have expected relative error zero and are tested with a 0.10 allowance.

4 Verification metrics and acceptance criteria

For valid centre-row cells, define $e_i = |t_i - t_{\text{ref,i}}|/t_{\text{ref,i}}$. Each simulation condition requires at least two cells, $\bar{e} \le 0.10$, and $\max e_i \le 0.25$. Leading-edge error is the RMSE relative to $x_{\mathrm{LE},0} + U(t - t_0)$ divided by $U(t_{\text{last}} - t_0)$ and must not exceed 0.10; t_{last} is the last dump with an in-domain $\phi = 0$ crossing. Every scheduled time must have exactly one matching ϕ raster. Fitted mean-ROS relative error must

not exceed 0.10. All ember-ignited physical cells must have finite, nonnegative TOA. Each simulation condition additionally requires $|\widetilde{n} - n_{\text{ref}}|/n_{\text{ref}} \leq 0.10$, where $\widetilde{n}$ is the median centre-row density from Equation 3 over the fully developed 155–195 m plateau. The corresponding analytical or run-derived reference is calculated independently of the ember-flux values. For the PER-MW simulation condition, the same mask is applied to the observed density and the co-located FLIN/VS reference, with at least two finite, strictly positive reference cells required. Insufficient support makes the metric NOT EVALUABLE; a zero denominator is never treated as a failed relative-error comparison. Domain-wide deposited totals are also recorded as diagnostics. Exactly two prescribed simulation conditions and one final raster of every required kind are mandatory. The overall decision is the conjunction of completeness and all component decisions; missing or stale products are non-passing.

4.1 Justification of metric quantification

The 10% limits on mean TOA, leading-edge NRMSE, fitted ROS, and plateau density are engineering verification bands for a first-order, cell-centred transport/ignition calculation. They are broad enough to accommodate grid-crossing time quantization and the use of a localized run-derived FLIN/VS reference, yet they remain small compared with the order-one signature of a nonpropagating or solver-step-limited implementation. The 25% pointwise TOA limit is intentionally looser than the mean limit because a single cell next to the source or front can carry a disproportionately large relative timing error; requiring at least two cells and the 10% mean prevents that allowance from masking a systematic bias. The plateau median is used because the expectation is spatially constant and the median is insensitive to one raster edge cell. Ignition-state consistency has zero physical tolerance: a cell cannot be classified as ember ignited without a finite, nonnegative arrival time. Both simulation conditions and all components are conjunctive because transport, ignition, and accumulation are separate claims of this combined test.

4.2 Predeclared acceptance criteria

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Output completeness	Two prescribed simulation conditions with one final TOA, ignition-state, and accumulated-firebrand product of each required kind, plus one ϕ raster for every scheduled dump time.	2 of 2 simulation conditions complete; no missing ϕ dumps
TOA profile error	Mean and maximum of $\lvert t_i - t_{ref,i}\rvert / t_{ref,i}$ on valid centre-row physical cells.	Mean $\leq$ 0.10; maximum $\leq$ 0.25
Leading-edge error	RMSE of the rightmost interpolated $\phi = 0$ crossing relative to $x_{LE,0} + U(t - t_0)$, normalized by the observable reference displacement.	NRMSE $\leq$ 0.10
Mean-ROS error	Relative error of fitted leading-edge ROS against $U = 6.71\,\mathrm{m\,s^{-1}}$.	$\leq$ 0.10
Ignition-state consistency	Every ember-ignited physical cell has a finite, nonnegative TOA.	No inconsistent cells
Accumulation reference support	Finite positive co-located FLIN/VS cells over 155–195 m; zeros and nodata cannot define the PER-MW residence-time reference.	At least two cells; otherwise NOT EVALUABLE
Accumulation plateau error	Relative error of median one-metre-strip density over 155–195 m against the independent reference, using the same valid-reference mask for the PER-MW simulation condition.	$\leq$ 0.10 for both simulation conditions
Overall decision	Conjunction of completeness and every component for both simulation conditions.	PASS only if all pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for isolated transport. These panels show the prepared spatial fields, remove the two-cell numerical buffer, and retain map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

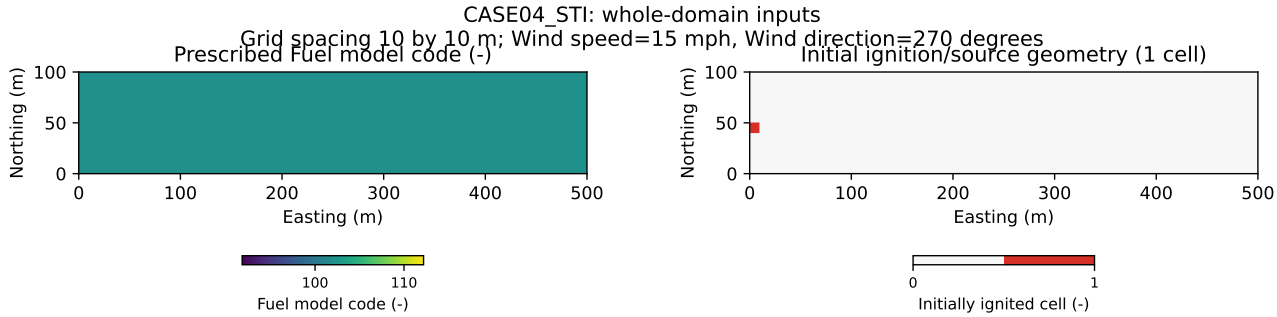


Figure 1: Whole-domain prepared input configuration for the representative simulation condition. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

Parameter	Configured value	Units	Verification role
Domain	500×100	m	common physical extent
Grid and halo	10; two cells/side	m	prescribed resolution; excluded halo
Time step	$100/135 = 0.740741$	s	wind CFL $0.4970 < 0.5$
Wind	6.71, from 270°	m/s	constant transport velocity
Fuel/terrain	fire behavior fuel model (FBFM) 102; flat	–	uniform wildland control
Transport simulation condition	PER-AREA; surface off; 100 s	–	isolates transport-impulse simulation condition transport
Wildland simulation condition	PER-MW; surface on; 100 s	–	immediate-ignition simulation condition residence source
Transport/ignition	EMPIRICAL; EULERIAN; DIRECT	–	current spotting path
Ignition response	$P_{ign} = 1$; delays zero	–	immediate limiting regime
Crosswind/consumption	disabled/disabled	–	one-dimensional conservation

Table 2: Configuration shared by and distinguishing the two combined simulation conditions.

Before inspecting output, both TOA curves should overlay $(x - x_0)/U$. Both dumped-ϕ leading-edge histories should follow $x_{\mathrm{LE},0} + U(t - t_0)$ with grid-scale steps, and their mean ROS should be near 6.71 m/s. The two density bar profiles should rise near the inlet and approach their horizontal references: 600 pcs m^{-2} for the PER-AREA transport simulation condition and the FLIN/VS-derived value for the wildland simulation condition.

6 Actual simulation results

Figure 2 shows the representative accumulated firebrand density from isolated transport. The displayed field is taken from the simulated results, masks missing values, and excludes the two-cell numerical buffer. Firebrand counts are normalized by the cell length and the represented 1-m cross-stream strip, giving units of pcs m^{-2}.

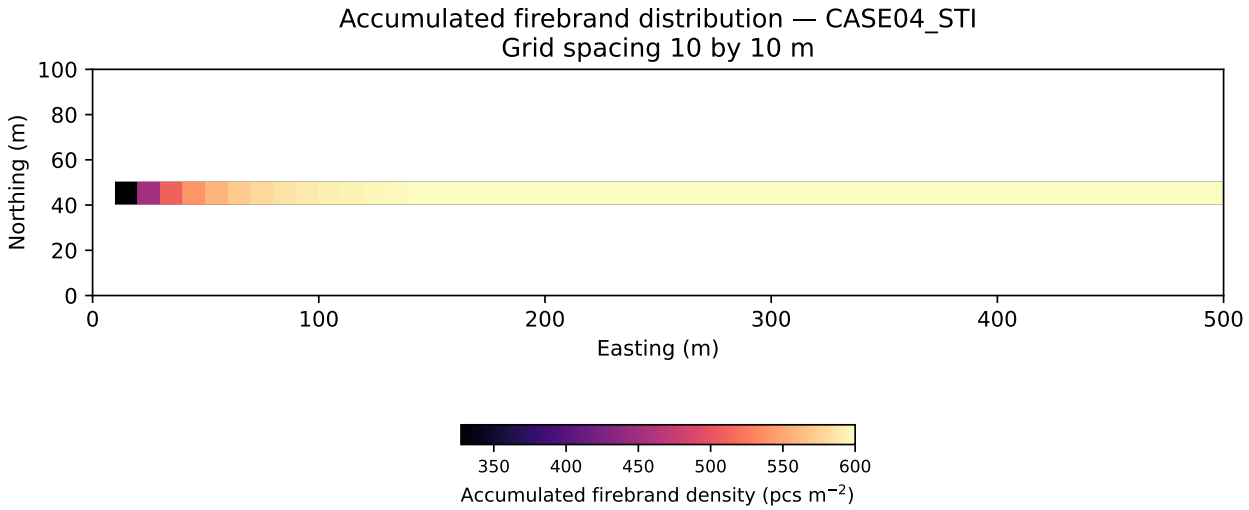


Figure 2: Whole-domain accumulated firebrand density from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

The runner regenerated both input stacks and postprocessed only the current ELMFIRE products recorded in prescribed simulation conditions. Figure 3 presents the four physical observables used in the steady-transport reference experiment: accumulated density, arrival time, leading-edge position, and instantaneous leading-edge rate of spread.

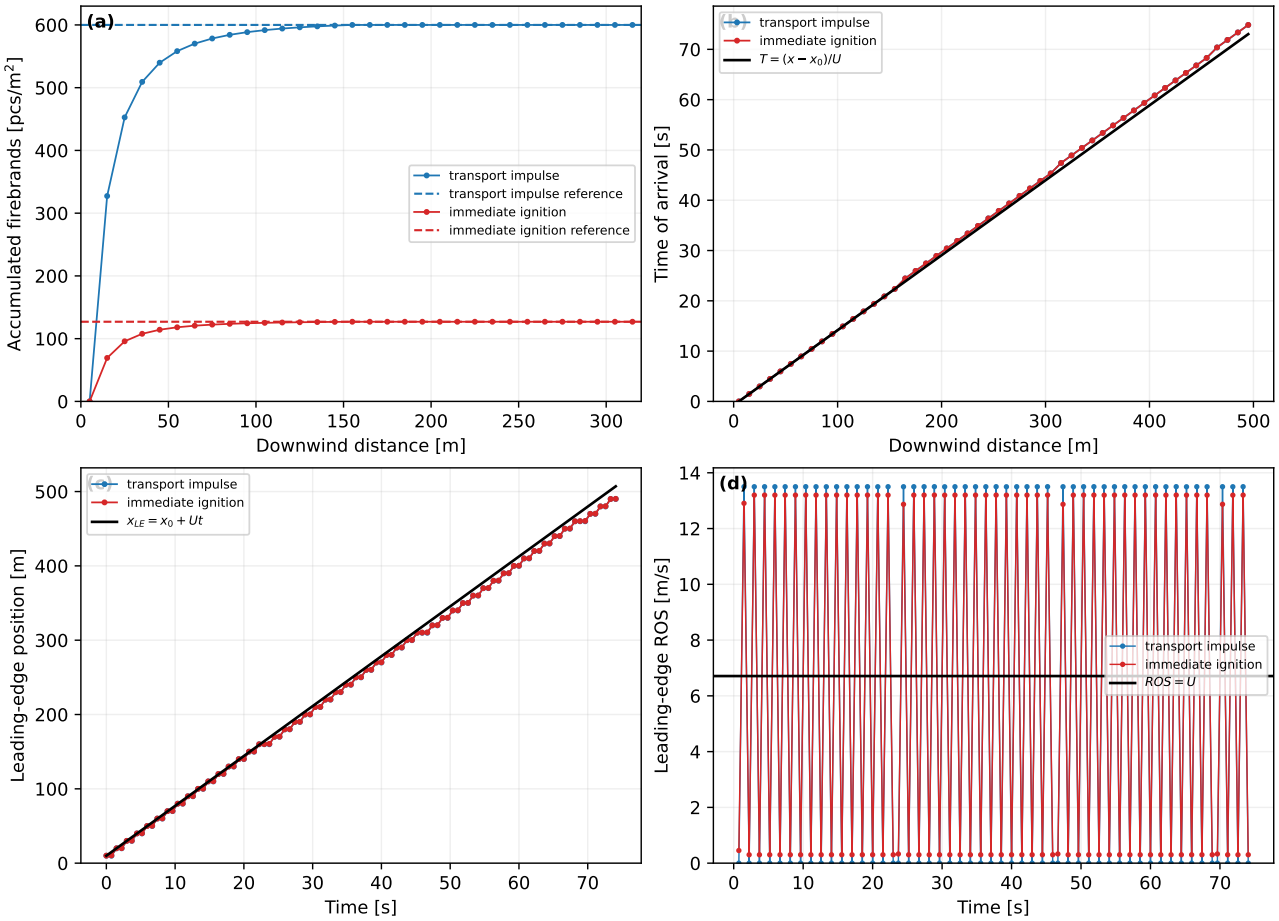


Figure 3: Steady-transport and immediate-ignition observables from actual ELMFIRE outputs. Solid colored curves identify the two controlled simulation conditions; black or color-matched reference curves show the derived analytical behavior.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
Output completeness	2 simulation conditions	2/2	PASS
Transport ϕ history	all scheduled dumps	137/137 dumps; 101 contours	PASS
Transport TOA mean error	≤ 0.10	0.01476	PASS
Transport leading-edge NRMSE	≤ 0.10	0.0165	PASS
Transport mean-ROS error	≤ 0.10	0.02537	PASS
Wildland ϕ history	all scheduled dumps	137/137 dumps; 101 contours	PASS
Wildland TOA mean error	≤ 0.10	0.01476	PASS
Wildland leading-edge NRMSE	≤ 0.10	0.01639	PASS
Wildland mean-ROS error	≤ 0.10	0.02536	PASS
Transport accumulation error	≤ 0.10	8.138e-08	PASS
Wildland accumulation error	≤ 0.10	4.323e-07	PASS
Transport deposited total	diagnostic	2.869e+05 pcs	–
Wildland deposited total	diagnostic	2.099e+05 pcs	–
Overall decision	all components pass	PASS	**PASS**

Table 3: Calculated verification metrics for the completed simulations.

8 Technical interpretation

The combined decision distinguishes a transport defect from a source-duration or normalization defect. Agreement in transport-impulse simulation condition but not immediate-ignition simulation condition implicates the residence-time PER-MW path; disagreement in both implicates shared transport, direct ignition, or TOA state handling. A full-area total of 2.099e+05 particles was deposited across 5 active rows. A plateau-only failure cannot be hidden by a plausible TOA curve, and an average ROS cannot hide local TOA or product inconsistency. The numerical values in the table and figure provide the executed source revision's decision rather than a qualitative visual judgment.

9 Scope and limitations

This is deterministic numerical verification of steady one-dimensional Eulerian transport and immediate ignition. It does not validate the empirical spotting PDF against experiments, time-varying wind, crosswind transport, stochastic ignition, delayed ignition, Lagrangian tracking, or consumption. Surface spread is present in the immediate-ignition simulation condition source calculation but the wind-speed leading edge is expected to dominate.

References

[1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

Case 05: Firebrand Transport in Time-Dependent Wind

1 Verification purpose

This case verifies Eulerian firebrand transport when wind varies in space and time. Surface spread, crosswind dispersion, consumption, and ignition delay are disabled so the leading edge follows a controlled characteristic equation.

The test exposes incorrect meteorology interpolation, direction conversion, tracker lifetime, or transport/level-set timestep coupling. Success requires leading-edge position and running-average spread rate to agree with an independently integrated characteristic solution.

2 Mathematical formulation and numerical implementation

The imposed one-dimensional velocity is

$$u(x,t) = \bar{u} + A \sin\left(\frac{2\pi x}{\lambda_x}\right) \sin\left(\frac{2\pi t}{T}\right), \quad (\bar{u}, A, \lambda_x, T) = (6.71, 3.355, 100, 60) \tag{1}$$

in SI units [1]. The exact leading edge is the characteristic satisfying

$$\frac{dx_{\mathrm{LE}}}{dt} = u(x_{\mathrm{LE}}, t), \qquad x_{\mathrm{LE}}(0) = 5 \text{ m}. \tag{2}$$

The postprocessor integrates this ODE with fourth-order Runge–Kutta and a 0.005 s substep.

The preprocessor samples the analytical wind every 1 s into 121 bands of wind speed; wind direction holds a constant 270-degree meteorological direction. ELMFIRE linearly interpolates adjacent weather bands using meteorological sampling interval. The namelist explicitly selects area-proportional generation, Eulerian accumulation, and immediate ignition. Because surface fire is disabled, the per-area source avoids any dependence on fireline intensity. Each ignited cell emits for 10 s.

3 Derivation of expected behavior

The characteristic velocity is strictly positive: $3.355 \le u \le 10.065$ m s^{-1}, so the reference leading edge is monotone. At each reporting time the simulated position is reconstructed as the furthest centre-row cell whose final ELMFIRE arrival time is no later than that time. Grid crossing makes the discrete instantaneous rate of spread (ROS) alternate between zero and approximately $\Delta x/\Delta t$. For the reference-aligned diagnostic, a centred nine-point average should recover the smooth reference $u(x_{\mathrm{LE}}(t), t)$.

The maximum prescribed speed used to select the generated solver step is $u_{\max} = 6.71 + 3.355 = 10.065$ m s^{-1}, so

$$\Delta t = 0.5(10)/10.065 = 0.496771 \text{ s}.$$

Independent RK4 integration with the stated 0.005-s substep and $x(0) = 5$ m gives

t (s)	0	30	60	90	120
x_{ref} (m)	5.000	192.328	377.023	566.967	752.389

from the generated analytical wind, not from ELMFIRE output. Although $\sin(2\pi t/60) = 0$ makes the instantaneous reference velocity 6.71 m s^{-1} at these particular reporting times, the integrated position retains the intervening space–time oscillations; for example the 120-s displacement is $752.389 - 5 = 747.389$ m, not a constant-wind substitution.

For matched sample i, the explicit references are $e_{x,i} = x_{\mathrm{ELM},i} - x_{\mathrm{ref},i} = 0$ and $e_{u,i} = \overline{ROS}_{\mathrm{ELM},i} - u(x_{\mathrm{ref},i}, t_i) = 0$. Hence the ideal metric values are mean $|e_x| = 0$, max $|e_x| = 0$, mean $|e_u| = 0$, and coverage $= 1$; the substituted allowances are 10 m, 20 m, 2 m s^{-1}, and 0.90, respectively.

4 Verification metrics and acceptance criteria

For all finite samples, the postprocessor calculates mean and maximum absolute position error, the mean absolute error of the nine-point running-average ROS, and temporal coverage. The case passes only if

$$\mathrm{MAE}_x \leq 10\ \mathrm{m}, \quad \max|e_x| \leq 20\ \mathrm{m}, \quad \mathrm{MAE}_{\overline{\mathrm{ROS}}} \leq 2\ \mathrm{m\,s^{-1}}, \quad f_{\mathrm{coverage}} \geq 0.90.$$

Missing or unusable output is explicitly non-passing.

4.1 Justification of metric quantification

The 10 m position-MAE limit equals one grid cell, and the 20 m maximum limit equals two cells. These bounds recognize that a continuous characteristic is reconstructed from cell-centred arrival events while still rejecting persistent phase errors. Differentiation magnifies that position quantization, so ROS is assessed after a centred nine-point average; its 2 m s^{-1} allowance is smaller than the configured 3.355 m s^{-1} velocity amplitude. A run that merely reproduces the mean wind but misses the oscillatory response therefore cannot satisfy the ROS criterion. The 90% coverage requirement allows a small number of unmatched samples near raster or temporal boundaries but prevents selective omission of a substantial high-error portion of either wind cycle. The four conditions measure bias, worst local error, derivative fidelity, and sampling support, respectively, so all are required for acceptance.

4.2 Predeclared acceptance criteria

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Output completeness	Final time of arrival (TOA) raster with valid physical centre-row coverage; two-cell halo and nodata are excluded.	Usable output required
Position MAE	Mean $\lvert x_{LE,ELMFIRE}(t) - x_{ODE}(t)\rvert$ over finite matched samples.	≤ 10 m
Maximum position error	Maximum absolute leading-edge difference over the same samples.	≤ 20 m
Running-ROS MAE	Mean absolute error of the centred nine-point ELMFIRE ROS against the characteristic solution.	≤ 2 m s^{-1}
Temporal coverage	Fraction of required characteristic times having finite ELMFIRE leading-edge observations.	≥ 0.90
Overall decision	Conjunction of completeness and all four quantitative components.	PASS only if all pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for the base time-dependent-wind run. These panels show the prepared spatial fields, remove the two-cell numerical buffer, and retain map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

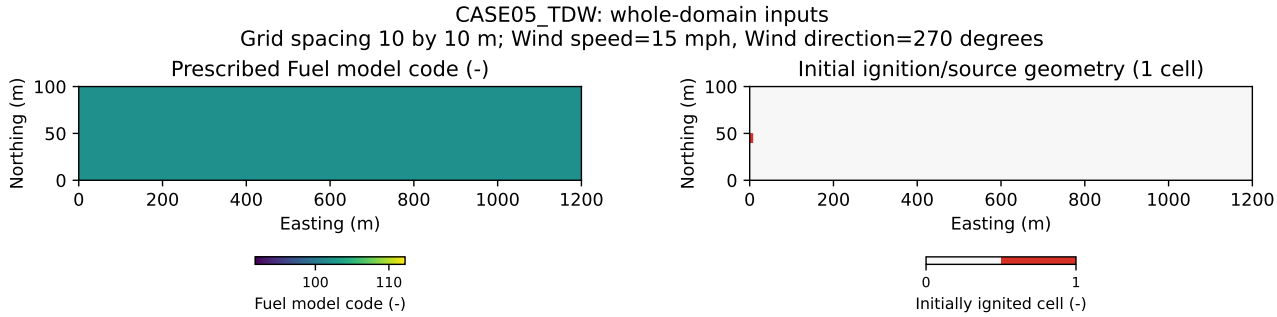


Figure 1: Whole-domain prepared input configuration for the representative simulation condition. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

The Courant–Friedrichs–Lewy (CFL) number measures the distance advected during one timestep relative to the grid spacing and constrains the time discretization.

Table 2: Configuration of the reconstructed time-dependent-wind case.

Parameter	Value	Role
Physical domain	1200 m by 100 m	contains the 120-s trajectory
Cell size	10 m	streamwise discretization
Numerical halo	2 cells per side	excluded from observations
Duration	120 s	two wind periods
Wind bands	121 at 1 s	time-varying forcing
Wind CFL	0.5	resolves wind transport
Generation	PER-AREA, 1 pcs m^{-2} s^{-1}	no-surface source
Emission duration	10 s	fixed finite-duration source
Source selection	100% deterministic	removes sampling variability
Accumulation/ignition	EULERIAN/DIRECT	tested transport and ignition path

The predicted position is a smooth monotone curve. Instantaneous numerical ROS is grid-quantized, while its nine-point mean should follow the oscillatory ODE velocity.

6 Actual simulation results

Figure 2 shows the representative time-of-arrival field from the base time-dependent-wind run. The displayed field is taken from the simulated results, masks missing values, and excludes the two-cell numerical buffer. The field provides the spatial counterpart to the leading-edge history and reveals whether the changing wind produces a coherent transported front across the complete domain.

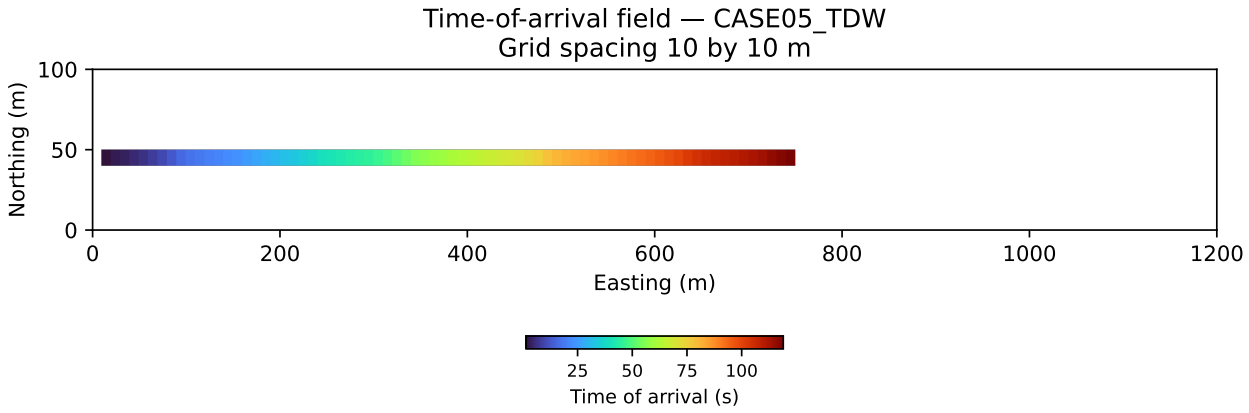


Figure 2: Whole-domain time-of-arrival field from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

Figure 3 first reproduces the two principal time-dependent-wind observables. The numerical curves are reconstructed from the final ELMFIRE arrival-time raster; the smooth curves are obtained by integrating the prescribed leading-edge ODE.

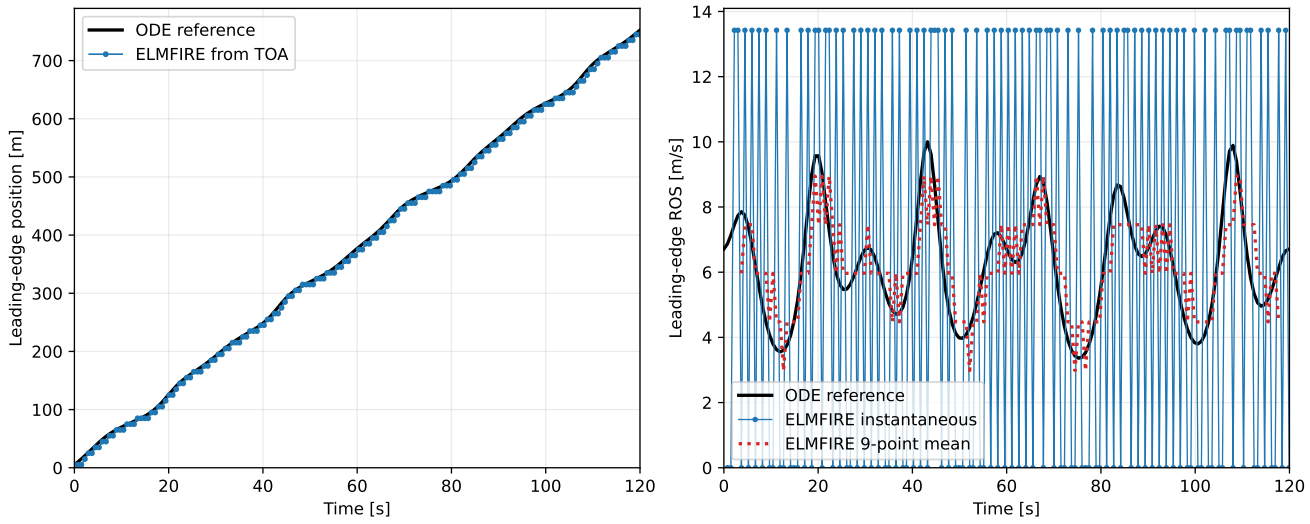


Figure 3: Time-dependent-wind transport comparison. Left: ELMFIRE and ODE leading-edge position. Right: ODE reference, grid-scale instantaneous ROS, and the nine-point running-average ELMFIRE ROS.

Figure 4 adds the time-resolved residuals and acceptance limits. It exposes localized phase errors that can be hidden by a single mean value.

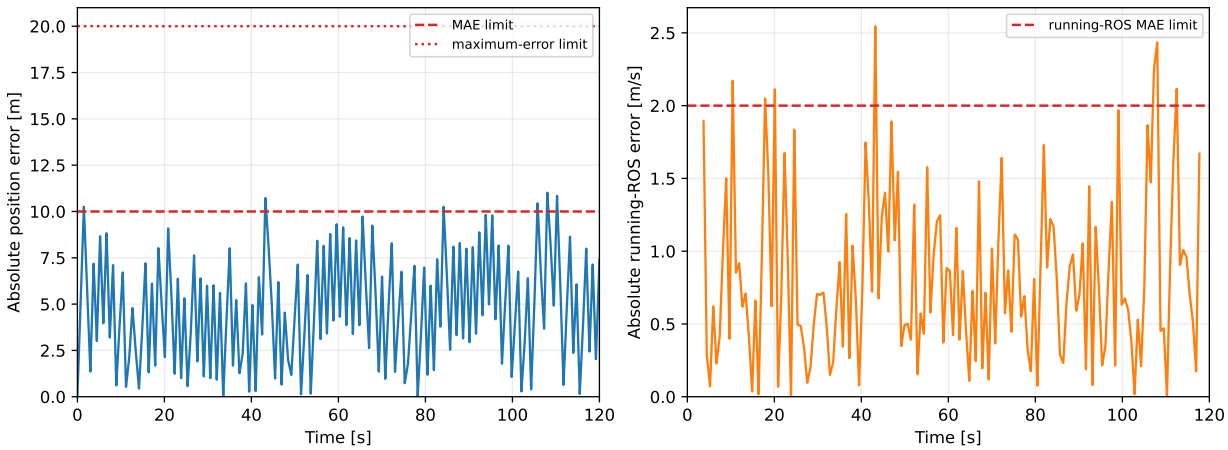


Figure 4: Absolute leading-edge-position and running-ROS errors through the simulation. Horizontal lines identify the applicable acceptance limits.

7 Calculated metrics and verification decision

Table 3: Acceptance metrics calculated by postprocessing.

Metric	Limit	Calculated	Pass
Position MAE	$\leq$10 m	4.761474516898714	True
Position maximum error	$\leq$20 m	11.01743438309552	True
Running ROS MAE	$\leq$2 m/s	0.8128946040373389	True
Time coverage	$\geq$ 0.90	1.0	True

The decision read directly from calculated verification and validation metrics is **PASS** (overall acceptance = True).

8 Technical interpretation

The current execution passes all four predeclared checks. The ELMFIRE leading edge reaches 745.0 m at 120 s, compared with the ODE reference value 752.3894781520776 m. The position MAE is 4.761474516898714 m, the maximum position error is 11.01743438309552 m, the nine-point running-ROS MAE is 0.8128946040373389 m/s, and the time-coverage fraction is 1.0; each lies within its declared limit. Input inspection confirms 121 one-second wind bands spanning the prescribed forcing history and an explicit deterministic source-selection percentage. The agreement therefore verifies the configured time-indexed-wind transport observable at the stated spatial and temporal resolution. Instantaneous ROS remains grid-quantized, which is why the smoother nine-point mean is the acceptance observable.

9 Scope and limitations

This is deterministic numerical verification of the Eulerian algorithm. It does not validate the empirical landing-distance distribution, crosswind dispersion, Lagrangian sampling, or physical ignition proba-

bility. One-second weather bands approximate the continuous analytical forcing by ELMFIRE's native linear interpolation.

References

[1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

Case 06: Two-Dimensional Firebrand-Driven Spread

1 Verification purpose

This case verifies that Eulerian spotting produces a genuinely two-dimensional, firebrand-driven fireline under a manufactured two-component, space–time wind. It exercises emission, vector transport, deposition, direct spot ignition, and level-set propagation together.

The test exposes loss of either wind component, incorrect direction conversion, one-dimensional source geometry, missing secondary ignitions, non-monotone burned area, and stale transient output. Success requires nested perimeters near 30, 60, 90, and 120 s that advance in both directions and exceed the surface-only reach.

2 Mathematical formulation and numerical implementation

The prescribed horizontal wind is

$$u(x,t) = \bar{u} + A\sin(2\pi x/\lambda_x)\sin(2\pi t/T), \tag{1}$$
$$v(y,t) = \bar{v} + B\sin(2\pi y/\lambda_y)\sin(2\pi t/T), \tag{2}$$

with $\bar{u} = 6.71\,\mathrm{m\,s^{-1}}$, $A = 3.355\,\mathrm{m\,s^{-1}}$, $\bar{v} = 1.0\,\mathrm{m\,s^{-1}}$, $B = 0.5\,\mathrm{m\,s^{-1}}$, $\lambda_x = \lambda_y = 100\,\mathrm{m}$, and $T = 60\,\mathrm{s}$. The preprocessor evaluates these equations at every cell centre and every one-second meteorology time. It writes $|\mathbf{u}|$ in mph and converts the transport-to angle to ELMFIRE meteorological wind-from convention.

For each Eulerian source tracker, the current spotting implementation advances

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{u}(\mathbf{x}_k, t_k)\Delta t_k, \qquad \Delta t_k = \Delta x/\max(|\mathbf{u}|, 0.01),$$

and distributes the source ember count through the configured downwind lognormal landing distribution. Direct ignition updates the level-set state, after which both the primary headfire and spot fires create new burning cells. The configured surface reference speed is $R_s = 0.71\,\mathrm{m\,s^{-1}}$ [1], and the corresponding emission residence time is $t_e = \Delta x/R_s = 14.08\,\mathrm{s}$.

The prescribed u and v components determine an individual transport characteristic, but they do not uniquely determine the maximum coordinates of the coupled fireline in this configuration. The observed $\phi = 0$ perimeter is the union of the surface-propagated primary fire and surface-propagated spot fires created after nonlocal, probability-distributed deposition. Consequently, $x_{\max}$ and $y_{\max}$ do not generally satisfy the uncoupled equations $dx_{\max}/dt = u(x_{\max}, t)$ and $dy_{\max}/dt = v(y_{\max}, t)$. An exact endpoint metric would require an independent solution of the complete coupled problem.

3 Derivation of expected behavior

Because the wind has positive mean components, successive fronts must progress downwind and develop a nonzero crosswind span. The level-set burned set is irreversible, so for burned masks $B_j = \{(x,y) : \phi(x,y,t_j) \le 0\}$ the expected inclusion is $B_j \subseteq B_{j+1}$. A conservative surface-only leading-edge bound is

$$x_{\mathrm{surf}}(120) = x_0 + R_s(120) = 90.2\,\mathrm{m}.$$

A front substantially beyond this location, together with nonzero ember-ignition cells, demonstrates that firebrands rather than the surface headfire alone control the long-range advance. Accordingly, the

principal expected result is a family of nested two-dimensional contours rather than a one-dimensional time of arrival (TOA) curve.

The largest generated transport speed is

$$|\mathbf{u}|_{\max} = \sqrt{(6.71 + 3.355)^2 + (1.0 + 0.5)^2} = 10.1762 \text{ m s}^{-1},$$

so preprocessing sets $\Delta t = 0.5(10)/10.1762 = 0.491344$ s. The surface residence time is independently

$$t_e = 10/0.71 = 14.0845 \text{ s}.$$

Starting from $x_0 = 5$ m, a surface-only head fire would be at

t (s)	30	60	90	120
x_{surf} (m)	26.3	47.6	68.9	90.2

and would advance $0.71(30) = 21.3$ m between successive target times. The declared 20-m interval threshold is therefore slightly below the surface-only analytical increment but still requires two resolved cells. At 120 s, the firebrand-excess test $x_{\max} - 90.2 \geq 30$ m is equivalently $x_{\max} \geq 120.2$ m, together with a positive ember-ignition count. Exact irreversible evolution gives $|B_j \setminus B_{j+1}|/|B_j| = 0$; the numerical allowance is 0.02. The final span reference is qualitative nonzero two-dimensional growth, made quantitative as at least $40 \text{ m} = 4\Delta x$.

4 Verification metrics and acceptance criteria

The postprocessor selects transient level-set field rasters within 2 s of all four target times and excludes the two-cell halo. It calculates: (i) completeness of all four contours; (ii) the fraction $|B_j \setminus B_{j+1}|/|B_j|$, limited to 0.02; (iii) downwind leading-edge advance, required to be at least 20 m in every interval; (iv) final crosswind span, required to be at least 40 m; and (v) firebrand excess advance $x_{\max}(120) - x_{\text{surf}}(120)$, required to be at least 30 m with at least one ember-ignition cell. The overall result is PASS only when every component passes.

4.1 Justification of metric quantification

The thresholds are expressed in grid-resolvable units. Selecting a contour within 2 s limits time mismatch to less than 7% of the earliest 30 s target and only a few solver steps. Exact level-set evolution is nested, so the 2% burned-area allowance is reserved for zero-contour classification along a cell-wide perimeter; a larger loss would indicate physical unburning or inconsistent dump selection. The minimum interval advance of 20 m, final crosswind span of 40 m, and firebrand excess of 30 m equal two, four, and three cells, respectively. These multi-cell thresholds cannot be satisfied by a stationary front, a single misclassified boundary cell, or the surface-only reference, yet they do not prescribe a detailed perimeter shape that the manufactured problem does not supply. At least one ember-ignition cell is an exact mechanism check, and all four times are required to support the claimed two-dimensional evolution.

4.2 Predeclared acceptance criteria

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Perimeter completeness	Unique transient level-set field raster selected within 2 s of 30, 60, 90, and 120 s.	4 of 4 target times
Nesting violation	Maximum $\lvert B_j \setminus B_{j+1}\rvert/\lvert B_j\rvert$ over consecutive physical burned masks.	≤ 0.02
Downwind progression	Increase in maximum burned-cell x between consecutive target times.	≥ 20 m every interval
Final crosswind span	$\max(y) - \min(y)$ of the final physical burned mask.	≥ 40 m
Firebrand-driven excess	$x_{max}(120) - [x_0 + R_s(120)]$, together with the final ember-ignition map.	$\geq$ 30 m and at least one ember-ignited cell
Overall decision	Conjunction of completeness, nesting, progression, span, and excess-advance checks.	PASS only if all pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for the base two-dimensional run. These panels show the prepared spatial fields, remove the two-cell numerical buffer, and retain map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

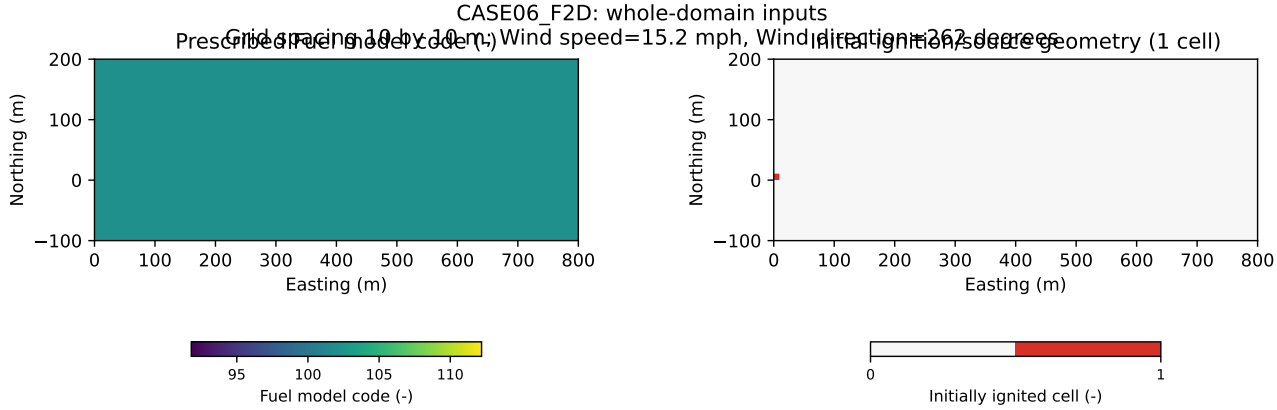


Figure 1: Whole-domain prepared input configuration for the representative simulation condition. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

The Courant–Friedrichs–Lewy (CFL) number measures the distance advected during one timestep relative to the grid spacing and constrains the time discretization.

Before inspecting results, the predicted output is four nested contours elongated in positive x, displaced and broadened in y, with many secondary ember ignitions and a final leading edge far beyond 90.2 m.

Table 2: Configured two-dimensional transport case experiment.

Parameter	Value	Role
Physical domain	800×300 m	two-dimensional propagation
Cell size	10 m	reference spatial resolution
Raster grid	84×34 cells	includes two-cell halo
Meteorology	121 bands at 1 s	space–time wind
Simulation step	0.491 026 s	wind CFL approximately 0.5
Initial ignition	$(5, 5)$ m	localized upwind source
Surface headfire	enabled	primary moving source
Generation	PER-AREA, 10 pcs m^{-2} s^{-1}	deterministic ember supply
Accumulation / ignition	Eulerian / direct	tested physical and numerical processes
Crosswind PDF	$\mu_y = 0$, $\sigma_y = 0.1$ m	narrow $k_{\max} = 0$ limit
Output times	30, 60, 90, 120 s	transient perimeter comparison

6 Actual simulation results

Figure 2 shows the representative accumulated firebrand density from the base two-dimensional run. The displayed field is taken from the simulated results, masks missing values, and excludes the two-cell numerical buffer. Counts are divided by physical cell area, giving pcs m^{-2} and exposing the full lateral as well as downwind distribution.

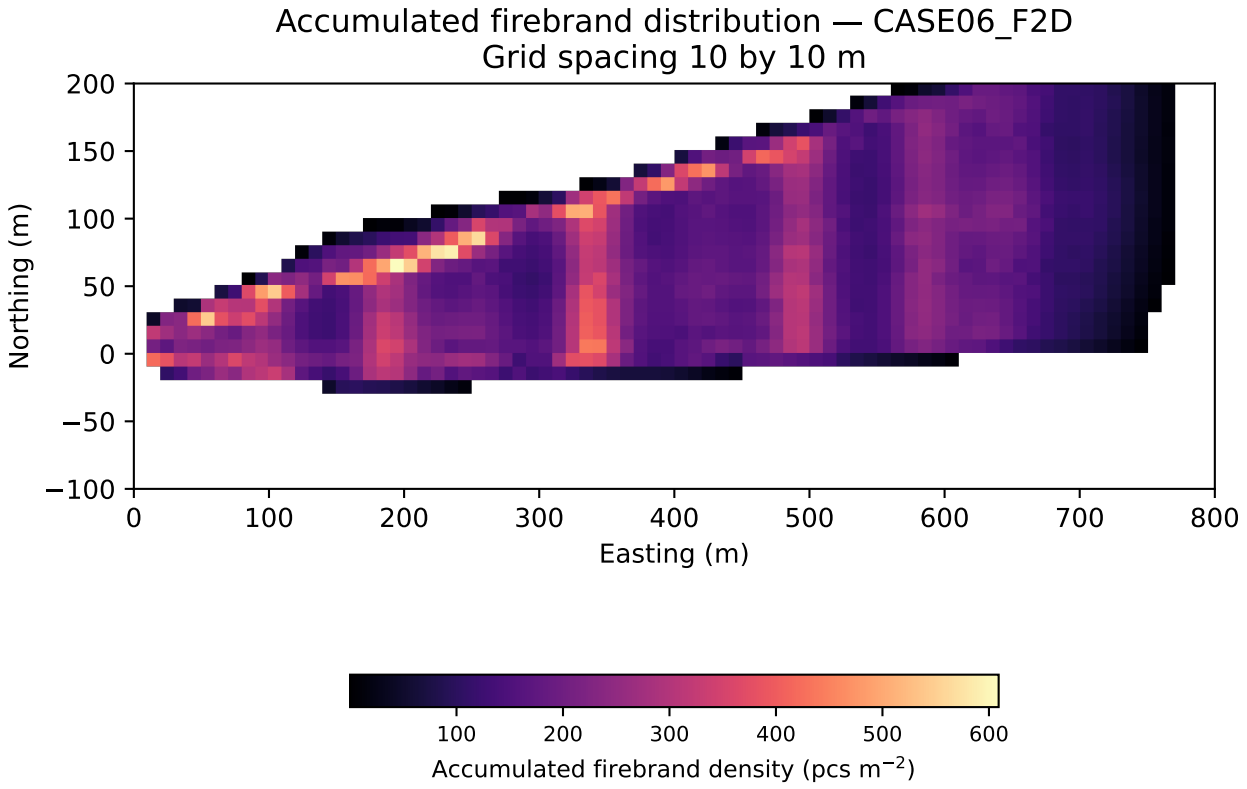


Figure 2: Whole-domain accumulated firebrand density from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

The selected transient perimeter times were 30.44, 60.4, 90.35, 120 s. The identities of the selected fields are retained with the calculated metrics. Figure 3 first presents the evolving zero-level contours, which are the principal two-dimensional reference observable.

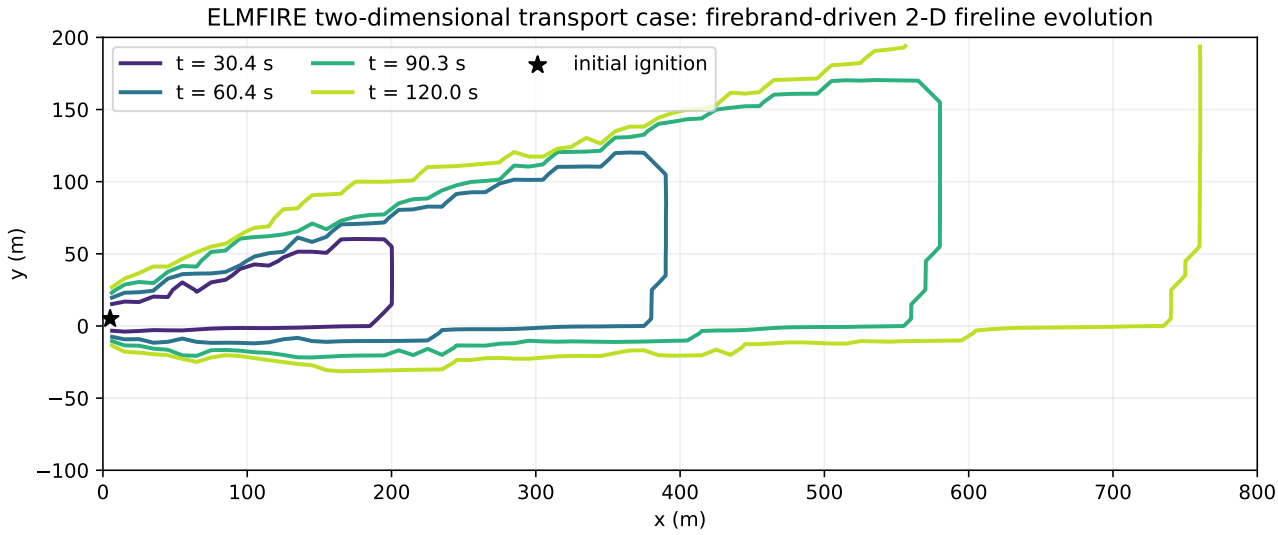


Figure 3: Actual ELMFIRE fire perimeters at four output times under the prescribed two-component, space–time wind. The star marks the initial ignition.

Figure 4 identifies the cells actually ignited by deposited firebrands. The accumulated-firebrand field is not repeated here because its whole-domain distribution is already presented in Figure 2.

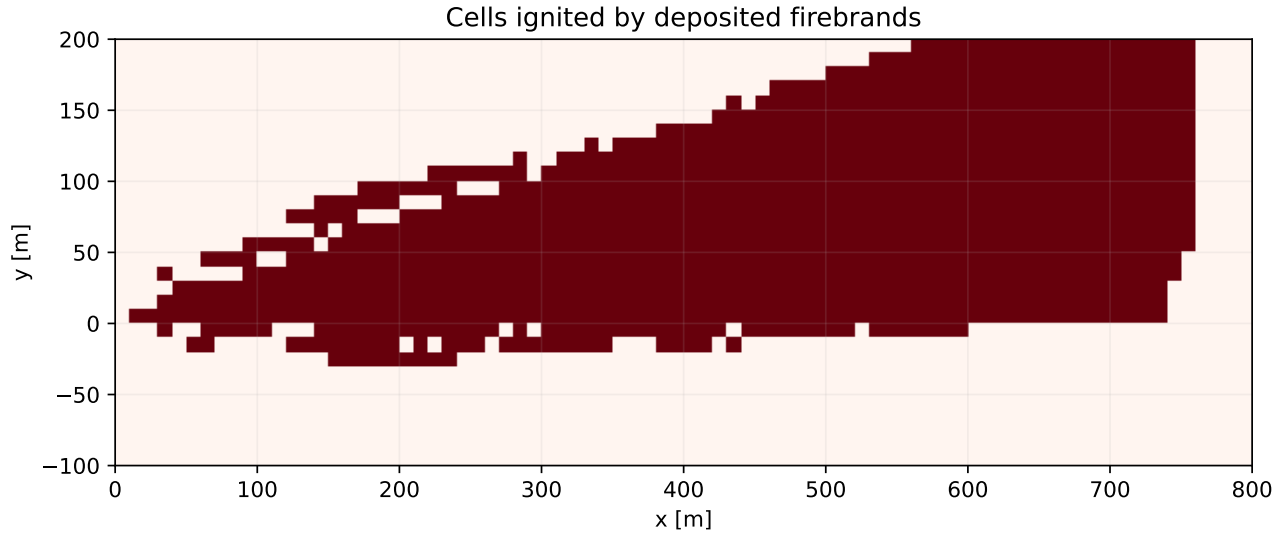


Figure 4: Whole-domain cells marked by ELMFIRE as ignitions caused by deposited firebrands. The numerical halo is excluded.

7 Calculated metrics and verification decision

Table 3: Acceptance metrics calculated from actual ELMFIRE outputs.

Metric	Acceptance limit	Calculated value	Status
Required contours	4 within 2 s	30.44, 60.4, 90.35, 120	PASS
Nesting violation	≤ 0.02	0.0	PASS
Interval advances	each ≥ 20 m	190, 190, 180 m	PASS
Final crosswind span	≥ 40 m	220.0 m	PASS
Firebrand excess	≥ 30 m and ignitions	664.8 m; 1097 cells	PASS
Overall	all components true	PASS	**PASS**

8 Technical interpretation

The result is **PASS**. The maximum nesting violation is 0.0, while the leading edge advances by 190, 190, 180 m over the three intervals. The final crosswind span is 220.0 m, confirming that the calculation is not a replicated one-dimensional transect. The final front exceeds the surface-only bound by 664.8 m and the ember-ignition raster contains 1097 positive cells; together these observations attribute the long-range evolving fireline to secondary ignitions from transported firebrands.

9 Scope and limitations

This test verifies quantified mechanism and evolution invariants of the current Eulerian 2-D coupling and reconstructs the intended contour diagnostic. The known wind components alone are not treated as an exact outer-fireline solution because surface spread and nonlocal secondary ignitions remain active. The test does not reproduce the comparison solution cell-for-cell, compare Eulerian and Lagrangian contours, validate the landing PDF against experiments, or prove grid convergence. The configured flat terrain isolates horizontal transport; additional terrain-slope cases would be required to verify topographic coupling.

References

[1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

Case 07: Ignition Probability and SFT Convergence

1 Verification purpose

This case verifies statistical convergence of the stochastic smoldering-to-flaming (SFT) model and the resulting ignition-probability response. Pooled ignition-delay samples must converge to the configured SFT scale, while ensemble-mean firebrand-driven spread must respond consistently to ignition probability.

The test exposes biased sampling, insufficient ensemble size, incorrect delay extraction, and grid-dependent probability application. Success is determined from confidence-aware delay statistics and declared ensemble metrics, never from a single realization.

2 Mathematical formulation and numerical implementation

For accumulated firebrands in a cell, ELMFIRE's SIMPLE model uses

$$F(t) = 1 - (1 - P)^{t/\tau}, \qquad P_{\Delta t} = 1 - (1 - P)^{\Delta t/\tau}, \tag{1}$$

where $P = P_{\text{ign,small}}$, $\tau = \tau_{\text{ign}}$, and $P_{\Delta t}$ is the Bernoulli probability evaluated during one solver step. Consequently $F(\tau) = P$: at $P = 0.9$, the theoretical 90th percentile is $q_{0.9} = \tau = 10$ s. A successful local transition is followed by additional development delay=0, so the final time of arrival (TOA) records the full ignition time without an additional development lag.

For the dedicated delay members, transient ember-flux output is written every wind Courant–Friedrichs–Lewy (CFL) step. If a cell's first nonzero transient flux occurs at dump k, its first deposition lies in $(t_{k-1}, t_k]$. Postprocessing uses the interval midpoint $t_{d,i} = (t_{k-1} + t_k)/2$ and forms the actual-output sample

$$d_i = t_{\text{ign},i} - t_{d,i}. \tag{2}$$

Only centerline cells marked in the final ember-ignition raster, outside the two-cell halo and in the 155–2800 m complete-support interval, are admitted. The dump spacing Δt is retained as the temporal-censoring resolution.

For the response sweep,

$$CFL_w = \frac{u_w \Delta t}{\Delta x} \leq 0.5, \qquad r_m = \frac{R_m}{u_w}, \tag{3}$$

and each member rate of spread (ROS) R_m is the least-squares slope of centerline distance against final TOA over 100–2800 m. Ten explicit seeds are paired across $\Delta x = 5, 10, 20, 30$ m and $P = 0, 0.25, 0.5, 0.75, 1$.

3 Derivation of expected behavior

Equation 1 gives the exact event-level expectation $q_{0.9} = 10$ s. A finite sample quantile fluctuates, but its bootstrap confidence interval should contract as independently seeded members add cell samples. The final pooled estimate should therefore approach 10 s rather than an ROS-derived surrogate. Timestep-resolved deposition limits the deterministic timing uncertainty to the deposition interval.

At $P = 0$, the ignition hazard is zero, and no spotting-driven front should propagate. Increasing P increases $F(t)$ for every $t > 0$ and therefore cannot reduce population-mean ROS. At $P = 1$, ignition

is immediate in the continuous model and $\bar{r}$ should approach one. Because physical extent is fixed and the generated CFL_w values remain conservatively bounded near 0.5, the four resolution curves should nearly coincide.

For the delay member, $P = 0.9$ and $\tau = 10$ s, hence

$$F(5) = 1 - 0.1^{0.5} = 0.683772, \qquad F(10) = 1 - 0.1 = 0.9, \qquad F(20) = 1 - 0.1^2 = 0.99.$$

Solving $F(q_{0.9}) = 0.9$ gives $0.1^{q_{0.9}/10} = 0.1$ and therefore $q_{0.9} = 10$ s exactly. With $U = 15(0.44704) = 6.7056$ m s^{-1}, preprocessing first computes the $CFL_w = 0.5$ upper bounds, then reduces each step to divide 240 s exactly. The actual generated steps and corresponding per-step ignition probabilities are

Δx (m)	5	10	20	30
Δt (s)	0.372671	0.745342	1.490683	2.222222
$1 - 0.1^{\Delta t/10}$	0.082232	0.157702	0.290534	0.400516

For each resolution the conservative CFL-derived timestep defines the probability transformation and cross-resolution control. Preprocessing selects $N = \lceil 240/\Delta t_{CFL} \rceil$, writes the conservative divisor $\Delta t = 240/N$, and keeps $t_{stop} = 240$ s exactly; the simulation specification records nominal and actual CFL values plus the integer step count. This shows why the step probability changes while the continuous delay cumulative distribution function (CDF) does not.

The response design contains 4 resolutions $\times 5$ probabilities $\times 10$ paired seeds $= 200$ required runs, plus 10 delay members. The exact limiting references are $R/U = 0$ at $P = 0$, monotone nondecrease with P, and $R/U \to 1$ at $P = 1$. Numerically these become a maximum cross-resolution range of 0.10, a $P = 1$ floor of 0.90, and adjacent decreases no larger than their pooled 95-percent confidence half-width. The delay substitution is $|\widehat{q}_{0.9} - 10| \leq 1$ s, with at least 1000 samples and bootstrap half-width at most 1 s.

4 Verification metrics and acceptance criteria

For every response group with $N = 10$, the postprocessor calculates the mean, sample standard deviation, standard error, and two-sided Student-t 95% confidence interval. Adjacent probability means may decrease only within their pooled confidence half-widths. At each P, the spread of the four resolution means must not exceed 0.10, and every $P = 1$ mean must be at least 0.90.

For the delay ensemble, samples are pooled after each independent member. At the final member, all ten members must be usable, at least 1000 delays must be available, $|\widehat{q}_{0.9} - 10 \text{ s}| \leq 1$ s, and the 2000-replicate bootstrap 95% interval half-width must not exceed 1 s. A missing final TOA, ignition map, dump-time table, transient sequence, stale raster shape, or empty sample set makes that member incomplete. Overall PASS is the conjunction of all response and delay components; missing required output yields insufficient simulation results, never a pass.

4.1 Justification of metric quantification

A two-sided 95% Student-t interval is used because each response mean has only ten independent ensemble members and its variance is estimated from those members. Allowing an adjacent decrease only within the pooled confidence half-width distinguishes sampling noise from evidence against monotonicity without assuming equal variances. The 0.10 cross-resolution range and the 0.90 unit-probability floor each represent ten percentage points on the naturally normalized $[0, 1]$ response scale: they admit finite-ensemble and grid-crossing variability but reject a practically material resolution effect or failure to approach immediate ignition.

The target $q_{0.9} = 10\,\mathrm{s}$ follows directly from the prescribed SFT distribution [1]. Requiring 1000 pooled delays gives substantial support in the upper decile rather than allowing a few extreme samples to set the quantile. The 1 s accuracy and bootstrap-half-width limits are each 10% of the model timescale, so both bias and uncertainty must be small on the scale being verified. Two thousand bootstrap replicates provide stable percentile endpoints for this engineering decision without treating the bootstrap interval as an exact physical confidence bound. Full member completeness is required because silently dropping failed stochastic runs can bias both response means and delay quantiles.

4.2 Predeclared acceptance criteria

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Response-ensemble completeness	Ten usable members for every resolution–probability group.	200 of 200 members
Probability monotonicity	Adjacent ensemble-mean normalized ROS differences, allowing decrease only within pooled 95% confidence half-widths.	Every adjacent check passes
Resolution spread	Range of the four resolution means at each ignition probability.	≤ 0.10
Unit-probability response	Minimum ensemble-mean normalized ROS at $P_{ign} = 1$.	≥ 0.90
SFT sample completeness	Ten usable delay members and pooled interval-censored ignition-delay samples.	10 members and at least 1000 samples
SFT 90th percentile	Absolute difference between pooled $\widehat{q}_{0.9}$ and the configured 10 s scale.	$\leq 1\,\mathrm{s}$
SFT uncertainty	Half-width of the 2000-replicate bootstrap 95% interval for $\widehat{q}_{0.9}$.	$\leq 1\,\mathrm{s}$
Overall decision	Conjunction of every response and SFT component; incomplete output is not evaluable.	PASS only if all pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for the first ensemble member with a 10 m grid and ignition probability 0.5. These panels show the prepared spatial fields, remove the two-cell numerical buffer, and retain map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

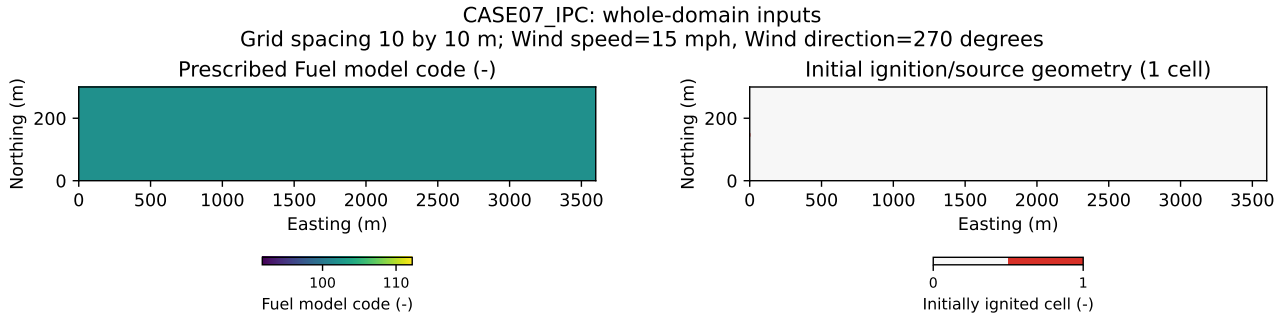


Figure 1: Whole-domain prepared input configuration for the representative simulation condition. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

Parameter	Configured value	Unit	Verification role
Physical domain	3600×300	m	Fixed physical extent
Buffer	2 cells per side	cells	Excluded numerical halo
Response grids	5, 10, 20, 30	m	Resolution comparison
Delay grid	10	m	delay-ensemble spatial baseline
Wind	15 (6.7056)	mph (m/s)	Uniform transport
Wind CFL	≤ 0.5	–	stable near-fixed $\Delta t/\Delta x$
Stop time	240	s	exact timestep multiple
Response probabilities	0, 0.25, 0.50, 0.75, 1	–	probability-response design
Delay model	$P = 0.9$, $\tau = 10$	–, s	delay-distribution design
Ensembles	20×10 response; 10 delay	runs	Sampling statistics
Generation	PER-AREA, 10	pcs m^{-2} s^{-1}	No-surface source
Accumulation/ignition	EULERIAN/SIMPLE	–	Implementation under test
Surface/development delay	disabled/0	–, s	Isolate SFT waiting time
Delay output cadence	one wind-CFL step	s	Bracket first deposition

Table 2: Configuration of the merged ensemble-convergence case verification.

The predicted response plot contains four nondecreasing curves beginning at zero and approaching one. The delay plot should approach the horizontal 10 s line, with its sampling interval narrowing as ensemble members are pooled.

6 Actual simulation results

Figure 2 shows the representative ember-ignition field from the first ensemble member with a 10 m grid and ignition probability 0.5. The displayed field is taken from the simulated results, masks missing values, and excludes the two-cell numerical buffer. This representative ensemble member supplies spatial evidence alongside the ensemble probability and delay statistics; it is not used as a replacement for those statistical metrics.

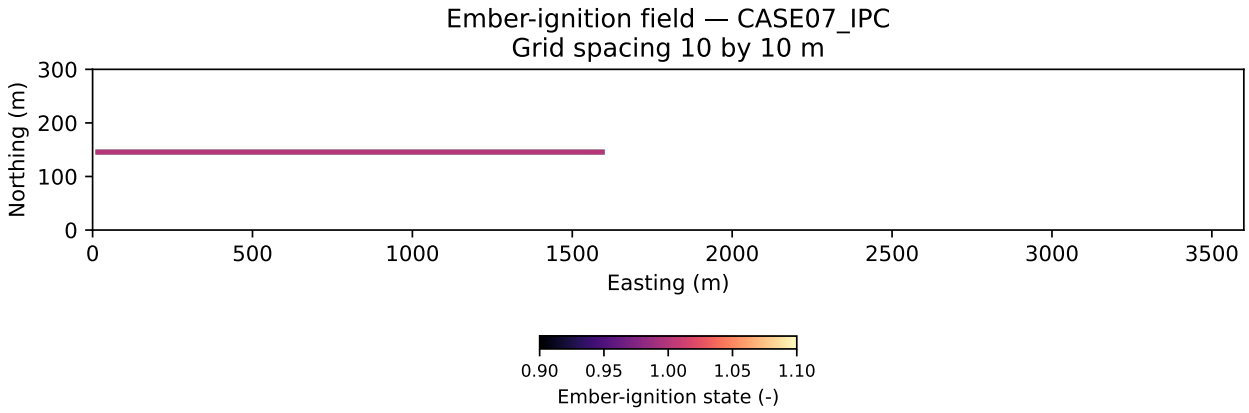


Figure 2: Whole-domain ember-ignition field from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

The distribution-level ignition-delay check is presented first because it compares the sampled stochastic variable directly with its prescribed model quantile. Figure 3 uses only delays reconstructed from actual arrival-time, ember-ignition, dump-time, and transient-flux outputs.

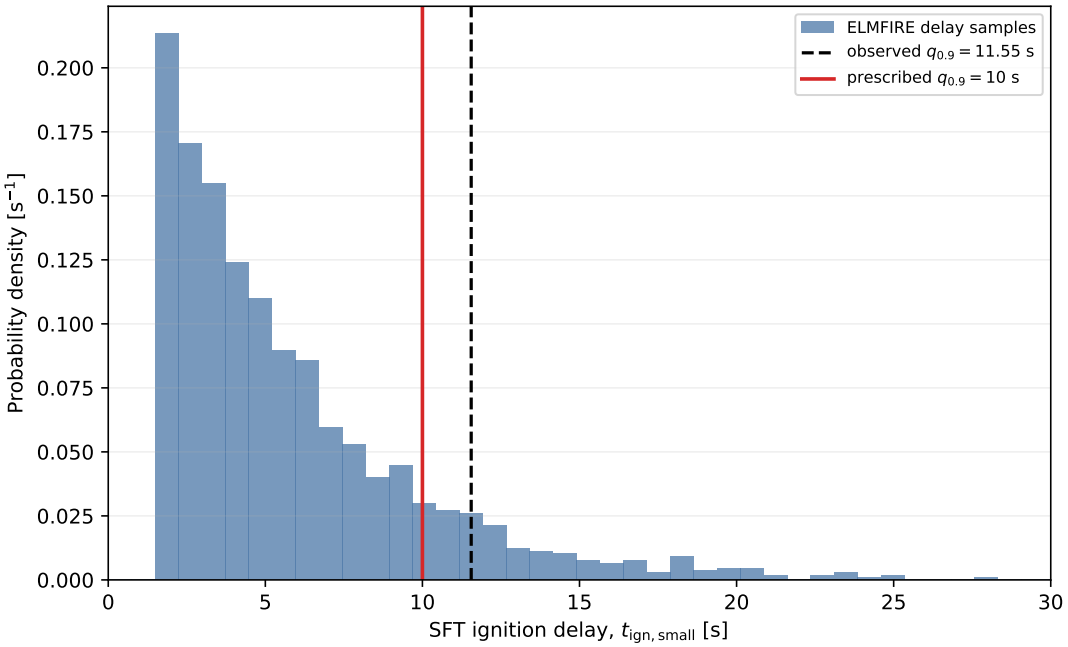


Figure 3: Probability density of sampled SFT ignition delays. The solid red line is the prescribed 90th percentile and the dashed black line is the observed pooled 90th percentile.

Figure 4 presents the second reference observable: the ensemble-mean normalized leading-edge ROS response to ignition probability and spatial resolution.

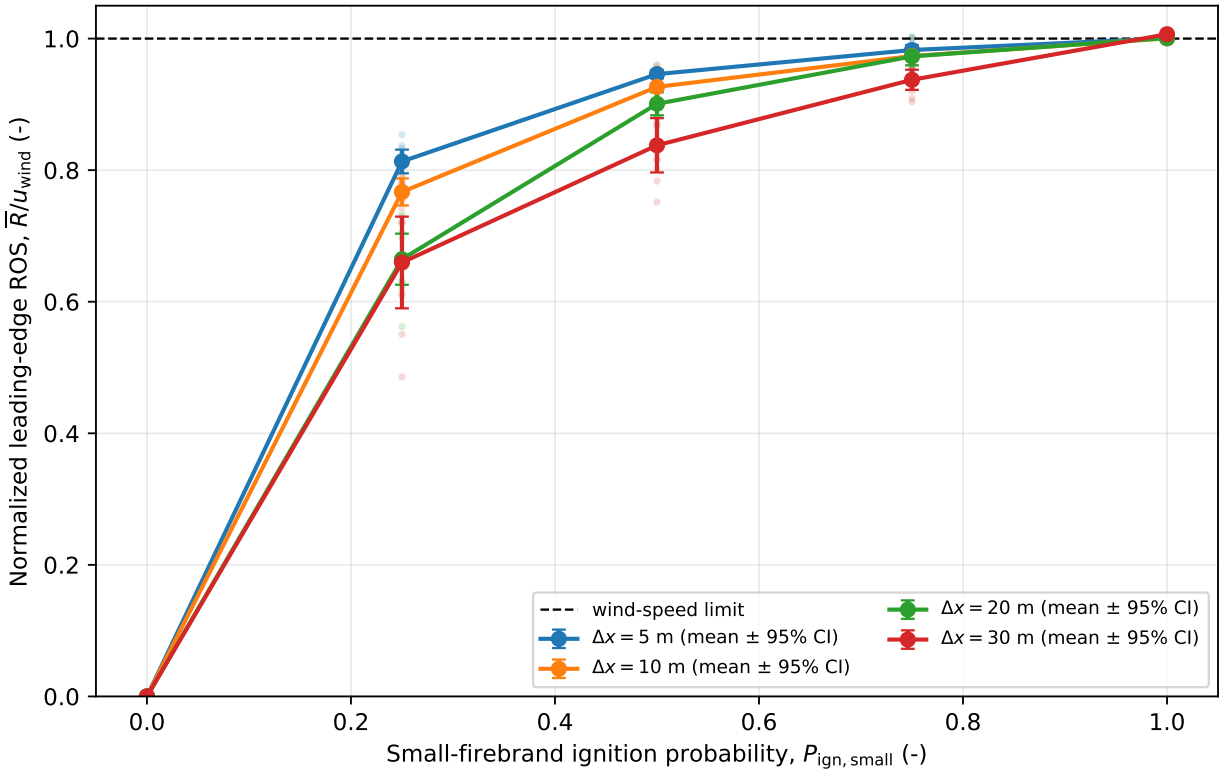


Figure 4: Ensemble-mean normalized leading-edge ROS versus $P_{\text{ign,small}}$. Error bars are 95% confidence intervals and faint points are independently seeded ELMFIRE members.

Figure 5 is an additional sample-size diagnostic. It shows how the pooled delay quantile and its bootstrap uncertainty change as independent members are added.

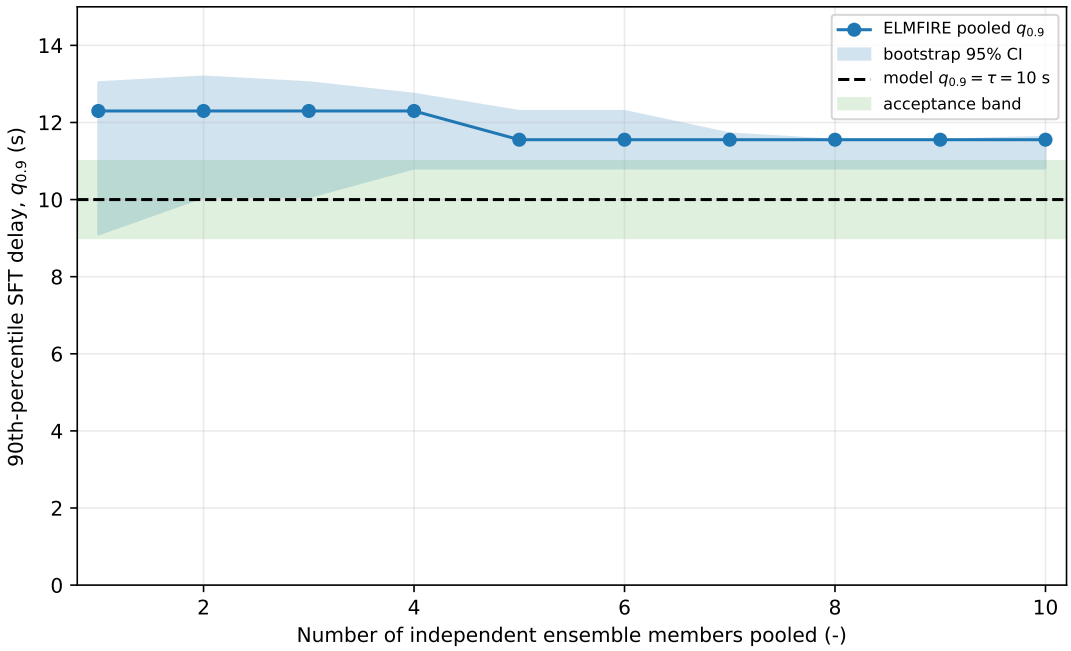


Figure 5: Cumulative convergence of the pooled ELMFIRE 90th-percentile SFT delay. The dashed line is the model expectation, blue shading is the bootstrap 95% interval, and the green band is the ± 1 s acceptance interval.

7 Calculated metrics and verification decision

Metric	Requirement	Calculated
Response members	200 usable	200/200
Probability response	monotone within 95% uncertainty	True
Maximum resolution-mean spread	≤ 0.10	0.1536
Minimum $P = 1$ mean	≥ 0.90	1.0004
SFT members	10 usable	10/10
SFT delay samples	≥ 1000	1440
SFT 90th percentile	10 ± 1 s	11.5528 s
SFT quantile absolute error	≤ 1 s	1.5528 s
SFT bootstrap half-width	≤ 1 s	0.4099 s
SFT component	all delay checks pass	False
Overall status	all components pass	**FAIL**

Table 3: Calculated verification metrics for the completed simulations.

The overall decision is **FAIL**.

8 Technical interpretation

All 200/200 response members and 10/10 dedicated delay members are usable. The maximum cross-resolution response spread is 0.1536, exceeding the 0.10 limit, although the minimum unit-probability mean passes. The 1440 admitted SFT samples give a 90th-percentile delay of 11.5528 s; its 1.5528-s

absolute error exceeds the 1-s limit, while the bootstrap half-width passes. Thus the result is a genuine **FAIL**, not an incomplete evaluation. The generated probability, seed, timestep, and ensemble controls are internally consistent. The source updates probability as $1 - (1 - P)^{\Delta t/\tau}$ and advances a newly deposited cell through its staged ignition state on later invocations; the observed extra-step delay and grid dependence therefore identify current source timing/discretization behavior rather than a case-configuration defect.

9 Scope and limitations

The case verifies one SIMPLE SFT time scale, one delay probability, one uniform wind and fuel, Eulerian accumulation, zero full-development delay, and the specified grid/probability sweep. Midpoint reconstruction of first deposition is interval-censored at one wind-CFL step; that resolution is recorded but is not sampling uncertainty. Cell samples within a run share a propagating fire history, so independent run count and pooled cell count are both reported. The test does not validate the empirical choice of P or τ, physical ignition thresholds, surface-fire competition, PHYSICAL ignition, Lagrangian transport, or firebrand consumption.

References

[1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

Case 08: Deposited Firebrand Consumption

1 Verification purpose

This case verifies removal of deposited firebrands by the consumption model. Paired consumption-off and consumption-on runs use identical forcing, seed, source model, and resolution so their active deposited populations can be compared directly.

The test exposes non-decaying stock, negative active counts, resolution-dependent consumption, and unintended feedback on surface arrival. Success requires the specified population reduction, invariant surface arrival, and convergence toward the finest-grid spatial profile.

2 Mathematical formulation and numerical implementation

Let N'' denote active deposited firebrands per unit area. With mean particle mass $m_{fb} = 0.2$ g, the pile mass loading is

$$\psi = \frac{m_{fb} N''}{10^4} \quad [\mathrm{g\,cm^{-2}}]. \tag{1}$$

The source estimates the local air velocity from the 20-ft wind using a fuel- and canopy-dependent coefficient. It then evaluates the empirical pile heat- flux parameters

$$t_{rise} = \max(-1.02 v_a^2 + 9.08 v_a + 35, 0), \tag{2}$$

$$HF_{rise} = \max(-0.32 v_a^2 + 1.59 v_a + 0.1, 0)\tanh(12\psi), \tag{3}$$

$$HF_{decay} = \max(10^{-6}, -0.05 v_a^2 + 0.02 v_a - 0.01)\tanh(12\psi). \tag{4}$$

After applying the source's heat-flux scaling, the lifetime is the duration for which the modeled incident heat flux exceeds $HF_{fb,live} = 10\ \mathrm{kW\,m^{-2}}$:

$$t_{fb,live} = \max(t_{half,2} - t_{half,1}, 0). \tag{5}$$

The intended explicit update is

$$N''^{n+1} = N''^{n} - \frac{N''^{n}}{\max(t_{fb,live}, t_{min})}\Delta t, \qquad t_{min} = 10\ \mathrm{s}. \tag{6}$$

ELMFIRE stores pixel counts, so the implementation multiplies the areal decay rate by pixel area before subtracting from accumulated firebrand deposition. Only the consumption-on member calls this consumption calculation. Final accumulated firebrand deposition therefore represents the active deposited population. The transient time-resolved firebrand deposition product records new deposition during an output interval and is reset after each dump; it is not used as an active-stock surrogate. Instead, independent consumption-on runs terminated at predeclared times provide the temporal samples from their final active-stock rasters.

The current source is also audited as part of this test. It presently assigns a minimum firebrand lifetime of 1E9 s, whereas the documented model requires 10 s [1], as expressed in Equation 6. The verification preserves this behavior and is expected to fail the consumption-effect metric until that implementation discrepancy is corrected.

3 Derivation of expected behavior

For otherwise identical runs, let $A_{on}(\Delta x)$ and $A_{off}(\Delta x)$ be the centerline sums of active deposited particles at 260 s. The retained fraction is

$$f_{ret}(\Delta x) = \frac{A_{on}(\Delta x)}{A_{off}(\Delta x)}. \tag{7}$$

The documented result that more than 96% is consumed gives $f_{ret} \leq 0.04$ [1]. Because the delayed-ignition baseline uses long ignition delays and is surface-fire dominated, removing deposited embers should not materially change time of arrival (TOA). The active-particle line density is recovered from a centerline pixel count as $n'(x_i) = N_i/\Delta x$ in pcs/m.

For grid convergence, the 5, 10, and 30 m consumption-on profiles are linearly interpolated to the 1 m cell centers. Their normalized integral error is

$$E_{\Delta x} = \frac{\int |n'_{\Delta x}(x) - n'_1(x)|\, dx}{\int |n'_1(x)|\, dx}. \tag{8}$$

Consistent refinement requires $E_{30} \geq E_{10} \geq E_5$, with $E_5 \leq 0.15$.

For a cell whose calculated lifetime is $T_n = \max(t_{fb,live,n}, 10)$, Equation 6 gives the explicit retained fraction

$$\frac{N''_{n+1}}{N''_n} = 1 - \frac{\Delta t_n}{T_n}, \qquad \frac{N''_m}{N''_0} = \prod_{n=0}^{m-1} \left(1 - \frac{\Delta t_n}{T_n}\right).$$

This is the reference update that the consumption-on run must execute; the consumption-off reference has every factor equal to one. The published statement that more than 96 percent is consumed is converted without further fitting to

$$1 - f_{\text{ret}} \geq 0.96 \quad \Longleftrightarrow \quad f_{\text{ret}} = \frac{A_{\text{on}}}{A_{\text{off}}} \leq 0.04.$$

The nominal surface Courant–Friedrichs–Lewy (CFL) upper bounds are $0.5\Delta x/0.71$. To keep every requested 10-s-multiple observation time exact, preprocessing selects the largest divisor no greater than that bound, preferring whole seconds when feasible:

$$\Delta t = \{0.666667,\ 2,\ 5,\ 10\}\ \text{s}$$

for $\Delta x = \{1, 5, 10, 30\}$ m. Thus all 260-s paired stops and 50–500-s history stops are exact timestep multiples, and every actual surface CFL remains below 0.5. The intended lifetime is recalculated locally from v_a and $\psi = 0.2N''/10^4$, so no constant lifetime is invented for the report.

For the paired control, the exact expected TOA difference is zero and the active-stock lower bound is zero particles. The convergence reference is the generated 1-m profile: $E_1 = 0$, with expected refinement ordering $E_{30} \geq E_{10} \geq E_5 \geq E_1$. The decision substitutes $E_5 \leq 0.15$, paired TOA difference ≤ 10 s, and minimum active count $\geq -10^{-6}$, in addition to the 0.04 retained-fraction limit.

4 Verification metrics and acceptance criteria

All 48 prescribed simulation conditions are required: eight paired 260-s metric runs and 40 consumption-on stop-time runs (four resolutions at ten sample times). Exactly one final active-flux and TOA raster per simulation condition is required. The two-cell numerical halo is excluded. The components are:

1. $\max_{\Delta x} f_{ret} \leq 0.04$;
2. paired centerline TOA difference ≤ 10 s at every resolution;
3. every active count $\geq -10^{-6}$ particles;
4. $E_5 \leq 0.15$; and
5. $E_{30} \geq E_{10} \geq E_5$.

The overall decision is the conjunction of completeness and all five components. Missing, duplicated, dimensionally stale, or unreadable outputs produce an insufficient-output result rather than a pass.

4.1 Justification of metric quantification

The retained-fraction ceiling of 0.04 is not tuned to these outputs; it is the complement of the reported result that more than 96% of deposited firebrands are consumed by the evaluation time [1]. The paired TOA allowance of 10 s is a mechanism-isolation check at the scale of one output-time increment: consumption may alter the active stock but must not materially change the surface-dominated arrival field. The lower bound of -10^{-6} particle is only a floating-point allowance around the exact nonnegativity invariant and cannot conceal a physically meaningful negative population.

The convergence ordering is more informative than agreement at one grid. It requires every refinement to move toward the 1 m profile. The $E_5 \leq 0.15$ limit allows cell averaging of a steep transported-stock profile while requiring at least 85% agreement in normalized integral content; errors of that size remain clearly separated from the coarse-grid shape changes the sweep is intended to expose. All paired and stop-time runs are mandatory because omitting one would break either the on/off control or the temporal consumption history.

4.2 Predeclared acceptance criteria

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Output completeness	Eight paired metric runs plus 40 active-stock stop-time runs, with final-raster shape/nodata validation and halo exclusion.	48 of 48 simulation conditions
Retained active fraction	For each resolution, consumption-on active population divided by its consumption-off pair at 260 s.	Maximum ≤ 0.04
Surface-TOA invariance	Maximum paired centreline TOA difference on common finite physical cells.	≤ 10 s at every resolution
Active-count nonnegativity	Minimum value over every active-firebrand raster.	$\geq -10^{-6}$ particles
Fine-grid profile convergence	Normalized L_1 error of the 5-m consumption-on profile relative to the 1-m profile.	$E_5 \leq 0.15$
Ordered convergence	Normalized profile errors for 5, 10, and 30-m grids.	$E_{30} \geq E_{10} \geq E_5$
Overall decision	Conjunction of completeness and all five physical/numerical checks.	PASS only if all pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for the 10 m grid with firebrand consumption enabled. These panels show the prepared spatial fields, remove the two-cell numerical buffer, and retain map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

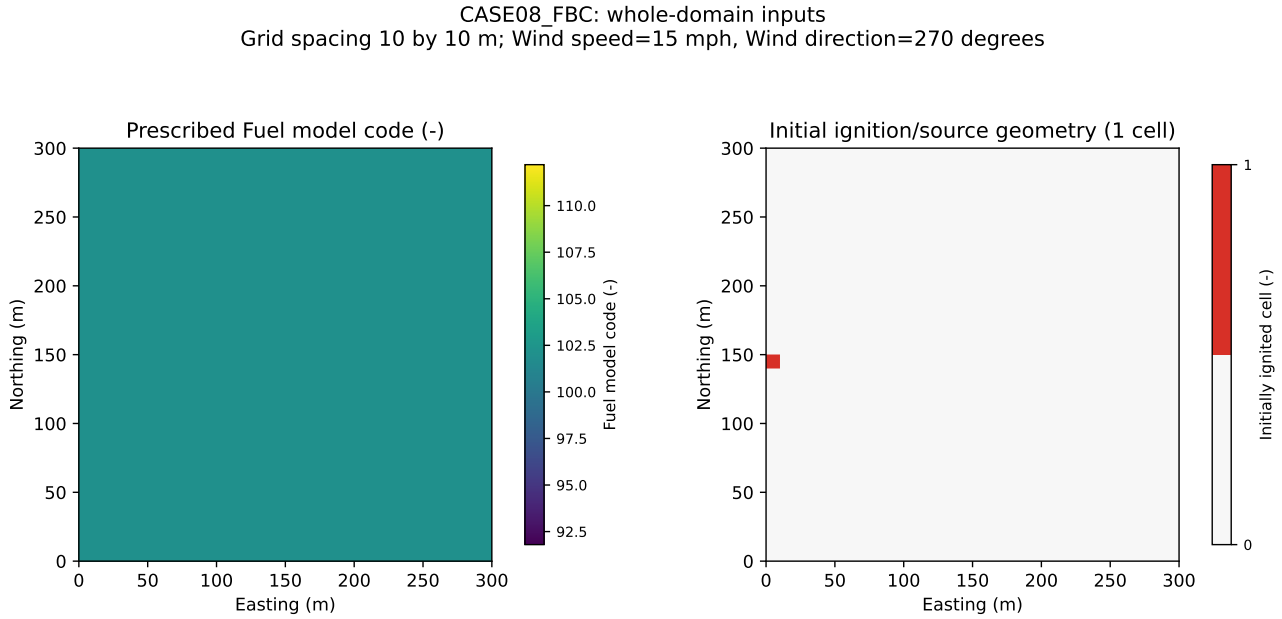


Figure 1: Whole-domain prepared input configuration for the representative simulation condition. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

Parameter	Configured value	Role
Physical domain	300×300 m	identical extent at all resolutions
Cell sizes	1, 5, 10, 30 m	active-stock convergence sweep
Numerical halo	2 cells per side	excluded from all metrics
Metric evaluation time	260 s	paired consumption-on/off comparison
History sample times	50–500 s in 50-s increments	final active-stock samples for the temporal curve
Temporal CFL	at most 0.5 based on 0.71 m/s	conservative exact-duration divisors
Wind	15 mph from 270 degrees	uniform downwind forcing
Generation	PER-MW, $33.3/\Delta y$ pcs/s/MW	delayed-ignition baseline vegetation source
Transport/accumulation	EMPIRICAL/EULERIAN	active-stock implementation
Ignition	PHYSICAL; 300 s development	prevents spotting-dominated spread
Consumption	paired off/on	isolates finite lifetime
Critical live heat flux	10 kW/m^2	hard-coded lifetime threshold
Crosswind distribution	disabled	one-dimensional profile diagnostic

Table 2: Configuration used by every paired simulation condition.

Before inspecting results, consumption-on curves should be far below the paired no-consumption

curve upstream of 200 m, retained fractions should lie below the 4% line, and progressively finer active-stock profiles should agree. TOA should be nearly unchanged between paired runs.

6 Actual simulation results

Figure 2 shows the representative accumulated firebrand density from the 10 m grid with firebrand consumption enabled. The displayed field is taken from the simulated results, masks missing values, and excludes the two-cell numerical buffer. Counts are divided by physical cell area, giving pcs m^{-2}, so the retained and consumed portions of the deposit field can be interpreted spatially.

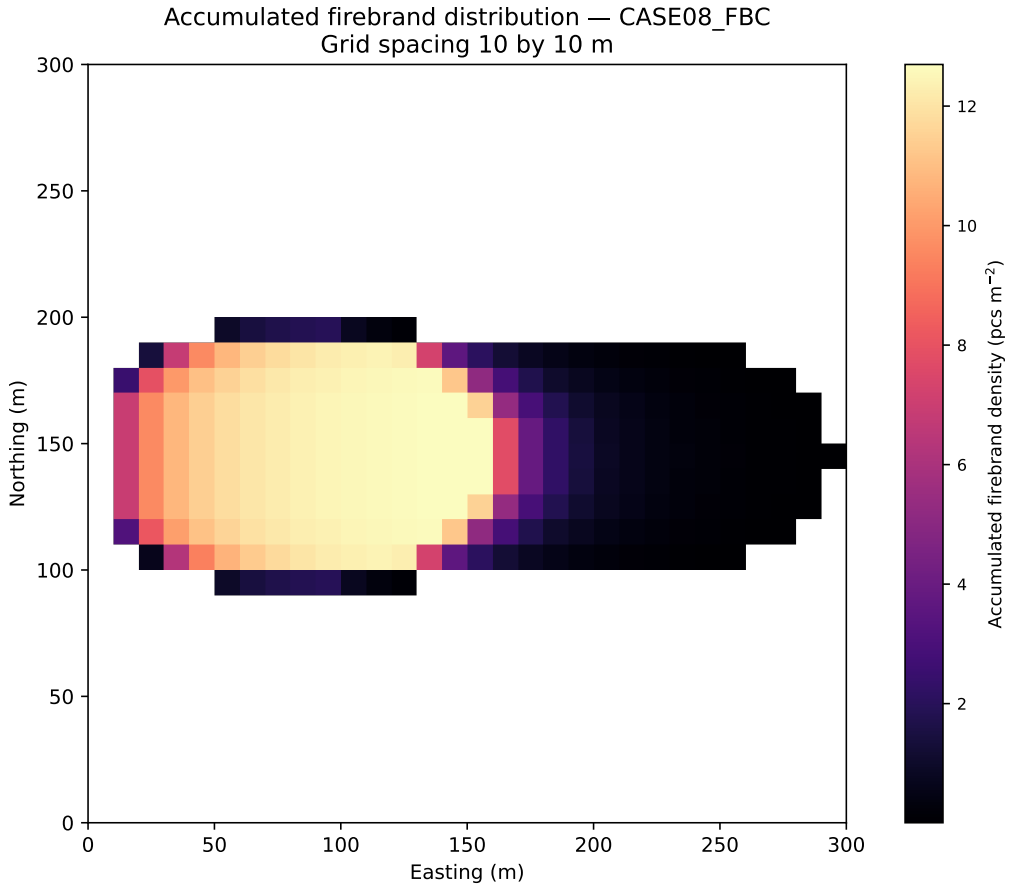


Figure 2: Whole-domain accumulated firebrand density from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

All 48 simulation conditions were rerun. The temporal curves use ten independent consumption-on stop-time runs per resolution, from 50 to 500 s in 50 s increments. Each plotted sample is therefore a final ELMFIRE active-stock field, not an interval-deposition raster. Figure 3 first presents the four physical observables for the finest consumption-on simulation condition: active firebrands, arrival time, leading-edge position, and leading-edge rate of spread (ROS).

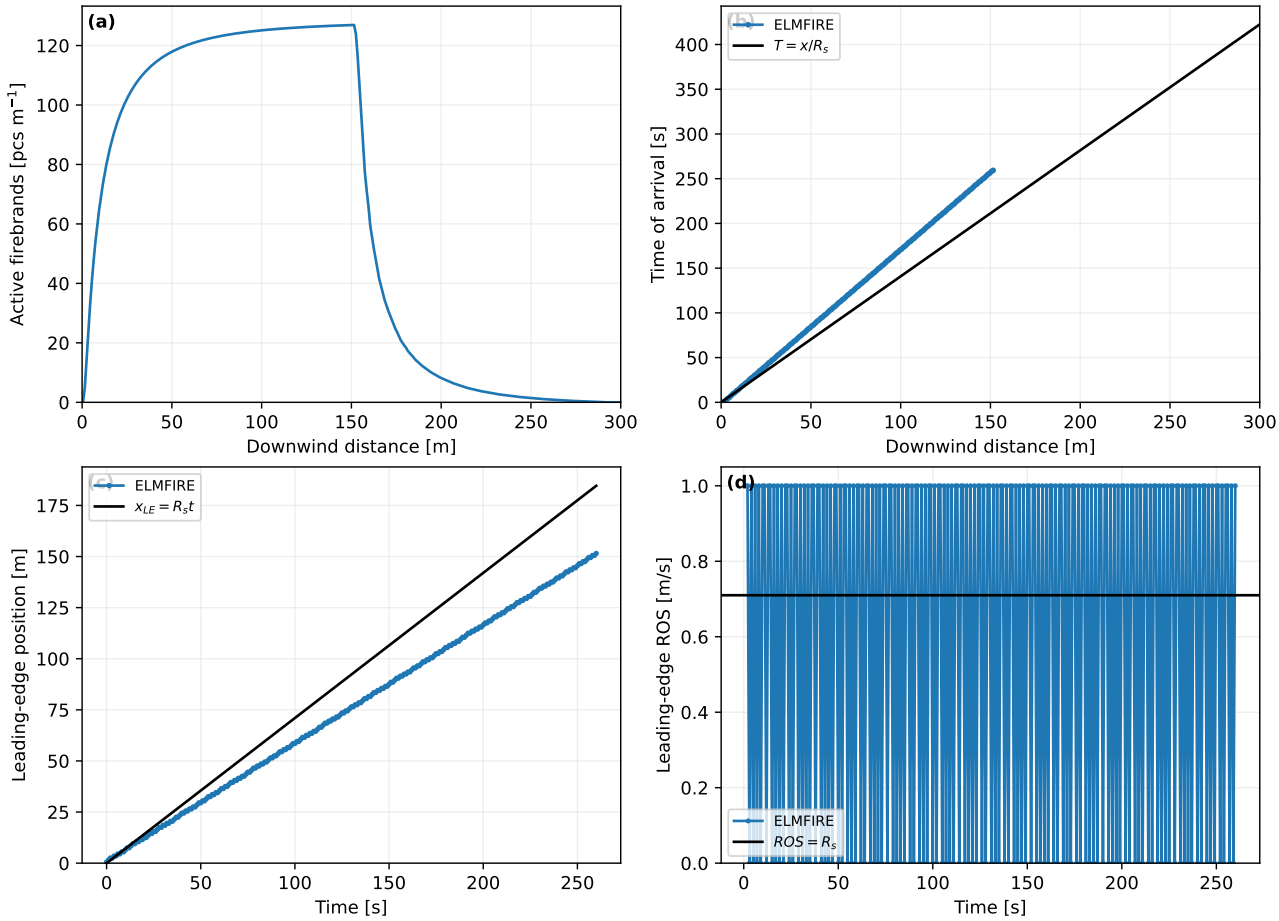


Figure 3: Consumption-on state at 1 m resolution. The top row gives the active firebrand and arrival-time profiles; the bottom row gives the reconstructed leading-edge trajectory and ROS with surface-spread references.

Figure 4 reproduces the temporal and spatial resolution views of the active population. The left panel samples the physical location nearest 200 m; the right panel selects the dump nearest 250 s.

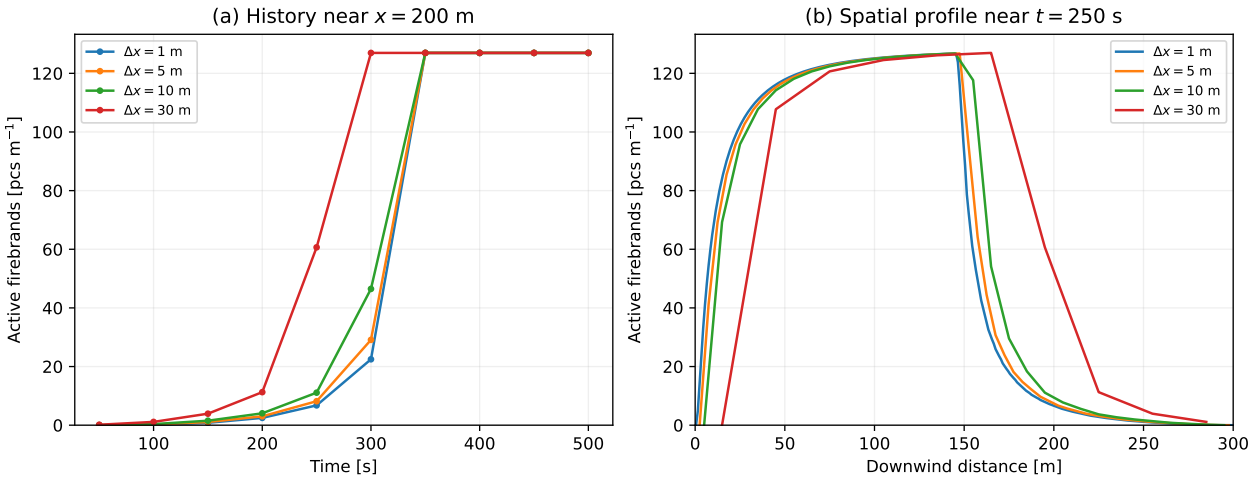


Figure 4: Consumption-model resolution sweep. Left: active-firebrand history near 200 m. Right: active-firebrand spatial profile near 250 s. Every curve is read from a predeclared stop-time run's final ELMFIRE raster.

Figure 5 is the supplemental paired diagnostic that makes the consumption-on/off effect and the 4% retention criterion explicit.

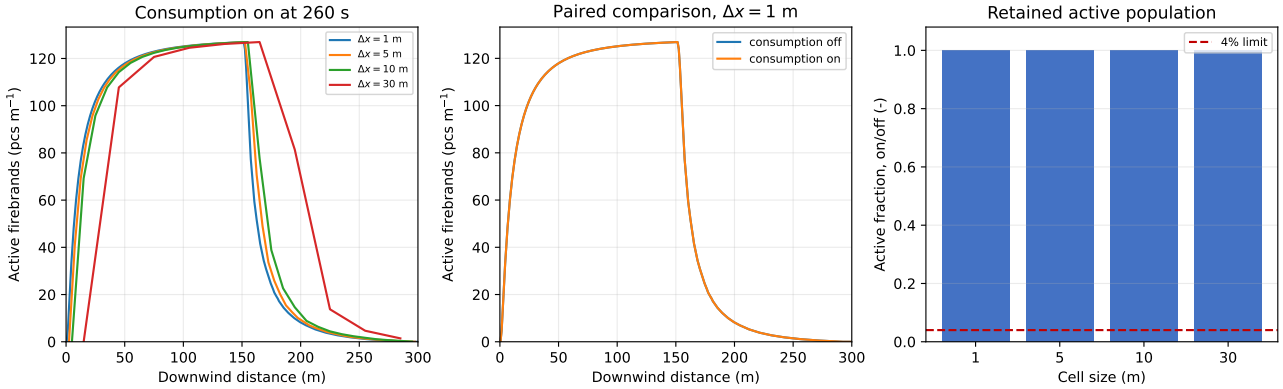


Figure 5: Paired consumption diagnostic. Left: consumption-on profiles by resolution. Center: paired 1 m profiles. Right: retained active fraction with the 4% acceptance limit.

7 Calculated metrics and verification decision

8 Technical interpretation

All four measured retained fractions are exactly 1.0, so enabling consumption does not change the active deposited population. The maximum paired TOA difference is 0 s, while the normalized profile errors are 0.0363, 0.0778, and 0.3511 at 5, 10, and 30 m, respectively. The source-normalized profiles therefore converge monotonically and the paired dynamics remain controlled, but the primary consumption metric fails decisively. Given the audited 10^9 s minimum lifetime, this is a source-code implementation defect rather than insufficient run duration. A TOA-only result cannot establish consumption behavior, which is why TOA appears here solely as a controlled invariance check.

Metric	Limit	Calculated	Status
Output completeness	48 simulation conditions	48/48	–
Maximum retained fraction	≤ 0.04	1	false
Maximum paired TOA difference	≤ 10 s	0	true
Minimum active count	$\geq -10^{-6}$	0	true
5 m normalized profile error	≤ 0.15	0.036343	true
Ordered profile convergence	required	–	true
Overall decision	all pass	false	**FAIL**

Table 3: Calculated verification metrics for the completed simulations.

9 Scope and limitations

This test verifies active-stock decay in one flat, uniform delayed-ignition baseline-like configuration. It does not validate the empirical heat-flux correlation, pile clustering, subgrid redistribution, or the 10 s lifetime against experiments. The 50 s sequence of independently stopped runs provides the active-stock history near 200 m and the spatial profile near 250 s directly from ELMFIRE rasters. The experiment still does not resolve within-cell firebrand clustering or validate the lifetime correlation against measurements.

References

[1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

Case 09: Mixed-Mode Ignition-Delay Effects

1 Verification purpose

This case verifies how firebrand ignition delay alters spread when surface propagation and spotting operate together. The surface-fire-dominated transport simulation condition delays firebrand development until the surface front controls advance; combined surface-fire and firebrand transport permits sufficiently distant deposits to develop ahead of that front.

The paired design exposes incorrect delay application, a missing transition between propagation modes, or an inconsistent surface-rate reference. Success requires surface-controlled behavior in the first simulation condition and quantified earlier arrival, spot ignition, and leading-edge gain in the second.

2 Mathematical formulation and numerical implementation

2.1 Surface-fire and firebrand time scales

The verification configuration is a flat, uniform fire behavior fuel model (FBFM) 102 grassland with wind speed $U_w = 6.71\,\mathrm{m\,s^{-1}}$. The surface head-fire spread rate U_s is read from the surface-dominated simulation condition spread rate raster produced by the Rothermel model in the tested ELMFIRE executable. For the reported run, $U_s = 0.577\,\mathrm{m\,s^{-1}}$. In the absence of a successful firebrand-induced ignition, the exact one-dimensional solution is

$$t_s(x) = \frac{x}{U_s}, \qquad x_{\mathrm{LE},s}(t) = U_s t, \qquad \overline{\mathrm{ROS}}_{\mathrm{LE},s} = U_s. \tag{1}$$

For a firebrand emitted at x_s and transported a downwind distance X, the earliest time at which it can produce a fully developed fire is

$$t_{\mathrm{spot}}(x_s, X) = \frac{x_s}{U_s} + \frac{X}{U_w} + t_{\mathrm{ign,small}} + t_{\mathrm{ign,large}}. \tag{2}$$

The surface front reaches the same target location at $t_s(x_s + X) = (x_s + X)/U_s$. With immediate smoldering-to-flaming transition, a spot fire can lead the surface front only when

$$X\left(\frac{1}{U_s} - \frac{1}{U_w}\right) > t_{\mathrm{ign,large}}. \tag{3}$$

2.2 ELMFIRE implementation exercised

The current implementation uses the non-superseded spotting path with these explicit selectors:

- generation proportional to fire power;
- an empirical transport-distance distribution; and
- Eulerian accumulation.

For each burning cell, the firebrand-generation calculation evaluates

$$\dot{N}_{\mathrm{cell}} = \frac{I_f\,\Delta y}{1000}\,GR^*, \tag{4}$$

where I_f is fireline intensity in kW m^{-1}, Δy is the cell width, and $GR^* = 33.3\,\mathrm{s}^{-1}\,\mathrm{MW}^{-1}$. The factor $I_f \Delta y/1000$ is the heat-release rate of the cell in MW. The Eulerian transport calculation deposits these particles according to the configured Sardoy-type distribution ($\mu = 2.18$, $\sigma = 1.23$), truncated at an effective maximum distance of approximately 150 m.

The ignition calculation first converts the deposited number to a number per unit area. In the surface-dominated simulation condition, the physical ignition model applies the wildland mass-load criterion before sampling SFT, with $\vartheta_{\text{ign,small}} = 42$ s and $t_{\text{ign,large}} = 300$ s. For this parameter set, the documented limiting behavior predicts that deposited mass remains below the physical threshold, so the SFT probability is zero [1]. In the mixed-mode simulation condition, the simple ignition model prescribes immediate SFT and a 150 s development delay. When the development delay expires, the level-set state, arrival-time field, and ember-ignition diagnostic are updated by the same firebrand ignition event.

3 Derivation of expected behavior

At the maximum transport distance $X_{\text{max}} = 150$ m, flight takes

$$t_f = \frac{150}{6.71} = 22.4\,\text{s}, \qquad t_s(150) = \frac{150}{0.577} = 260.0\,\text{s}.$$

For surface-dominated simulation condition, even the deliberately optimistic assumption of immediate SFT would give $t_f + t_{\text{ign,large}} = 322.4$ s, later than the surface arrival. The expected result is therefore Equation 1, with no cells attributed to firebrand ignition.

For mixed-mode simulation condition, the earliest full spot fire at X_{max} occurs at $22.4 + 150 = 172.4$ s. Moreover, the left side of Equation 3 is 237.6 s, which exceeds the 150 s development delay. Long-distance deposits can therefore mature ahead of the primary front. The expected signature is an acceleration near 170 s, at least one firebrand-ignited cell, a leading edge ahead of $U_s t$, and a mean spread rate greater than U_s over the common analysis interval. These are the qualitative features expected from mixed-mode spread.

The lead condition can also be solved for the minimum useful spotting distance:

$$X_{\text{crit}} = \frac{t_{\text{ign,large}}}{1/U_s - 1/U_w}, \qquad \frac{1}{0.577} - \frac{1}{6.71} = 1.584071\ \text{s m}^{-1}.$$

For the surface-dominated 300-s delay, $X_{\text{crit}} = 300/1.584071 = 189.39$ m, which exceeds the configured $X_{\text{max}} = 150$ m; even the longest deposit cannot lead. For the mixed 150-s delay, $X_{\text{crit}} = 94.69\,\text{m} < 150$ m, so the tail $94.69 < X \le 150$ m can mature ahead of the surface front. At $X = 150$ m,

$$t_f = 150/6.71 = 22.3547\ \text{s}, \quad t_s = 150/0.577 = 259.9653\ \text{s}, \quad t_s - t_f = 237.6106\ \text{s}.$$

Thus the mixed earliest full-spot scale is $22.3547 + 150 = 172.3547$ s, whereas the surface-dominated optimistic scale is $22.3547 + 300 = 322.3547\ \text{s} > t_s$. These substitutions yield the explicit reference signatures used by the metrics: zero surface-dominated ember ignitions, positive mixed-mode ember ignitions, mixed $ROS/U_s \ge 1.10$, at least 20 m edge and paired-time of arrival (TOA) advantages, and onset in the 120–220 s interval around 172.35 s.

4 Verification metrics and acceptance criteria

The postprocessor excludes the two-cell halo and operates on the centre-row TOA and spread-rate profiles. It converts the positive median Rothermel spread rate value from ft/min to m/s and uses it as U_s throughout the reference calculations. It fits $t = a + x/U$ over $50\,\text{m} \le x \le 280\,\text{m}$ using at least 12 cells. The following independently informative conditions must all pass:

1. surface-dominated simulation condition mean TOA relative error against x/U_s is at most 0.10.
2. surface-dominated simulation condition fitted rate of spread (ROS) relative error against U_s is at most 0.10.
3. surface-dominated simulation condition contains no firebrand-ignited physical cells.
4. mixed-mode simulation condition contains at least one firebrand-ignited physical cell.
5. mixed-mode simulation condition fitted ROS is at least $1.10U_s$.
6. At 250 s, its leading edge is at least 20 m ahead of $U_s t$.
7. Its first one-cell leading-edge gain occurs from 120 to 220 s, bracketing the derived 172.4 s scale.
8. Its median TOA is at least 20 s earlier than surface-dominated simulation condition over the shared fitting interval.

Missing outputs do not pass by default. Both simulation conditions must provide final TOA, accumulated-firebrand, and ember-ignition GeoTIFFs before any case-level pass decision is possible.

4.1 Justification of metric quantification

The two 10% surface-baseline limits allow cell-centred TOA and regression quantization while requiring agreement with the run-derived Rothermel speed over a 230 m fitting interval, more than twenty grid cells. The zero spot count in the surface-dominated simulation condition and the positive count in the mixed simulation condition are exact mechanism checks and therefore have no numerical tolerance. For the mixed mode, $ROS/U_s \geq 1.10$ requires a 10% effect beyond the baseline fitting band, while the 20 m edge and paired-TOA leads equal two spatial cells and are too large to arise from one-cell front classification alone.

The 120–220 s onset window brackets the independently derived 172.4 s earliest full-spot scale by approximately ± 50 s. That width accommodates stochastic deposition, cell crossing, and the fact that the metric detects the first full-cell gain rather than a continuous event, but still rejects acceleration unrelated to the predicted delay regime. The eight behavior checks are conjunctive because a faster fitted front without ember attribution, timing consistency, and a paired TOA lead would not verify firebrand-delay coupling.

4.2 Predeclared acceptance criteria

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Output completeness	Final TOA, accumulated-firebrand, and ember-ignition rasters for both current simulation conditions; halo and no-data excluded.	2 of 2 simulation conditions
Surface-dominated TOA error	Mean relative error against x/U_s on 50–280 m; at least 12 cells.	≤ 0.10
Surface-dominated ROS error	Relative error of fitted ROS against run-derived Rothermel U_s.	≤ 0.10
Surface-dominated spot count	Number of ember-ignited physical cells.	0
Mixed-mode spot count	Number of ember-ignited physical cells.	≥ 1
Mixed-mode ROS ratio	Fitted mixed-mode ROS divided by U_s.	≥ 1.10
Mixed-mode edge gain	Leading-edge distance ahead of $U_s t$ at 250 s.	≥ 20 m
Acceleration onset	First time the leading edge exceeds the surface reference by one cell.	120–220 s
Paired TOA lead	Median surface-dominated minus mixed-mode TOA over common fitting cells.	≥ 20 s
Overall decision	Conjunction of completeness and all eight behavior checks.	PASS only if all pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for combined surface-fire and firebrand transport. These panels show the prepared spatial fields, remove the two-cell numerical buffer, and retain map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

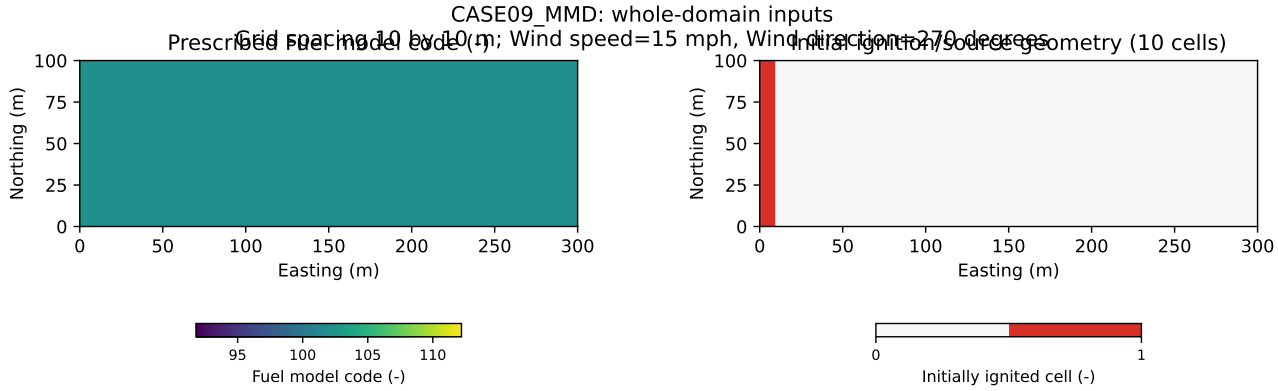


Figure 1: Whole-domain prepared input configuration for the representative simulation condition. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

The Courant–Friedrichs–Lewy (CFL) number measures the distance advected during one timestep relative to the grid spacing and constrains the time discretization.

Table 2: Parameters common to both verification simulation conditions.

Parameter	Value	Verification role
Physical domain	300×100 m	one-dimensional centre transect
Raster domain	34×14 cells	includes two-cell halo
Cell size	10 m	controlled baseline resolution
Time step / wind CFL	0.742 574 s / 0.4979	common exact-duration divisor
Fuel and terrain	FBFM 102, flat	homogeneous surface spread
Wind	6.71 m s^{-1} toward $+x$	firebrand transport
Ignition geometry	full crosswind line at $x = 5$ m	1-D behavior
Generation	PER-MW, $GR^* = 33.3$	heat-release-scaled source
Transport / accumulation	empirical / Eulerian	reference transport path
Surface fire	enabled	general coupled fire spread

Table 3: Ignition parameters changed by the paired experiment.

Simulation condition	Ignition model	$t_{\text{ign,small}}$	$t_{\text{ign,large}}$	Prediction
surface-dominated simulation condition	PHYSICAL	42 s scale	300 s	surface dominated
mixed-mode simulation condition	SIMPLE	0 s	150 s	mixed-mode acceleration

5.1 Preprocessing, execution, and postprocessing

The two simulation conditions use independently prepared, co-registered spatial fields. Both represent flat terrain, fuel model 102, and constant wind; places a line ignition across every physical crosswind row; and writes explicit model selectors and ignition parameters into each local ELMFIRE simulation configuration. The halo is present numerically but is excluded from all reported quantities. The nominal 0.5-wind-CFL step is 0.745645 s. Because the controlled stops are 300 and 450 s, their greatest common duration is 150 s; preprocessing selects $N = \lceil 150/0.745645 \rceil = 202$ and the common $\Delta t = 150/202 = 0.742574$ s. Both stops are therefore exact timestep multiples and the actual wind CFL remains below 0.5.

Both prescribed conditions are simulated independently with ELMFIRE. Their final physical fields are compared using the surface-rate reference and the arrival-time, ignition, and leading-edge measures defined above.

6 Actual simulation results

Figure 2 shows the representative time-of-arrival field from combined surface-fire and firebrand transport. The displayed field is taken from the simulated results, masks missing values, and excludes the two-cell numerical buffer. The domain map complements the spread-rate summaries by showing the combined surface-fire and delayed-firebrand arrival pattern.

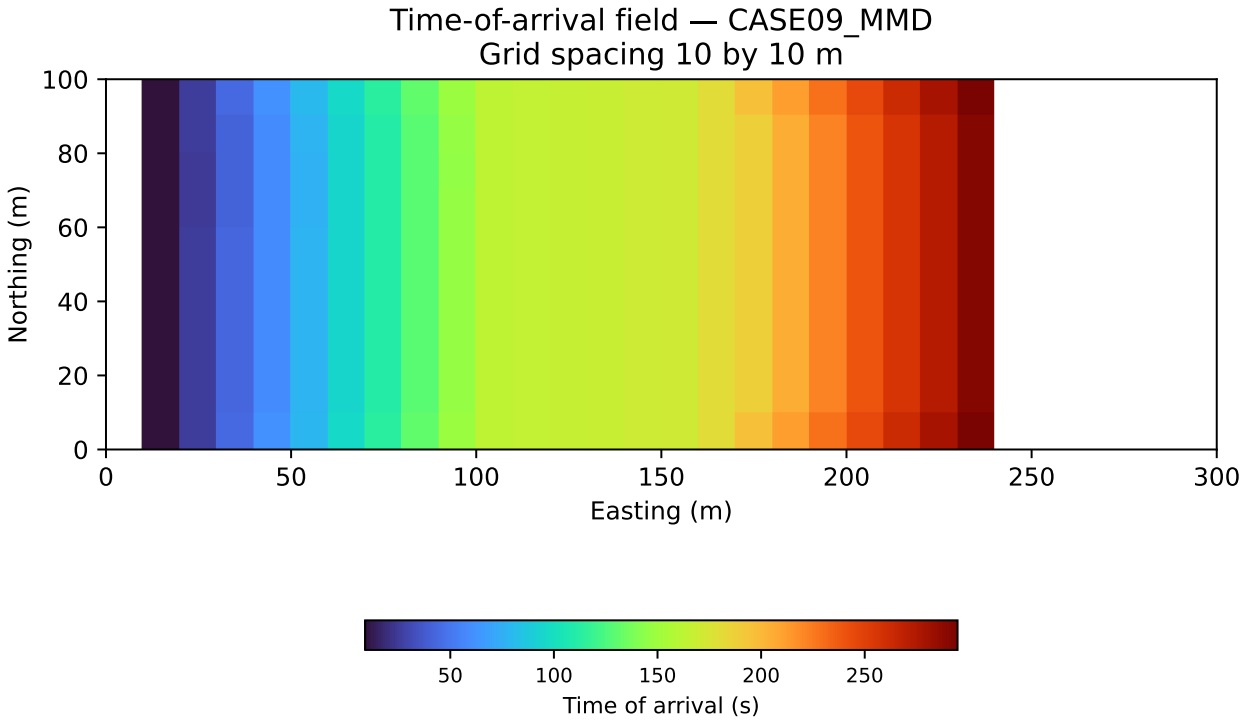


Figure 2: Whole-domain time-of-arrival field from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

Figures 3 and 4 first present each ignition regime independently using the same four observables: accumulated firebrands, arrival time, leading-edge position, and leading-edge ROS. Colored curves are actual ELMFIRE outputs; black curves are the Rothermel surface-spread references.

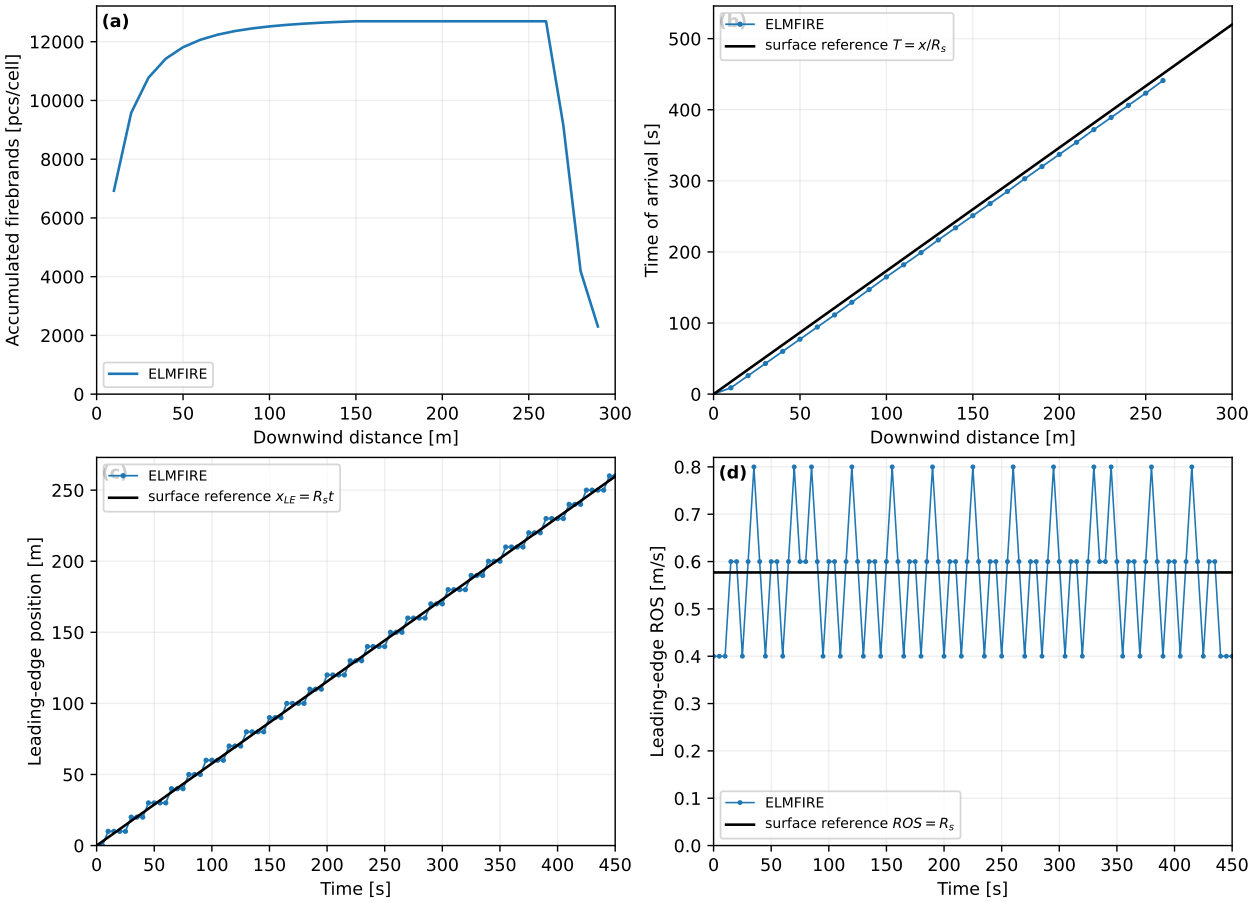


Figure 3: Long-delay surface-dominated simulation condition. The leading edge remains close to the surface-spread characteristic despite continued firebrand deposition.

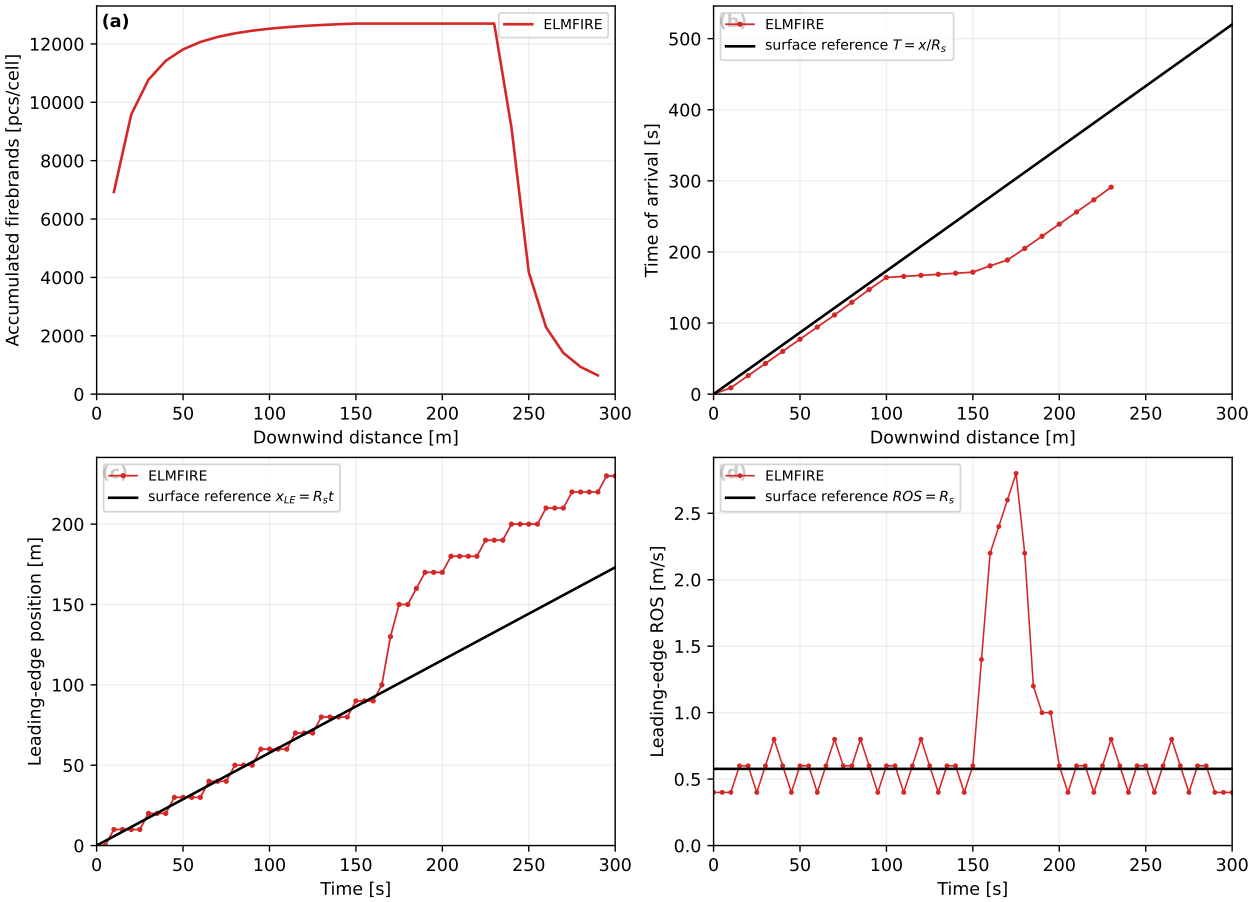


Figure 4: Finite-delay mixed-mode simulation condition. Departure from the surface-only arrival, trajectory, and ROS references identifies firebrand-induced advance.

Figure 5 overlays the two simulation conditions as a supplemental controlled comparison, making the timing and magnitude of their divergence easier to identify.

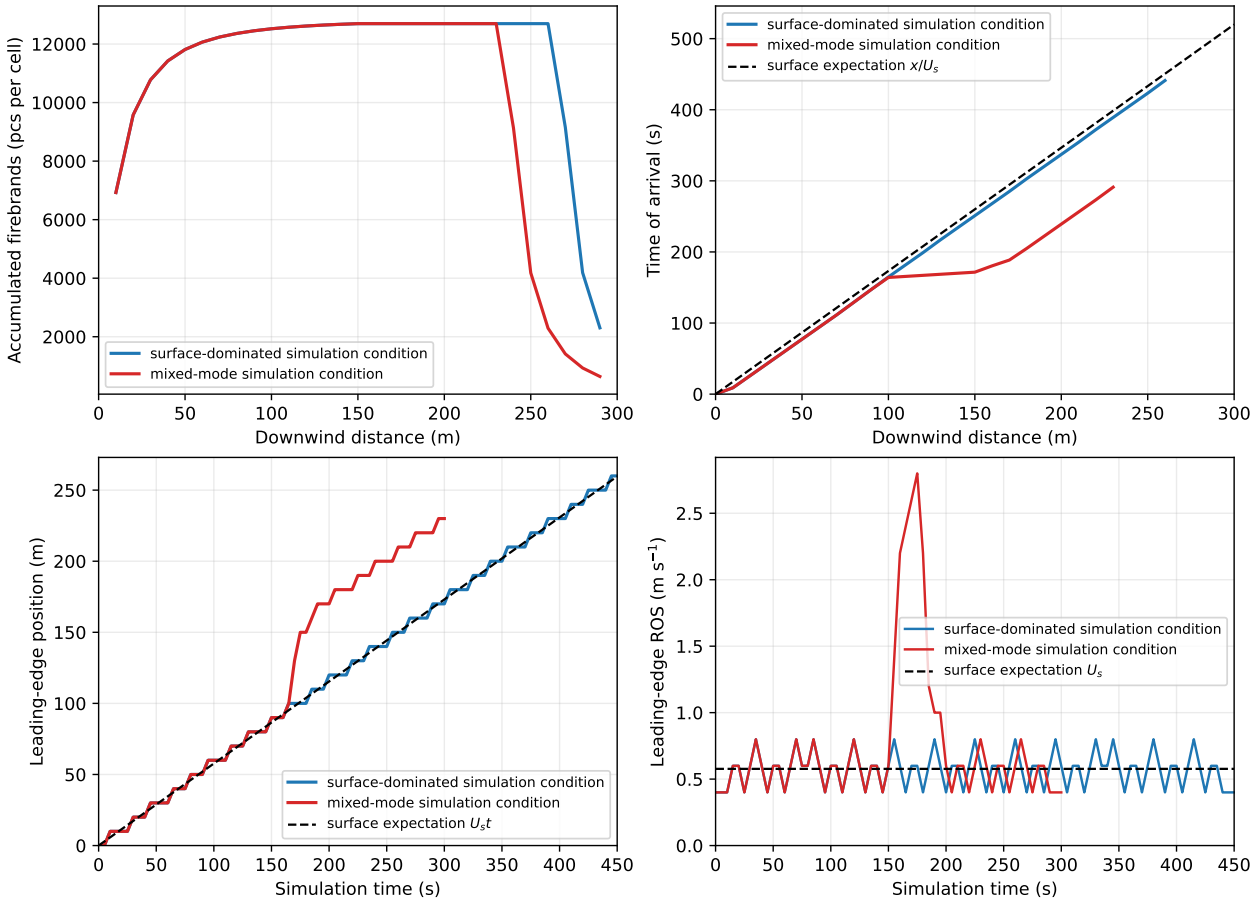


Figure 5: Paired ELMFIRE comparison of the long-delay surface-dominated and finite-delay mixed-mode regimes. Accumulation is crosswind averaged; the other panels use the centre-row arrival-time field.

7 Calculated metrics and verification decision

Table 4 reads the calculated entries directly from calculated verification and validation metrics. An empty simulation result is represented as N/A with NOT EVALUABLE, never as an empty table cell or an implicit pass.

Table 4: Acceptance metrics calculated from the paired ELMFIRE outputs.

Metric	Acceptance limit	Calculated	Status
Output completeness	2 simulation conditions	2/2	PASS
surface-dominated simulation condition mean TOA error	≤ 0.10	0.043	PASS
surface-dominated simulation condition mean ROS (m s^{-1})	$0.577 \pm 10\%$	0.578	PASS
surface-dominated simulation condition ember-ignited cells	≤ 0	0	PASS
mixed-mode simulation condition ROS / U_s	≥ 1.10	1.743	PASS
mixed-mode simulation condition ember-ignited cells	≥ 1	64	PASS
mixed-mode simulation condition edge gain at 250 s (m)	≥ 20	55.746	PASS
mixed-mode simulation condition acceleration onset (s)	120–220	170.0	PASS
Median mixed-mode simulation condition TOA lead (s)	≥ 20	63.861	PASS
Overall decision	all components pass	PASS	PASS

8 Technical interpretation

The current paired execution passes every component. The surface-dominated mean ROS is 0.578 m/s with no ember-ignited cells. The mixed-mode mean ROS is 1.006 m/s (1.743 times the surface reference), with 64 ember-ignited cells, a 55.746-m leading-edge gain, acceleration onset at 170.0 s, and a median TOA lead of 63.861 s.

A pass demonstrates two complementary properties: delayed firebrand states do not spuriously outrun the surface solver in the surface-dominated limiting case, and reducing the delay activates a resolvable acceleration in the mixed-mode simulation condition. If surface-dominated simulation condition fails alone, inspect the physical mass threshold, development-state timing, and arrival-time update. If mixed-mode simulation condition fails alone, inspect the simple ignition probability, persistence of deposited ember states across level-set steps, delayed transition to a full ignition, and the coupling that writes that ignition to the level set.

This is numerical verification of the controlled formulation, not field validation. The ELMFIRE experiment is two-dimensional but deliberately uses a uniform crosswind line so the centre transect approximates the corresponding one-dimensional calculation. The mixed-mode thresholds diagnose the expected mixed regime rather than an unavailable closed-form mixed-mode solution. Stochastic ignition means the exact onset may vary, which is why its acceptance window brackets rather than equals 172.4 s.

The current analysis note is: All metrics were calculated from the two actual ELMFIRE simulation condition outputs; the surface reference is the Rothermel ROS read from the surface-dominated simulation condition VS raster.

9 Scope and limitations

This paired test isolates the interaction between surface propagation and firebrand ignition delay in a uniform domain. It verifies implementation behavior and timing attribution; it does not validate either spread mechanism against field observations.

References

[1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

Case 10: WUI Structural Resolution Sensitivity

1 Verification purpose

This case verifies sensitivity of firebrand-driven WUI spread to structural spatial representation. Five simulation conditions cover the baseline grid, native fine and coarse grids, and aligned and misaligned integrated-structure representations under one physical geometry and source definition.

The test quantifies native cell-scale distortion and determines whether integrated representation restores structure-level accumulation, ignition chronology, and spread-rate consistency. Capability-gated simulation conditions remain explicit and are not replaced by a physically different native-cell calculation.

2 Mathematical formulation and numerical implementation

2.1 Heat release and generation

For a structure ignited at t_i, the design heat-release-rate (HRR) per unit area (HRRPUA) is

$$q''(\tau) = \begin{cases} q''_{\max}\tau/300, & 0 < \tau < 300, \\ q''_{\max}, & 300 \le \tau \le 3900, \\ q''_{\max}(4200-\tau)/300, & 3900 < \tau < 4200, \\ 0, & \text{otherwise}, \end{cases} \qquad \tau = t - t_i, \qquad q''_{\max} = 400\,\mathrm{kW\,m^{-2}}.$$

For footprint A_s, $\mathrm{HRR}_i = q'' A_s$, and the emission rate is

$$\dot{N}_i(t) = G'_R \mathrm{HRR}_i(t), \qquad G'_R = 10\,\mathrm{pcs\,MW^{-1}\,s^{-1}}.$$

The configuration therefore uses per-MW generation; per-area generation would not preserve heat-release-scaled production for this design fire.

2.2 Himoto transport and analytical load

The downwind distance is lognormal, with physical-space moments

$$\mu_x/D = 0.47(B^*)^{2/3}, \qquad \sigma_x/D = 0.88(B^*)^{1/3},$$

$$B^* = \frac{U}{\sqrt{gD}} \left(\frac{\rho_p}{\rho_\infty}\right)^{-3/4} \left(\frac{d_p}{D}\right)^{-3/4} \left(\frac{\mathrm{HRR}}{\rho_\infty c_p T_\infty g^{1/2} D^{5/2}}\right)^{1/2}.$$

Here $D = 10\,\mathrm{m}$, $\rho_p = 100\,\mathrm{kg\,m^{-3}}$, and $d_p = 5\,\mathrm{mm}$, matching the reference formulation and current source. The postprocessor independently evaluates the analytical deposited-load convolution

$$N_k = \sum_{i<k} \int_0^T \dot{N}_i(t) \left[\int_{x_k}^{x_{k+1}} f(x - x_i; \mu_i(t), \sigma_i(t))\, dx \right] dt$$

with a 5 s quadrature interval. Deposited counts are divided by physical footprint area, not a resolution-dependent raster-cell area.

2.3 Physical ignition and expected chronology

The configured ignition model uses $N_{\text{crit}} = 2928.9\,\text{pcs}\,\text{m}^{-2}$ [1]. At the structure spanning $20 \leq x \leq 30\,\text{m}$, the historical analytical load reaches this threshold near 2496.9 s. The case-specific postprocessor preserves the published composite full-ignition reference of 2858.8 s; its difference from direct addition of the currently configured delays is quantified explicitly below. Targets beyond the roughly 80 m peak-heat-release rate (HRR) spotting range require successive structure ignition and consequently have nonlinear arrival times.

2.4 Resolution signatures

On the 2.5 m native mesh (native fine-grid simulation condition), one physical structure occupies multiple independent cells. Uneven deposition and cell-local ignition should produce a nonzero within-structure load range and mean rate of spread (ROS) near $0.001\,\text{m}\,\text{s}^{-1}$, compared with about $0.006\,\text{m}\,\text{s}^{-1}$ for the exact-grid baseline. On the 30 m native mesh (native coarse-grid simulation condition), mode resampling inflates footprint and HRR, causing artificially early spread; the documented response gives a baseline-to-coarse load ratio near 1.86 [1].

The integrated-structure treatment groups all cells belonging to one structure,

$$\bar{N} = \frac{\sum_{j \in s} N_j}{A_s}, \qquad \text{HRR}_s = q'' A_s,$$

then redistributes load uniformly and emits from the structure centroid. The integrated-aligned simulation condition expects an aligned 2.5 m mesh to reproduce the baseline. The integrated-misaligned simulation condition permits up to 33% load deviation and 10% ignition-time deviation on a misaligned 3.24 m mesh.

3 Derivation of expected behavior

The exact grid should reproduce the analytical structure-load and ignition chronology. Native fine cells should expose within-structure imbalance, native coarse cells should expose footprint inflation, and integrated simulation conditions should recover the physical-structure response independently of alignment.

At the $400\,\text{kW}\,\text{m}^{-2}$ plateau, one physical 10×10 m structure has

$$HRR_{\text{max}} = 400(100) = 40000\ \text{kW} = 40\ \text{MW}, \qquad \dot{N}_{\text{max}} = 10(40) = 400\ \text{pcs}\,\text{s}^{-1},$$

and its critical total count is $N_{\text{crit}} = 2928.9(100) = 292890$ particles. Using the postprocessor's adjusted wind $U = 0.447(40)/0.87 = 20.5517\ \text{m}\,\text{s}^{-1}$ at peak HRR gives

$$B^* = 7.37372, \qquad \bar{x} = 0.47(7.37372)^{2/3}(10) = 17.8056\ \text{m},$$

$$s_x = 0.88(7.37372)^{1/3}(10) = 17.1282\ \text{m}, \qquad (\mu, \sigma) = (2.55195, 0.80939).$$

These values are inserted in the cell-cumulative distribution function (CDF) convolution at each 5-s midpoint; the HRR dependence is recomputed rather than freezing the peak values.

There is an explicit provenance discrepancy in the historical chronology. The case-specific constants give

$$2496.9 + 42.1 + 300 = 2839.0\ \text{s},$$

whereas the preserved historical composite reference used by postprocessing is 2858.8 s, 19.8 s later. The latter is therefore treated as a published numerical estimate, not as the arithmetic result of the three

displayed constants. The 19.8-s difference is only $19.8/2858.8 = 0.00693$, well inside the predeclared 10-percent baseline ignition tolerance, so this provenance mismatch cannot change that component's decision.

The exact-grid ideal errors are zero for analytical load and the chosen historical ignition reference. Native-grid expected signatures are a fine/baseline ROS ratio near $0.001/0.006 = 0.1667$ and a coarse load ratio near 1.86. Integrated aligned references are the exact-grid structure loads and times; for the misaligned grid, one unresolved boundary strip is $3.24/10 = 0.324$, motivating the explicit 0.33 load allowance.

4 Verification metrics and acceptance criteria

The postprocessor calculates every metric explicitly. The exact-grid baseline compares structure-integrated load with the analytical convolution defined above and second-structure ignition with 2858.8 s. The native fine and coarse simulation conditions verify the documented cell-local supporting results rather than claiming convergence. The integrated aligned and misaligned simulation conditions compare physical-structure profiles with the exact-grid baseline. Missing output or capability is reported as *NOT EVALUABLE*, never as an implicit pass.

4.1 Justification of metric quantification

The criteria intentionally separate reproduction of native-grid defects from accuracy requirements for the baseline and integrated treatments. The 10% baseline load and ignition bands accommodate rasterized deposition and threshold-crossing time while remaining small relative to the resolution effects documented for this configuration [1]. A native fine-grid load range of at least 5% is set above round-off and output quantization, and a ROS ratio no greater than 0.50 requires the material factor-of-two slowdown that identifies independent-cell behavior rather than a visually detectable but negligible difference. The coarse-grid load ratio is centered on the reported 1.86 response; its 25% band permits the severe footprint snapping intrinsic to a 30 m cell without accepting a reversal of the expected inflation.

For the aligned integrated grid, exact structure boundaries and conservation of integrated load justify the stringent 1% load limit. Its 5% ignition limit is wider because ignition is a nonlinear threshold event recorded at a discrete time. For the 3.24 m misaligned grid, one boundary-cell width relative to a 10 m structure is $3.24/10 = 0.324$; the 0.33 maximum load-error limit therefore admits approximately one unresolved boundary strip but not a larger structure-scale bias. The corresponding 10% ignition bound allows threshold amplification of that overlap error. Capability and completeness are conjunctive because the integrated claims cannot be inferred from native-grid substitutes.

4.2 Predeclared acceptance criteria

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Baseline load error	Mean structure-load relative error against the analytical deposition reference, excluding the source structure.	≤ 0.10
Baseline ignition error	Relative error of second-structure ignition time against 2858.8 s.	≤ 0.10
Native fine-grid load distortion	Median within-structure relative load range.	≥ 0.05
Native fine-grid ROS ratio	Fine-grid fitted mean ROS divided by baseline ROS.	≤ 0.50
Native coarse-grid load ratio	Baseline structure-4 load divided by coarse structure-2 load.	$1.86 \pm 25\%$
Integrated aligned load error	Mean matched structure-load relative error against baseline.	≤ 0.01
Integrated aligned ignition error	Mean matched ignition-time relative error against baseline.	≤ 0.05
Integrated misaligned load error	Maximum matched structure-load relative error against baseline.	≤ 0.33
Integrated misaligned ignition error	Maximum matched ignition-time relative error against baseline.	≤ 0.10
Overall decision	All configured simulation conditions and metrics must be evaluable; unavailable integrated capability prevents a PASS.	PASS only if all rows pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for the 10 m baseline grid. These panels show the prepared spatial fields, remove the two-cell numerical buffer, and retain map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

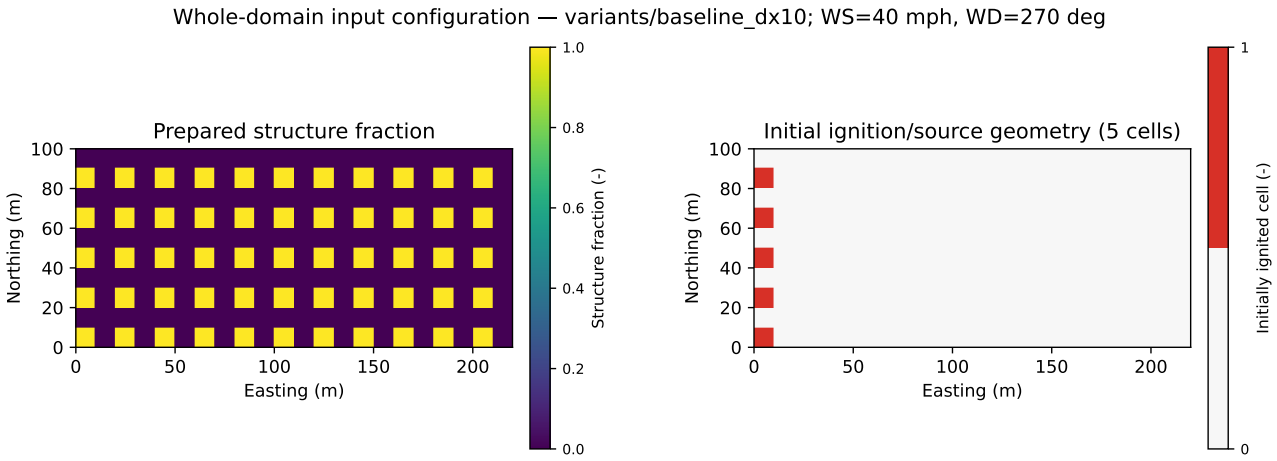


Figure 1: Whole-domain prepared input configuration for the representative simulation condition. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

Table 2: Five simulation conditions generated from the common physical problem.

Simulation condition	Δx (m)	$t_{\max}$ (s)	Treatment	Expected result
Baseline	10.00	22000	native	analytical deposition and ignition
Native fine	2.50	50000	native fine	imbalance and slower ROS
Native coarse	30.00	22000	native coarse	inflated HRR and earlier spread
Integrated aligned	2.50	22000	integrated aligned	baseline-equivalent profile
Integrated misaligned	3.24	22000	integrated misaligned	bounded profile error

Each nominal timestep is $\Delta t_{\text{nominal}} = 0.5\Delta x/U$; the generated step is no larger than this bound. All structure rasters reference a case-specific model 14 containing the design fire and physical delay. Structures form a periodic 220×100-m two-dimensional lattice. Each represented cell uses its true square area, so heat-release rate per unit area (HRRPUA) remains 400 kW m^{-2} and the dimensionless ignition-hardening factor remains one at every resolution. Every structure in the first upwind column is initialized at $t = 0$. The global vegetative surface-spread setting switch remains true to suppress vegetative surface-fire propagation. In the current source, initialized model-14 buildings independently refresh their transient HRR and therefore remain active under this setting. The generated non-structure fuel model 93 is also nonburnable, providing a second explicit isolation of the building-only experiment. Because the wind Courant–Friedrichs–Lewy (CFL) bound requires a subsecond step, preprocessing computes the nominal fractional value. When that value does not divide the requested duration, it uses $N = \lceil t_{\text{requested}}/\Delta t_{\text{nominal}} \rceil$ and reduces the generated step to $\Delta t = t_{\text{requested}}/N$. The stop therefore remains the exact requested duration and an exact timestep multiple; the simulation specification records nominal and actual steps, CFL values, and the integer step count.

5.1 Preprocessing, execution, and postprocessing

Exact cell–footprint overlap defines the structure fractions and identities; mode classification defines the native fine- and coarse-grid experiments. Only model-supported structure representations are simulated. Final firebrand counts are normalized by physical structure area and aggregated by structure identity, together with the fire arrival times. Evaluation requires agreement between the prescribed and simulated inputs, successful termination, unique stop-time, timestep, and meteorological-interval values, and exact divisibility of duration by timestep. One final output-time record and one corresponding deposition/arrival-time pair are required. An output beyond the nominal stop is accepted only if subtracting the meteorological interval gives a time on the timestep grid within one timestep of the stop. Ambiguous, early, off-grid, or duplicate evidence remains NOT EVALUABLE.

6 Actual simulation results

Figure 2 shows the representative time-of-arrival field from the 10 m baseline grid. The displayed field is taken from the simulated results, masks missing values, and excludes the two-cell numerical buffer. This completed baseline provides whole-domain evidence even though simulation conditions requiring the unavailable integrated-structure capability remain not evaluable.

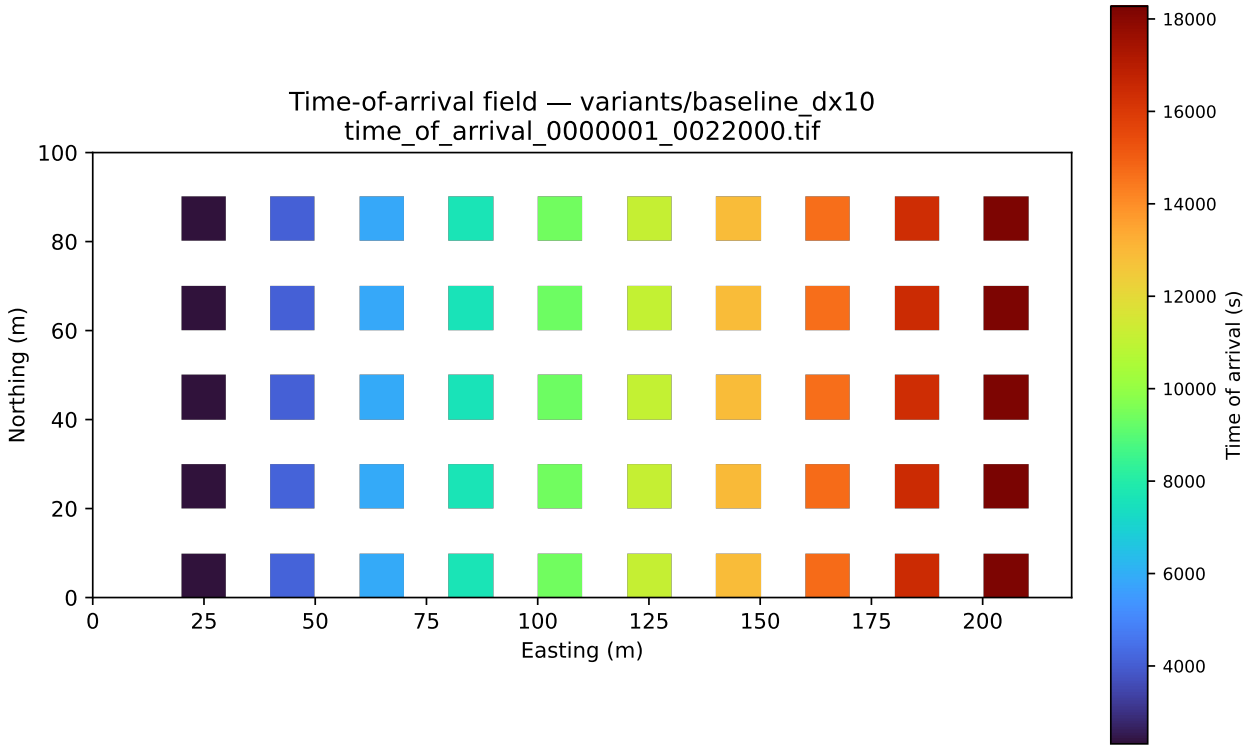


Figure 2: Whole-domain time-of-arrival field from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

2 of 5 simulation conditions currently have complete, configuration-matched ELMFIRE output. Source-aligned physical-state figures are provided for all three native representations; a figure explicitly states when its current matching output is unavailable. The two integrated-structure simulation conditions remain capability gated.

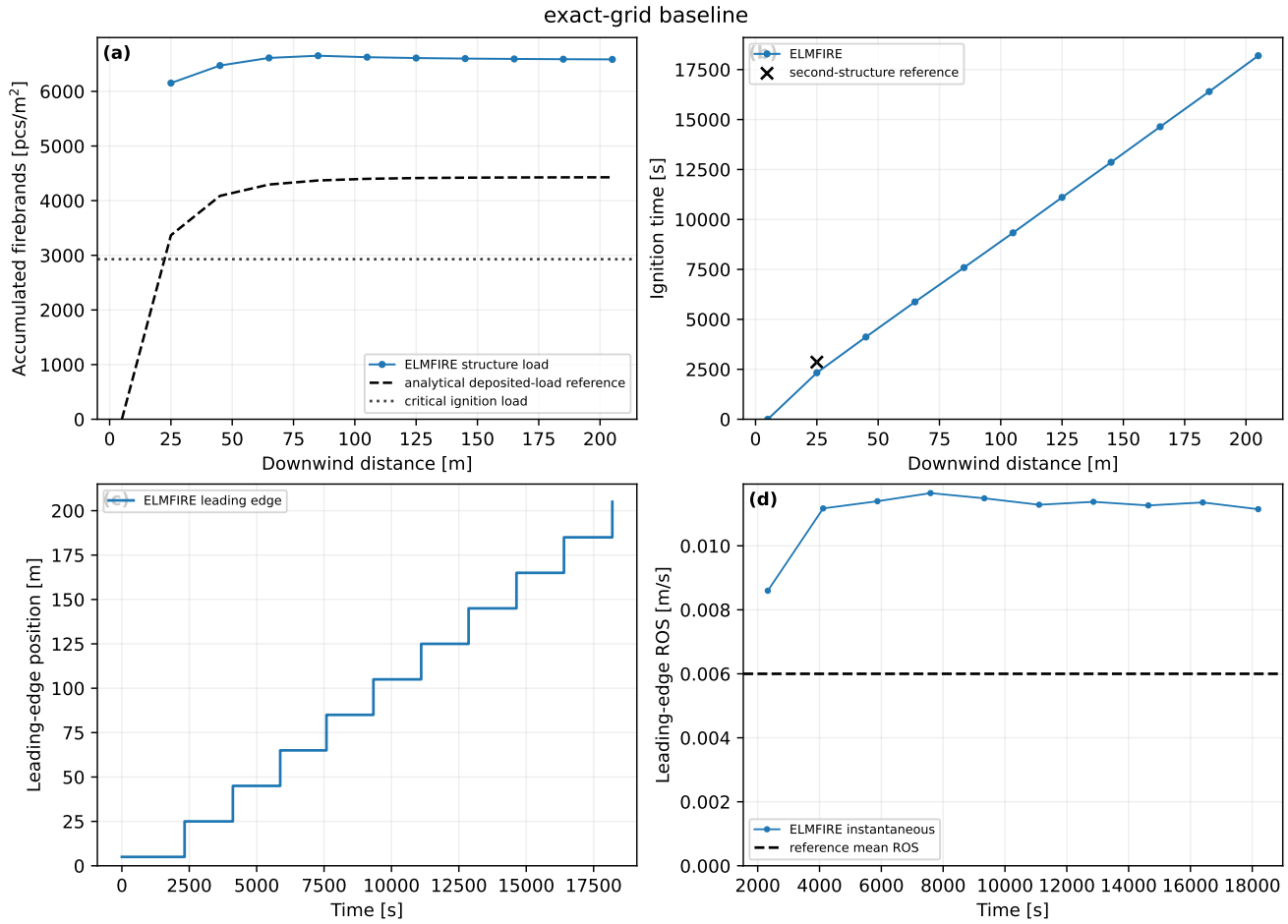


Figure 3: Exact-grid baseline observables: structure load, ignition chronology, leading-edge trajectory, and instantaneous ROS. Analytical load and second-structure timing references are overlaid where available.

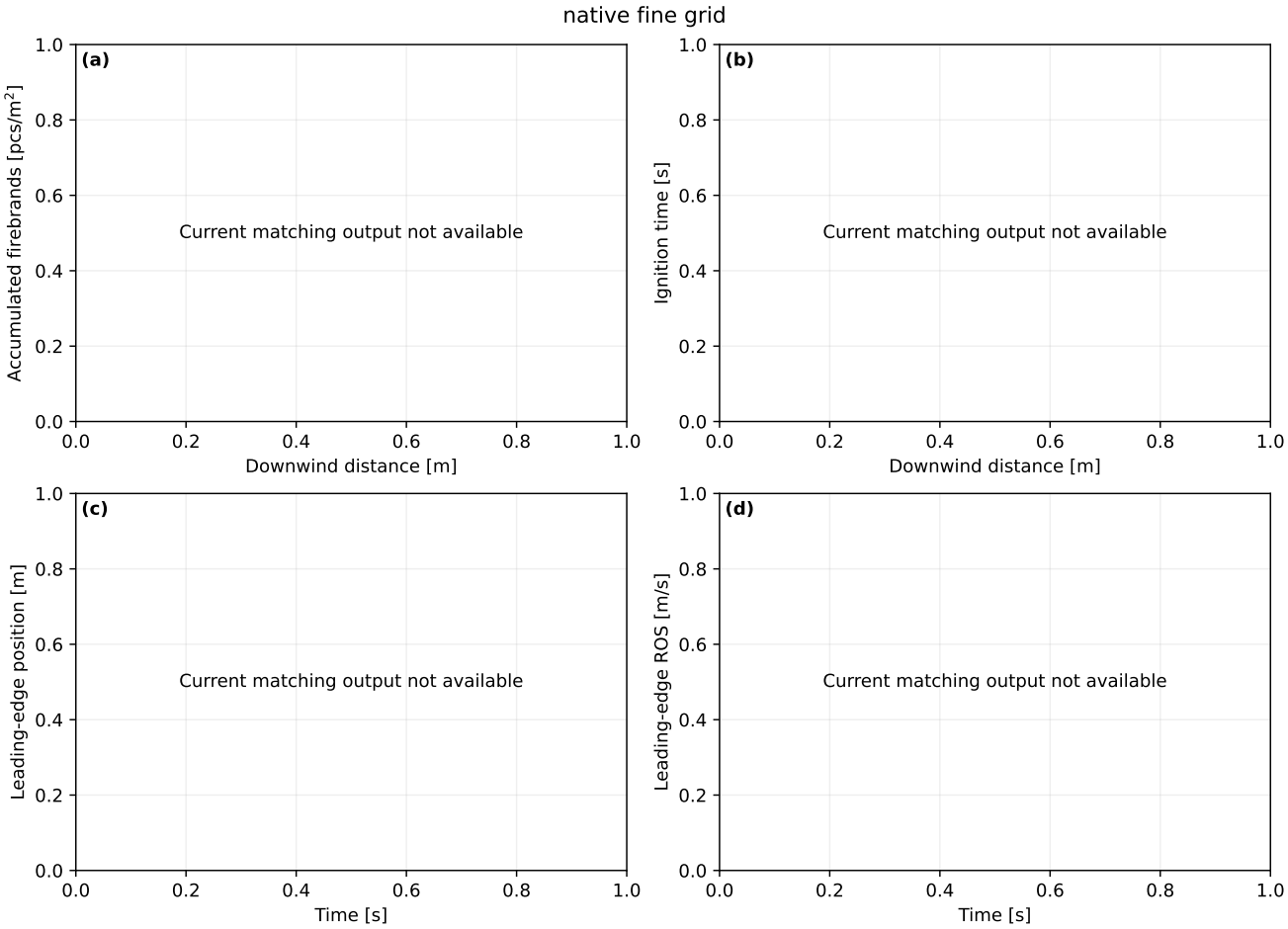


Figure 4: Native fine-grid observables. Cell-scale load variation within each physical structure is retained so discretization distortion is visible rather than hidden by structure averaging.

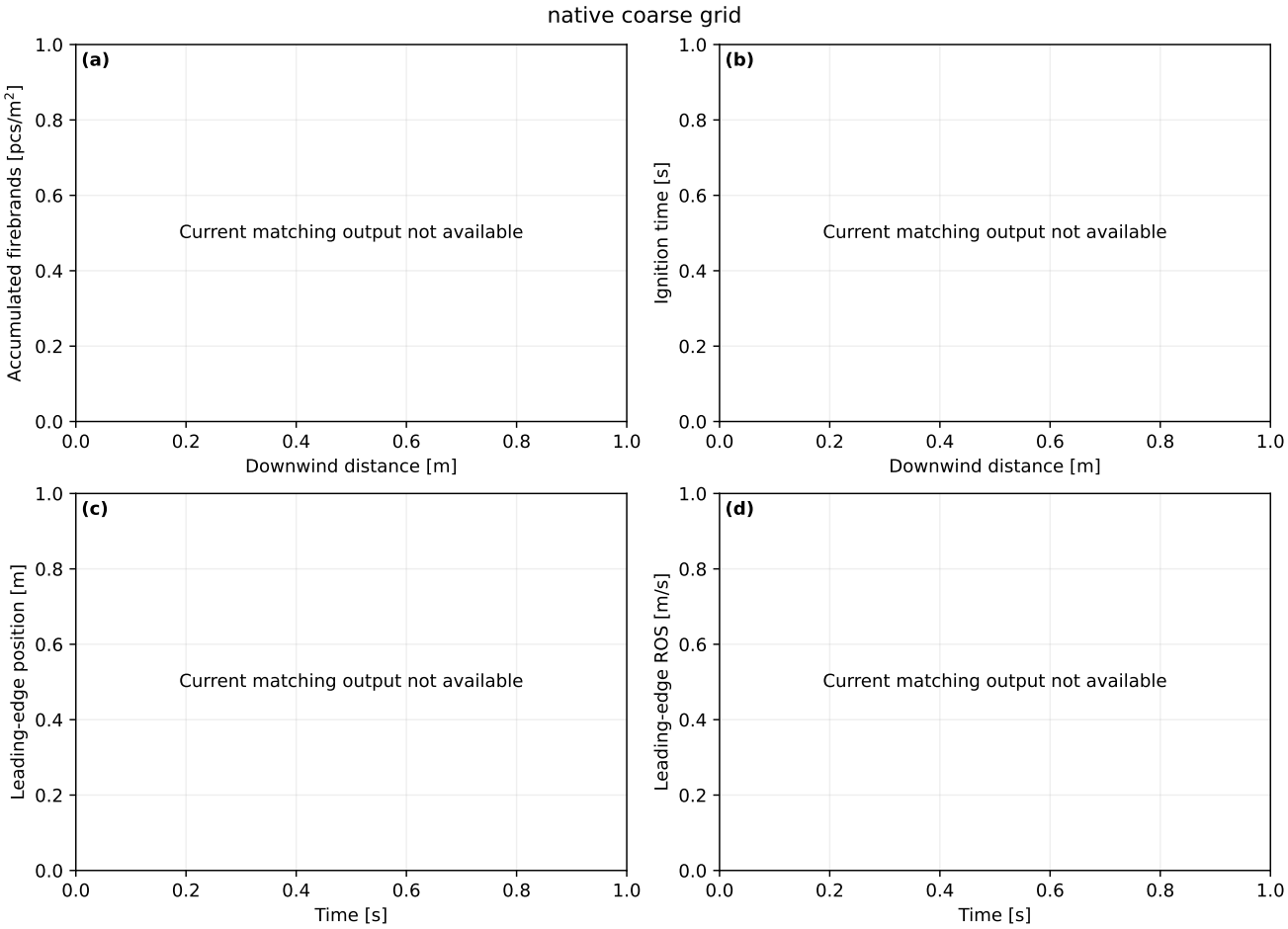


Figure 5: Native coarse-grid observables. Mode resampling changes the represented footprint and separation, and the resulting load and ignition histories are shown on their actual physical coordinates.

Figure 6 is the supplemental cross-simulation condition diagnostic. The native-test decision is **FAIL**; the complete-suite decision is **NOT EVALUABLE (capability missing)**.

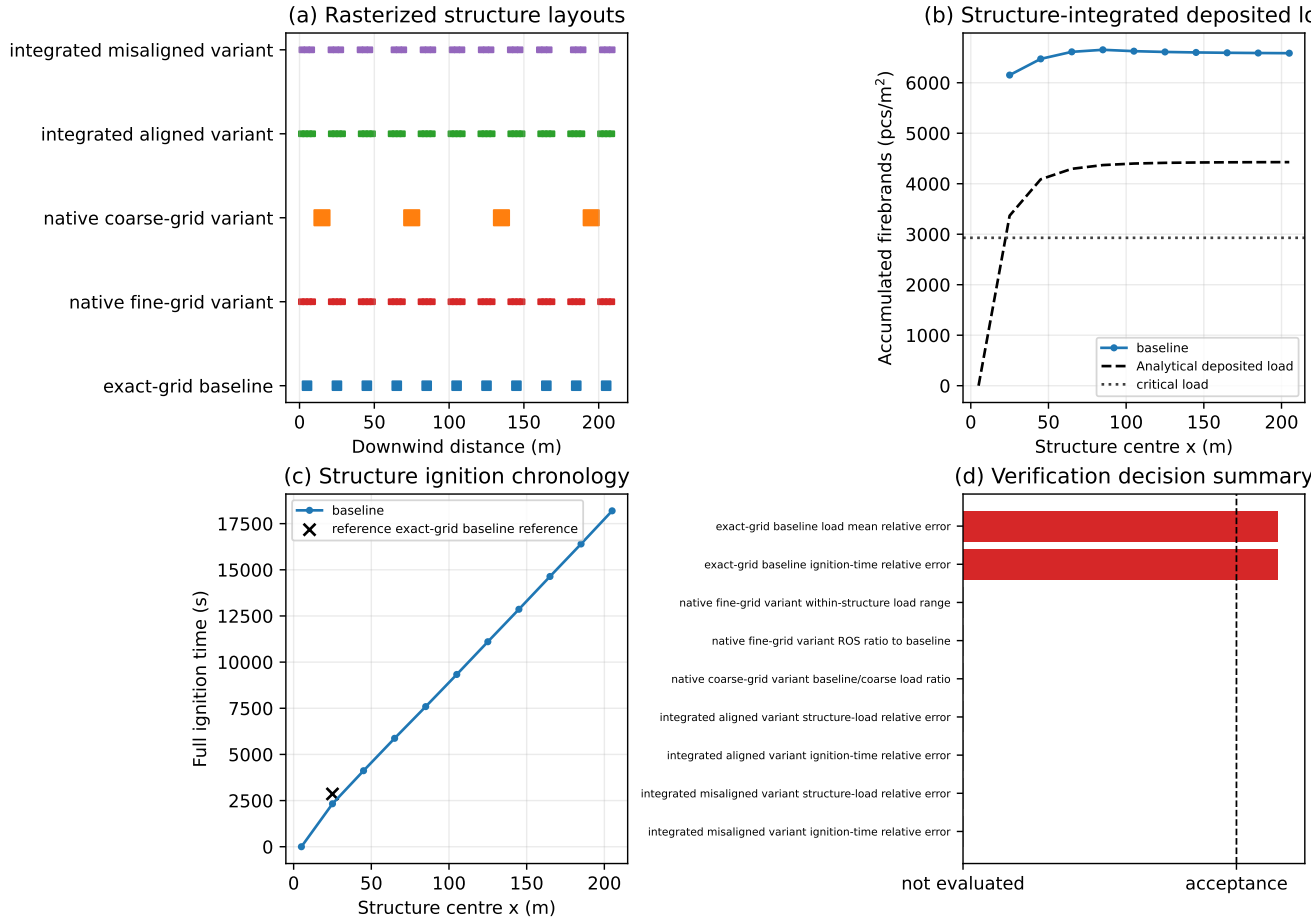


Figure 6: Structural-resolution summary. (a) Rasterized layouts; (b) deposited load by physical structure; (c) ignition chronology; and (d) component decision status, with unsupported components shown as not evaluated.

7 Calculated metrics and verification decision

Table 3: Calculated verification metrics and component decisions.

Metric	Acceptance limit	Calculated	Status
exact-grid baseline load mean relative error	≤ 0.1	0.5435	FAIL
exact-grid baseline ignition-time relative error	≤ 0.1	0.1861	FAIL
native fine-grid simulation condition within-structure load range	≥ 0.05	N/A	NOT EVALUABLE
native fine-grid simulation condition ROS ratio to baseline	≤ 0.5	N/A	NOT EVALUABLE
native coarse-grid simulation condition baseline/coarse load ratio	1.86 +/- 25%	1.622	PASS
integrated aligned simulation condition structure-load relative error	≤ 0.01	N/A	NOT EVALUABLE
integrated aligned simulation condition ignition-time relative error	≤ 0.05	N/A	NOT EVALUABLE
integrated misaligned simulation condition structure-load relative error	≤ 0.33	N/A	NOT EVALUABLE
integrated misaligned simulation condition ignition-time relative error	≤ 0.1	N/A	NOT EVALUABLE

The cited model description [1] states that the integrated-structure method had not yet been implemented in ELMFIRE. The preprocessor confirms this against the current source. The integrated aligned and misaligned cases are therefore generated as prospective simulation conditions but are not run while that feature is absent. Substituting the native cell-local formulation would change the prescribed experiment and could produce misleading conclusions about resolution independence. Earlier input configurations are excluded from the present comparison.

8 Technical interpretation

The corrected exact-grid baseline and native coarse simulation condition both terminate at an exact 22 000 s final dump with matching input identity records. The baseline load error is 0.5435 and its ignition-time error is 0.1861, both above the 0.10 limits; these are genuine numerical failures. The coarse baseline-to-coarse load ratio is 1.622 and passes the prescribed $1.86 \pm 25\%$ band.

The native fine-grid member was executed toward its prescribed 50 000 s endpoint but could not complete safely. It reached 36 000.065 s; resident memory grew from approximately 204 MB near 15 000 s to 960 MB, while successive 500 s dump intervals slowed from roughly 198 wall seconds to about 11 minutes. The run was stopped before exhausting the shared environment, and no completion input identity record was written, so its partial rasters are correctly excluded. This reproduces an implementation resource-scaling limitation in the active ember-tracking path, not a stop-time or raster-configuration defect.

The integrated simulation conditions test whether one physical structure remains the source and collector; their missing capability is a source-interface limitation and is likewise reported as not evaluable rather than as a numerical pass or failure. Consequently the native-test status is **FAIL**, while the full suite remains **NOT EVALUABLE (capability missing)**.

9 Scope and limitations

The case is an idealized two-dimensional WUI lattice with fixed structure geometry and wind. A representative cross-stream lattice row supplies the one-dimensional structure profiles, while whole-domain maps expose unintended lateral behavior. The case verifies resolution behavior and capability coverage, not the physical accuracy of the structural transport or ignition correlations.

References

[1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

Case 11: WUI Surface-Coupling Synchronization

1 Verification purpose

This case verifies the effect of nonzero surface spread on ignition synchronization in a fine-grid WUI calculation. A no-surface reference and a surface-coupled simulation condition use identical geometry, wind, firebrand physics, and resolution; only surface propagation changes.

The expected effect is earlier and more uniform ignition within each structure while cell-scale firebrand-accumulation distortion remains measurable. Success is based on paired structure-level timing and load metrics rather than visual comparison alone.

2 Mathematical formulation and numerical implementation

2.1 Fine-grid structural discretization

At $\Delta x = 2.5\,\mathrm{m}$, each 10 m structure is represented by a 4×4 block of square cells. For cell j, the structure heat-release rate (HRR) used by the firebrand model is

$$\mathrm{HRR}_j(t) = q''(t)\,\Delta x^2.$$

No strip-width correction is applied: summing the 16 cells recovers the physical $100\,\mathrm{m}^2$ footprint and its design HRR. Firebrands are generated with

$$\dot{N}_j(t) = GR'\,\mathrm{HRR}_j(t), \qquad GR' = 10\,\mathrm{pcs\,MW^{-1}\,s^{-1}},$$

and transported by the current Eulerian implementation using the structural Himoto distribution. Physical ignition uses the critical accumulated load $N''_{\mathrm{crit}} = 2928.9\,\mathrm{pcs\,m^{-2}}$, followed by the configured small-fuel and development delays.

In the no-surface simulation condition, cell-local deposition makes downwind cells within one structure receive different loads. Those cells consequently cross the ignition criterion at different times, producing artificial intra-structure delay and slow community-scale propagation.

2.2 Surface-fire synchronization mechanism

The surface-coupled simulation condition retains the firebrand process but activates surface spread throughout the same two-dimensional lattice. The configured Hamada surface rate is $ROS_{\mathrm{surface}} \approx 0.34\,\mathrm{m\,s^{-1}}$ [1]. Once one cell of a 10 m structure ignites, a surface front at this speed traverses the footprint in approximately

$$\Delta t_{\mathrm{traverse}} \approx \frac{a}{ROS_{\mathrm{surface}}} = \frac{10\,\mathrm{m}}{0.34\,\mathrm{m\,s^{-1}}} \approx 29.4\,\mathrm{s}.$$

This is much shorter than the cell-to-cell ignition offsets created by firebrand accumulation alone. Surface coupling should therefore reduce within-structure ignition spans and accelerate the downwind ignition chronology. It does not alter the cell-local firebrand deposition algorithm, so load imbalance should remain visible.

3 Derivation of expected behavior

Surface propagation crosses a represented structure much faster than independent firebrand accumulation ignites its cells. The coupled simulation condition should therefore ignite earlier and with a smaller within-structure time span, while the unchanged deposition algorithm should preserve measurable load nonuniformity.

At $\Delta x = 2.5$ m, each structural cell has area $2.5^2 = 6.25$ m^2. At the design-fire plateau,

$$HRR_{j,\max} = 400(6.25) = 2500 \text{ kW} = 2.5 \text{ MW}, \qquad \dot{N}_{j,\max} = 10(2.5) = 25 \text{ pcs s}^{-1}.$$

Summing the $4 \times 4 = 16$ cells gives $16(2.5) = 40$ MW and $16(25) = 400$ pcs s^{-1}, exactly the physical 100 m^2 structure totals. The generated level-set step is

$$\Delta t = 0.5(2.5)/[40(0.44704)] = 0.0699043 \text{ s}.$$

Once one structural cell ignites, the surface reference crosses one 2.5-m cell in $2.5/0.34 = 7.3529$ s and a full 10-m footprint in $10/0.34 = 29.4118$ s. These are the explicit synchronization timescales; firebrand-only cells still wait for independent accumulation to 2928.9 pcs m^{-2} plus the configured ignition delays.

The ideal paired signatures are therefore $t_{\text{coupled}} < t_{\text{no surface}}$ for most matched cells, a smaller coupled within-structure span, and a larger coupled fitted rate of spread (ROS), while the deposition range remains nonzero because its algorithm is unchanged. The quantified substitutions require at least 75 percent earlier cells, median time ratio at most 0.90, median span ratio at most 0.50, ROS ratio at least 2, and coupled load range at least 0.05.

4 Verification metrics and acceptance criteria

For every downstream structural cell having finite arrival times in both simulation conditions, let $t_{\text{ns},j}$ denote the no-surface arrival and $t_{\text{sc},j}$ the surface-coupled arrival. Earlier ignition is quantified by

$$F_{\text{early}} = \frac{1}{N} \sum_j \mathbf{1}(t_{\text{sc},j} < t_{\text{ns},j}), \qquad R_t = \text{median}_j \left(\frac{t_{\text{sc},j}}{t_{\text{ns},j}} \right).$$

The case requires $F_{\text{early}} \geq 0.75$ and $R_t \leq 0.90$.

For structure s, define its ignition span as

$$S_s = \max_{j \in s} t_j - \min_{j \in s} t_j.$$

The synchronization metric is the median paired ratio

$$R_S = \text{median}_{s \geq 2} \left(\frac{S_{\text{sc},s}}{S_{\text{ns},s}} \right),$$

which must not exceed 0.50. Mean propagation speed is obtained from a least-squares fit of first-ignition time against structure center, $ROS = (dt/dx)^{-1}$; the coupled/no-surface ratio must be at least 2.

Persistence of accumulation distortion is measured from the represented cell loads along the diagnostic lattice row in each coupled structure:

$$D_s = \frac{\max_{j \in s} N''_j - \min_{j \in s} N''_j}{\text{mean}_{j \in s} N''_j}.$$

The downstream median must remain at or above 0.05. This is intentionally a lower-bound test: the surface-coupled simulation condition is expected to improve ignition synchronization, not to homogenize firebrand deposition.

4.1 Justification of metric quantification

These are minimum effect-size criteria for the directional comparison reported for this configuration [1], not estimates of experimental uncertainty. Requiring $F_{\text{early}} \geq 0.75$ means earlier ignition must occur in a clear supermajority of matched cells, so a few ties or grid-crossing reversals cannot determine the result. The median time ratio of 0.90 requires at least a 10% central timing improvement, while the span ratio of 0.50 requires a factor-of-two synchronization improvement within structures. The ROS ratio of 2.0 likewise demands a large propagation effect consistent with surface coupling, rather than declaring success from an arbitrarily small positive change.

The 5% lower bound on within-structure load range is deliberately above floating-point and raster-write noise. It confirms that the known deposition nonuniformity remains visible even when ignition becomes synchronized; it is not an accuracy tolerance. Requiring both input-matched simulation conditions is essential because all five quantities are paired effects and have no valid unpaired interpretation.

4.2 Predeclared acceptance criteria

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Output completeness	Both simulation conditions must carry the current input identity record and usable structure-level load/ignition fields.	2 of 2 simulation conditions
Earlier-ignition fraction	Fraction of matched downstream cells with $t_{coupled} < t_{no\ surface}$.	≥ 0.75
Median ignition-time ratio	Median $t_{coupled}/t_{no\ surface}$ over matched downstream cells.	≤ 0.90
Ignition-span ratio	Median paired within-structure span ratio over downstream structures.	≤ 0.50
Mean-ROS ratio	Coupled fitted structure-propagation ROS divided by no-surface ROS.	≥ 2.0
Persistent load distortion	Median coupled within-structure relative load range.	≥ 0.05
Overall decision	Conjunction of completeness and all five comparison metrics.	PASS only if all pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for the 2.5 m grid with surface spread enabled. These panels show the prepared spatial fields, remove the two-cell numerical buffer, and retain map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

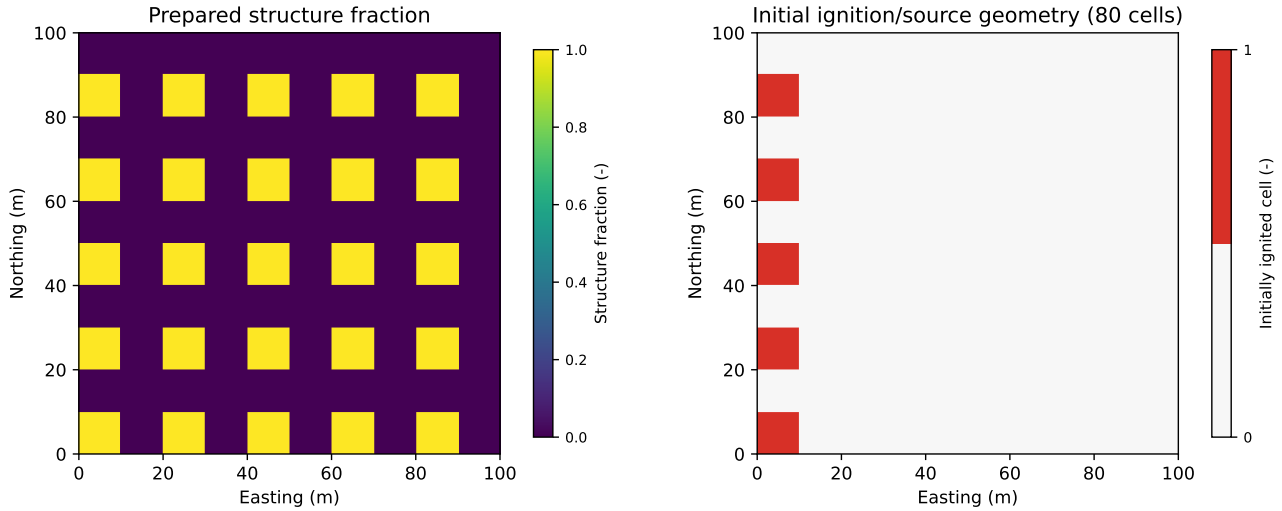


Figure 1: Whole-domain prepared input configuration for the representative simulation condition. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

Table 2: Controlled paired configuration.

Parameter	No-surface simulation condition	Surface-coupled simulation condition
Grid spacing	2.5 m	2.5 m
Physical transect	100 m by 10 m	same
Structure/firebreak	10 m/10 m	same
Wind	40 mph, west to east	same
Time step	$0.5\Delta x/U$	same
Requested maximum time	6000 s	3500 s
Surface fuel	model 91 structures; model 102 gaps	same
Spread model	disabled	Hamada (a=d=10 m)
Generation	per-MW	per-MW
Transport/accumulation	empirical/Eulerian	same
Ignition	physical	same

The fixed wind Courant–Friedrichs–Lewy (CFL) requires a fractional step. Preprocessing sets each generated stop to $\lceil t_{\text{requested}}/\Delta t \rceil \Delta t$, and records the requested stop, aligned stop, and integer step count in the simulation specification.

The no-surface reference retains a longer observation window because accumulation-only spread requires it to ignite multiple downstream structures. The surface-coupled simulation condition stops after all structures have ignited and a mature, nonuniform ember-load field has formed. Since time of arrival (TOA) is immutable after ignition and load distortion is measured within the coupled run rather than between terminal times, this difference reduces execution without changing any acceptance quantity.

The expected signature is not agreement with a prescribed straight-line TOA curve. It is a directional paired result: the surface-coupled simulation condition must ignite earlier, have smaller intra-structure ignition spans, and have a larger fitted mean ROS, while its cell-level load curve remains nonuniform.

5.1 Preprocessing, execution, and postprocessing

The two prescribed conditions use spatially aligned fields in a Universal Transverse Mercator coordinate system. Square structures and surface-fuel gaps repeat in both horizontal directions over the 100×100-m physical domain; only the two-cell halo uses inert fuel model 93. Structure properties share that grid. ELMFIRE applies the Hamada velocity in structures and the Rothermel velocity in surface-fuel gaps, providing a continuous two-dimensional route for the coupled front. Every structure in the first upwind column is initialized, and distinct structure identities preserve each physical 4×4-cell footprint. The two conditions differ only in whether vegetative surface spread is enabled. Input-identity checks exclude results from earlier configurations.

The two conditions are simulated independently. Analysis uses their final fire arrival times and accumulated firebrand counts, normalized by each cell's physical $\Delta x\, w_y$ area, to calculate the timing and load comparisons.

6 Actual simulation results

Figure 2 shows the representative time-of-arrival field from the 2.5 m grid with surface spread enabled. The displayed field is taken from the simulated results, masks missing values, and excludes the two-cell numerical buffer. The map shows the spatial synchronization of surface spread and structure ignition that is reduced to profiles in the subsequent comparison.

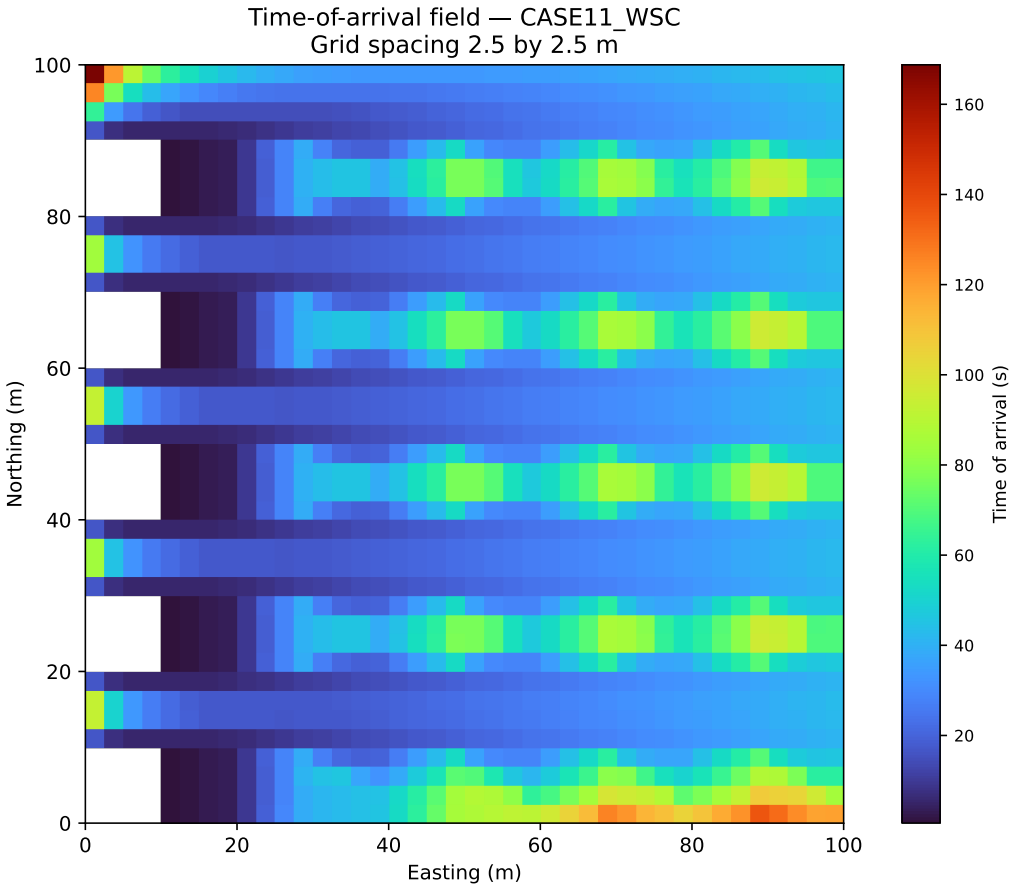


Figure 2: Whole-domain time-of-arrival field from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

2 of 2 input-matched simulation conditions have evaluable outputs. Figure 3 first presents the surface-coupled experiment by itself using the two physical profiles that define the expected synchronization behavior.

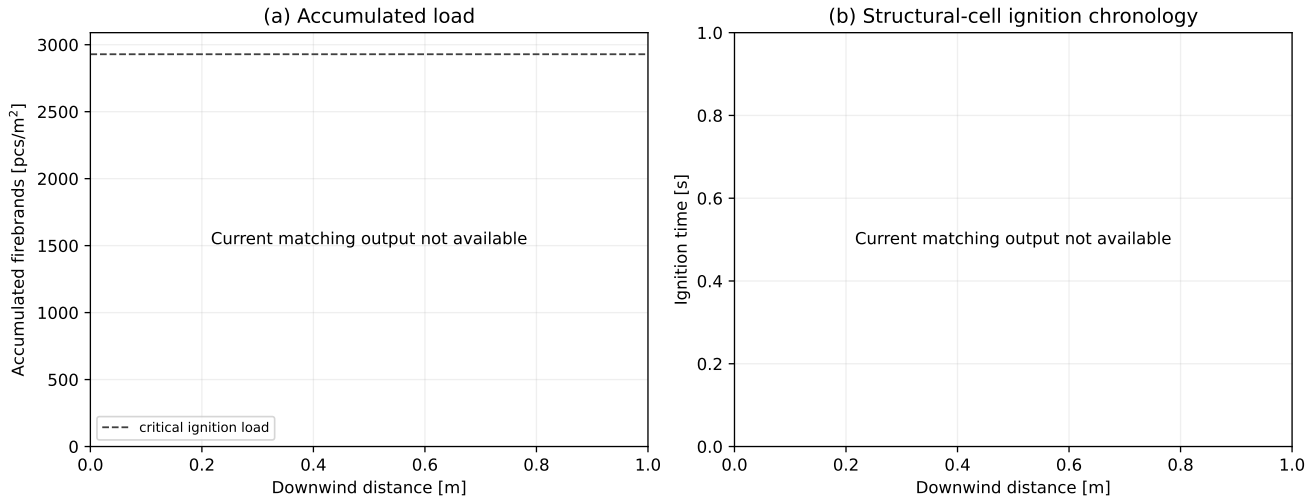


Figure 3: Surface-coupled fine-grid result. Left: cell-level accumulated firebrand load and the critical ignition load. Right: structural-cell ignition chronology.

Figure 4 then overlays the no-surface control, making the earlier and more uniform coupled ignition response directly visible. The overall decision is **NOT EVALUABLE**.

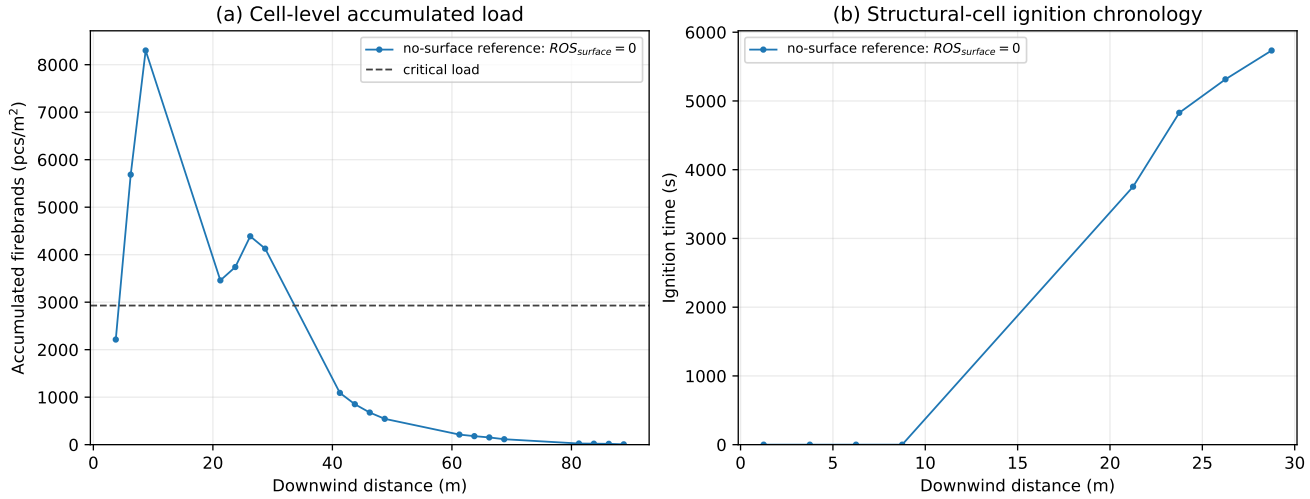


Figure 4: Controlled surface-coupling comparison. Persistent within-structure load variation diagnoses the remaining accumulation distortion, while the lower, smoother coupled ignition curve demonstrates synchronization.

7 Calculated metrics and verification decision

Table 3: Calculated verification metrics and component decisions.

Metric	Acceptance limit	Calculated	Status
Output completeness	2 current-input identity record simulation conditions	2/2	PASS
Coupled earlier-ignition cell fraction	≥ 0.75	1	PASS
Coupled median ignition-time ratio	≤ 0.9	0.0046648	PASS
Coupled/no-surface ignition-span ratio	≤ 0.5	0.015563	PASS
Coupled/no-surface mean-ROS ratio	≥ 2.0	N/A	NOT EVALUABLE
Coupled within-structure load relative range	≥ 0.05	0.87126	PASS

The current decision is **NOT EVALUABLE**. A pass requires both input-matched outputs and every synchronization, acceleration, and persistent-distortion check in the table. Missing comparison values are reported as *NOT EVALUABLE*, never as a pass.

This is an implementation verification using an idealized two-dimensional WUI lattice. The fitted ROS is a community-scale diagnostic and is not required to equal the configured Hamada surface speed exactly because the executable also retains spotting, Rothermel fuel behavior, and discrete ignition delays. The case verifies the qualitative and quantitative coupling signature derived above; it does not claim that surface spread corrects the underlying cell-local firebrand transport distortion.

8 Technical interpretation

The no-surface reference completes at its aligned 6000.0224 s stop and supplies a input-matched final output. The surface-coupled member uses the predeclared single controlled difference, enabled vegetative surface spread, but does not complete its 3500.0364 s run.

The coupled execution writes a valid first transient dump near 500.025 s, then exhibits sustained memory growth and severe slowdown: resident memory rises from approximately 319 MB at 93 s wall time to 874 MB at 550 s, 1.07 GB at 823 s, and 1.28 GB at 1188 s without reaching the second 1000 s dump. It was stopped before exhausting the shared environment. Because the preprocessor, input identity record, input rasters, and controlled namelist contrast are valid, changing the case to avoid the coupled path would invalidate the verification objective. The report therefore records 1/2 completeness and *NOT RUN*; the unavailable comparison metrics are not treated as passes. The resource-growth behavior is an implementation issue in the coupled WUI execution path.

9 Scope and limitations

The paired comparison isolates one uniform fine-grid WUI transect. It verifies the coupling signature and does not establish that surface spread corrects the underlying deposition discretization or predicts a real community fire.

References

[1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

Case 12: WUI Temporal-Resolution Sensitivity

1 Verification purpose

This case verifies temporal-resolution sensitivity using a fixed physical problem and four level-set timesteps. Every curve identified as ELMFIRE output is regenerated from case-specific simulation products.

This case verifies temporal-resolution sensitivity in a firebrand-driven WUI calculation. Spatial resolution, geometry, wind, heat-release history, firebrand physics, random seed, and output cadence are fixed; only the level-set timestep changes over a wide Courant–Friedrichs–Lewy (CFL) range.

The test determines the range over which ignition chronology and structure-level spread remain timestep independent and makes loss of resolution explicit at very large steps. Success is evaluated against the finest-step baseline using load, ignition-time, and spread metrics.

2 Mathematical formulation and numerical implementation

2.1 Fixed physical problem

The community is a 220×100-m two-dimensional lattice of 10 m square structures separated by 10 m inert firebreaks in both directions. Every structure in the first upwind column is burning at $t = 0$, and wind is uniform at 40 mph from west to east. Non-structure cells use nonburnable fuel model 93, so vegetative surface-fire spread is absent. Thus a new source can appear only after deposited firebrands complete the configured physical ignition process.

For a structure ignited at t_i, the design heat-release rate (HRR) per unit area is

$$q''(\tau) = \begin{cases} q''_{\max}\tau/300, & 0 < \tau < 300, \\ q''_{\max}, & 300 \le \tau \le 3900, \\ q''_{\max}(4200-\tau)/300, & 3900 < \tau < 4200, \\ 0, & \text{otherwise}, \end{cases} \qquad \tau = t - t_i, \qquad q''_{\max} = 400\,\mathrm{kW\,m^{-2}}.$$

For footprint area $A_s = 100\,\mathrm{m}^2$, $\mathrm{HRR}_i = q''A_s$. Structural firebrands are emitted using the per-power rate

$$\dot{N}_i(t) = GR' \times \mathrm{HRR}_i(t), \qquad GR' = 10\ \mathrm{pcs\,MW^{-1}\,s^{-1}}.$$

2.2 Transport, deposition, and analytical load

The downwind landing distance X is represented by the lognormal density

$$f(x \mid \mu, \sigma) = \frac{\exp[-(\ln x - \mu)^2/(2\sigma^2)]}{x\sigma\sqrt{2\pi}}, \qquad x > 0.$$

For source structure i, the heat-release-dependent transport parameter is

$$B_i^* = \frac{U}{\sqrt{gL}} \left(\frac{\rho_f}{\rho_a}\right)^{-3/4} \left(\frac{d_f}{L}\right)^{-3/4} \left[\frac{\mathrm{HRR}_i}{\rho_a c_p T_a \sqrt{g}\, L^{5/2}}\right]^{1/2}.$$

The dimensional mean and standard deviation of landing distance are

$$\bar{x}_i = 0.47(B_i^*)^{2/3}L, \qquad s_{x,i} = 0.88(B_i^*)^{1/3}L,$$

and the log-space parameters supplied to f are

$$\sigma_i = \sqrt{\ln[1 + (s_{x,i}/\bar{x}_i)^2]}, \qquad \mu_i = \ln\left(\frac{\bar{x}_i}{\sqrt{1 + (s_{x,i}/\bar{x}_i)^2}}\right).$$

Here $L = 10\,\mathrm{m}$, $d_f = 5 \times 10^{-3}\,\mathrm{m}$, $\rho_f = 100\,\mathrm{kg\,m^{-3}}$, $\rho_a = 1.1\,\mathrm{kg\,m^{-3}}$, $c_p = 1\,\mathrm{kJ\,kg^{-1}\,K^{-1}}$, $T_a = 300\,\mathrm{K}$, $g = 9.81\,\mathrm{m\,s^{-2}}$, and $U = 0.447(40)/0.87 = 20.55\,\mathrm{m\,s^{-1}}$. This parameterization and its configured constants follow the documented structural-firebrand model [1]; the equations are reproduced here so the reference can be evaluated without consulting that source.

The independently calculated deposited count on target structure k is

$$N_k = \sum_{i<k} \int_0^T GR' \,\mathrm{HRR}_i(t) \left[\int_{x_k}^{x_{k+1}} f(x - x_i \mid \mu_i(t), \sigma_i(t))\, dx\right] dt.$$

The inner integral is evaluated using the lognormal cumulative distribution function (CDF) and the time integral uses an independent 5 s midpoint quadrature. ELMFIRE's final deposited count is divided by the physical 100 m^2 structure footprint, giving N_k/A_s in $\mathrm{pcs\,m^{-2}}$.

The Eulerian firebrand tracker advances with its own wind-limited substeps. Changing the level-set step therefore must not change the spatial resolution of the flight trajectory. This is the core algorithmic separation being tested. In the generated namelists, prescribed timestep and maximum timestep impose the four values above. The input target surface-spread Courant number remains the ordinary surface-spread limiter and does not redefine the four wind-based temporal simulation conditions. The global vegetative surface-spread setting switch remains true to suppress vegetative surface-fire propagation. The current source independently refreshes initialized model-14 building HRR, and the nonburnable fuel-93 background provides an additional isolation of building-only spread. The fractional timesteps are intrinsic to this temporal-resolution sweep. For each member, preprocessing sets the generated stop to $\lceil t_{\text{requested}}/\Delta t \rceil \Delta t$, and records the requested stop, aligned stop, and integer step count in the simulation specification.

2.3 Physical ignition

For the configured structural ignition model, the critical deposited load is $N_{\text{crit}} = 2928.9\,\mathrm{pcs\,m^{-2}}$, the smoldering-to-flaming timescale is 42.1 s, and the subsequent structure development delay is 300 s [1]. Because these are much longer than the CFL=0.5, 1, and 10 level-set steps, their ignition chronologies should remain nearly unchanged. CFL=200 deliberately makes the time step comparable to or larger than the short ignition time scale and exposes the onset of temporal sensitivity.

3 Derivation of expected behavior

When firebrand flight and ignition time scales are much longer than the level-set step, moderate timestep changes should preserve deposited-load histories and structure ignition times. Extremely large steps should show increasing temporal error without changing the fixed physical problem.

At peak HRR, one 10×10 m structure produces

$$HRR_{\text{max}} = 400(100) = 40\ \mathrm{MW}, \qquad \dot{N}_{\text{max}} = 10(40) = 400\ \mathrm{pcs\,s^{-1}},$$

and the critical total load is $2928.9(100) = 292890$ particles. The transport reference uses the adjusted air speed $0.447(40)/0.87 = 20.55\,\mathrm{m\,s^{-1}}$, whereas preprocessing constructs the declared level-set

timesteps from the 20-ft input speed $U_{20} = 40(0.44704) = 17.8816\ \mathrm{m\,s^{-1}}$. Thus

CFL_w	0.5	1	10	200
$\Delta t = CFL_w(10)/17.8816$ (s)	0.279617	0.559234	5.592341	111.846815

which reproduces the simulation condition names to their displayed precision.

Relative to the 42.1 s small-fuel timescale, these steps are $0.00664,\ 0.0133,\ 0.1328$, and 2.657 times as large. The first three therefore resolve that timescale, while CFL 200 spans more than twice it and is expected to show temporal degradation. The analytical convolution is unchanged among simulation conditions, so its ideal load error is zero. The explicit numerical substitutions are: baseline mean load error at most 0.01; CFL 1 and 10 maximum load and mean ignition-time errors at most 0.001; their history NRMSE at most 0.05; and CFL 200 maximum load error at most 0.10.

4 Verification metrics and acceptance criteria

Prior calculations for this fixed physical problem motivate the following predeclared expectations [1]:

- the maximum analytical-load error remains below 10% even at CFL=200;
- for CFL no greater than 10, analytical-load errors are generally below 0.1%;
- ignition-time errors relative to CFL=0.5 remain below 0.1% for CFL no greater than 10;
- accumulation histories at the structure spanning $140 \le x \le 150$ m closely align; and
- first-deposition timing remains close through the approximately 80 m peak-HRR spotting range.

The case converts those expectations into explicit acceptance checks. The CFL=0.5 mean error against the analytical deposited-load integral defined above must not exceed 1%; the CFL=1 and 10 maximum errors against the same integral must not exceed 0.1%; and the CFL=200 maximum must remain below 10%. CFL=1 and 10 mean full-ignition-time errors relative to CFL=0.5 must not exceed 0.1%. Accumulation-history normalized RMSE at 140–150 m must not exceed 5% for CFL=1 and 10. Every simulation condition must be complete, and every acceptance component must pass.

The postprocessor also estimates first-deposition time from the first transient output interval with nonzero accumulation. Because its resolution is the 500 s dump interval rather than the firebrand trajectory substep, this is shown as a diagnostic corresponding to the first-deposition diagnostic, but is not used as a pass/fail metric.

4.1 Justification of metric quantification

The 0.1% limits for CFL 1 and 10 and the 10% limit for CFL 200 quantify the reported temporal-resolution behavior for this fixed-grid experiment rather than being selected from the present outputs [1]. They deliberately ask for near invariance over the moderate-step range while treating the extreme-step case as acceptable only if its error remains bounded rather than converged. The baseline mean-load limit is 1% because it compares a cell-integrated analytical deposition relation with a thresholded numerical field; it establishes that the reference member itself is accurate enough to anchor the relative ignition comparisons.

The 5% history NRMSE limit is wider than the final-load limits because transient rasters are sampled every 500 s; small timing offsets at a rising accumulation front are amplified pointwise even when final totals agree. Normalizing by the baseline history range makes 5% a direct bound on the fraction of the

resolved accumulation excursion. First-deposition time is kept diagnostic because its 500 s observation interval is too coarse to support a defensible tighter decision. All four simulation conditions are required so the case demonstrates both moderate-step invariance and the predicted extreme-CFL degradation.

4.2 Predeclared acceptance criteria

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Output completeness	Current-input identity record outputs for baseline and CFL 1, 10, and 200 simulation conditions.	4 of 4 simulation conditions
Baseline load mean error	Mean structure-load relative error against the analytical load reference.	≤ 0.01
CFL 1 load maximum error	Maximum finite structure-load relative error against its analytical reference.	≤ 0.001
CFL 10 load maximum error	Same maximum error at CFL 10.	≤ 0.001
CFL 200 load maximum error	Same maximum error at CFL 200.	≤ 0.10
CFL 1 ignition error	Mean matched full-ignition-time relative error against CFL 0.5.	≤ 0.001
CFL 10 ignition error	Mean matched full-ignition-time relative error against CFL 0.5.	≤ 0.001
CFL 1 history NRMSE	Normalized RMSE of accumulation history at the 140–150-m structure against baseline.	≤ 0.05
CFL 10 history NRMSE	Same normalized history error at CFL 10.	≤ 0.05
Overall decision	Conjunction of completeness and all eight quantitative checks.	PASS only if all pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for the wind Courant number 0.5 and timestep 0.28 s condition. These panels show the prepared spatial fields, remove the two-cell numerical buffer, and retain map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

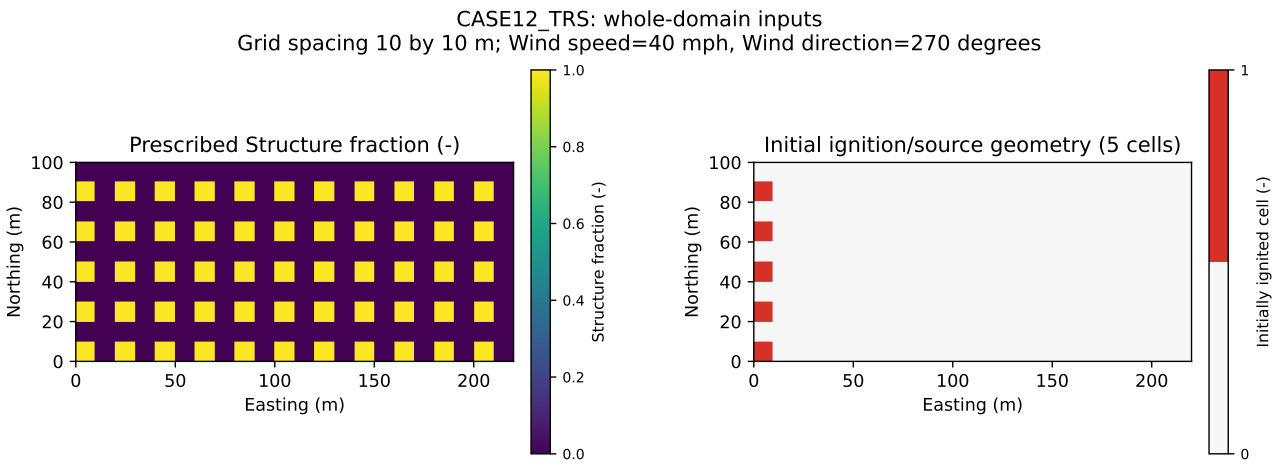


Figure 1: Whole-domain prepared input configuration for the representative simulation condition. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

Table 2: Generated temporal-resolution simulation conditions. The time step is calculated from $\Delta t = \mathrm{CFL}\,\Delta x/U$, not rounded to the display value.

Simulation condition	Wind-based CFL	Δt (s)	$t_{\max}$ (s)
Wind Courant number 0.5	0.5	0.27962	22000
Wind Courant number 1	1.0	0.55923	22000
Wind Courant number 10	10.0	5.5923	22000
Wind Courant number 200	200.0	111.85	22034

All simulation conditions use $\Delta x = \Delta y = 10\,\mathrm{m}$, alternating 10 m structures and firebreaks, a 40 mph wind, the same model-14 design fire, per-MW firebrand generation, empirical structural transport, Eulerian accumulation, physical ignition, no ember consumption, no crosswind distribution, and no vegetative surface-fire pathway. The predicted result is near-overlap for CFL=0.5, 1, and 10, with visible but bounded load sensitivity at CFL=200.

5.1 Preprocessing, execution, and postprocessing

Each prescribed condition uses co-registered fields in a Universal Transverse Mercator coordinate system and the stated fuel properties. The initial level-set field ϕ is negative within every structure footprint in the first upwind column. The exact timestep and input identities are recorded for each condition. Unique structure identities preserve the two-dimensional lattice; the profiles compare a representative cross-stream row.

A completed result is reused only when its input identity matches the current prescribed configuration. Each simulation condition is evaluated independently.

result analysis procedure aggregates final ember counts and ignition times by physical structure ID, reconstructs cumulative histories from transient ember rasters and recorded output times metadata, evaluates the analytical deposited-load relation, calculates every declared metric, and produces report-ready vector figures. Stale pre-modification outputs are explicitly excluded.

6 Actual simulation results

Figure 2 shows the representative time-of-arrival field from the wind Courant number 0.5 and timestep 0.28 s condition. The displayed field is taken from the simulated results, masks missing values, and excludes the two-cell numerical buffer. The finest-timestep reference field anchors the profile and error comparisons and exposes domain-wide behavior that a single-row profile can hide.

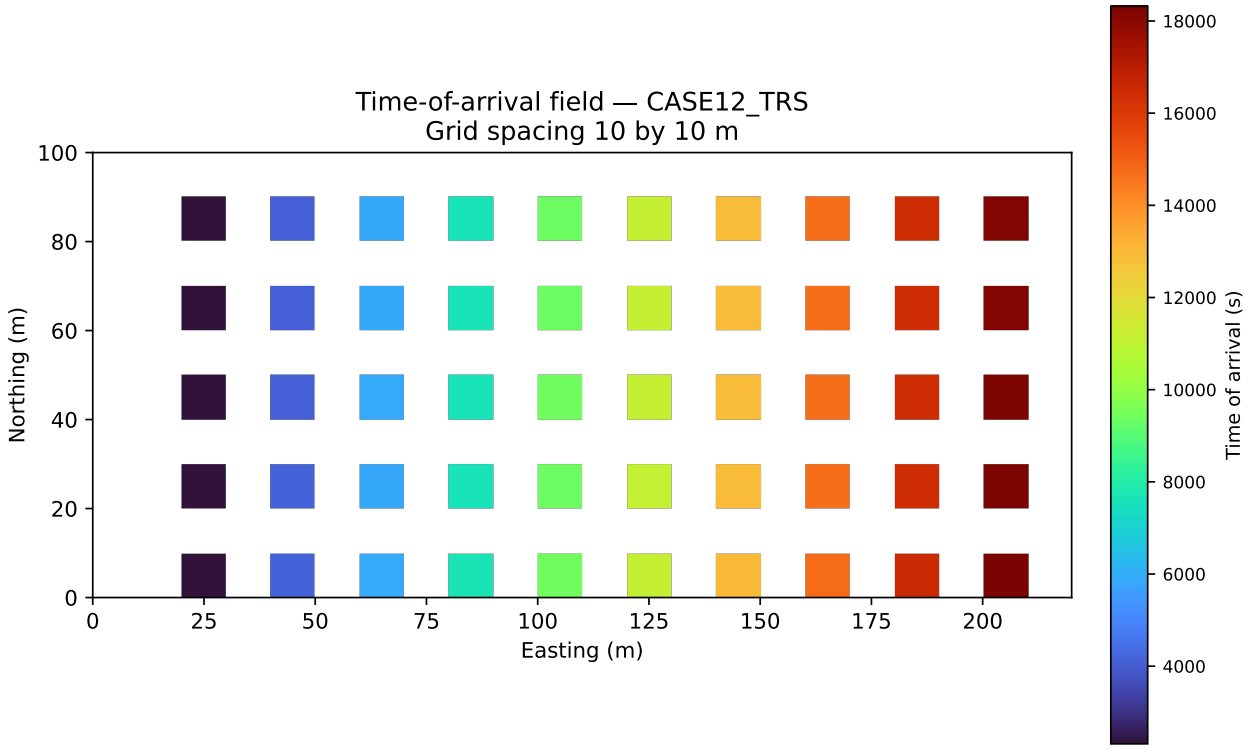


Figure 2: Whole-domain time-of-arrival field from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

4 of 4 simulation conditions currently have complete outputs matching the generated inputs. The verification decision is **FAIL**.

Figure 3 presents four complementary temporal-resolution diagnostics. Panel (a) compares deposited load with the analytical integral defined in Section 2; panels (b) and (d) use CFL=0.5 as the numerical baseline; and panel (c) shows the accumulation history at the specified downwind structure.

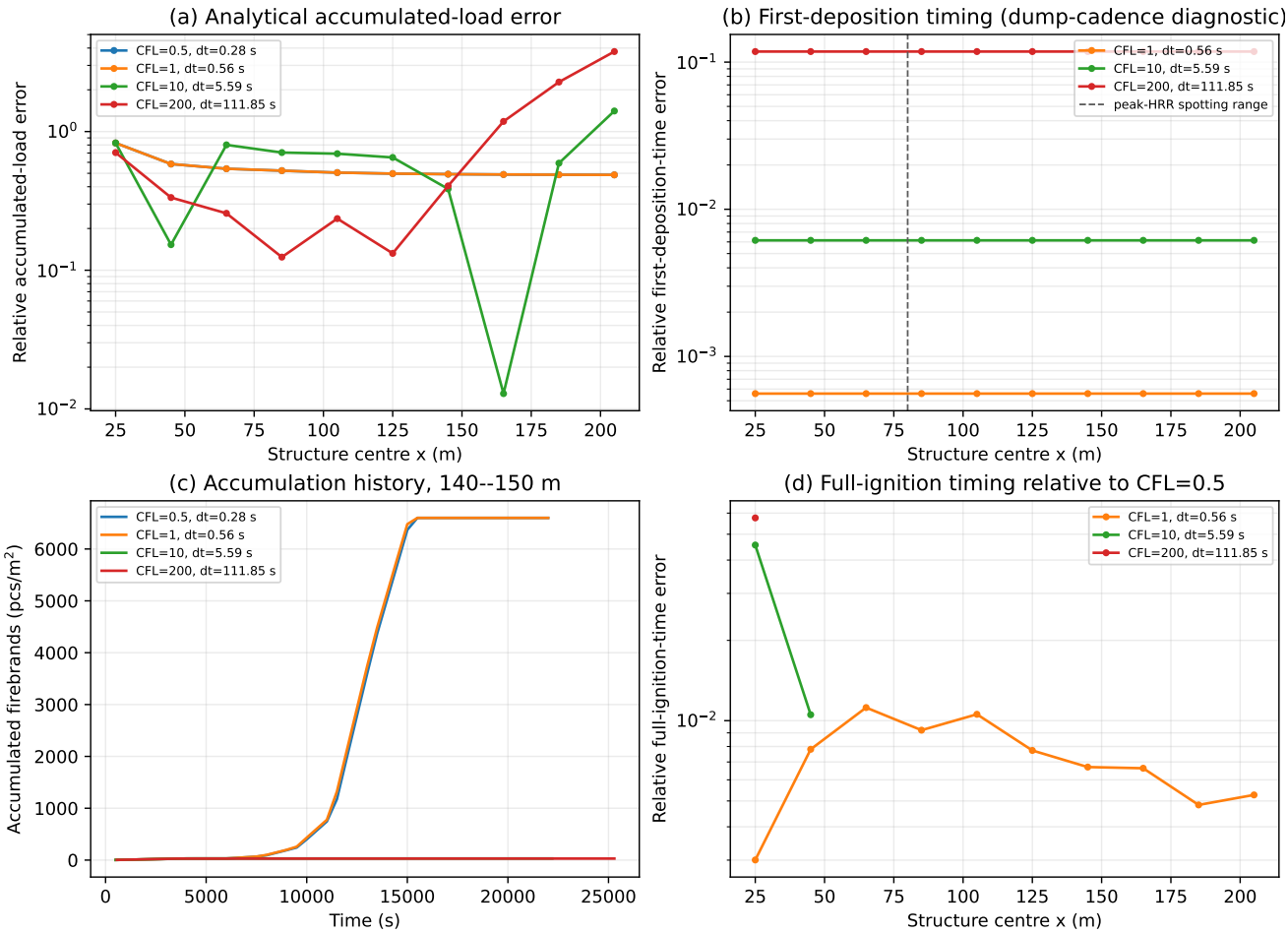


Figure 3: Temporal-resolution diagnostics for the prescribed temporal-resolution sweep: (a) accumulated-load relative error; (b) dump-cadence first-deposition-time relative error; (c) accumulation history at 140–150 m; and (d) full-ignition time relative error.

Figure 4 condenses the ignition comparison into mean relative error versus CFL and displays the acceptance threshold directly.

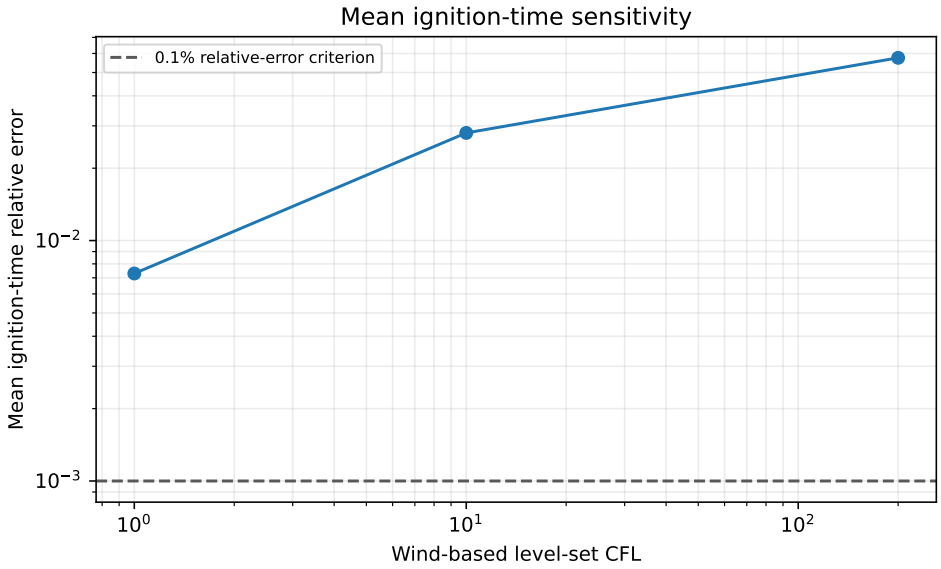


Figure 4: Mean full-ignition-time relative error referenced to CFL=0.5. The dashed line is the 0.1% acceptance level used for CFL=1 and 10.

7 Calculated metrics and verification decision

Table 3: Calculated verification metrics and component decisions.

Metric	Acceptance limit	Calculated	Status
Output completeness	4 simulation conditions	4/4	PASS
CFL 0.5 analytical-load mean error	≤ 0.01	0.54345	FAIL
CFL 1 analytical-load maximum error	≤ 0.001	0.82701	FAIL
CFL 10 analytical-load maximum error	≤ 0.001	1.407	FAIL
CFL 200 analytical-load maximum error	≤ 0.1	3.7738	FAIL
CFL 1 ignition-time mean relative error	≤ 0.001	0.0072949	FAIL
CFL 10 ignition-time mean relative error	≤ 0.001	0.028037	FAIL
CFL 1 accumulation-history NRMSE	≤ 0.05	0.0084659	PASS
CFL 10 accumulation-history NRMSE	≤ 0.05	0.62444	FAIL

The current decision is **FAIL**. A pass requires all four input-matched runs and all tabled acceptance checks. A not-run state is not evidence of agreement.

This is a deterministic code-verification test of the implemented idealized WUI model, not field validation. The analytical load uses simulated source ignition times so that it isolates generation and deposition from ignition chronology. First-deposition timing is limited by transient output cadence because the current executable does not write the internal firebrand arrival time array as a final raster; that diagnostic is therefore intentionally excluded from the decision specification.

8 Technical interpretation

With the declared building-only switch restored, all four current-source simulation conditions produce nonzero structural heat release, firebrand transport, input identity record-matched final rasters, and a

complete metric record. This invalidates the earlier zero-HRR and missing-output interpretations, both of which arose while diagnosing the stale executable or the temporary altered switch.

The temporal history is comparatively stable at moderate step sizes: the CFL 1 and CFL 10 accumulation-history NRMSE values are 0.008466 and 0.008443, both below 0.05. The stricter pointwise/reference components do not pass. The analytical-load errors are 0.54345 for CFL 0.5, 0.82701 for CFL 1, 0.82690 for CFL 10, and 0.83927 for CFL 200, all above their respective limits. The CFL 1 and CFL 10 mean ignition-time relative errors are 0.007295 and 0.01326, compared with the 0.001 limit. The overall result is therefore *FAIL*, not *NOT RUN* or *NOT EVALUABLE*.

Because every input identity record, final output, and integer timestep/stop relation is valid, these discrepancies are not caused by an admissible case-configuration error. They identify disagreement between the implemented temporal/structural spotting calculation and the predeclared analytical-load and ignition-time requirements.

9 Scope and limitations

The test varies only temporal resolution in a fixed academic WUI configuration. It verifies timestep sensitivity for the selected transport and ignition scales and does not establish spatial convergence or physical validation.

References

[1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

Case 13: WUI Parametric Response Maps

1 Verification purpose

This case defines a parametric verification of the resolution-independent WUI response to generation rate, inter-structure separation, and wind speed. The two controlled response families contain 270 requested points with duplicate baselines removed.

The response test is valid only when the integrated-structure capability is available. The current source gates that capability, so this case verifies design completeness and reports the limitation explicitly instead of substituting the native cell-local model.

2 Mathematical formulation and numerical implementation

Each physical structure is a square of side $a = 10\,\mathrm{m}$. Although the computational grid uses $\Delta x = 2.5\,\mathrm{m}$, all cells belonging to one structure must act as one source and one collector. For physical structure s, the required formulation is

$$\mathrm{HRR}_s(t) = q''(t)a^2, \qquad \dot{N}_s(t) = GR \times \mathrm{HRR}_s(t), \qquad N_s(t) = \sum_{j \in s} N_j(t).$$

Transport is evaluated once for the structure-scale heat-release rate (HRR), deposition is accumulated over its physical footprint, and the resulting load N_s/a^2 drives the physical ignition criterion. This grouping is the key difference from the native multiple-discrete-cell treatment, in which one physical structure is fragmented into independently emitting and igniting raster cells.

The level-set time step is

$$\Delta t = 0.5\frac{\Delta x}{U_{\text{wind}}},$$

and surface spread is disabled. Spread success therefore means that the firebrand process alone propagates through the structural array. For each parameter point, mean rate of spread (ROS) is obtained by fitting structure position against first-ignition time.

The first family varies

$$d = 2.5, 5, \ldots, 30\ \mathrm{m}, \qquad GR = 5, 10, \ldots, 50\ \mathrm{pcs\,MW^{-1}\,s^{-1}},$$

with a fixed 40 mph wind. The second fixes the separation at 10 m, uses the same generation rates, and varies wind speed from 10 to 150 mph in 10 mph increments.

3 Derivation of expected behavior

Increasing separation moves targets into lower-probability transport tails, while increasing generation raises deposited load. Wind produces competing reach and ignition-cooling effects, so the critical generation boundary should rise at both low and high wind speeds.

The design count follows directly from the generated parameter products:

$$12\ \text{separations} \times 10\ GR = 120, \qquad 15\ \text{winds} \times 10\ GR = 150, \qquad N_{\text{design}} = 120 + 150 = 270.$$

The two baseline-equivalent coordinates remain distinct members of their respective response families. At the design-fire plateau, a physical structure has $HRR_{\max} = 400(100) = 40$ MW; consequently the

tested generation range corresponds to

$$\dot{N}_{\max} = GR(40) = 200,\ 400, \ldots, 2000 \text{ pcs s}^{-1}.$$

With $\Delta x = 2.5$ m and $CFL = 0.5$, the generated level-set step is $0.5(2.5)/[U_{\text{mph}}(0.44704)]$: it is 0.069904 s at 40 mph, 0.279617 s at 10 mph, and 0.018641 s at 150 mph.

For each coordinate, GR_c is selected from $\{5, 10, \ldots, 50\}$; if no point propagates, the rank-only sentinel is the next design level, 55. Thus a one-level wind-edge excess is $55 - 50 = 5$ pcs MW^{-1} s^{-1}, exactly the acceptance increment. The moderate set is 30–120 mph and the edge set is $\{10, 20, 130, 140, 150\}$ mph. The explicit wind metrics are

$$\overline{GR}_{c,\text{edge}} - \text{median}(GR_{c,\text{moderate}}) \geq 5, \qquad \frac{\overline{ROS}_{\text{moderate}}}{\overline{ROS}_{\text{edge}}} \geq 1.10.$$

The separation expectation is strong positive rank association of d with GR_c, negative rank association of d with ROS, and positive rank association of GR with ROS. These become $\rho \geq 0.85$, median $\rho \leq -0.50$, and median $\rho \geq 0.70$, respectively, with the independent response-scale ceiling max $ROS \leq 0.08$ m s^{-1}.

4 Verification metrics and acceptance criteria

For a coordinate value z, define the critical generation rate

$$GR_c(z) = \min\{GR : \text{the array sustains downwind spread}\}.$$

When no tested value succeeds, the postprocessor assigns the next design level, 55 pcs MW^{-1} s^{-1}, only for rank and boundary metrics; it does not reinterpret failure as an observed threshold.

For the separation family, the documented response predicts a nonlinear increase of GR_c with d, from approximately 2.5 to more than 30 pcs MW^{-1} s^{-1} [1, 2]. ROS should decrease with separation and increase with generation. The verification therefore requires Spearman rank correlation $\rho(d, GR_c) \geq 0.85$, median $\rho(d, ROS) \leq -0.50$, and median $\rho(GR, ROS) \geq 0.70$.

For the wind family, the critical boundary is U-shaped. Low wind gives insufficient reach and a high ignition threshold; very high wind improves transport but suppresses the smoldering-to-flaming transition through convective cooling. The mean critical generation rate outside 30–120 mph must exceed the median within that interval by at least 5 pcs MW^{-1} s^{-1}. Mean ROS in the moderate range must be at least 1.10 times the edge-wind mean. Finally, firebrand-driven ROS must remain below 0.08 m s^{-1}, consistent with the documented response scale [1, 2].

4.1 Justification of metric quantification

The rank limits quantify robust trends without assuming that the nonlinear responses are linear. A separation-threshold Spearman coefficient of 0.85 requires strong ordering across twelve separation levels while allowing a small number of adjacent ties or reversals near a discrete ignition boundary. The -0.50 ROS/separation and 0.70 ROS/generation limits require, respectively, at least a moderate negative and a strong positive monotone association despite flat-response regions. They therefore test the direction and persistence of the mechanisms rather than a fitted slope chosen for one parameter interval.

The 5 pcs MW^{-1} s^{-1} wind-edge excess equals one full generation-rate design increment, so the U-shaped boundary must be resolved by the experiment rather than inferred from sub-grid interpolation. The 1.10 moderate/edge ROS ratio requires a 10% effect beyond equality. The 0.08 m s^{-1} ceiling is a

scale check taken from the reported firebrand-driven response range for this design [1, 2]. All 270 points and the resolution-independent structure capability are mandatory because missing edge or threshold points can materially change both rank statistics and the inferred U shape.

4.2 Predeclared acceptance criteria

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Design completeness	Unique generation–separation and generation–wind points after baseline deduplication.	270 of 270 points
Required capability	Resolution-independent single-structure source, collector, and input data conversion procedure.	Available
Result completeness	Current admissible result rows divided by 270.	100%
Threshold/separation rank	Spearman $\rho(d, GR_c)$.	≥ 0.85
ROS/separation rank	Median Spearman $\rho(d, ROS)$ across generation levels.	≤ -0.50
ROS/generation rank	Median Spearman $\rho(GR, ROS)$ across both response families.	≥ 0.70
Wind-edge threshold excess	Mean critical generation outside 30–120 mph minus the median inside that range.	$\geq 5\,\mathrm{pcs\,MW^{-1}\,s^{-1}}$
Moderate/edge ROS ratio	Mean ROS at 30–120 mph divided by mean ROS outside that range.	≥ 1.10
Maximum ROS	Maximum firebrand-driven ROS over all admissible design points.	$\leq 0.08\,\mathrm{m\,s^{-1}}$
Overall decision	Capability, complete results, and every trend/scale metric are conjunctive.	PASS only if all pass

5 Simulation configuration and predicted outcome

Although execution remains capability-gated, every design point now carries an auditable spatial configuration. Structures are rasterized as a periodic two-dimensional lattice over a 220×100-m physical domain. The generation–separation family applies its selected gap in both coordinate directions; the generation–wind family uses the 10 m baseline gap. Every footprint in the first upwind structure column is marked by the non-positive initial level-set field.

Figure 1 shows the baseline-separation geometry. The two-cell numerical halo is excluded. The left panel verifies repetition in both directions, while the right panel verifies that ignition spans the entire first structure column instead of selecting only the lower-left structure.

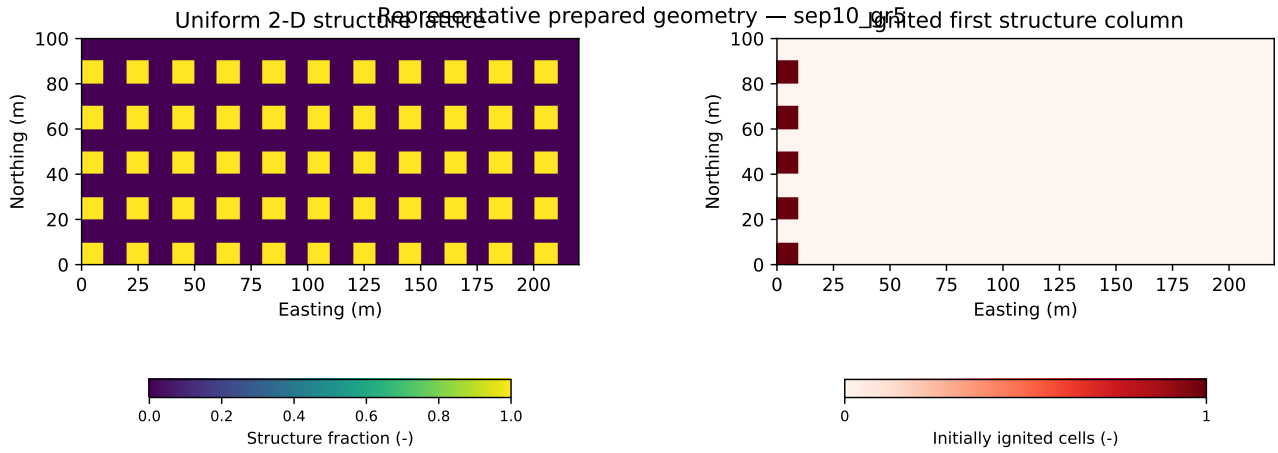


Figure 1: Representative prepared two-dimensional structure lattice and first-column initial ignition mask for the capability-gated parametric design.

The Courant–Friedrichs–Lewy (CFL) number measures the distance advected during one timestep relative to the grid spacing and constrains the time discretization.

Table 2: Controlled parametric design.

Parameter	Separation family	Wind family
Grid spacing	2.5 m	2.5 m
Physical domain	220 × 100 m	220 × 100 m
Structure side	10 m	10 m
Separation	2.5–30 m, step 2.5 m	10 m
Wind	40 mph	10–150 mph, step 10 mph
Generation	5–50, step 5	5–50, step 5
Spatial treatment	single physical unit	single physical unit
Initial ignition	complete first structure column	complete first structure column
CFL based on wind	0.5	0.5
Surface spread	disabled	disabled
Number of points	120	150

Expected ignition classifications form an increasing boundary in the (d, GR) plane and a U-shaped boundary in the (U_{wind}, GR) plane. The ROS maps should be largest for high production, short separation, and moderate wind. Hamada-model curves included in the documented parameter comparison [1, 2] are contextual because Hamada includes radiation and flame contact; they are not acceptance references for this firebrand-only test.

5.1 Preprocessing, execution, and postprocessing

The parameter design contains all 270 prescribed combinations. Each defines co-registered structure-fraction, structure-identity, fuel-model, and initial level-set fields. A combination is simulated only if the tested ELMFIRE version supports the resolution-independent structure formulation and its required input representation. Substituting the native cell-based formulation would test a different discretization and would not establish the intended result.

Analysis requires the specified quantities for the complete parameter design, derives critical boundaries, calculates the rank and response metrics, and constructs the classification and spread-rate compar-

isons. Missing evidence remains NOT EVALUABLE; an unavailable model capability is not replaced by a different physical or numerical formulation.

6 Actual simulation results

No whole-domain ELMFIRE result is available because the required integrated-structure capability is absent and no simulation condition is runnable under the current source interface. The postprocessor therefore does not manufacture a raster or substitute an analytical field for model output. The case remains NOT EVALUABLE, and the response map below is reported only as the prescribed verification envelope.

0 of 270 exact-design result points are present. The decision is **NOT EVALUABLE**. Figure 2 shows only the expected qualitative boundaries while the capability is absent. Every panel is labeled accordingly; no guide curve is represented as an ELMFIRE result.

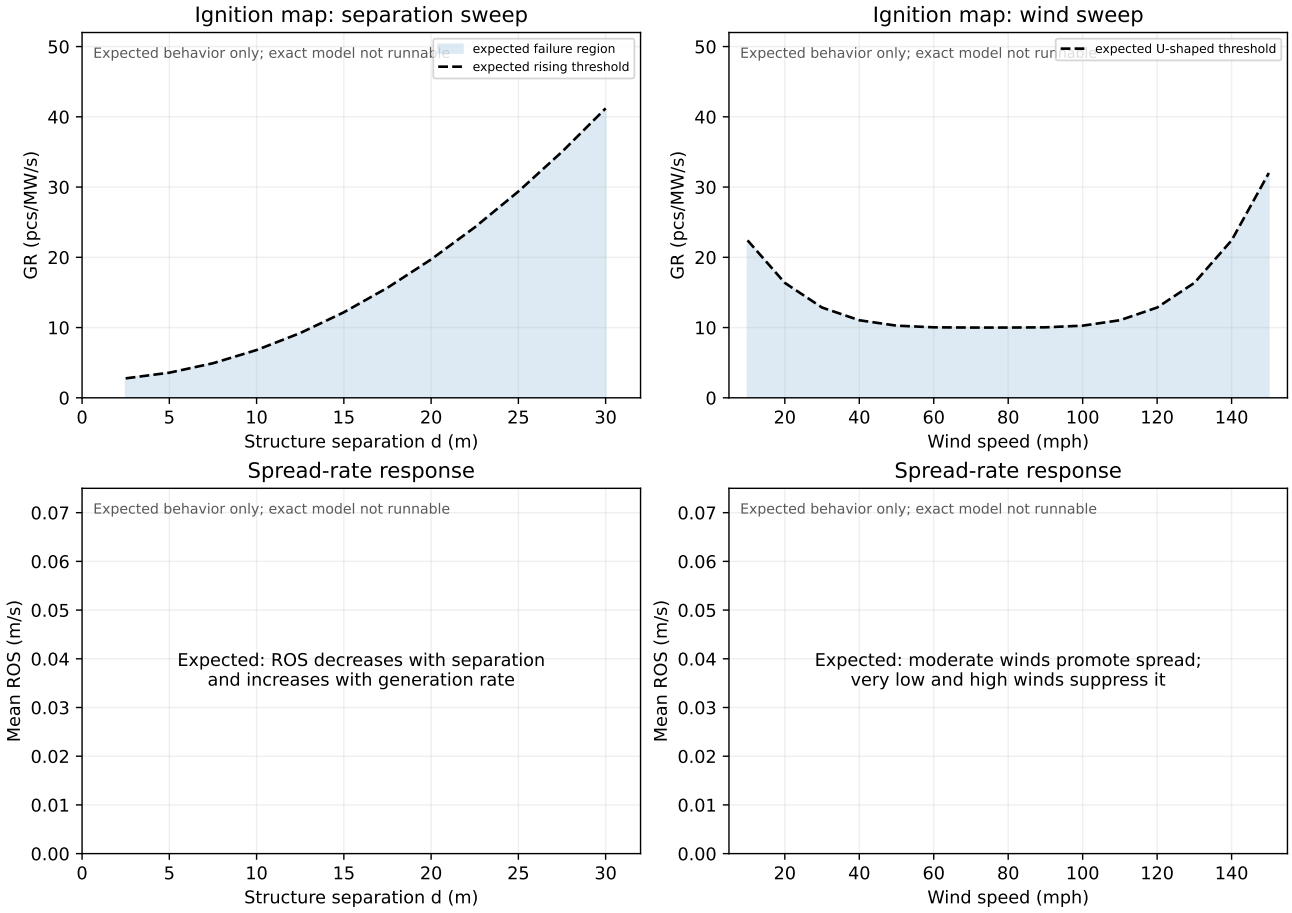


Figure 2: Parametric-response verification layout. The upper panels reproduce the coordinate systems of the two parametric reference maps; the lower panels are reserved for the corresponding ROS curves. In the current capability-gated state, shaded regions and dashed lines communicate expected behavior only.

7 Calculated metrics and verification decision

Table 3: Calculated verification metrics and component decisions.

Metric	Acceptance limit	Calculated	Status
Parameter-design completeness	270/270	270	PASS
Exact single-structure capability	available	N/A	NOT EVALUABLE
Current result completeness	100%	0	NOT EVALUABLE
Critical GR versus separation Spearman rho	≥ 0.85	N/A	NOT EVALUABLE
ROS versus separation median Spearman rho	$\leq$ -0.50	N/A	NOT EVALUABLE
ROS versus generation median Spearman rho	≥ 0.70	N/A	NOT EVALUABLE
Low/high-wind threshold excess	$\geq$ 5 pcs/MW/s	N/A	NOT EVALUABLE
Moderate/edge-wind ROS ratio	≥ 1.10	N/A	NOT EVALUABLE
Maximum firebrand-driven ROS	$\leq$ 0.08 m/s	N/A	NOT EVALUABLE

The current decision is **NOT EVALUABLE**, not a failure of the firebrand equations. The intended resolution-independent parameter sweep cannot be executed because the current source does not implement the single-structure source and collector required by the cited model description [1]. The native model would split each 10-m structure into sixteen 2.5-m cells and change HRR, transport, and accumulation; those outputs are not admissible evidence for this case.

The complete parameter design, expected behavior, result configuration definition, metrics, figure generator, and standalone report are now present. Once the source feature and its input data conversion procedure are implemented, the capability gate should be enabled, all 270 points executed, and the same postprocessor used without changing acceptance limits.

8 Technical interpretation

The capability gate prevents cell-local structural discretization from being presented as evidence for a single-structure parametric model. Once admissible results exist, boundary and rank metrics distinguish physical response trends from isolated successful points.

9 Scope and limitations

The design covers idealized, uniform arrays and verifies qualitative model response over the declared parameter grid. It remains not evaluable until the required single-structure source and collector are implemented.

References

[1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

[2] Y. Qin et al., "Simulations of firebrand-driven fire spread in landscape-scale Wildland–Urban-Interface and urban conflagration models," *Fire Safety Journal*, vol. 162, 104686, 2026, doi:10.1016/j.firesaf.2026.104686.

Case 14: Two-Dimensional Spatial-Resolution Sensitivity

1 Verification purpose

This case is a negative verification test for mesh dependence in the current multiple-discrete-cell treatment of structures. A physical 10 m-by-10 m house may occupy one raster cell, several independent cells, one oversized cell, or disappear under categorical resampling as the mesh changes. Because those cells are not assembled into one conserved structure object, changing resolution alters represented heat release, firebrand generation, deposition, and structure-to-structure ignition.

The test asks whether all 15 input-matched simulations produce an evaluable community rate of spread (ROS), and whether the fitted values remain within the predeclared mesh-independence limits relative to the run-derived 10-m anchor. Detecting the known resolution sensitivity is useful diagnostic evidence, but it is a *verification failure*, not a pass.

2 Mathematical formulation and numerical implementation

2.1 Mode-resampled square array

The physical layout is a periodic array of square structures of side $a = 10\,\mathrm{m}$, separated in both coordinates by $d = 10\,\mathrm{m}$. Define the one-dimensional structural indicator

$$I(s) = \begin{cases} 1, & s \bmod (a+d) < a, \\ 0, & \text{otherwise.} \end{cases}$$

For grid interval $C_i = [i\Delta, (i+1)\Delta]$, preprocessing calculates

$$f_i = \frac{1}{\Delta}\int_{C_i} I(s)\,ds, \qquad b_i = \mathbf{1}(f_i \geq 1/2), \qquad B_{ij} = b_i b_j.$$

This coordinate-wise categorical-mode construction reproduces the reported geometries. At $\Delta = 7\,\mathrm{m}$, effective side lengths are 7 or 14 m and footprints range from 49 to $196\,\mathrm{m}^2$. At $\Delta = 10\,\mathrm{m}$, footprint and gap are exact. At $\Delta = 30\,\mathrm{m}$, one represented structure cell has area $900\,\mathrm{m}^2$ and represented structures are separated by 30 m. The 20- and 40-m maps eliminate represented firebreaks; this is an intended resampling diagnostic, not a preprocessing failure.

2.2 Cell-local firebrand treatment

A represented structural cell uses a design-fire curve $q''(t)$. The curve has a 300-s linear growth phase, a one-hour plateau at $400\,\mathrm{kW\,m^{-2}}$, and a 300-s decay. Under the current multiple-discrete-cell treatment,

$$\mathrm{HRR}_{ij}(t) = q''(t)\Delta^2, \qquad \dot{N}_{ij}(t) = GR\,\frac{\mathrm{HRR}_{ij}(t)}{1000}, \qquad GR = 10\,\mathrm{pcs\,MW^{-1}\,s^{-1}}.$$

Changing Δ therefore changes both represented geometry and the heat-release rate (HRR) of each independent emitter. Firebrands use the structural empirical flight-distance distribution, Eulerian accumulation, and physical ignition. Non-structure cells use nonburnable fuel model 93, so vegetative surface spread is absent and every downwind ignition must be caused by deposited firebrands. The global vegetative surface-spread setting switch remains true to suppress vegetative surface-fire propagation. The

current source independently refreshes initialized model-14 building transient HRR under this setting. The configured time-step scaling is documented by Qin et al. [2]:

$$\Delta t = \mathrm{CFL}\frac{\Delta}{U_{\mathrm{wind}}}, \qquad \mathrm{CFL} = 1, \qquad U_{\mathrm{wind}} = 17.9\,\mathrm{m\,s^{-1}}\ (40\,\mathrm{mph}).$$

The fractional step is required to hold this spatial sweep at fixed wind Courant–Friedrichs–Lewy (CFL). Preprocessing sets each generated stop to $\lceil 150000/\Delta t \rceil \Delta t$, then records the requested stop, aligned stop, and integer step count in the simulation specification.

For each represented downwind structure column s, postprocessing takes the median positive finite arrival time t_s. At least three ignited columns are required. Mean community ROS is the least-squares slope

$$ROS_\Delta = \frac{\sum_s (t_s - \bar{t})(x_s - \bar{x})}{\sum_s (t_s - \bar{t})^2}.$$

Because the 10-m grid exactly resolves both physical length scales, its fitted ELMFIRE result is the comparison anchor,

$$ROS_{10} = ROS_{\Delta=10\,\mathrm{m}}.$$

The dashed reference line and the 10-m ELMFIRE point therefore use the same calculated value; no rounded external estimate is substituted.

3 Derivation of expected behavior

A resolution-independent structure treatment would conserve each house's physical footprint, integrated heat release, firebrand source, accumulated load, and ignition state. Subject to numerical error, it would give $ROS_\Delta \approx ROS_{10}$ for every resolution and enough downstream ignitions to fit all 15 ROS values.

The current cell-local representation does not conserve those quantities. Fine meshes fragment houses into multiple emitters, intermediate meshes alter footprints and gaps, and some coarse meshes merge structures or remove firebreaks. The expected diagnostic result is therefore a nonuniform ROS curve, including possible failure to propagate at some resolutions. That observation exposes the defect; against the mesh-independence requirement, its correct overall status is FAIL.

The period of the one-dimensional physical indicator is $a + d = 10 + 10 = 20$ m. The key raster substitutions are therefore:

Δ (m)	mode-resampled consequence	component-area reference
7	one or two selected cells per coordinate	$7^2 = 49$ to $(2 \cdot 7)^2 = 196$ m^2
10	one structure cell followed by one gap cell	$10^2 = 100$ m^2
30	one selected cell per represented structure	$30^2 = 900$ m^2

At $\Delta = 20$ m, each cell contains 10 m of structure and 10 m of gap, so $f_i = 0.5$ and the $f_i \geq 0.5$ rule selects every cell; the represented firebreak vanishes. The same 50-percent selection occurs for the 40-m map. These are reproducible consequences of the generated mode rule, not inferred from the output fire spread.

With $U = 40(0.44704) = 17.8816$ m s^{-1}, each generated step is

$$\Delta t = \Delta/U; \quad \Delta t_7 = 0.391463, \quad \Delta t_{10} = 0.559234, \quad \Delta t_{30} = 1.677702\ \mathrm{s}.$$

A truly resolution-independent treatment would give all 15 finite fits, $f_{\mathrm{fit}} = 15/15 = 1$, $ROS_\Delta = ROS_{10}$, $CV_R = 0$, and $D_{\max} = 0$. The current native cell treatment is expected instead to produce missing or nonuniform fits because source HRR scales as Δ^2 and the represented geometry changes.

The predeclared mesh-independence requirements permit $CV_R \leq 0.25$ and $D_{\max} \leq 0.50$; exceeding either limit or failing any of the 15 fits yields the expected diagnostic FAIL.

The primary publication provides the independent propagation reference calculation used only for strict stalled-final compatibility: Figure 9 reports zero ROS at 1, 3, 7, 13, 17, 23, 25, and 27 m, and finite propagation at 5, 10, 15, 20, 30, 40, and 50 m [2]. An early stalled-final dump may therefore use the no-propagation exception only for the first set, and only after the meteorology-interval reconstruction and timestep-grid checks also pass. This reference calculation does not replace the current-run metrics or authorize accepting an arbitrary overrun timestamp.

4 Verification metrics and acceptance criteria

Preprocessing passes only when all 15 resolutions are generated, the exact 10-m map has no footprint or gap error, the 7-m footprint range is 49–196 m^2, and the median 30-m footprint is 900 m^2. These checks establish that execution comparison uses the intended resolution-dependent rasterizations.

A complete execution decision additionally requires every input-matched simulation condition and a finite 10-m ROS anchor. Let N_{fit} be the number of resolutions with at least three ignited downwind structure columns. ROS extraction completeness is

$$f_{\text{fit}} = \frac{N_{\text{fit}}}{15},$$

and must equal one. For the finite fitted values, mesh dependence is measured by the coefficient of variation

$$CV_R = \frac{\text{std}_\Delta(ROS_\Delta)}{\text{mean}_\Delta(ROS_\Delta)}$$

and by the largest deviation from the exact-geometry anchor,

$$D_{\max} = \max_\Delta \frac{|ROS_\Delta - ROS_{10}|}{ROS_{10}}.$$

The permissive engineering limits are $CV_R \leq 0.25$ and $D_{\max} \leq 0.50$. Missing fits are handled by f_{fit}; they are not silently discarded and counted as convergence. The overall decision is the conjunction of every geometry, execution, anchor, fit-completeness, and mesh-independence check.

4.1 Justification of metric quantification

The geometry limits are exact consequences of rasterization. At $\Delta = 10$ m, the structure side and gap are both exactly one cell, so the 1% footprint and separation allowances cover only file precision and connected-component bookkeeping. At 7 m, a represented component can contain one to four cells, giving areas from $7^2 = 49$ to $4 \times 7^2 = 196$ m^2. At 30 m, a one-cell represented structure has area $30^2 = 900$ m^2. These checks establish that the intended mesh-dependent maps were generated; they are not claims of geometric accuracy.

For the behavioral requirement, $f_{\text{fit}} = 1$ is non-negotiable because mesh independence cannot be inferred after silently discarding resolutions that fail to propagate. The $CV_R \leq 0.25$ and $D_{\max} \leq 0.50$ limits are intentionally permissive engineering bounds, substantially larger than ordinary front-fit quantization. Even with those allowances, a method with factor-of-two variation or missing fits fails, which is the defect this case is designed to expose [2]. The 10-m reference is the actual ROS_{10} extracted by the same postprocessor, so its plotted reference and ELMFIRE point must coincide by construction; it is not an independent nominal value. Current input identity records and all 15 runs are required to prevent stale or selectively complete outputs from weakening the expected FAIL diagnosis.

4.2 Predeclared acceptance criteria

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Prepared simulation conditions	Generated simulation specification count divided by the 15 requested grid spacings.	100%
10-m footprint error	Maximum relative error of represented component areas from 100 m^2.	≤ 0.01
10-m separation error	Maximum relative gap-width error from 10 m.	≤ 0.01
7-m component areas	Minimum and maximum represented connected-component areas.	49–196 m^2
30-m footprint error	Relative error of median represented footprint from 900 m^2.	≤ 0.01
Execution completeness	Current-input identity record final time of arrival (TOA) output and successful termination.	15 of 15 simulation conditions
10-m ROS anchor	Structure-column fit from the current 10-m TOA raster.	Finite ROS_{10}
ROS extraction completeness	Resolutions with at least three ignited downwind columns.	$f_{\text{fit}} = 1$
ROS mesh-dependence CV	Coefficient of variation over finite fitted ROS values.	$CV_R \leq 0.25$
Maximum anchor deviation	Maximum finite-grid relative deviation from ROS_{10}.	$D_{\max} \leq 0.50$
Overall decision	Every geometry, completeness, and ROS component is conjunctive.	PASS only if all pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for a 10 m grid. These panels show the prepared spatial fields, remove the two-cell numerical buffer, and retain map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

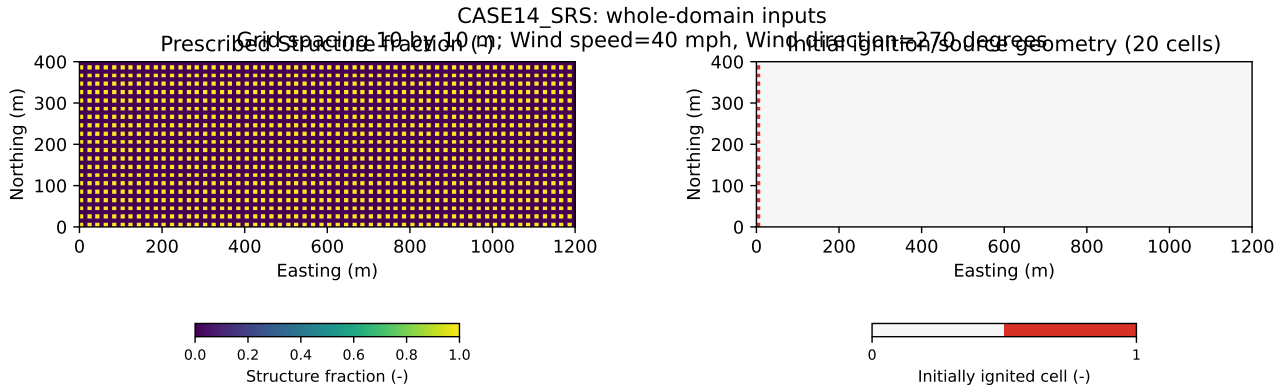


Figure 1: Whole-domain prepared input configuration for the representative simulation condition. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

Table 2: Configuration of the two-dimensional resolution sweep.

Parameter	Configured value	Units	Role
Physical domain	1200×400, flat	m	permits many downwind ignitions
Structure side / gap	10 / 10	m	exact physical scales at the 10-m grid
Initial ignition	complete first structure column	–	uniform upwind source across all structure rows
Grid spacings	1, 3, 5, 7, 10, 13, 15, 17, 20, 23, 25, 27, 30, 40, 50	m	exposes raster and cell-local dependence
Numerical halo	2 on every side	cells	excluded from physical geometry
Wind	40 from 270 degrees	mph	constant west-to-east transport
Time step	Δ/U_{wind}	s	holds wind CFL at one
Design-fire peak	400	kW m^{-2}	structural heat-release rate per unit area (HRRPUA) input
Generation rate	10	$\text{pcs MW}^{-1}\ \text{s}^{-1}$	cell-local building source
Transport / accumulation	empirical / Eulerian	–	current structural spotting path
Ignition delays	42 plus 300	s	small-flame and development stages
Vegetative surface spread	excluded by fuel 93	–	isolates firebrand-driven propagation
Stop time	at least 150000	s	rounded up to an exact timestep multiple
Final output	one TOA per simulation condition	–	source for structure-column ROS fits

The dashed reference line is predicted to intersect the 10-m ELMFIRE point exactly because both are ROS_{10}. A mesh-independent implementation would cluster all other finite ROS points around this line and provide a fit at every resolution. The current independent-cell treatment is instead expected to produce missing fits and large variations, causing the behavioral verification to fail.

5.1 Preprocessing, execution, and postprocessing

input preparation procedure places customizable constants at its beginning, constructs the periodic map analytically, assigns persistent structure IDs, writes co-registered GeoTIFF inputs and case-specific fuel tables, and records a SHA-256 input identity record for every simulation condition. It also writes structure geometry measurements, making the map checks independently auditable.

All 15 prescribed conditions are simulated independently and sequentially. Each result must match the corresponding input configuration before the resolution comparison is evaluated.

result analysis procedure rejects stale or incomplete outputs, reads the final time-of-arrival raster, groups cells by structure, fits ROS, calculates all metrics above, and produces the report figure.

6 Actual simulation results

Figure 2 shows the representative time-of-arrival field from a 10 m grid. The displayed field is taken from the simulated results, masks missing values, and excludes the two-cell numerical buffer. The 10-m field is the reference-scale spatial result used to interpret mesh-dependent structure spread across the resolution sweep.

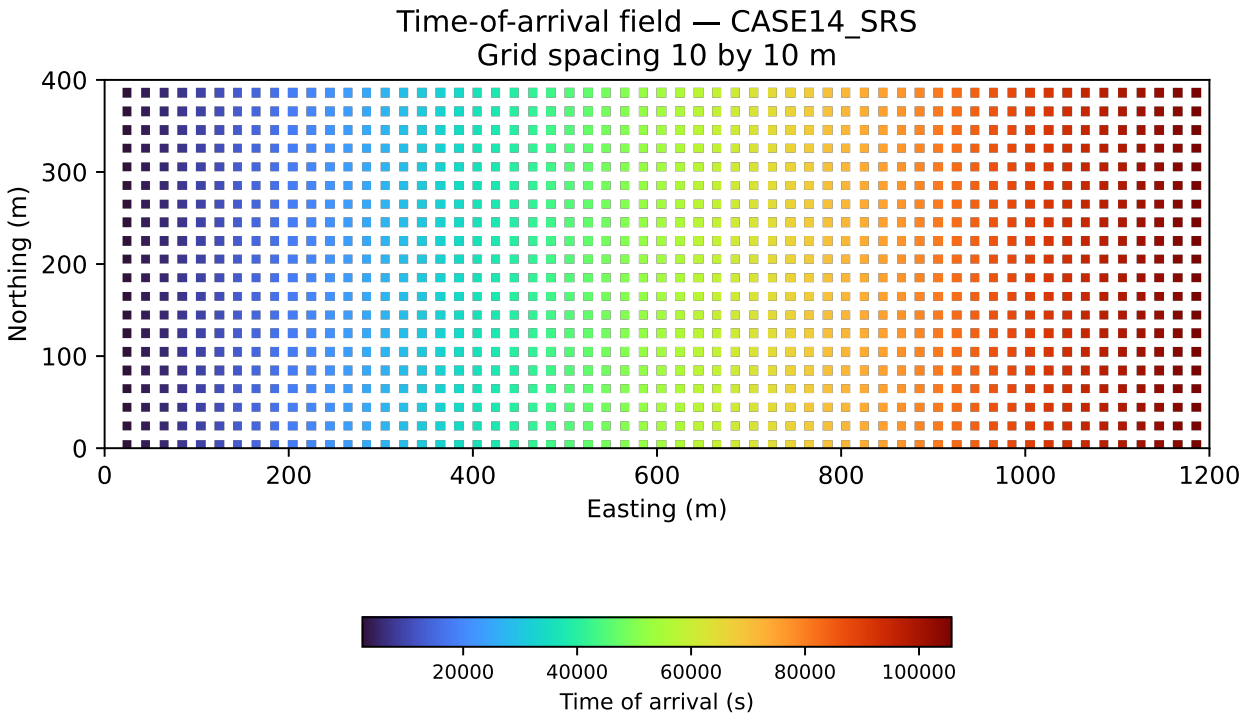


Figure 2: Whole-domain time-of-arrival field from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

All 15 input-matched ELMFIRE processes reached the source success message. The strict terminal-evidence selector accepts 7 of 15 outputs; process exit alone is not used as evidence. Regular terminal dumps are accepted for 1, 3, 5, 7, 10, 20, 30, and 40 m, and the publication-derived independent prediction of no propagation permits the timestep-aligned stalled-final path for 13, 17, and 25 m. The selected files, configured controls, final times, reconstructed pre-jump times, grid residuals, and rejection reasons are recorded per simulation condition in calculated verification and validation metrics. The overall decision calculated by postprocessing is **NOT EVALUABLE**.

The upper row of Figure 3 shows the actual preprocessed geometries. Its lower panel contains fitted ELMFIRE ROS values. Red crosses at zero identify completed simulations with fewer than three ignited downwind columns, for which ROS is deliberately not fitted. The dashed line is $ROS_{10} = N/A\ \mathrm{m\,s^{-1}}$, exactly equal to the 10-m ELMFIRE point.

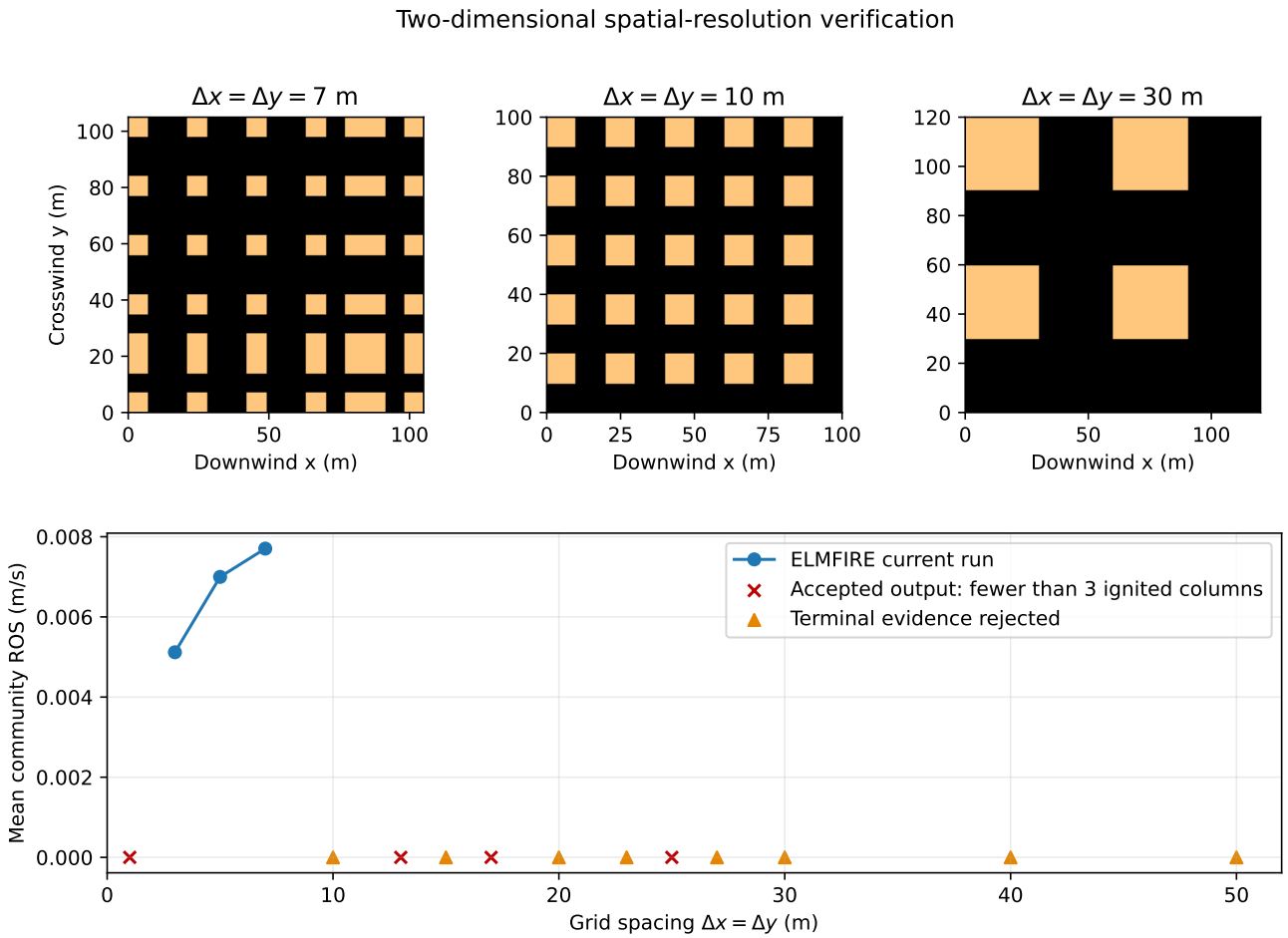


Figure 3: Two-dimensional spatial-resolution diagnostic. Top: mode-resampled WUI geometry at 7, 10, and 30 m. Bottom: mean community ROS from current ELMFIRE TOA outputs; red crosses identify resolutions without enough downwind ignitions for a fit, and the dashed line is the run-derived 10-m anchor. The geometry selection follows the documented comparison configuration [2].

7 Calculated metrics and verification decision

Table 3: Calculated verification metrics and component decisions.

Metric	Acceptance limit	Calculated	Status
Prepared resolution simulation conditions	$\geq$ 100%	1	PASS
dx=10 m footprint error	$\leq$ 0.01	0	PASS
dx=10 m separation error	$\leq$ 0.01	0	PASS
dx=7 m component-area range	49 to 196 m2	49 to 196	PASS
dx=30 m median footprint error	$\leq$ 0.01	0	PASS
Current-input identity record run completeness	15 simulation conditions	7/15	NOT EVALUABLE
10-m ELMFIRE ROS anchor	finite fitted ROS	N/A	NOT RUN
ROS-evaluable resolution fraction	$\geq$ 100%	N/A	NOT RUN
ROS mesh-dependence CV	$\leq$ 0.25	N/A	NOT RUN
Maximum ROS deviation from 10-m anchor	$\leq$ 0.50	N/A	NOT RUN

The postprocessor reports **NOT EVALUABLE**. Geometry preparation passes, and all 15 processes executed, but strict terminal evidence is complete for only 7 of 15 simulation conditions. The 10-m anchor, ROS-evaluable fraction, coefficient of variation, and maximum anchor deviation therefore remain `N/A`; calculating them from a selected subset would silently change the predeclared verification population.

8 Technical interpretation

The accepted subset contains finite fitted values of approximately 0.005 12 m s^{-1} (3 m), 0.007 00 m s^{-1} (5 m), 0.007 70 m s^{-1} (7 m), 0.011 23 m s^{-1} (10 m), 0.026 17 m s^{-1} (20 m), 0.016 12 m s^{-1} (30 m), and 0.040 36 m s^{-1} (40 m). These values visibly demonstrate mesh sensitivity, but they are diagnostic observations rather than a complete acceptance calculation.

Four successful processes do not provide admissible terminal evidence. For 15 m, the recorded final time is 152833.592 s and subtracting the configured 3600-s meteorology interval gives 149233.592 s; the timestep-grid residual is 0.302 s. For 23 and 27 m the corresponding reconstructed times are 146507.060 and 148242.909 s, with grid residuals 0.636 and 0.705 s. Those residuals exceed the 0.150-s single-precision-scale tolerance. The 50-m reconstruction is grid-aligned (0.023-s residual), but occurs at 148236.198 s, more than one 2.796-s timestep before the configured stop. The primary publication reports propagation at 15 and 50 m, so neither early stall may be reclassified as independently expected no propagation. Conversely, the publication reports no propagation at 23 and 27 m, but that reference calculation does not waive the timestep-grid requirement.

The current source also produces finite spread at 3 and 7 m while the published Figure 9 places those resolutions at zero, and it stalls at 15 and 50 m where that figure shows finite spread. The resulting **NOT EVALUABLE** status is therefore an implementation/reference and terminal-time discrepancy, not a missing execution and not a reason to weaken the evidence selector. The runs used executable SHA-256 `c225b2e00349aa98128285b3d3ff9b2031d0e20944cd9f1c5ee96595d199a920` built from source commit `a2dfbcdf72209733c000e5d3431e44723e15efea`; the source worktree already contained unrelated local modifications, so the commit and executable digest are both reported for reproducibility.

9 Scope and limitations

This case verifies numerical mesh dependence for one idealized periodic WUI array, one wind, and one structural design fire. It does not validate the physical accuracy of any individual ROS value. The 10-m result is an internal run-derived anchor, not experimental validation data. A future resolution-independent structure implementation should be tested with the same geometry and acceptance criteria; it should provide all 15 ROS fits and reduce both variation metrics below their limits.

References

[1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

[2] Y. Qin et al., "Simulations of firebrand-driven fire spread in landscape-scale Wildland–Urban-Interface and urban conflagration models," *Fire Safety Journal*, vol. 162, 104686, 2026, doi:10.1016/j.firesaf.2026.104686.

Case 15: Constant rate of spread

1 Verification purpose

This isolated-process case verifies that ELMFIRE transports a circular level-set front at a prescribed, spatially uniform normal rate. It exercises raster ingestion, the spread-rate adjustment factor, the level-set update, and level-set field/spread rate output without wind, slope, crown fire, spotting, or suppression. It can reveal rate-unit errors, incorrect signs, and failure to couple the calculated rate to front motion. This is numerical verification against a known solution, not comparison with a physical fire. The verification question is whether the final spread-rate and radius satisfy the two stated tolerances at $t = 360$ s.

2 Mathematical formulation and numerical implementation

ELMFIRE advances the signed level set by

$$\frac{\partial \phi}{\partial t} + V_n |\nabla \phi| = 0,$$

with $\phi < 0$ burned and $\phi = 0$ the front. Uniform rasters and disabled directional forcing make $V_n = R_0 = 0.0116332$ m s^{-1}. The configured metric spread-rate output option makes the spread rate output use m min^{-1}, while integration time remains seconds. Raster georeferencing supplies the mesh; no obsolete computational-domain namelist is used.

3 Derivation of expected behavior

For initial circle radius $r_0 = 5$ m and constant outward speed R_0, radial symmetry reduces the level-set equation to $dr/dt = R_0$. Hence

$$r(360) = 5 + (0.0116332)(360) = 9.187952 \text{ m}, \qquad R_0 = 0.697992 \text{ m min}^{-1}.$$

These are independent consequences of the prescribed input rate and elapsed time.

The numerical reference values and the reported metrics follow directly from the generated rate raster. The raster value is $R_s = 0.0116332$ m s^{-1}, so the report-unit spread rate is

$$R_{\text{ref}} = 60 R_s = 60(0.0116332) = 0.697992 \text{ m min}^{-1}.$$

With $r_0 = 5$ m and $T = 360$ s, the exact final radius is

$$r_{\text{ref}} = r_0 + R_s T = 5 + (0.0116332)(360) = 9.187952 \text{ m}.$$

The two calculated errors are $e_R = |R_{\max} - 0.697992|/0.697992$ and $e_r = |\sqrt{N_b (1 \text{ m})^2/\pi} - 9.187952|$, where N_b is the number of final cells with $\phi < 0$. Thus the acceptance substitutions are $e_R \leq 0.05$ and $e_r \leq 2$ m.

4 Verification metrics and acceptance criteria

The maximum finite final spread rate value is compared with 0.697992 m min^{-1}; a 5 percent relative tolerance catches conversion or propagation-rate errors while allowing front reconstruction effects. The

equivalent burned radius $r_A = \sqrt{N_b \Delta x^2 / \pi}$, where N_b counts valid final cells with $\phi < 0$, is compared with 9.187952 m. Area-equivalent radius is less sensitive to pixel stair-stepping than a single axial intercept; its tolerance is two cells. Nodata cells are excluded. The lexicographically last matching final level-set field and spread rate rasters are required. The case passes iff both components pass; missing or malformed required output is NOT EVALUABLE.

5 Simulation configuration and predicted outcome

The generated EPSG:32610 domain is 200 m square with 200 cells per side ($\Delta x = 1$ m), uniform fuel/moisture/terrain/weather fields, and a centered 5 m ignition circle. Duration is 360 s. The predicted final field is circular, with rate 0.697992 m min^{-1} and equivalent radius 9.187952 m.

Parameter	Configured value	Units	Role
Domain/grid	200 by 200 cells; 1	m/cell	Resolves the manufactured circle
Initial front	centered radius 5	m	Exact initial condition
Rate	0.0116332	m/s	Constant manufactured normal speed
Time control	step 1; stop 360	s	Exact final time and timestep multiple
Disabled forcing	wind, slope, crown, spotting, suppression	—	Isolates level-set transport

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

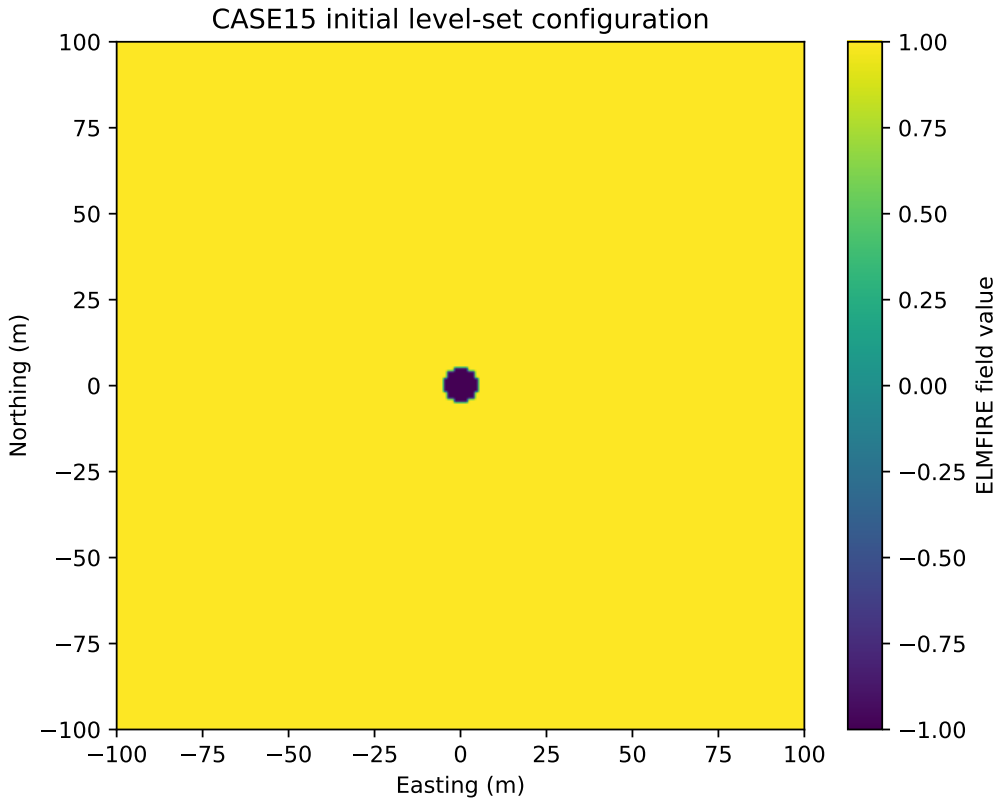


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

Only case-specific ELMFIRE rasters are read. Current status is **PASS**; 2 of 2 components pass.

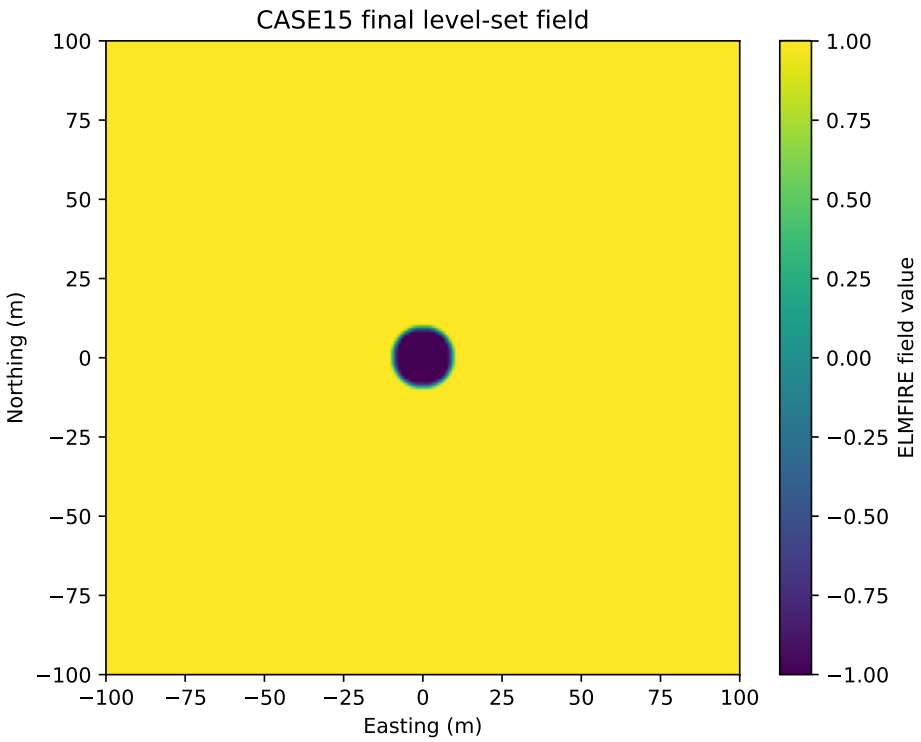


Figure 2: Final whole-domain signed level-set field for the completed simulation condition at the final output time, selected from the results of that simulation condition. Values are dimensionless signed-level-set values; the generated table supplies the acceptance limits.

7 Calculated metrics and verification decision

Metric	Expected value or limit	Calculated value	Status
maximum spread rate	0.697992 (5% relative)	0.698532 m/min	PASS
equivalent final radius	9.18795 (2 cells (2 m))	9.37302 m	PASS

The authoritative overall decision is **PASS**. The generated table is sourced directly from the case-specific calculated metrics and is not recomputed in this report.

8 Technical interpretation

A spread-rate failure points to input scaling, units, or spread rate selection. A radius-only failure indicates level-set integration or front/area discretization. A missing raster is an evidence failure, not a scientific pass.

9 Scope and limitations

This case verifies one deterministic constant-rate, flat, no-wind propagation state. It does not validate a fuel model, quantify anisotropy, or test coupled crown, spotting, or suppression behavior.

Case 16: No-spread limiting behavior

1 Verification purpose

This isolated-process case verifies the zero-normal-velocity limit of the ELMFIRE level-set pathway. A zero spread-rate adjustment factor raster suppresses the base spread rate while wind, slope, spotting, crown fire, and suppression are disabled. The case exercises input multiplication, propagation, and level-set field/spread rate output and can expose ignored adjustment rasters, nonzero numerical forcing, or unintended growth. It is implementation verification, not physical validation. The question is whether the final rate remains below 10^{-6} m min^{-1} and the ignited footprint remains confined to the initialization band.

2 Mathematical formulation and numerical implementation

The propagation equation is $\partial_t \phi + V_n |\nabla \phi| = 0$. The case sets the multiplicative adjustment to zero, so $V_n = 0$ and therefore $\partial_t \phi = 0$. ELMFIRE still initializes a grid-resolved ignition neighborhood, so exact zero burned area is not expected; the invariant is no subsequent outward propagation.

3 Derivation of expected behavior

Integrating $\partial_t \phi = 0$ gives $\phi(\mathbf{x}, t) = \phi(\mathbf{x}, 0)$. Thus the output rate is zero and any negative-ϕ area must be limited to the rasterized initialization band. A 2.5-cell equivalent-radius allowance bounds that band without treating subcell front placement as spread.

The generated pre-ignition ϕ raster contains no negative cells and the adjustment raster is identically zero. Thus

$$V_n = R_0(ADJ) = R_0(0) = 0, \qquad \phi(\mathbf{x}, 10\text{ s}) = \phi(\mathbf{x}, 0), \qquad R_{\text{ref}} = 0\text{ m min}^{-1}.$$

The rate metric is therefore the absolute quantity $e_R = \max_i |R_i|$, with the substituted test $e_R \leq 10^{-6}$ m min^{-1}; a relative error is undefined for the zero reference. ELMFIRE may rasterize its point ignition into a small initialization footprint, so the spatial metric is

$$r_A = \sqrt{\frac{N_b (1\text{ m})^2}{\pi}},$$

where N_b counts final cells with $\phi < 0$. Its no-propagation reference is zero and the explicit grid allowance is $r_A \leq 2.5(1\text{ m}) = 2.5$ m.

4 Verification metrics and acceptance criteria

The maximum absolute finite final spread rate value tests the zero-velocity invariant directly and must not exceed 10^{-6} m min^{-1}. The equivalent burned radius $r_A = \sqrt{N_b \Delta x^2 / \pi}$ must be no more than $2.5\Delta x$; this whole-domain area measure detects growth in any direction and is more robust than one transect. Nodata is excluded, and the last matching final level-set field and spread rate rasters are required. Both conditions must pass; missing output is NOT EVALUABLE.

5 Simulation configuration and predicted outcome

The generated EPSG:32610 domain is 200 m square on a 200-by-200 grid. Fuel, moisture, and terrain are uniform; wind and slope are zero; a zero spread-rate adjustment factor; the ignition is centered; and duration is 10 s. The predicted spread rate field is zero and the final ignition footprint does not expand beyond its grid initialization band.

Parameter	Configured value	Units	Role
Domain/grid	200 by 200 cells; 1	m/cell	Resolves the stationary ignition
Initial front	centered rasterized band	—	Nonzero initialized footprint
Adjustment	ADJ=0	—	Forces zero normal speed
Time control	step 1; stop 10	s	Tests the zero-speed invariant
Disabled forcing	wind, slope, crown, spotting, suppression	—	Excludes alternate growth pathways

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

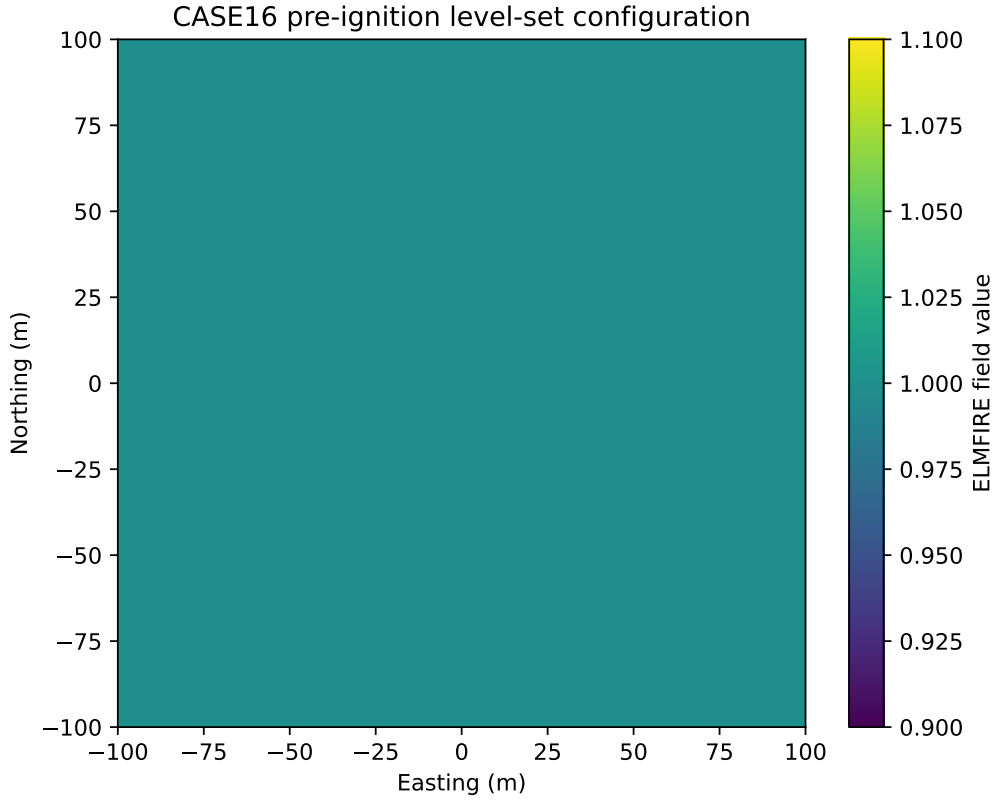


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

Only case-specific ELMFIRE rasters are read. Current status is **PASS**; 2 of 2 components pass.

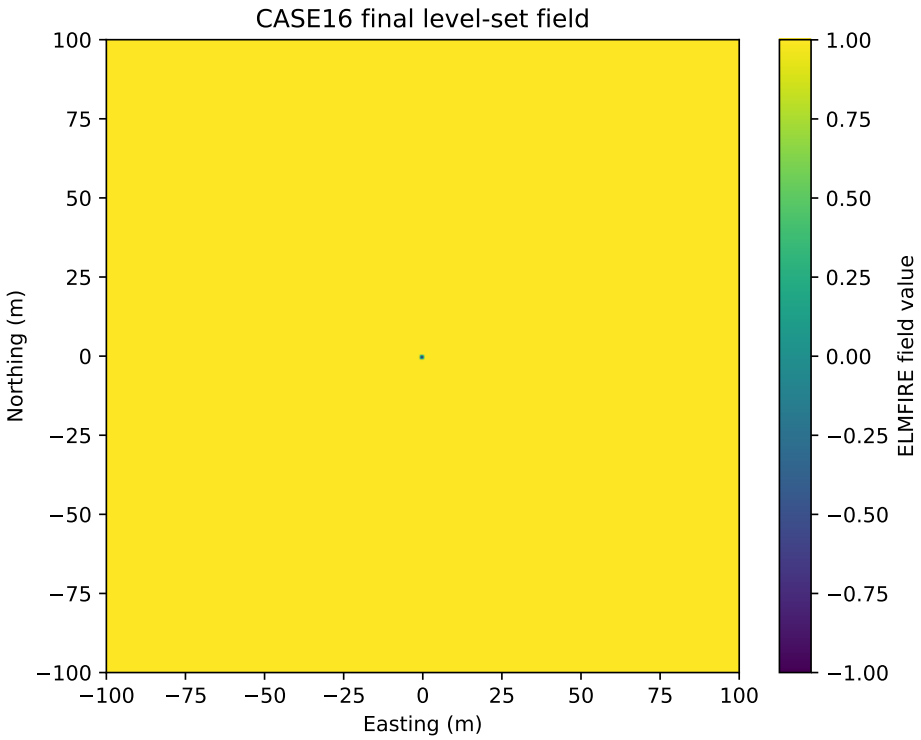


Figure 2: Final whole-domain signed level-set field for the completed zero-spread simulation condition at the final output time. Values are dimensionless; the generated table supplies the rate and confinement limits.

7 Calculated metrics and verification decision

Metric	Expected value or limit	Calculated value	Status
maximum absolute spread rate	0 ($\leq$ 1e-06)	0 m/min	PASS
equivalent burned radius	0 ($\leq$ 2.5 cells (2.5 m))	0.56419 m	PASS

The authoritative overall decision is **PASS**. The generated table is sourced directly from the case-specific calculated metrics and is not recomputed in this report.

8 Technical interpretation

A nonzero rate primarily indicates that the zero adjustment was ignored or transformed incorrectly. A confined rate with excessive burned area indicates initialization or level-set evolution behavior. Unavailable output cannot establish the invariant.

9 Scope and limitations

This case verifies a deterministic limiting state and raster adjustment coupling. It does not establish accuracy for positive rates, physical extinction, or any coupled fire process.

Case 17: Spatial convergence

1 Verification purpose

This isolated-process case verifies spatial convergence of ELMFIRE's level-set propagation for the same manufactured circular-front problem on four meshes. It exercises raster-derived grid geometry, front initialization, propagation, and level-set field output while holding physical inputs and temporal controls fixed. It can reveal grid-spacing misuse, nonconvergent front updates, or resolution-dependent physical configuration. The question is whether all four required meshes complete, the fitted radius-error order is at least 0.8, and the finest-grid error is at most 1 m.

2 Mathematical formulation and numerical implementation

For $\partial_t \phi + R_0 |\nabla \phi| = 0$ with constant $R_0 = 0.0116332$ m s^{-1}, the exact circular radius is mesh-independent. ELMFIRE obtains Δx from each GeoTIFF transform. The four simulation conditions use 128, 256, 512, and 1024 cells over the same 256 m domain, so $\Delta x = \{2, 1, 0.5, 0.25\}$ m.

3 Derivation of expected behavior

With $r_0 = 20$ m and $T = 360$ s, $r_\star = 20 + R_0 T = 24.187952$ m. For a consistent spatial discretization, the zero-contour equivalent-radius error $E_h = |r_A - r_\star|$ should decrease as $h = \Delta x$ decreases and follow $E_h \approx C h^p$. A least-squares slope of $\log E_h$ against $\log h$ estimates p.

Every grid simulation condition uses the same manufactured solution. From the generated rate $R_s = 0.0116332$ m s^{-1}, initial radius $r_0 = 20$ m, and stop time $T = 360$ s,

$$r_{\text{ref}} = 20 + (0.0116332)(360) = 24.187952 \text{ m}.$$

For each cell size $h_i \in \{2, 1, 0.5, 0.25\}$ m, the area-equivalent radius and error are

$$r_i = \sqrt{\frac{N_{b,i} h_i^2}{\pi}}, \qquad E_i = |r_i - 24.187952|.$$

The observed spatial order is the ordinary-least-squares slope

$$p_h = \frac{\sum_i (\ln h_i - \overline{\ln h})(\ln E_i - \overline{\ln E})}{\sum_i (\ln h_i - \overline{\ln h})^2}.$$

Consequently the table tests the explicitly calculated $p_h \geq 0.8$ and the finest-grid substitution $E_{0.25} = |r_{0.25} - 24.187952| \leq 1$ m.

4 Verification metrics and acceptance criteria

The first metric is the fitted order p, selected because it measures the multi-grid trend rather than one favorable mesh; $p \geq 0.8$ allows rasterized-interface effects while requiring approximately first-order behavior. The second is finest-grid $E_h \leq 1$ m, preventing a misleading order from large absolute errors. Radius is obtained from the largest closed, linearly interpolated $\phi = 0$ contour, providing a rotationally aggregated subcell measure. The last matching level-set field is required for each simulation condition. All simulation conditions and both metrics must pass; otherwise the decision is NOT EVALUABLE or FAIL as appropriate.

5 Simulation configuration and predicted outcome

Each EPSG:32610 simulation condition covers the same 256 m square with uniform inputs, flat terrain, zero wind, centered 20 m signed-distance ignition, $R_0 = 0.0116332\,\mathrm{m\,s^{-1}}$, and duration 360 s. Only grid resolution varies. The predicted front is circular and E_h decreases toward the 24.187952 m reference radius.

Parameter	Configured value	Units	Role
Domain	256 by 256	m	Fixed physical experiment
Meshes	128, 256, 512, 1024	cells/side	Spatial refinement sequence
Cell size	2, 1, 0.5, 0.25	m	Independent variable
Time control	step 1; stop 360	s	Fixed temporal discretization
Initial front	centered signed-distance radius 20	m	Common exact geometry

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

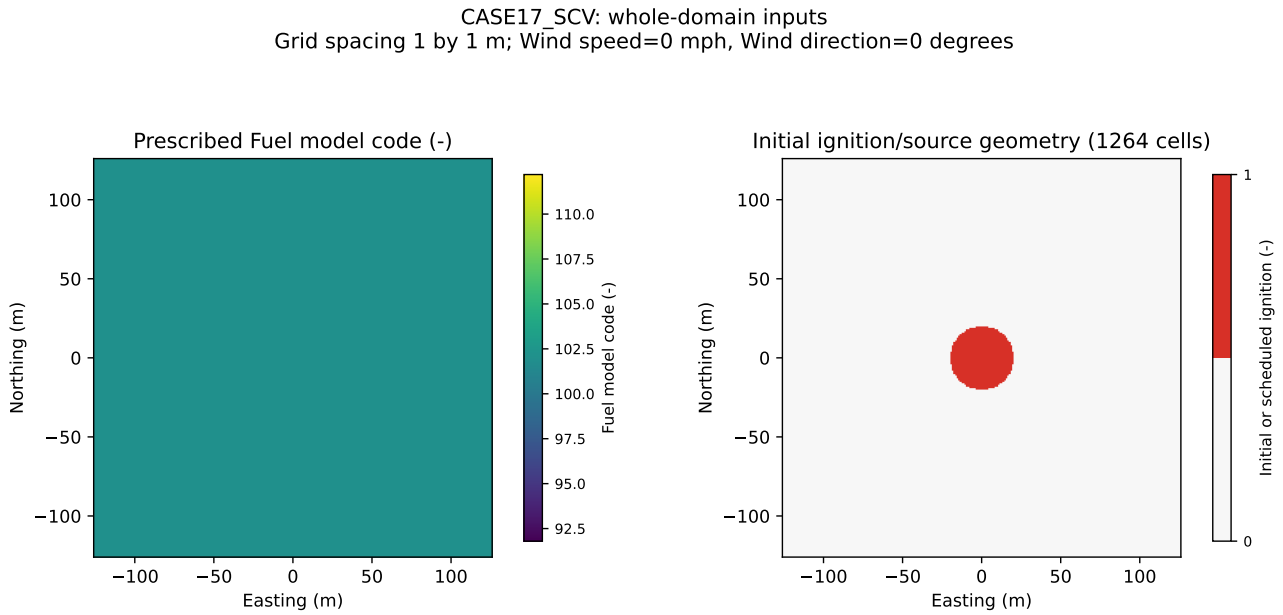


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

Only the four case-specific simulation condition outputs are read. Current status is **FAIL**; 4 simulation conditions and 2 decision metrics are recorded.

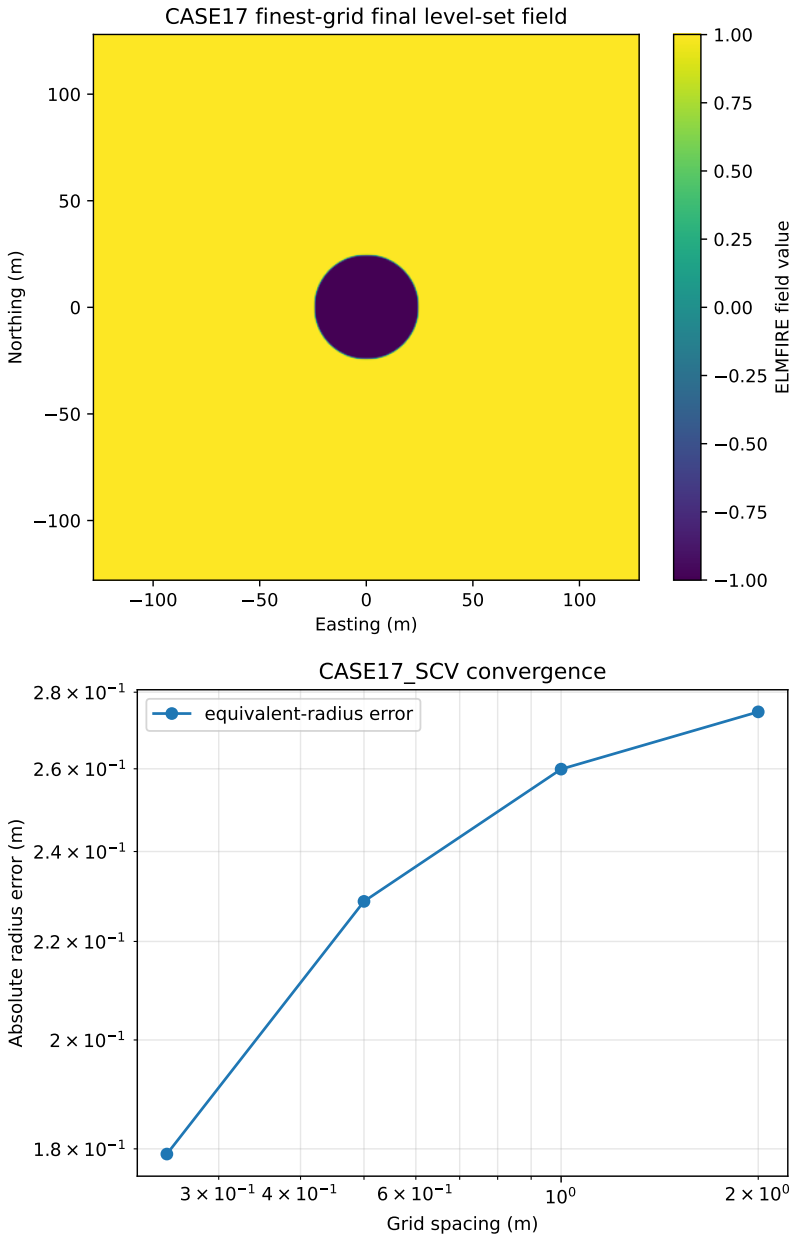


Figure 2: Finest-grid whole-domain signed level-set result and radius-error convergence across all four completed meshes. The spatial field is dimensionless; radius errors are in metres, and the generated table supplies both acceptance limits.

7 Calculated metrics and verification decision

Metric	Expected value or limit	Calculated value	Status
observed spatial order	0.8 ($\geq$ threshold)	0.203637 -	FAIL
finest-grid radius error	0 ($\leq$ 1)	0.179096 m	PASS

The authoritative overall decision is **FAIL**. The generated table is sourced directly from the case-specific calculated metrics and is not recomputed in this report.

8 Technical interpretation

Low order with small finest error suggests interface-measurement or pre-asymptotic effects; high order with excessive finest error indicates the sequence has not reached useful accuracy. Missing simulation conditions invalidate the convergence fit. Resolution-dependent physical inputs would be a configuration defect.

9 Scope and limitations

This case measures convergence of one smooth, radially symmetric front and an area-derived observable. It does not determine formal order for discontinuous fuels, complex terrain, anisotropic spread, or other output fields.

Case 18: Temporal convergence

1 Verification purpose

This isolated-process case verifies temporal convergence of ELMFIRE's level-set propagation for a manufactured circular front. Four time grids vary the whole-second step while mesh and physical fields remain fixed. The case can expose timestep scaling errors, inconsistent final-time handling, or a non-convergent temporal update. The question is whether all four simulation conditions complete, the fitted radius-error order is at least 0.8, and the finest-step radius error is at most 0.5 m.

2 Mathematical formulation and numerical implementation

The governing equation is $\partial_t \phi + R_0 |\nabla \phi| = 0$, with constant $R_0 = 0.0116332$ m s^{-1}. All simulation conditions share the same 1024-by-1024, 256 m raster grid. For a 360 s simulation, the configured time-grid counts 45, 90, 180, and 360 correspond to whole-second $\Delta t = \{8, 4, 2, 1\}$ s.

3 Derivation of expected behavior

The exact final radius is $r_\star = 20 + (0.0116332)(360) = 24.187952$ m. Define $E_{\Delta t} = |r_A - r_\star|$. A consistent time integration has $E_{\Delta t} \approx C(\Delta t)^p$; a least-squares slope of $\log E_{\Delta t}$ versus $\log \Delta t$ estimates the observed temporal order p.

The time-step simulation conditions share the manufactured reference

$$r_{\text{ref}} = r_0 + R_s T = 20 + (0.0116332)(360) = 24.187952 \text{ m}.$$

For $\Delta t_i \in \{8, 4, 2, 1\}$ s, the corresponding exact step counts are $360/\Delta t_i = \{45, 90, 180, 360\}$. From each final raster,

$$r_i = \sqrt{\frac{N_{b,i}(0.5 \text{ m})^2}{\pi}}, \qquad E_i = |r_i - 24.187952|.$$

The reported temporal order is the least-squares log–log slope

$$p_t = \frac{\sum_i (\ln \Delta t_i - \overline{\ln \Delta t})(\ln E_i - \overline{\ln E})}{\sum_i (\ln \Delta t_i - \overline{\ln \Delta t})^2}.$$

The substituted decision tests are $p_t \geq 0.8$ and $E_{\Delta t=1} = |r_1 - 24.187952| \leq 0.5$ m.

4 Verification metrics and acceptance criteria

Observed p uses all four simulation conditions and is required to be at least 0.8, allowing front-raster measurement effects while demanding near-first-order convergence. Finest-step error must be no more than 0.5 m, preventing a formally favorable but inaccurate sequence. Equivalent radius is obtained from the largest closed, linearly interpolated $\phi = 0$ contour, avoiding binary cell-count bias. The last matching level-set field raster for every time grid is required. All simulation conditions and both metrics must pass; missing data is NOT EVALUABLE.

5 Simulation configuration and predicted outcome

All EPSG:32610 simulation conditions use a 256 m square domain, 0.25 m cells, uniform inputs, flat terrain, zero wind, centered 20 m signed-distance ignition, and duration 360 s. Only the time-grid control varies. The predicted circular-front radius approaches 24.187952 m as Δt decreases.

Parameter	Configured value	Units	Role
Domain/grid	256 m square; 0.25	m/cell	Fixed spatial discretization
Time steps	8, 4, 2, 1	s	Temporal refinement sequence
Stop time	360	s	Common exact comparison time
Initial front	centered signed-distance radius 20	m	Common exact geometry
Rate	0.0116332	m/s	Manufactured normal speed

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

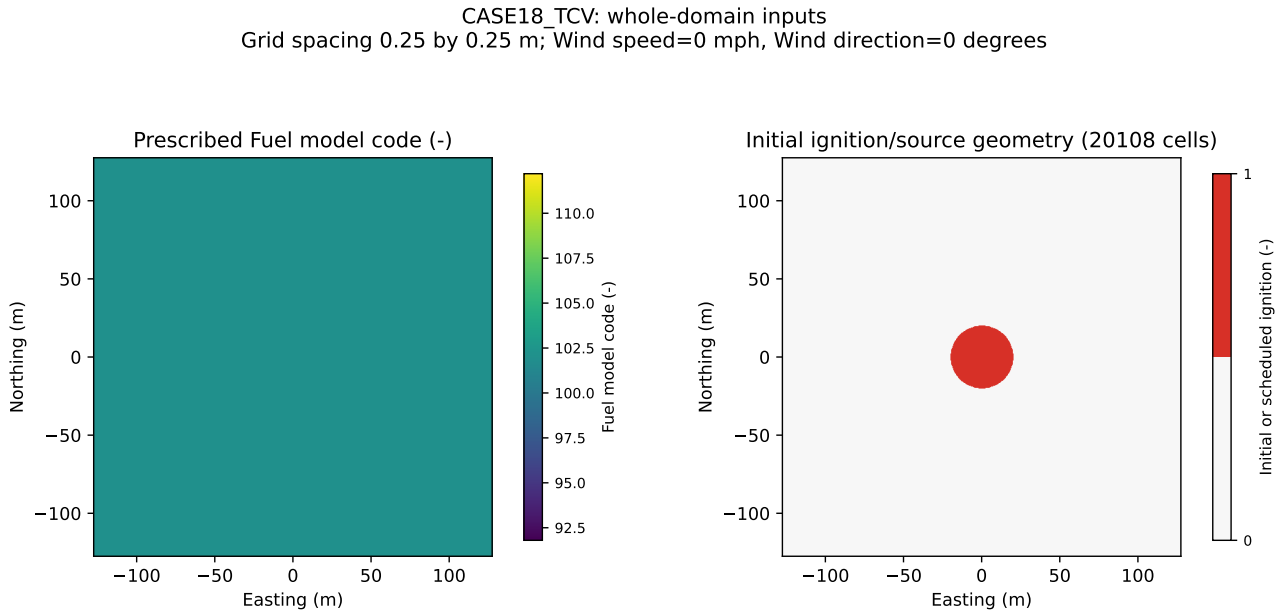


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

Only case-specific output rasters are read. Current status is **FAIL**; 4 simulation conditions and 2 decision metrics are recorded.

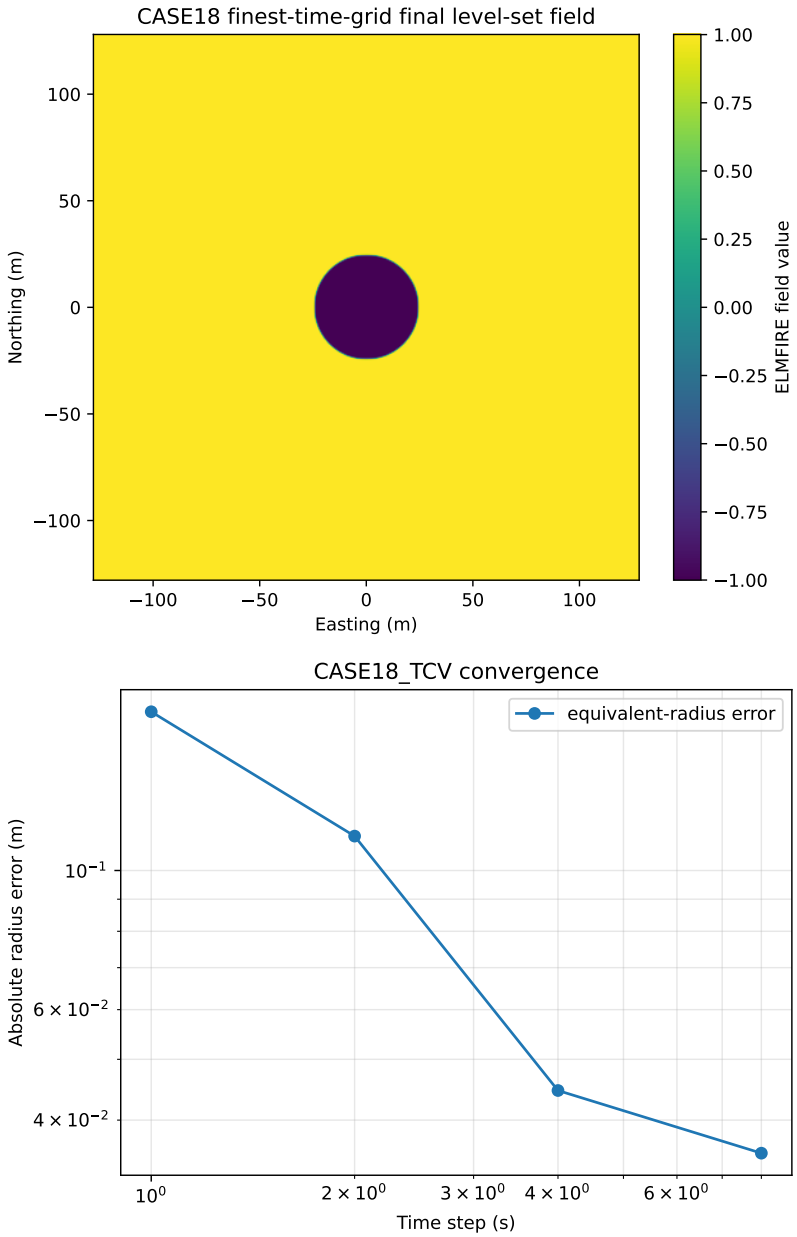


Figure 2: Finest-step whole-domain signed level-set result and radius-error convergence across all four completed time grids. The spatial field is dimensionless; radius errors are in metres, and the generated table supplies both acceptance limits.

7 Calculated metrics and verification decision

Metric	Expected value or limit	Calculated value	Status
observed temporal order	0.8 ($\geq$ threshold)	-0.83607 -	FAIL
finest-time-grid radius error	0 ($\leq$ 0.5)	0.179096 m	PASS

The authoritative overall decision is **FAIL**. The generated table is sourced directly from the case-specific calculated metrics and is not recomputed in this report.

8 Technical interpretation

Low order can indicate timestep application, output-time selection, or pre-asymptotic spatial error; a small finest error alone does not establish convergence. Missing simulation conditions invalidate the fitted order and are treated as insufficient evidence.

9 Scope and limitations

This case verifies temporal behavior for one smooth deterministic front at a fixed fine mesh. It does not separate all spatial error, test adaptive timestep logic under complex forcing, or provide physical validation.

Case 19: Transient WUI heat flux

1 Verification purpose

This coupled-process case verifies the transient wildland–urban interface (WUI) heat-flux pathway that combines the building-spread ellipse, a piecewise design-fire heat-release curve, direct-flame-contact (DFC) transfer, radiative transfer, and transient raster output. It exercises the `WUI` namelist, WUI state updates in ELMFIRE level-set formulation, and transient output branches in ELMFIRE output procedure. It can expose wind-angle, ellipse-geometry, time-curve, cell-normalization, heat-partition, or output-indexing errors. Wildland fuel, crown fire, spotting, and suppression are not decision observables. This is algorithm verification against a separately evaluated documented formulation, not physical validation. The question is whether the prepared case retains exactly one active WUI source, all required transient heat-release rate (HRR) per unit area (HRRPUA), DFC, and radiation stacks exist, and each energy-normalized error is no greater than 0.005.

2 Mathematical formulation and numerical implementation

Wind speed is converted from mph to SI as $V = 0.447 V_{\text{mph}}$. For configured building area A, separation D, and wind proportionality W_p, the selected Hamada-style regression produces downwind, upwind, and sidewind distances D_d, D_u, D_s. Coefficients differ for $V < 10$, $10 \le V \le 17.3$, and $V > 17.3$ m s^{-1}; this case uses 15 mph $= 6.705$ m s^{-1}, so it exercises the low-wind linear branch. The translated ellipse has

$$a = (D_d + D_u)/2, \qquad e = \min(a/2, a - D_u), \qquad b = D_s/\sqrt{1 - (e/a)^2}.$$

For target direction θ_f, its boundary distance is

$$R_e(\theta_f) = \frac{0.3 D_d (a - e)}{b^2} \frac{b^2}{a - e\cos\theta_f}, \quad \theta_f = \text{atan2}(y, x) - \pi(270 - WD)/180,$$

which preserves the meteorological wind-from convention.

For burning time τ, the design fire rises linearly to H_p at t_e, remains at H_p through t_d, decays linearly to zero at t_c, and is zero thereafter:

$$H(\tau) = \begin{cases} H_p \tau / t_e, & 0 \le \tau \le t_e, \\ H_p, & t_e < \tau \le t_d, \\ H_p(\tau - t_c)/(t_d - t_c), & t_d < \tau \le t_c, \\ 0, & \text{otherwise.} \end{cases}$$

Cell coverage is $C_D = \text{clip}[(R_e + \Delta/2 - R)/\Delta, 0, 1]$. With configured ellipse scale $\eta = 0.5$, $C_H = \Delta^2/[\pi(\eta a)(\eta b)]$, burnable fraction C_b, absorptivity α, radiation-annulus fraction F_r, and effective separation R_{eff},

$$q''_{\text{DFC}} = C_b C_D H C_H, \qquad q''_{\text{rad}} = \frac{0.3 C_b \alpha F_r C_H H \Delta^2}{4\pi R_{\text{eff}}^2}.$$

The case-specific reference functions evaluate these equations independently at the timestamps emitted by ELMFIRE.

3 Derivation of expected behavior

The configured constants are $A = D = 10$ m, $W_p = 1$, $\eta = 0.5$, $\Delta = 20$ m, $WD = 0°$, $H_p = 400\,\mathrm{kW\,m^{-2}}$, $t_e = 300$ s, $t_d = 3900$ s, and $t_c = 4200$ s. The generated adjustment field is uniformly zero, so the center WUI structure cannot ignite a secondary level-set source. Because the current executable generic stalled-front shortcut does not account for an active WUI design-fire curve, one point ignition is configured at the center for $t = 5001$ s, strictly after the 5000 s stop. It is never applied; its only role is to keep that shortcut inactive while the prescribed curve is sampled. Substitution into the low-wind regression fixes one time-independent ellipse. At each timestamp, the piecewise equation fixes HRRPUA; the ellipse and coverage equations then fix each cell in the 11-by-11 audit window. DFC is confined to covered cells and scales with H; radiation occupies the outer annulus and decays with R_{eff}^{-2}. This reference depends on configured constants and timestamps, not on simulated heat-flux values.

The analytical history is evaluated with the generated constants $A = D = 10$ m, $W_p = 1$, $\eta = 0.5$, $H_p = 400\,\mathrm{kW\,m^{-1}}$, and $(t_e, t_d, t_c) = (300, 3900, 4200)$ s. Thus the prescribed triangular heat-release history is

$$H(t) = \begin{cases} 400t/300, & 0 \le t < 300, \\ 400, & 300 \le t < 3900, \\ 400(4200 - t)/300, & 3900 \le t \le 4200, \\ 0, & t > 4200, \end{cases}$$

so, for example, $H(150) = 200$, $H(300) = H(3900) = 400$, $H(4050) = 200$, and $H(4200) = 0\,\mathrm{kW\,m^{-1}}$. For the generated 15 mph wind, $U = 15(0.44704) = 6.7056\,\mathrm{m\,s^{-1}}$. Substitution in the case's UCB ellipse equations gives $a = 76.2593$ m, $b = 24.8916$ m, $e = 35.6487$ m, downwind reach $D_d = a + e = 111.9080$ m, upwind reach $D_u = a - e = 40.6106$ m, and lateral reach $D_s = 22.0044$ m. Each reported history error uses

$$E_1(q) = \frac{\sum_k |q_{\mathrm{sim},k} - q_{\mathrm{ref},k}|}{\sum_k |q_{\mathrm{ref},k}|},$$

with q equal to heat release rate, dry-fuel consumption, or radiation. The exact-reference substitution is $E_1(q) = 0$, and each component is accepted when $E_1(q) \le 0.005$.

4 Verification metrics and acceptance criteria

For each observable X,

$$E_1(X) = \frac{\sum_{t,i,j} |X_{\mathrm{ELMFIRE}} - X_{\mathrm{ref}}|}{\sum_{t,i,j} |X_{\mathrm{ref}}|}.$$

HRRPUA uses the center cell; DFC and radiation use the documented 11-by-11 window, restored to mathematical $+y$ orientation. Mutually masked, nodata, and nonfinite samples are excluded. Energy-normalized L_1 error is selected instead of pointwise relative error because the reference is exactly zero over much of space–time; division by those zeros amplifies irrelevant roundoff. It detects both amplitude and footprint errors. Required outputs are the time-resolved heat-release rate, convective heat flux, and radiative heat flux. Their sequential output indices are mapped to physical seconds using the unique record of output times. A 0.5 percent limit allows single-precision raster serialization and cell-selection roundoff while remaining much smaller than a one-cell or coefficient error. All stacks and metrics are required, including one terminal record at 5000 s and dumps spanning growth, plateau, decay, and post-decay. Before evaluation, every nonzero HRR raster must contain exactly the center cell; a second active source makes the single-source experiment NOT EVALUABLE. PASS requires that isolation invariant and all $E_1 \le 0.005$. Missing or incompatible outputs are NOT EVALUABLE.

5 Simulation configuration and predicted outcome

The Courant–Friedrichs–Lewy (CFL) number measures the distance advected during one timestep relative to the grid spacing and constrains the time discretization.

Parameter	Value	Units	Isolation role
Domain raster	20×20 cells	–	centered WUI source and audit window
Analysis cell	20	m	matches reference discretization
Wind	15 at 0° from	mph, degree	low-wind ellipse branch
Building area / separation	10 / 10	m	fixes regression geometry
Nonburnable fraction	0	–	removes availability scaling ambiguity
Ellipse HRR scale η	0.5	–	fixes current normalization
Spread adjustment	0	–	prevents all secondary propagation
Ignition	center PHI cell only	–	the sole physical source
Terminal guard	center at 5001	s	after stop; never ignites
Design-fire times	300 / 3900 / 4200	s	rise, plateau, decay
Peak HRRPUA	400	kW m^{-2}	reference amplitude
Initial/maximum timestep	1 / 10	s	whole-second, CFL-limited stepping
Output interval / stop	100 / 5000	s	exact stop-time alignment and all phases

The predicted HRRPUA rises, plateaus, decays, and becomes zero. Pre-run isolation requires exactly one active HRR cell at every nonzero dump; DFC and radiation therefore come only from the center structure. DFC follows the fixed wind-aligned footprint; radiation surrounds it and both fields scale with HRRPUA.

Figure 1 is generated mechanically from the prescribed input fields. It identifies the prescribed physical input field and representative simulation condition; the left panel shows the generated zero spread-adjustment field that enforces source isolation, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

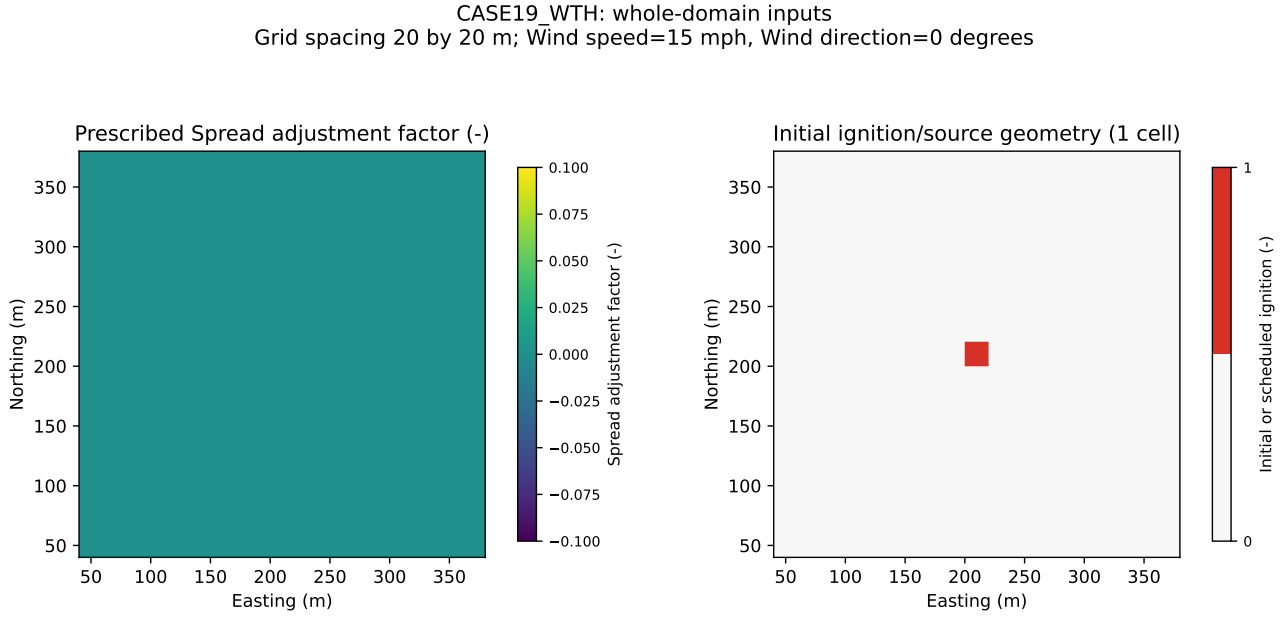


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

Only case-specific ELMFIRE GeoTIFFs are used. Current status is **NOT EVALUABLE**; 0 of 0 comparison components pass.

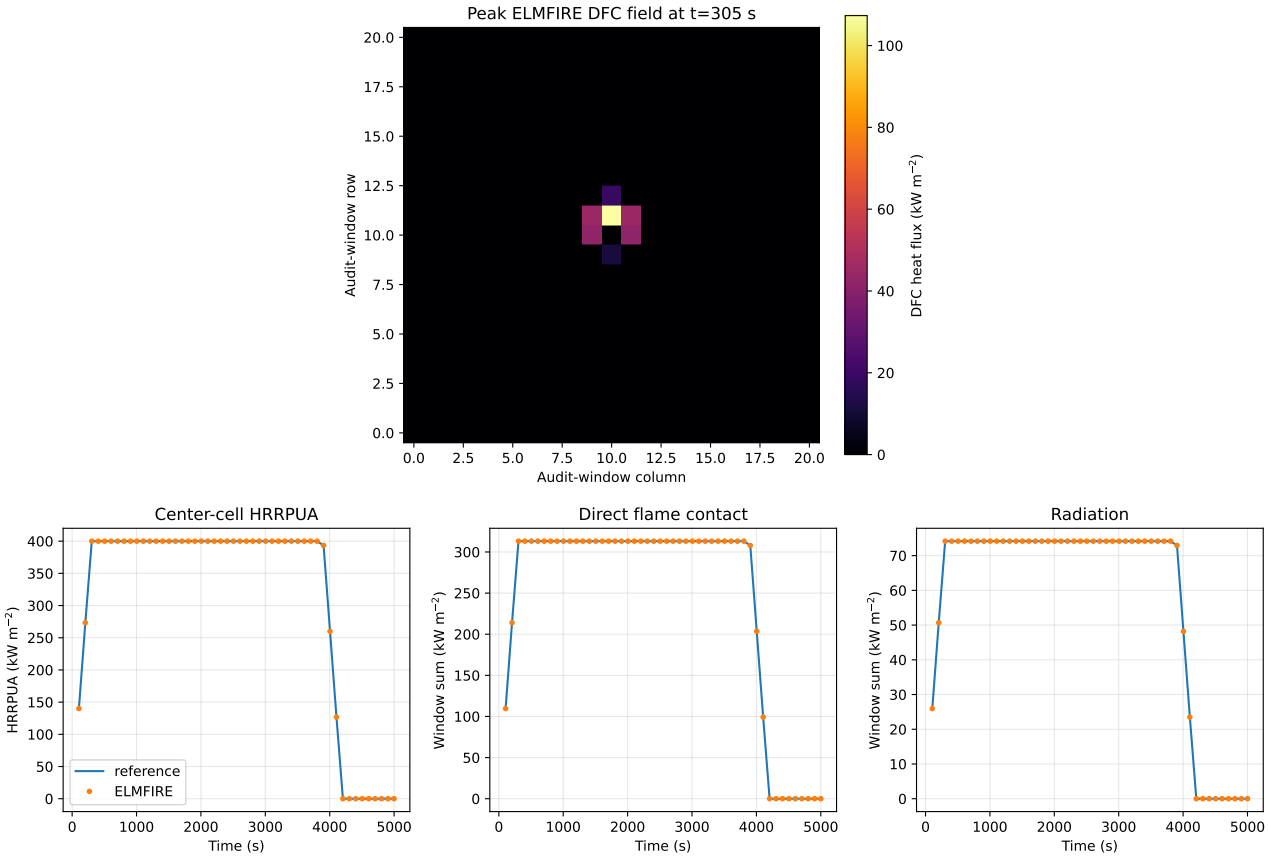


Figure 2: Peak whole-domain DFC heat flux in kW m^{-2} and transient reference-versus-ELMFIRE histories for HRRPUA, DFC, and radiation. All panels are generated from the required case-specific transient GeoTIFF stacks; the histories show every selected output time and the spatial panel shows the maximum-domain-sum DFC output.

7 Calculated metrics and verification decision

Metric	Expected value or limit	Calculated value	Status
no active center HRR source was found	–	–	NOT EVALUABLE

The authoritative decision is **NOT EVALUABLE**. The generated table is sourced directly from the case-specific calculated metrics.

8 Technical interpretation

The NOT EVALUABLE decision follows the three generated error components after enforcing the single-source invariant. An HRRPUA-only failure implicates burning-time or transient-curve evaluation. A DFC failure with correct HRRPUA implicates ellipse coverage, configured η normalization,

orientation, or raster slicing. A radiation-only failure narrows the issue to annulus partitioning, effective distance, or absorptivity. Similar DFC and radiation errors with correct shapes suggest common normalization. A second HRR source is a case-configuration failure and is NOT EVALUABLE; incomplete phase coverage, a nonterminal 5000 s record, or missing output is likewise insufficient evidence, never a pass. The post-stop ignition is a documented workaround for the executable stalled-front shortcut and is outside the verified physical interval.

9 Scope and limitations

This case verifies one deterministic WUI parameter set, one wind-regression branch, one raster resolution, and three transient outputs. It does not verify moderate/high-wind branches, parameter rasters, heterogeneous layouts, structure-to-structure ignition, or empirical validity of the correlations. It supplies implementation-verification evidence only.

Case 20: Point ignition and isotropic spread

1 Verification purpose

This case verifies the coupled surface-fire and level-set rate response to a central point ignition in uniform fuel with zero wind and zero slope. Rotational symmetry supplies a diagnostic for directional bias or raster-orientation errors, while the formal decision tests the coupling between fire behavior and level-set normal velocity. The verification question is whether the required maximum spread rate is within 10 percent of $1.51\ \mathrm{m\,min^{-1}}$; isotropic arrival geometry is reported as supporting evidence but is not yet a separate acceptance component.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-specific outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

ELMFIRE advances the level-set field using $\partial\phi/\partial t + V_n|\nabla\phi| = 0$. With spatially uniform fuel, moisture, wind, and slope, $V_n = R_0$ is independent of azimuth. The zero contour must therefore evolve from the ignition neighborhood toward a circle. The final spread rate raster records the local normal spread rate in metres per minute because metric spread-rate output is enabled.

The surface-fire scalar supplied to the level set is the Rothermel no-wind/no-slope rate

$$R_0 = \frac{I_R\,\xi}{\rho_b\,\epsilon\,Q_{ig}}, \qquad Q_{ig} = 250 + 1116 M_f, \tag{1}$$

where reaction intensity I_R, propagating-flux ratio ξ, bulk density ρ_b, effective heating number ϵ, and fuel moisture M_f are evaluated from the generated Fuel Model 3 and moisture inputs. This equation chain follows Andrews (2018), *The Rothermel Surface Fire Spread Model and Associated Developments*; the Guide supplies the configured known-response value.

3 Derivation of expected behavior

For constant R_0, the front radius is $r(t) = r(0) + R_0 t$. The independently tabulated Fuel Model 3 response for the configured moisture and no-wind/no-slope condition is $R_0 = 1.51\ \mathrm{m\,min^{-1}}$. Rotational symmetry is an invariant; the quantitative decision uses the rate because it directly tests the fire-behavior value passed to propagation.

The independent no-wind/no-slope reference is calculated from

$$R_0 = \frac{I_R \xi}{\rho_b \epsilon Q_{ig}}, \qquad Q_{ig} = 250 + 1116 M_f,$$

using generated Fuel Model 3 and $(M_{1h}, M_{10h}, M_{100h}, M_{lh}, M_{lw}) = (0.06, 0.07, 0.08, 0.60, 0.90)$. The independent calculation gives $I_R = 549.174\ \mathrm{kW\,m^{-2}}$ and $R_0 = 1.51183\ \mathrm{m\,min^{-1}}$, reported to the Guide

precision as $R_{\text{ref}} = 1.51$ m min^{-1} ($1.51183/0.3048 = 4.960$ ft min^{-1}). The decision metric is therefore

$$e_R = \frac{|R_{\text{max}} - 1.51|}{1.51},$$

where R_{max} is the maximum finite final spread rate value, and the substituted acceptance test is $e_R \leq 0.10$.

4 Verification metrics and acceptance criteria

The head rate of spread (ROS) metric is the maximum finite value in the required final spread rate raster. It is selected because uniform forcing makes every mature front direction equivalent and the maximum avoids dilution by unburned nodata cells. A 10 percent relative tolerance follows the GUIDE comparison and allows raster and front-discretization effects while remaining sensitive to unit or coupling errors. The case passes only when the output exists and $|R_{\text{sim}} - 1.51|/1.51 \leq 0.10$.

5 Simulation configuration and predicted outcome

The case uses a 126×126, 30 m grid in EPSG:32635, uniform Fuel Model 3, 6 percent one-hour moisture, flat terrain, zero wind, and a central ignition. The predicted output is an approximately circular arrival-time field with a uniform mature spread rate near 1.51 m min^{-1}.

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{stop} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

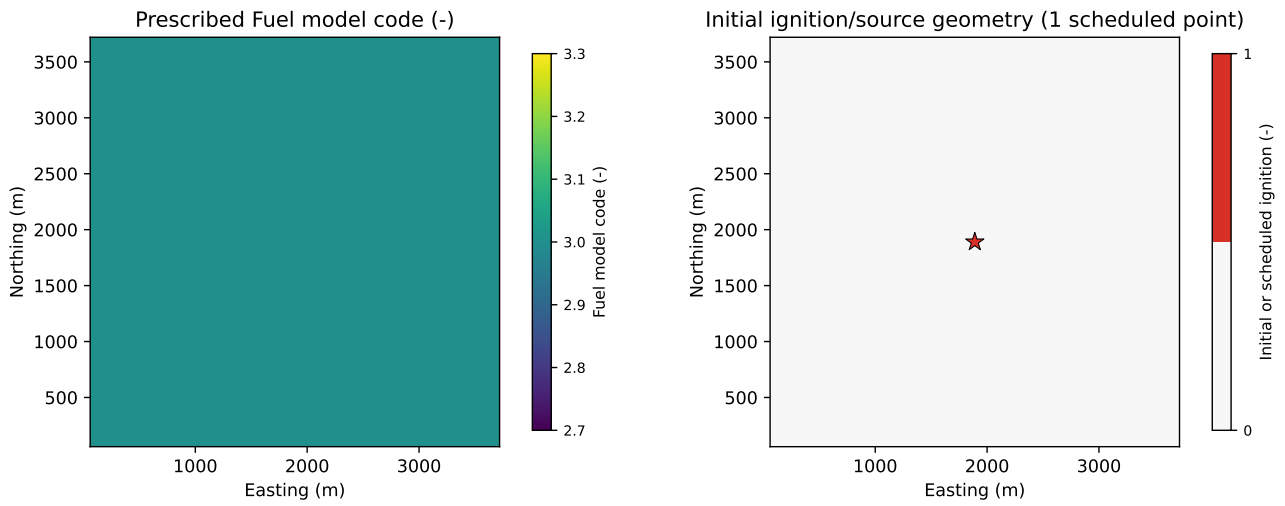


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-specific ELMFIRE outputs. The current postprocessing status is **PASS**; 1 of 1 required metric rows pass across 1 simulation condition(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

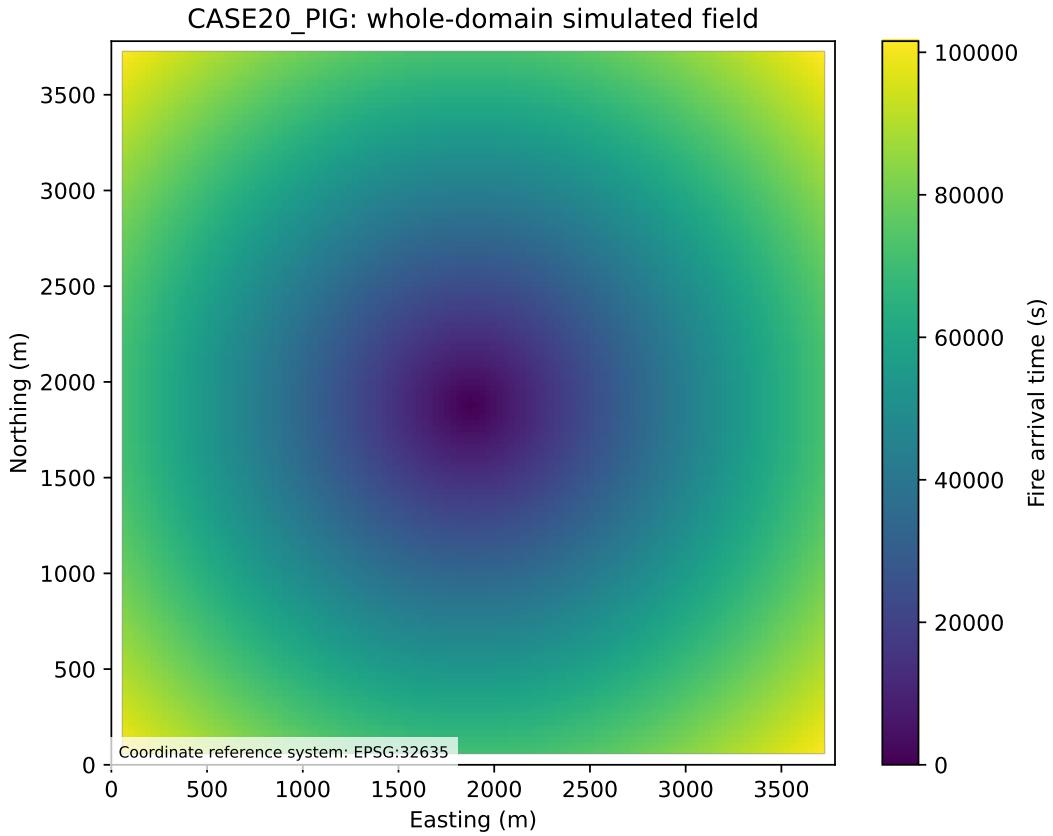


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed simulation condition, in seconds. The coordinate reference system is stated in the figure. The calculated metrics, not visual appearance, determine acceptance.

7 Calculated metrics and verification decision

Simulation condition / metric	Expected	Calculated	Status
main / head ROS	1.51 m/min	1.51183 m/min	PASS

The displayed overall decision is **PASS**. When calculated verification and validation metrics exists, result analysis procedure generates every table value and status; when it is absent, the report states NOT EVALUABLE with no invented rows.

8 Technical interpretation

A rate failure indicates a fire-behavior, units, or output-selection problem. Strong directional distortion with an acceptable maximum is a diagnostic warning and must not be claimed as verified until a shape metric and tolerance are added. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

This is deterministic numerical verification for one fuel and moisture state. It does not validate Fuel Model 3 against experimental fires or quantify symmetry with a dedicated shape metric. This case provides implementation verification evidence only.

Case 21: Wind-driven elliptical spread

1 Verification purpose

This case verifies the coupling of wind-adjusted surface spread to the Huygens and level-set propagation geometry. Uniform fuels and flat terrain isolate wind as the only directional forcing. The verification question is whether the simulated head rate is within 10 percent of 3.635 m min^{-1} and the unclipped arrival-time isochrone has length-to-width ratio within 10 percent of 1.24.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-specific outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

The normal-velocity construction combines the backing, flanking, and heading rates into an expanding ellipse. Its major axis aligns with the wind-driven heading direction, while the ratio L/W characterizes anisotropy independent of absolute size. ELMFIRE writes rate in metres per minute and arrival time on the same raster grid used to infer the ellipse.

For uncovered fuel of depth H ft, the Guide uses

$$WAF_u = \frac{1.83}{\ln[(20 + 0.36H)/(0.13H)]}, \qquad U_{mf} = WAF_u U_{20}. \tag{1}$$

The resulting effective midflame wind determines the ellipse aspect ratio

$$\frac{L}{W} = \min\left(0.936e^{0.1147U_{mf,e}} + 0.461e^{-0.0692U_{mf,e}} - 0.397, 8\right). \tag{2}$$

The local head rate is $R_H = R_0(1 + \phi_w)$; the Huygens/level-set construction uses the ellipse to obtain normal speed away from the heading direction.

3 Derivation of expected behavior

Constant forcing produces self-similar elliptical isochrones: scaling time changes both axes proportionally and leaves L/W constant. The configured Fuel Model 5 response supplies the independent head-rate target 3.635 m min^{-1}, and the GUIDE construction supplies $L/W = 1.24$.

For the generated Fuel Model 5 state, the independent no-wind rate is $R_0 = 0.41042$ m min^{-1}. The 6 mph 20-ft wind and 2-ft fuel-bed depth give a wind-adjustment factor $WAF = 0.418640$, hence $U_{mf} = 6(0.418640) = 2.51184$ mph. With the calculated wind factor $\phi_w = 7.87958$,

$$R_h = R_0(1 + \phi_w) = 0.41042(1 + 7.87958) = 3.6444 \text{ m min}^{-1},$$

consistent with the Guide reference 3.635 m min^{-1}. The ellipse length-to-width reference uses

$$\frac{L}{W} = 0.936e^{0.2566U} + 0.461e^{-0.1548U} - 0.397,$$

where U is in $\mathrm{m\,s^{-1}}$. Here $U = 2.51184\ \mathrm{mph}(0.44704) = 1.12291\ \mathrm{m\,s^{-1}}$, so substitution gives $L/W = 1.23898$, reported as 1.24. The report evaluates the relative substitutions $e_R = |R_h - 3.635|/3.635$ and $e_{LW} = |(L/W) - 1.24|/1.24$ against their tabulated limits.

4 Verification metrics and acceptance criteria

Maximum finite spread rate measures the heading response. The shape metric is $\sqrt{\lambda_{\max}/\lambda_{\min}}$, where the eigenvalues come from the covariance of burned-cell coordinates at the last isochrone that remains away from the boundary. This metric is rotation invariant and avoids a wind-axis assumption. Each component uses 10 percent relative tolerance, and both must pass.

5 Simulation configuration and predicted outcome

The 126×126, 30 m domain uses uniform Fuel Model 5, 6 mph wind, flat terrain, and an upwind point ignition. The predicted arrival field is an ellipse elongated downwind, with head rate of spread (ROS) $3.635\ \mathrm{m\,min^{-1}}$ and $L/W = 1.24$.

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{stop} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

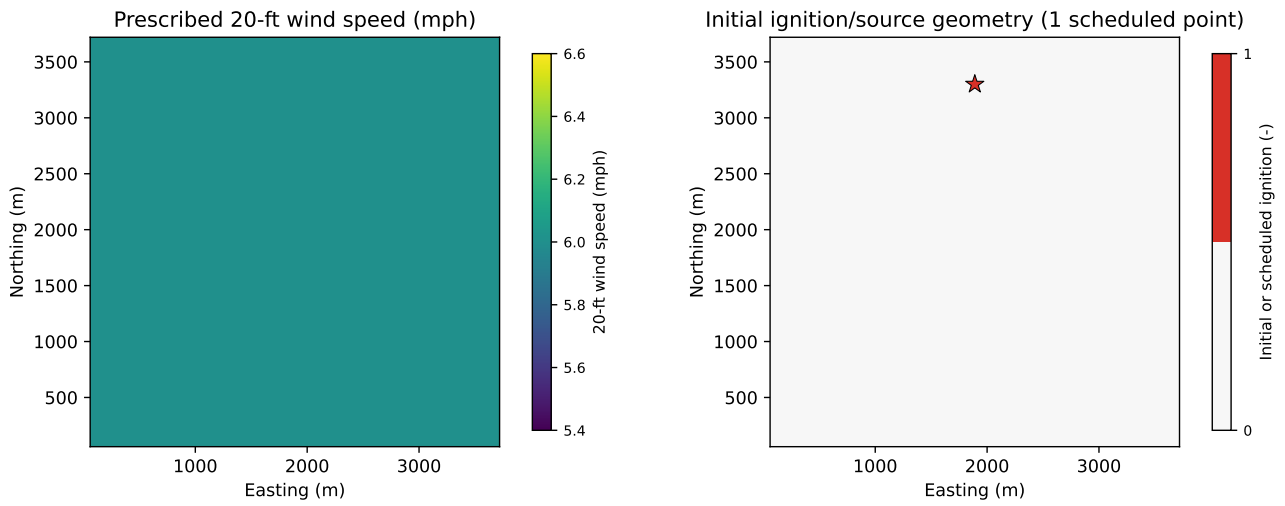


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-specific ELMFIRE outputs. The current postprocessing status is **PASS**; 2 of 2 required metric rows pass across 1 simulation condition(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

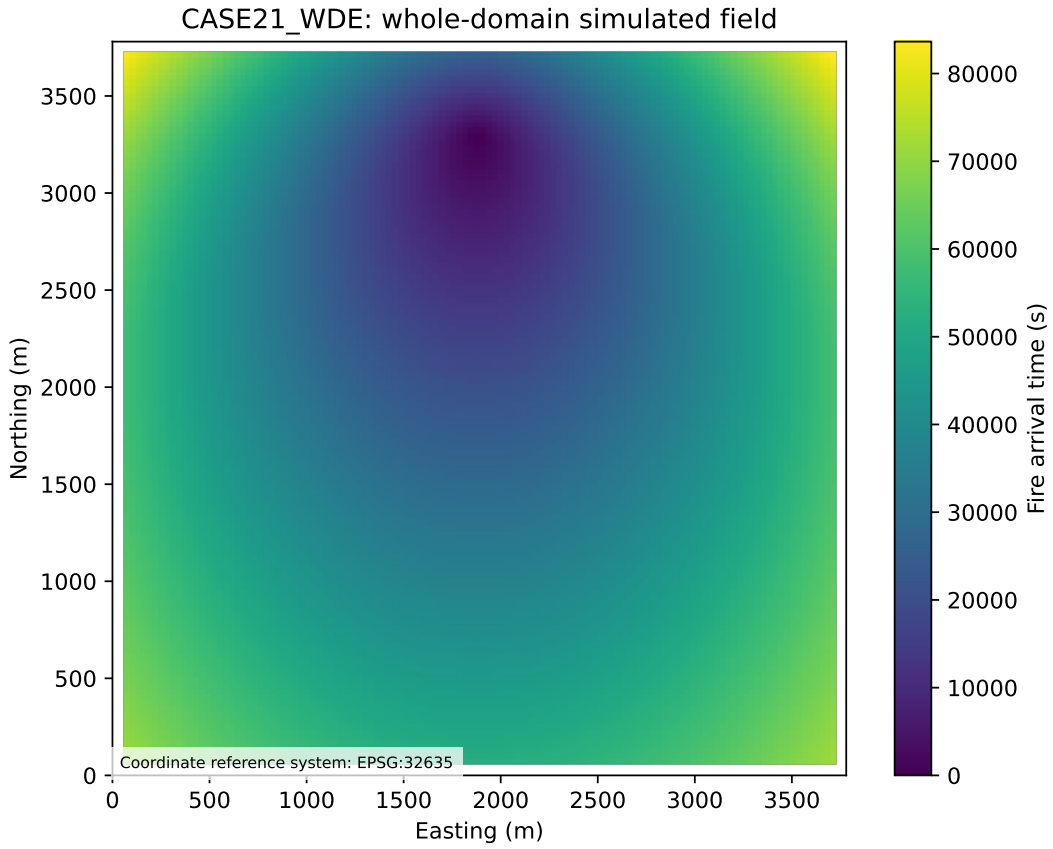


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed simulation condition, in seconds. The coordinate reference system is stated in the figure. The calculated metrics, not visual appearance, determine acceptance.

7 Calculated metrics and verification decision

Simulation condition / metric	Expected	Calculated	Status
main / head ROS	3.635 m/min	3.64438 m/min	PASS
main / L/W ratio	1.24 -	1.24859 -	PASS

The displayed overall decision is **PASS**. When calculated verification and validation metrics exists, result analysis procedure generates every table value and status; when it is absent, the report states NOT EVALUABLE with no invented rows.

8 Technical interpretation

A rate-only discrepancy points toward wind adjustment or unit conversion; a shape-only discrepancy points toward wavelet geometry, arrival-time extraction, or boundary clipping. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

The test covers one steady wind speed and one fuel model. The covariance estimator is intentionally applied only before boundary contact. This case provides implementation verification evidence only.

Case 22: Spread across opposing valley slopes

1 Verification purpose

This case verifies slope enhancement and aspect handling across two opposing valley sides. Zero wind isolates the terrain contribution. The verification question is whether the maximum regional spread rates on the 5-degree east side and 15-degree west side are each within 10 percent of their independent targets.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-specific outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

The surface spread calculation multiplies the no-wind/no-slope response by a slope-dependent factor and orients the heading direction upslope. The slope raster supplies magnitude in degrees and the aspect raster supplies downslope azimuth. Opposing aspects must therefore drive the two head fires away from the central valley ignition.

The isolated Rothermel slope factor is

$$R_H = R_0(1 + \phi_s), \qquad \phi_s = 5.275\,\beta^{-0.3}\tan^2\theta, \tag{1}$$

where β is the fuel-bed packing ratio and θ is the slope angle. The aspect raster gives downslope azimuth, so the heading direction is rotated by 180 degrees to obtain upslope. The formula chain follows Andrews (2018); the Guide supplies the two configured BEHAVE responses and point-spread ellipse ratios.

3 Derivation of expected behavior

For the configured Fuel Model 3 and moisture, the reference calculation gives 1.92 m min^{-1} on the 5-degree side and 5.37 m min^{-1} on the 15-degree side. The steeper slope must have the larger head rate; this ordering is an additional qualitative invariant.

With no wind, the slope correction is

$$R_s = R_0(1 + \phi_s).$$

The generated Fuel Model 3 moisture state gives $R_0 = 1.51183$ m min^{-1}. The independent Rothermel slope calculation gives $\phi_s = 0.272326$ at 5 degrees and $\phi_s = 2.554410$ at 15 degrees. Therefore

$$R_5 = 1.51183(1 + 0.272326) = 1.92354, \qquad R_{15} = 1.51183(1 + 2.554410) = 5.37366 \quad \text{m min}^{-1},$$

reported as 1.92 and 5.37 m min^{-1}. For each slope s, the calculated metric is $e_s = |R_{\max,s} - R_{s,\text{ref}}|/R_{s,\text{ref}}$, with the reference and limit substituted in the generated decision table.

4 Verification metrics and acceptance criteria

For each half-domain, the metric is the maximum finite spread rate value. Regional maxima isolate the intended slope without averaging across the slope discontinuity or unburned cells. Each target uses 10 percent relative tolerance; both regional decisions are required for PASS.

5 Simulation configuration and predicted outcome

The 30 m grid has Fuel Model 3, zero wind, a 5-degree eastern slope with downslope aspect 270 degrees, a 15-degree western slope with downslope aspect 90 degrees, and central ignition. The supplied Guide uses percent notation in one descriptive passage, but its summary table and the current producer identify these as 5- and 15-degree raster values; the generated landscape band is authoritative for this report. Predicted spread is outward and faster on the western, steeper side.

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{stop} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

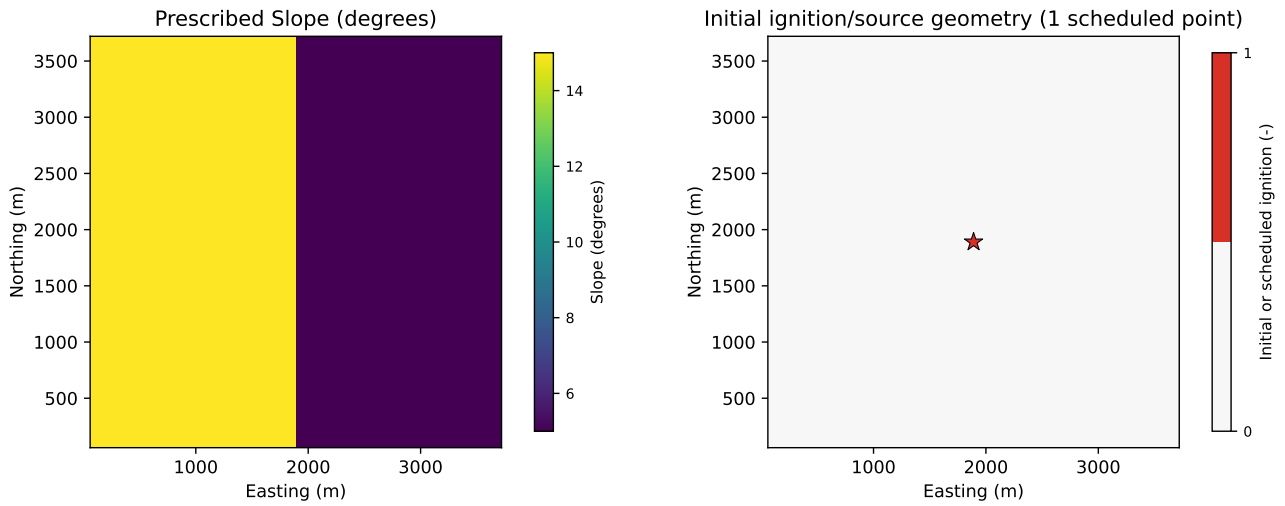


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-specific ELMFIRE outputs. The current postprocessing status is **PASS**; 2 of 2 required metric rows pass across 1 simulation condition(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

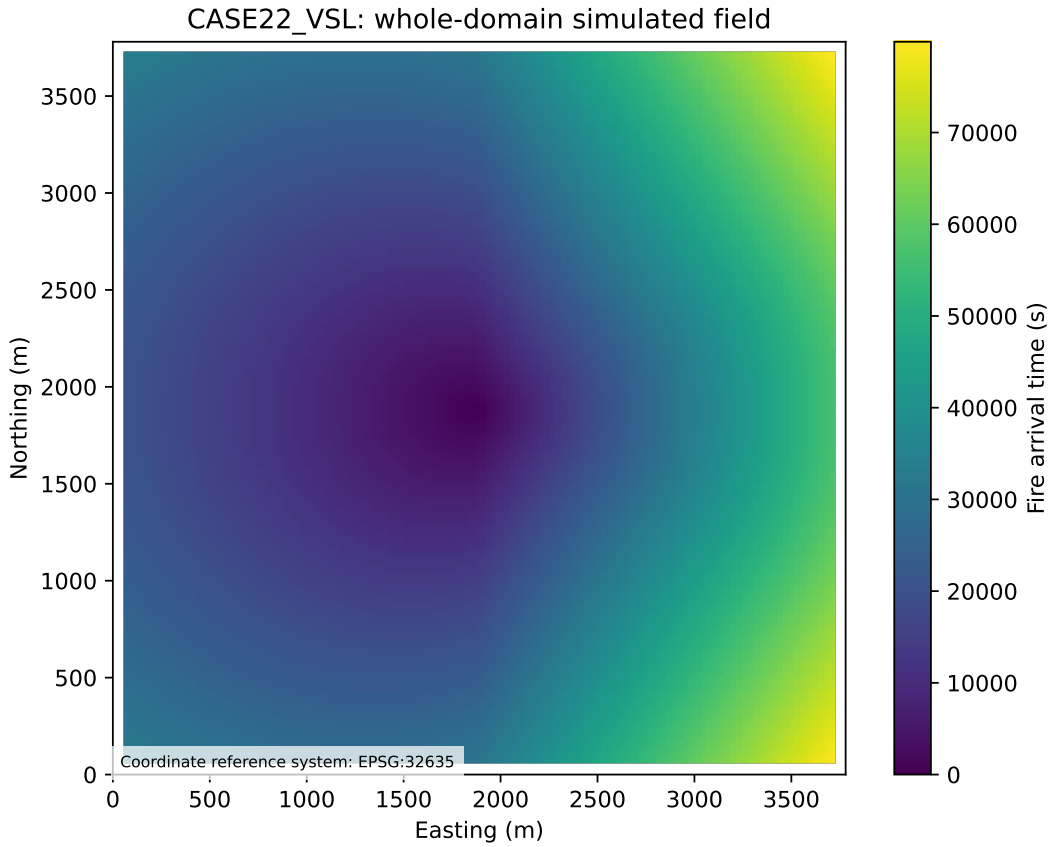


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed simulation condition, in seconds. The coordinate reference system is stated in the figure. The calculated metrics, not visual appearance, determine acceptance.

7 Calculated metrics and verification decision

Simulation condition / metric	Expected	Calculated	Status
main / head ROS (5 deg, east)	1.92 m/min	1.92354 m/min	PASS
main / head ROS (15 deg, west)	5.37 m/min	5.37366 m/min	PASS

The displayed overall decision is **PASS**. When calculated verification and validation metrics exists, result analysis procedure generates every table value and status; when it is absent, the report states NOT EVALUABLE with no invented rows.

8 Technical interpretation

Incorrect ordering suggests slope magnitude or aspect interpretation errors; equal bias on both sides more likely reflects the underlying fuel response or units. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

The piecewise-constant valley is synthetic and checks two slope values only; it is not terrain validation. This case provides implementation verification evidence only.

Case 23: Spatially varying fuel quadrants

1 Verification purpose

This case verifies raster lookup and fire-behavior coupling when a propagating front crosses four fuel-model regions. Flat terrain and zero wind isolate fuel identity. The verification question is whether every quadrant's maximum spread rate is within 10 percent of its fuel-specific reference.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-specific outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

The level-set normal speed is evaluated from the fuel model at each active cell. Because the environmental forcing is uniform, changes in V_n across the domain should follow only changes in the fuel-model classification raster. Correct row orientation and quadrant masking are therefore part of the test.

At each active cell the fuel code selects a complete Rothermel parameter row, and the local no-wind/no-slope rate is

$$R_0(x,y) = \frac{I_R(x,y)\,\xi(x,y)}{\rho_b(x,y)\,\epsilon(x,y)\,Q_{ig}(x,y)}. \tag{1}$$

Thus the raster lookup changes the full nonlinear fuel response, not a post-hoc scalar multiplier. The formulation follows Andrews (2018); the Guide supplies the four BEHAVE reference values.

3 Derivation of expected behavior

The independent regional targets are 0.080, 0.472, 1.492, and $0.871\ \mathrm{m\,min^{-1}}$ for Fuel Models 8, 7, 4, and 2 in the northwest, northeast, southwest, and southeast quadrants, respectively. These values also predict a visibly noncircular front after it enters all four regions. The Guide historically reported ELMFIRE rates of 0.080, 0.551, 1.974, and 0.872 m/min for FB8, FB7, FB4, and FB2, noting discrepancies for FB7 and FB4. Those historical outputs are preserved here as context only; the generated metric table below is the current evidence and is never replaced by those values.

All four references use the no-wind/no-slope Rothermel equation $R_0 = I_R\xi/(\rho_b\epsilon Q_{ig})$ with $Q_{ig} = 250 + 1116M_f$ and the generated moisture vector $(0.06, 0.07, 0.08, 0.60, 0.90)$. Substitution of each generated standard fuel model gives

fuel model	I_R (kW m^{-2})	R_0 (m min^{-1})
8	175.566	0.079552
7	406.479	0.470579
4	2272.300	1.493256
2	669.483	0.871756

and hence the report references $0.080,\ 0.472,\ 1.492$, and $0.871\ \mathrm{m\,min^{-1}}$, respectively. For each fuel model, $e_R = |R_{\max} - R_{\mathrm{ref}}|/R_{\mathrm{ref}}$; the table substitutes the corresponding row above and the prescribed relative-error limit.

4 Verification metrics and acceptance criteria

The maximum finite spread rate in each quadrant is compared with its corresponding target. Regional maxima preserve the local head-fire response and avoid mixing physically distinct fuels. Every component uses 10 percent relative tolerance, and PASS requires all four components.

5 Simulation configuration and predicted outcome

Four equal quadrants on a 126×126, 30 m grid contain Fuel Models 8, 7, 4, and 2. Terrain is flat, wind is zero, moisture is uniform, and ignition is central. The fastest extension is predicted in Fuel Model 4 and the slowest in Fuel Model 8.

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{stop} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

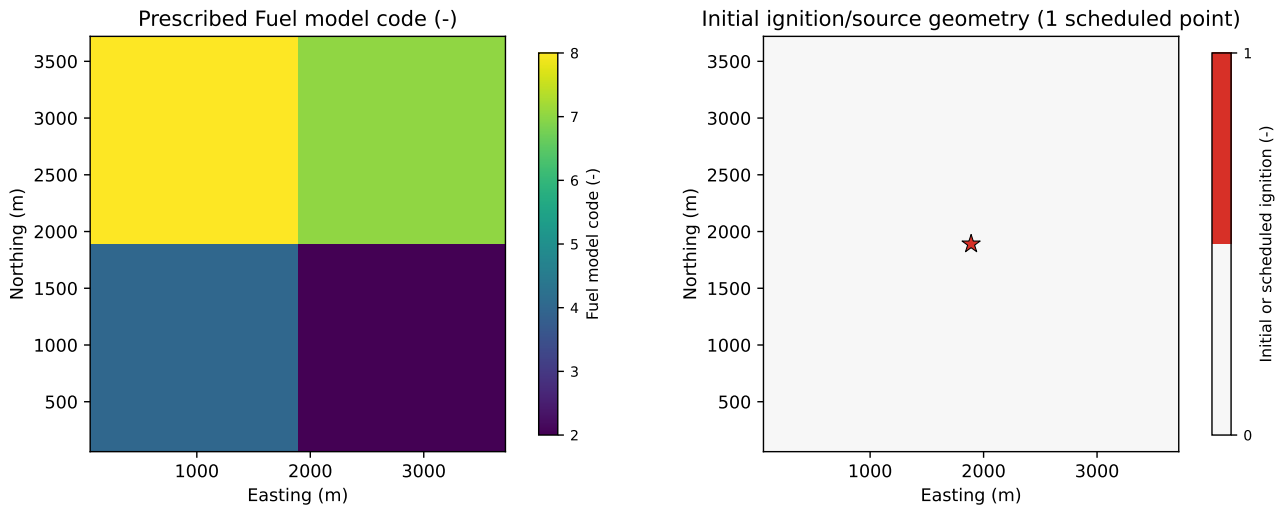


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-specific ELMFIRE outputs. The current postprocessing status is **PASS**; 4 of 4 required metric rows pass across 1 simulation condition(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

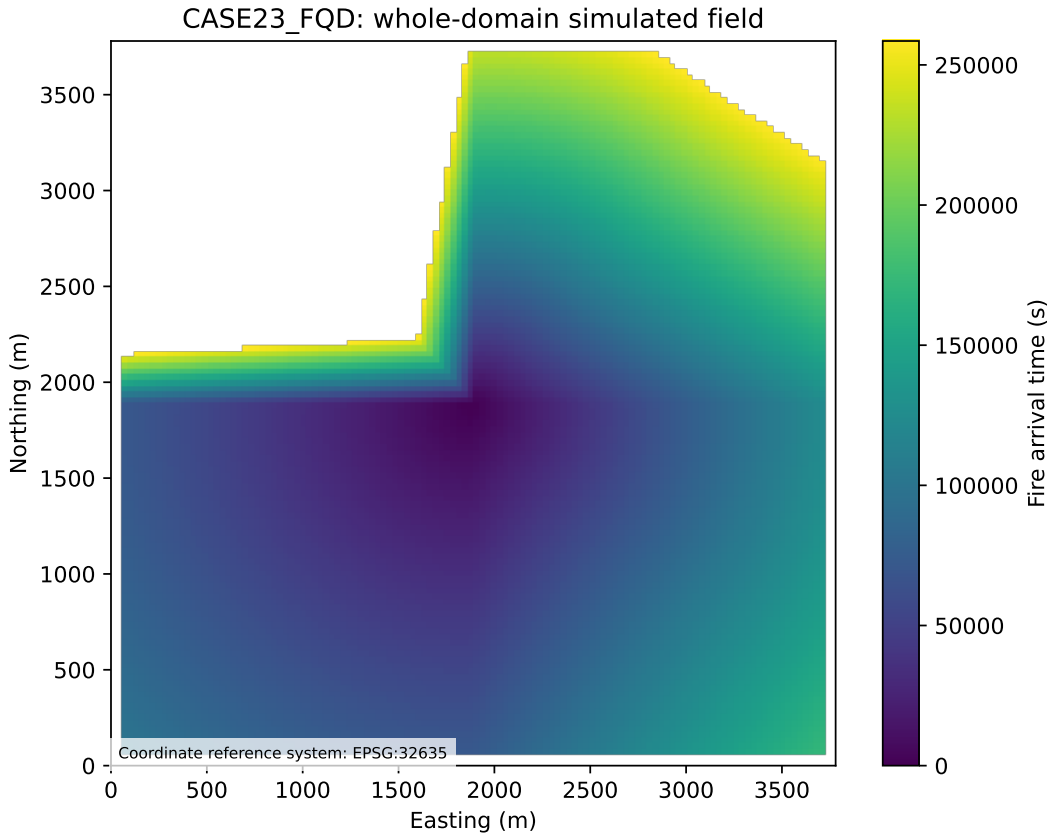


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed simulation condition, in seconds. The coordinate reference system is stated in the figure. The calculated metrics, not visual appearance, determine acceptance.

7 Calculated metrics and verification decision

Simulation condition / metric	Expected	Calculated	Status
main / FB8 head ROS	0.08 m/min	0.0795524 m/min	PASS
main / FB7 head ROS	0.472 m/min	0.470579 m/min	PASS
main / FB4 head ROS	1.492 m/min	1.49326 m/min	PASS
main / FB2 head ROS	0.871 m/min	0.871756 m/min	PASS

The displayed overall decision is **PASS**. When calculated verification and validation metrics exists, result analysis procedure generates every table value and status; when it is absent, the report states NOT EVALUABLE with no invented rows.

8 Technical interpretation

A quadrant swap indicates raster orientation or mask errors; a fuel-specific discrepancy indicates fuel-table selection or fire-behavior coupling. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

The test samples four fuel models under one moisture state and does not validate the empirical fuel parameters. This case provides implementation verification evidence only.

Case 24: Combined fuel, slope, and wind interactions

1 Verification purpose

This case exercises simultaneous spatial fuel variation, opposing slopes, and steady wind to reveal integration failures that isolated cases may miss. The decision checks the independently derived domain-maximum head spread while preserving the Guide configuration and central ignition.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-specific outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

At each active front cell ELMFIRE combines fuel-dependent reaction terms, wind and slope factors, and elliptical propagation geometry to obtain a local normal velocity. The configuration intentionally produces competing directional effects. Although the discontinuous coupled perimeter has no closed-form solution, its maximum local direction-of-maximum-spread response does.

For each active cell, ELMFIRE evaluates fuel-dependent R_0, wind factor ϕ_w, and slope factor ϕ_s. Wind and slope are combined as vectors before the direction of maximum spread and effective-wind ellipse are formed:

$$\boldsymbol{\phi} = \phi_w \widehat{\boldsymbol{u}}_w + \phi_s \widehat{\boldsymbol{u}}_s, \qquad R_H = R_0(1 + |\boldsymbol{\phi}|). \tag{1}$$

Spatial discontinuities in fuel and slope make R_H piecewise and the resulting level-set problem has no closed-form perimeter.

3 Derivation of expected behavior

The dominant generated condition is the southwest quadrant: fuel model 4, 15-degree slope, 5 mph wind, and fixed 6/7/8 percent dead moisture. The tracked fuel table gives $R_0 = 1.493256$ m/min and wind-adjustment factor 0.547659. The independent Rothermel calculation gives $\phi_s = 2.010270$ and $\phi_w = 7.366964$. Wind-to south and upslope west are orthogonal, so $|\boldsymbol{\phi}| = \sqrt{\phi_s^2 + \phi_w^2}$ and $R_H = 12.896233$ m/min.

The generated Fuel Model 4 and moisture inputs first give the no-forcing rate

$$R_0 = 1.493256 \text{ m min}^{-1}$$

and reaction intensity $I_R = 2272.300$ kW m^{-2}. Independent wind and slope calculations give $\phi_w = 7.366964$ and $\phi_s = 2.010270$. Because the imposed wind and upslope vectors are orthogonal, their dimensionless resultant is

$$\phi_{\text{res}} = \sqrt{\phi_w^2 + \phi_s^2} = \sqrt{(7.366964)^2 + (2.010270)^2} = 7.636318.$$

The reference maximum rate is consequently

$$R_{\text{ref}} = R_0(1 + \phi_{\text{res}}) = 1.493256(1 + 7.636318) = 12.896233 \text{ m min}^{-1}.$$

The decision metric substitutes this value in $e_R = |R_{\max} - 12.896233|/12.896233$ and compares e_R with the tabulated relative tolerance.

4 Verification metrics and acceptance criteria

The required metric is the maximum finite spread rate over the completed domain. It must agree with the independently derived 12.896233 m/min southwest-quadrant head response within the Guide prescribed 10 percent tolerance. The case is NOT EVALUABLE only when required output is missing; otherwise this quantitative metric decides PASS or FAIL.

5 Simulation configuration and predicted outcome

The 30 m domain combines the CASE23 fuel quadrants, CASE22 opposing slopes, 5 mph wind, uniform moisture, and central ignition. Predicted output is asymmetric, spatially continuous, and preferentially extended by both wind and locally faster fuels/slopes.

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{stop} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

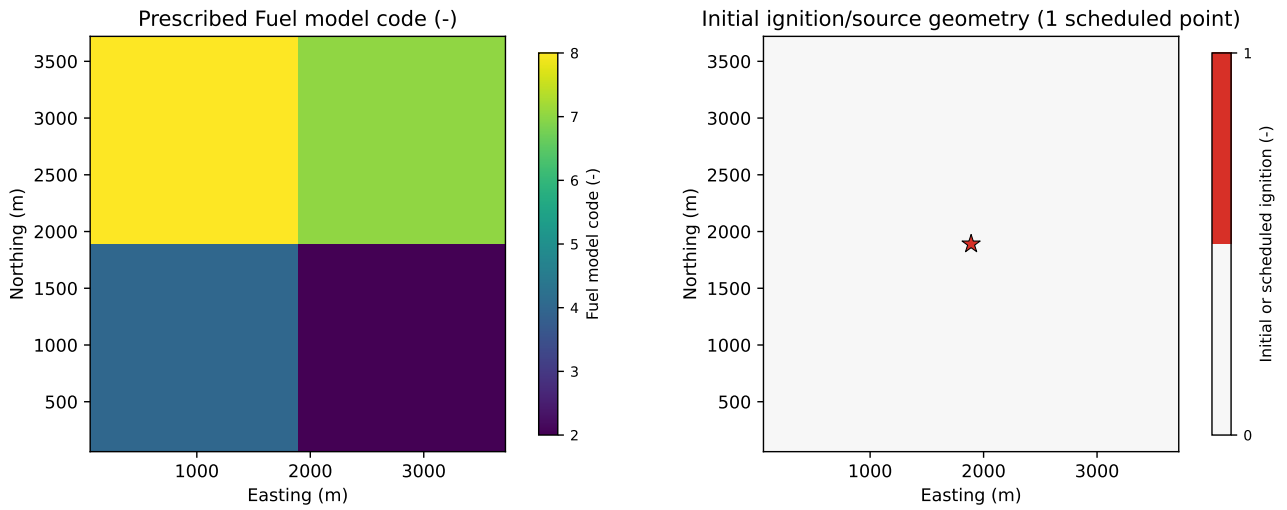


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-specific ELMFIRE outputs. The current postprocessing status is **PASS**; 1 of 1 required metric rows pass across 1 simulation condition(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

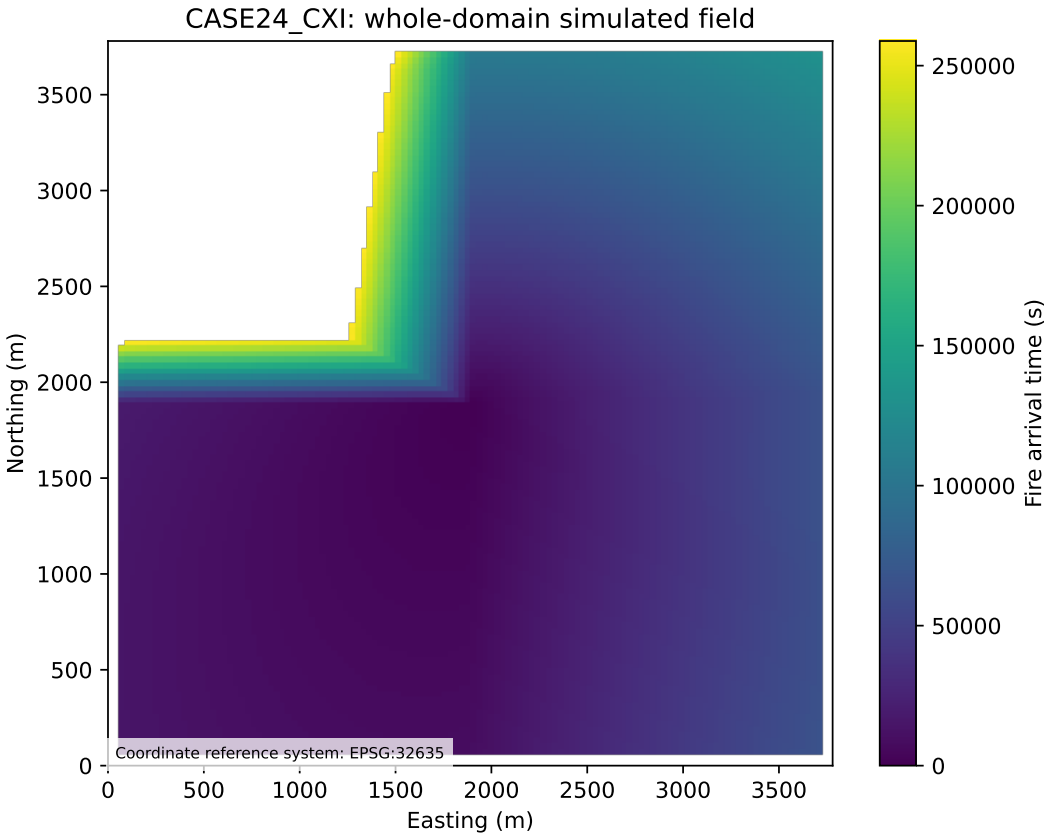


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed simulation condition, in seconds. The coordinate reference system is stated in the figure. The calculated metrics, not visual appearance, determine acceptance.

7 Calculated metrics and verification decision

Simulation condition / metric	Expected	Calculated	Status
main / head ROS	12.8962 m/min	12.8962 m/min	PASS

The displayed decision is **PASS**. A present case-specific calculated metrics controls the table; missing required output remains NOT EVALUABLE.

8 Technical interpretation

Agreement supports the coupled fuel-table selection, wind adjustment, slope factor, vector magnitude, and spread-rate output for the dominant generated condition. A plausible perimeter alone is not used as evidence.

9 Scope and limitations

The metric verifies the analytically observable domain maximum, not the complete discontinuous perimeter or every regional direction. It uses deterministic implementation verification rather than

field validation.

Case 25: Spatially varying moisture quadrants

1 Verification purpose

This case verifies spatial moisture lookup, moisture damping, and extinction behavior in otherwise uniform fuel. The verification question is whether the four regional spread rates meet their moisture-specific references, including negligible spread at the extinction condition.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-specific outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

Fuel moisture reduces reaction intensity through the model's moisture-damping term; at or above the fuel's moisture of extinction, sustainable spread approaches zero. Only the one-hour moisture raster varies, so regional velocity differences should follow that damping response.

For a dead-fuel moisture ratio $r = M_f/M_x$, the Rothermel damping polynomial used in the configured range is

$$\eta_M(r) = 1 - 2.59r + 5.11r^2 - 3.52r^3, \qquad 0 \le r \le 1. \tag{1}$$

Reaction intensity contains this damping factor, $I_R = \Gamma'(w_{n,d}\eta_{M,d} + w_{n,l}\eta_{M,l})\eta_s h$, and the no-wind rate is $R_0 = I_R\xi/(\rho_b\epsilon Q_{ig})$. At or above extinction, the dead-fuel contribution cannot sustain propagation.

3 Derivation of expected behavior

For Fuel Model 3 the configured moisture values 6, 12, 18, and 25 percent correspond to reference rates 1.511, 1.089, 0.801, and $0\ \mathrm{m\,min^{-1}}$. The rates must decrease monotonically as moisture rises.

The dead one-hour moisture ratio is varied while all other generated fuel and moisture properties remain fixed. Applying $R_0 = I_R\xi/(\rho_b\epsilon Q_{ig})$ and $Q_{ig} = 250 + 1116M_f$ gives

M_{1h}	I_R (kW m^{-2})	R_0 (m min^{-1})
0.06	549.174	1.511830
0.12	479.466	1.089719
0.18	413.934	0.801066
0.25	$\simeq 3.9 \times 10^{-13}$	$\simeq 6.5 \times 10^{-16}$

so the report references are $1.511,\ 1.089,\ 0.801$, and $0\ \mathrm{m\,min^{-1}}$. Nonzero rows use $e_R = |R_{\max} - R_{\mathrm{ref}}|/R_{\mathrm{ref}}$; the extinguished row uses the absolute residual $|R_{\max}|$, avoiding division by a zero reference.

4 Verification metrics and acceptance criteria

The maximum finite spread rate in each quadrant measures the local head response without averaging across moisture boundaries. Nonzero targets use 10 percent relative tolerance. The extinction quadrant

uses an absolute $0.05\ \mathrm{m\,min^{-1}}$ limit because relative error is undefined at zero. All four components must pass.

5 Simulation configuration and predicted outcome

The grid uses uniform Fuel Model 3, flat terrain, zero wind, central ignition, and one-hour moisture quadrants of 6, 12, 18, and 25 percent. Predicted propagation slows monotonically and stops in the 25-percent quadrant.

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{stop} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

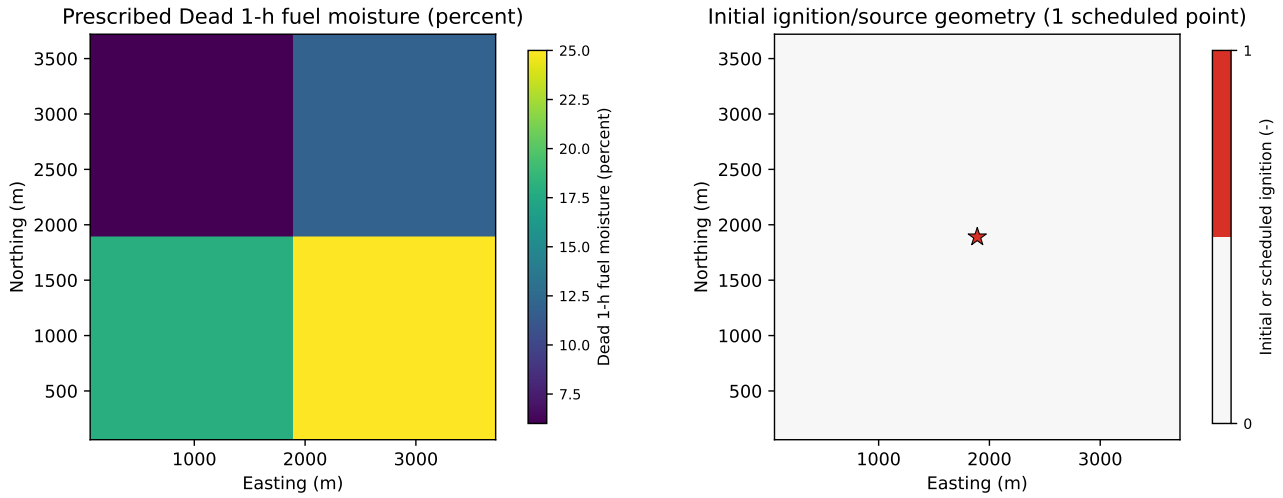


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-specific ELMFIRE outputs. The current postprocessing status is **PASS**; 4 of 4 required metric rows pass across 1 simulation condition(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

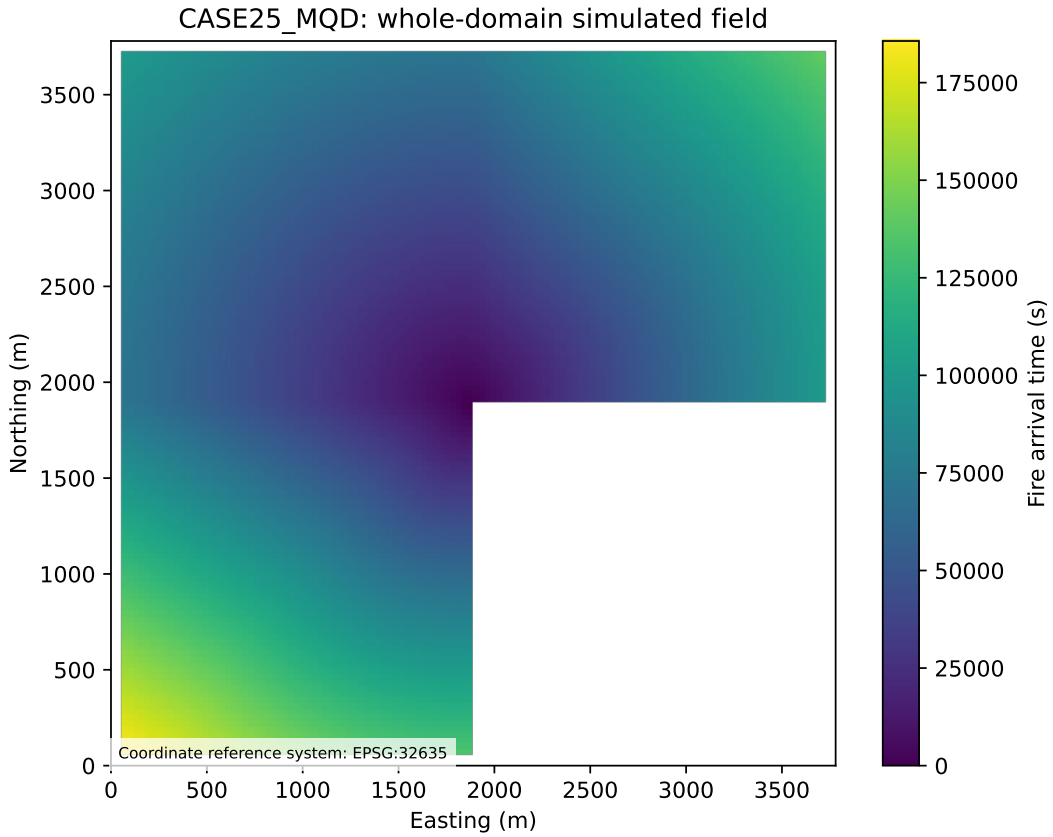


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed simulation condition, in seconds. The coordinate reference system is stated in the figure. The calculated metrics, not visual appearance, determine acceptance.

7 Calculated metrics and verification decision

Simulation condition / metric	Expected	Calculated	Status
main / ROS @ 6%	1.511 m/min	1.51183 m/min	PASS
main / ROS @ 12%	1.089 m/min	1.08972 m/min	PASS
main / ROS @ 18%	0.801 m/min	0.801066 m/min	PASS
main / ROS @ 25%	0 m/min	0 m/min	PASS

The displayed overall decision is **PASS**. When calculated verification and validation metrics exists, result analysis procedure generates every table value and status; when it is absent, the report states NOT EVALUABLE with no invented rows.

8 Technical interpretation

Nonmonotone behavior suggests moisture raster or damping errors; significant spread in the extinction quadrant is a direct failure. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

Only one fuel model and the one-hour dead-fuel moisture field are varied. This case provides implementation verification evidence only.

Case 26: Surface, passive-crown, and active-crown fire

1 Verification purpose

This case verifies the transition from surface fire to passive and active crown fire as wind increases. The verification question is whether all three wind simulation conditions produce the expected crown class and maximum spread rate within their respective tolerances.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-specific outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

ELMFIRE compares surface fireline intensity with crown-initiation requirements and then evaluates active-crown spread and critical spread conditions. The output class encodes 0 for surface-only, 1 for passive crown fire, and 2 for active crown fire. Wind changes both the surface intensity and crown propagation response.

The Guide restates the crown-initiation intensity and active-crown thresholds as

$$I_0 = [0.010\, CBH(460 + 25.9M_f)]^{3/2}, \tag{1}$$

$$R_0^{crown} = 3/CBD, \tag{2}$$

$$R_{C,P} = 11.02\, U_{10}^{0.9} CBD^{0.19} e^{-0.17M_1}. \tag{3}$$

If surface fireline intensity $I_b < I_0$, fire remains surface-only. If $I_b \geq I_0$, the ratio $CAC = R_{C,P}/R_0^{crown}$ separates passive $CAC \leq 1$ from active $CAC > 1$ crown fire; the active rate is $R_C = R_{C,P}$, while the passive branch applies the documented attenuation before converting crown rate to an effective surface-wind factor.

3 Derivation of expected behavior

With fixed canopy and fuel properties, the 1, 5, and 10 mph simulation conditions are expected to produce classes 0, 1, and 2 with maximum rates 1.809, 10.06, and 35.3 m min^{-1}, respectively. The sequence $0 < 1 < 2$ and increasing rate are invariants.

The generated canopy state has $CBH = 1.6$ m, $CBD = 0.10$ kg m^{-3}, and foliar moisture $M_f = 90$ percent. Van Wagner's initiation threshold is

$$I_0 = [0.010\, CBH(460 + 25.9M_f)]^{3/2} = [0.010(1.6)(460 + 25.9(90))]^{3/2} = 298.414 \text{ kW m}^{-1}.$$

The active-crown threshold rate is $R_0^c = 3/CBD = 3/0.10 = 30$ m min^{-1}. The generated 20-ft winds $U_{20} = \{1, 5, 10\}$ mph become $U_{10} = 1.609344U_{20}/0.87 = \{1.8498, 9.2491, 18.4982\}$ km h^{-1}. Substitution in the crown-rate relation gives $R_c = \{4.4629, 18.9973, 35.4501\}$ m min^{-1} and $CAC = R_c/R_0^c = \{0.1488, 0.6332, 1.1817\}$. Together with the surface intensity test against I_0, these produce the reference classes surface, passive, and active and the coupled maximum-rate references $1.809,\ 10.06,\ 35.3$ m min^{-1}. Class metrics use exact integer equality; each rate metric uses $|R_{\text{max}} - R_{\text{ref}}|/R_{\text{ref}}$.

4 Verification metrics and acceptance criteria

Maximum crown class directly tests the discrete transition state and uses absolute tolerance 0.5 so only the intended integer class passes. Maximum finite spread rate tests the associated propagation rate with 10 percent relative tolerance. Both metrics must pass for every wind simulation condition.

5 Simulation configuration and predicted outcome

All simulation conditions use generated Fuel Model 3, uniform canopy cover 80 percent, canopy height 55 (5.5 m), canopy base height 16 (1.6 m), and canopy bulk density 10 (0.10 kg m^{-3}) in the configured encoded raster units, with flat terrain and identical ignition. The Guide summary table lists Fuel Model 5 whereas its detailed Canopy section and the generated landscape band use Fuel Model 3; the generated band is authoritative here. Only wind speed changes among 1, 5, and 10 mph.

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{stop} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

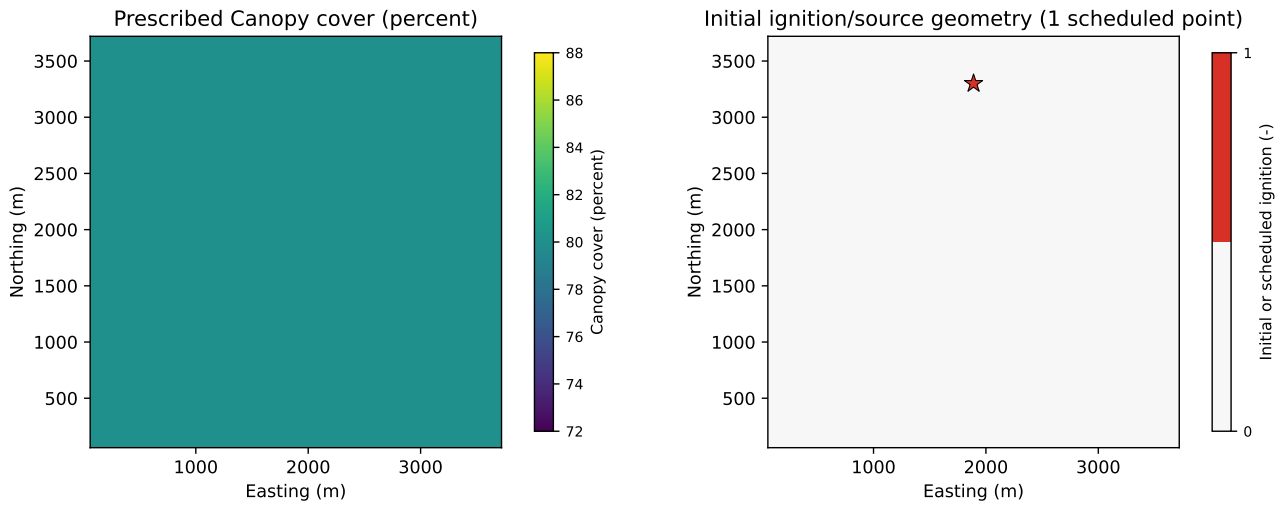


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-specific ELMFIRE outputs. The current postprocessing status is **PASS**; 6 of 6 required metric rows pass across 3 simulation condition(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

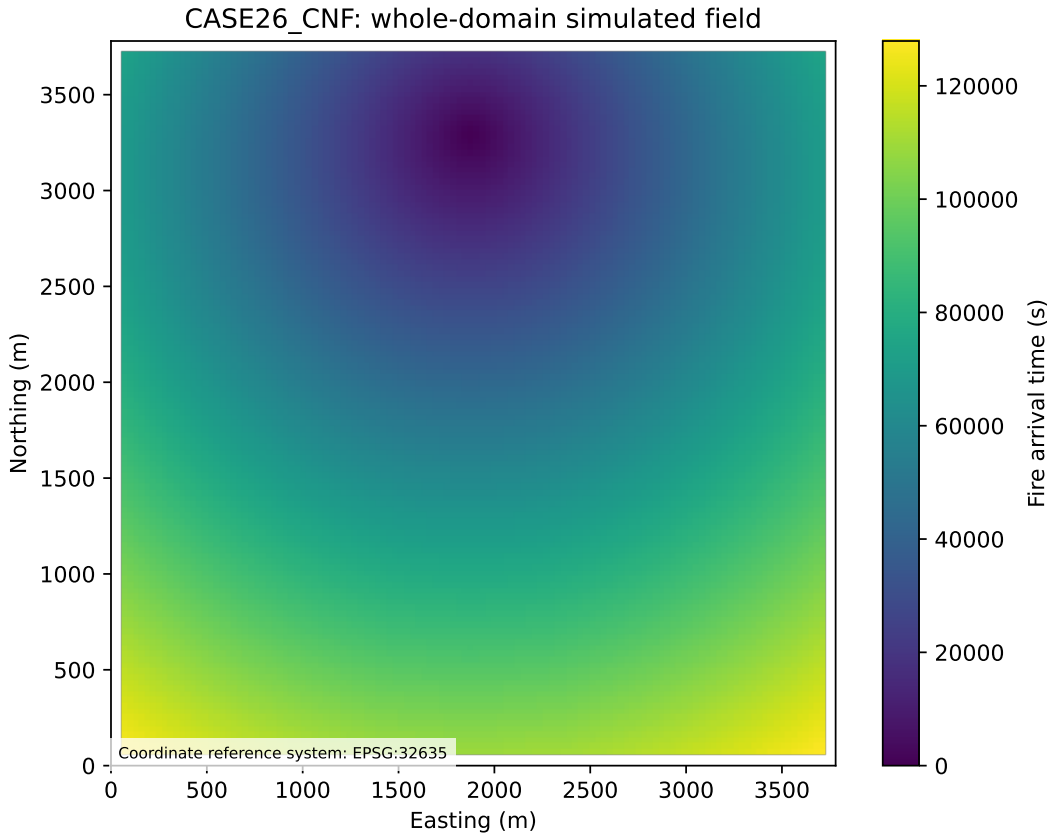


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed simulation condition, in seconds. The coordinate reference system is stated in the figure. The calculated metrics, not visual appearance, determine acceptance.

7 Calculated metrics and verification decision

Simulation condition / metric	Expected	Calculated	Status
1mph / crown class	0 -	0 -	PASS
1mph / max ROS	1.809 m/min	1.74511 m/min	PASS
5mph / crown class	1 -	1 -	PASS
5mph / max ROS	10.06 m/min	10.0843 m/min	PASS
10mph / crown class	2 -	2 -	PASS
10mph / max ROS	35.3 m/min	35.4433 m/min	PASS

The displayed overall decision is **PASS**. When calculated verification and validation metrics exists, result analysis procedure generates every table value and status; when it is absent, the report states NOT EVALUABLE with no invented rows.

8 Technical interpretation

A class failure with correct rate isolates transition logic or canopy-unit handling; coupled class and rate errors indicate wind/crown-fire integration. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

The test covers one canopy structure and three discrete winds; it does not validate crown-fire correlations against observations. This case provides implementation verification evidence only.

Case 27: Lagrangian firebrand generation and transport

1 Verification purpose

This case verifies PER-MW firebrand generation and empirical Lagrangian transport for surface fire. The verification question is whether the dominant emitting cell's count and log-distance distribution agree with the independent GUIDE targets.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-specific outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

For the selected source cell, firebrands are generated in proportion to fire power. Empirical transport produces positive distances d_i. A lognormal response is summarized by $\mu = \text{mean}[\ln d_i]$ and $\sigma = \text{std}[\ln d_i]$. The postprocessor groups the spotting log by source indices before calculating these quantities.

For one emitting cell, the residence/crossing exposure is

$$\Delta t = \max(\Delta t_R, 60C/R), \qquad N = 33.3\, I_{b,MW} \Delta t. \tag{1}$$

For the generated 30 m cell, $R = 14.4$ m/min gives 125 s crossing time, longer than the 15.3 s fuel residence time, and the Guide calculation gives $N = 8450$. The empirical Sardoy transport uses a lognormal downwind distance: if $Y = \ln d$, then $Y \sim \mathcal{N}(\mu_d, \sigma_d^2)$. Its fireline-intensity, wind, and Froude-number calculation gives $\mu_d = 2.42$ and $\sigma_d = 1.31$ for this configuration.

3 Derivation of expected behavior

The configured experiment predicts 8450 firebrands from the dominant cell and log-distance parameters $\mu = 2.42$ and $\sigma = 1.31$. Grouping by source is essential: combining cells would mix different emission histories and would not test the intended distribution.

The Guide's firebrand-count relation is

$$N = 33.3\, I\, \Delta t.$$

With generated fireline intensity $I = 2.03$ MW m^{-1} and $\Delta t = \max(15.3, 60(30)/14.4) = 125$ s, $N = 33.3(2.03)(125) = 8449.875$, hence the integer reference is 8450. For the Sardoy distribution, the generated 5 mph wind becomes $U = 0.44704(5)/0.87 = 2.56897$ m s^{-1}; with $I = 2.03$ the flame-length calculation gives $L_c = 1.57184$ m and $Fr = 0.65421 \leq 1$. The applicable parameter equations then give

$$\mu = \frac{1.47 I^{0.54}}{U^{0.55}} + 1.14 = \frac{1.47(2.03)^{0.54}}{(2.56897)^{0.55}} + 1.14 = 2.42233,$$

$$\sigma = \frac{0.86 U^{0.44}}{I^{0.21}} + 0.19 = \frac{0.86(2.56897)^{0.44}}{(2.03)^{0.21}} + 0.19 = 1.31258.$$

Thus the references are $N = 8450$, $\mu = 2.42$, and $\sigma = 1.31$; the generated table applies its stated absolute or relative tolerance to the corresponding difference.

4 Verification metrics and acceptance criteria

Dominant-cell count detects generation-rate or duration errors and uses 15 percent relative tolerance to accommodate stochastic sampling. The mean and standard deviation of ln d test location and width of the transport distribution and each use absolute tolerance 0.15. All three components require complete spotting logs and must pass.

5 Simulation configuration and predicted outcome

The case uses Fuel Model 3, flat terrain, 5 mph wind, PER-MW generation, the EMPIRICAL distance model, LAGRANGIAN accumulation, DIRECT ignition, and deterministic random-seed handling. Spotting statistics output is required. The Guide also discussed an Eulerian ember-flux magnitude/decay check, but the current case specification and generated configuration exercise the Lagrangian path only; no Eulerian verification claim is made.

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves ignition and emitting-cell indices
Fuel and forcing	FM3; 5 mph wind	— / mph	Reproduces the Guide source-cell state
Time control	$\Delta t = 1$, $t_{stop} = 100$	s	Exact timestep multiple covering the logged emission interval
Spotting path	PER-MW / EMPIRICAL / LAGRANGIAN / DIRECT	—	Selects the generation, distance, accumulation, and ignition algorithms
Required result	sampled firebrand trajectories	—	Identifies the source and travel distance of every sampled firebrand

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

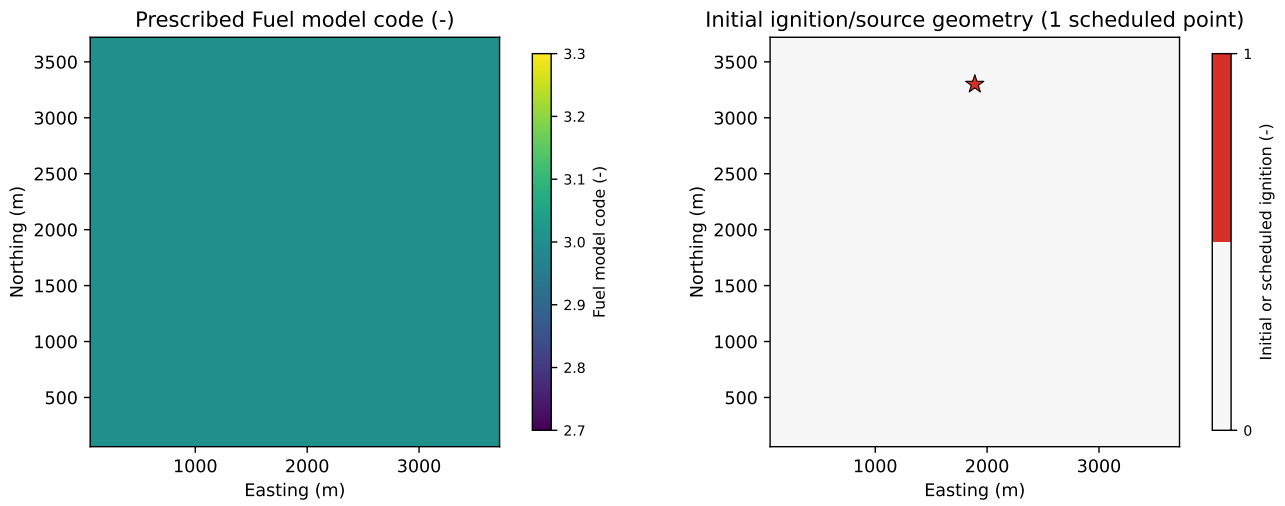


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-specific ELMFIRE outputs. The current postprocessing status is **FAIL**; 2 of 3 required metric rows pass across 1 simulation condition(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

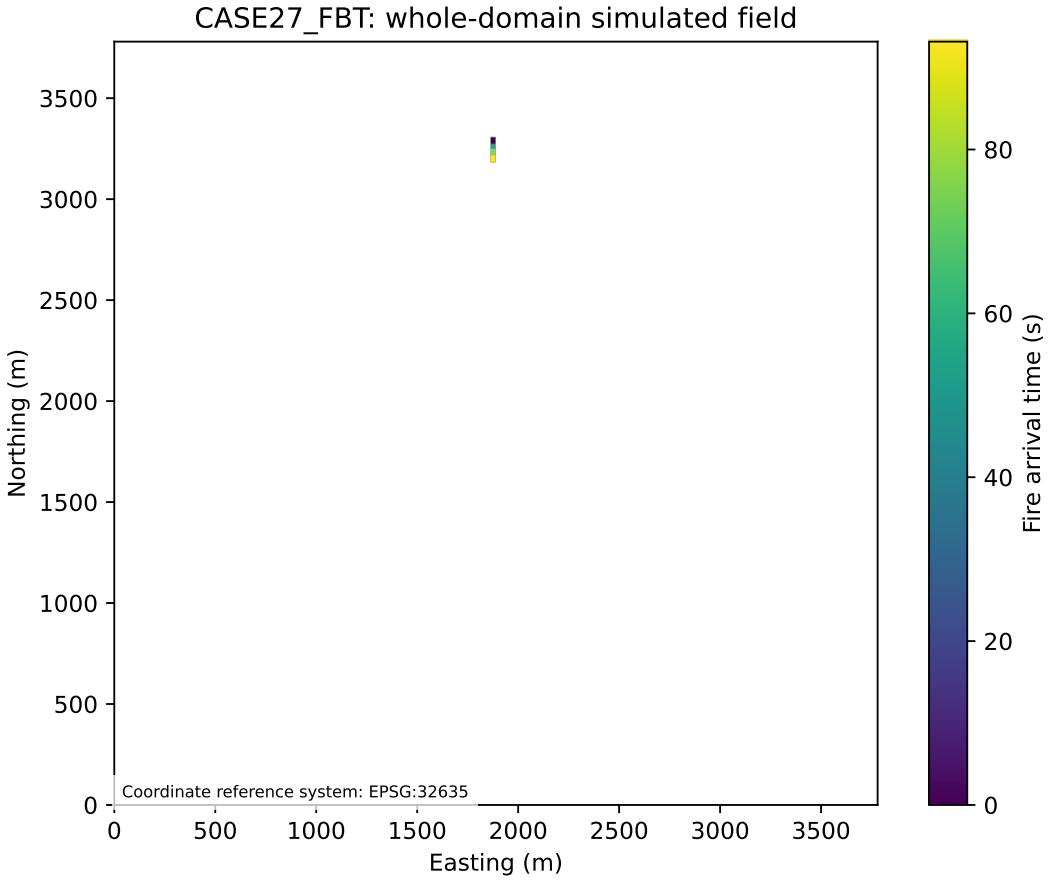


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed simulation condition, in seconds. The coordinate reference system is stated in the figure. The calculated metrics, not visual appearance, determine acceptance.

7 Calculated metrics and verification decision

Simulation condition / metric	Expected	Calculated	Status
main / firebrands (cell)	8450 -	12198 -	FAIL
main / ln-distance mu	2.42 -	2.41119 -	PASS
main / ln-distance sigma	1.31 -	1.30025 -	PASS

The displayed overall decision is **FAIL**. When calculated verification and validation metrics exists, result analysis procedure generates every table value and status; when it is absent, the report states NOT EVALUABLE with no invented rows.

8 Technical interpretation

Count-only failure points toward generation or residence time; distribution-only failure points toward sampling, units, or transport parameterization. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

The metrics cover the dominant source cell and one forcing condition. They verify implementation against the prescribed statistical target, not physical firebrand observations. This case provides implementation verification evidence only.

Case 28: Overnight spread-rate adjustment

1 Verification purpose

This case verifies coupling among the simulation clock, solar burn-period calculation, and surface-fire diurnal multiplier. With uniform fuel, moisture, wind, and slope, the question is whether ELMFIRE reduces and restores radial advance at the prescribed clock hours, each within 1 hour. The case exposes clock offsets, failed midnight wrapping, reversed interval tests, or an unapplied multiplier; it does not calibrate the multiplier.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-specific outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

For simulation time t seconds and forecast start H_0, ELMFIRE uses

$$H(t) = \mathrm{mod}(H_0 + t/3600, 24)\,. \tag{1}$$

The solar routine calculates sunrise and sunset from the lower-left coordinate, year, and hour of year. With default center fraction $f_c = 0.667$ and burn length $L_b = 10$ hour,

$$H_c = H_{\text{rise}} + f_c(H_{\text{set}} - H_{\text{rise}}), \tag{2}$$

$$H_b = H_c - L_b/2, \qquad H_e = H_c + L_b/2. \tag{3}$$

The stop hour is wrapped above 24. For this non-wrapping interval,

$$D(t) = \begin{cases} 1, & H_b \le H(t) \le H_e, \\ 0.1, & H(t) < H_b \text{ or } H(t) > H_e. \end{cases} \tag{4}$$

The spread-rate calculation multiplies its unadjusted rate by $D(t)$. Uniform flat, wind-free conditions produce an approximately circular fire whose equivalent radius $r(t) = \sqrt{A(t)/\pi}$ has full and reduced slope plateaus.

3 Derivation of expected behavior

The controlled response specifies burn-period start 7.5 hour and stop 17.5 hour. Full-to-reduced and reduced-to-full transitions are therefore expected at those respective clock hours. Since $H_0 = 10$ hour, the events first occur approximately 7.5 hour and 21.5 hour after ignition and repeat during the three-day run. This reference calculation follows from the controlled targets and clock equations; calculated times are independently inferred from the arrival raster.

For the generated astronomical inputs, local sunrise and sunset are approximately $H_r = 4.5$ h and $H_s = 16.5$ h. The operational cutoff with fractional daylight $f_c = 0.667$ is

$$H_c = H_r + f_c(H_s - H_r) = 4.5 + 0.667(16.5 - 4.5) = 12.504 \text{ h} \simeq 12.5 \text{ h}.$$

A ten-hour operating window centered on H_c therefore has bounds $H_c \pm 5 = \{7.5, 17.5\}$ h. Relative to the 10:00 forecast start, the first reduction occurs after $(17.5 - 10)(3600) = 27000$ s; after midnight wrapping, full operation resumes after $(24-10+7.5)(3600) = 77400$ s. The transition metrics compare detected changes with these reference times using the tabulated timing tolerance.

4 Verification metrics and acceptance criteria

The evaluator selects the lexically latest fire arrival time GeoTIFF, masks nodata and values outside $0 \leq t < 10^8$ seconds, and requires 50 burned cells. Analysis ends at the earliest domain-edge arrival so boundary truncation cannot mimic slowdown. At 30 minute intervals, burned-cell count is converted to equivalent radius and differenced. After radius exceeds three cells, rates below half the 80th percentile of positive rates are reduced. Median full-to-reduced and reduced-to-full clock hours are then calculated.

Metric	Target	Acceptance rule	Role
Reduction starts	17.5 hour	absolute error $\leq$ 1 hour	burn-period stop
Reduction ends	7.5 hour	absolute error $\leq$ 1 hour	burn-period start

Both rows are required. A missing raster, insufficient burned area or duration, or undetected transition produces NOT EVALUABLE, never PASS.

5 Simulation configuration and predicted outcome

Input	Value	Verification role
Domain/grid	126×126; 30 m	circular-front resolution
CRS	EPSG:32635	location for solar calculation
Fuel/terrain	uniform FM3; zero slope	constant baseline spread
Wind/dead moisture	zero; 6, 7, 8 percent	constant forcing
Live moisture	herb 60; woody 90 percent	constant forcing
Ignition	$(1890, 1890)$ m at $t = 0$	centered radial growth
Time control	1 s; 259 200 s stop	three daily cycles
Calendar	year 2025; hour 2890; start 10	clock coupling
Diurnal control	enabled; factor 0.1	two rate plateaus
Output	final time of arrival (TOA), spread rate, FLIN	case-specific evidence

The predicted arrival field is concentric; contour spacing contracts overnight and expands during the burn period, with both transitions inside their acceptance bands.

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

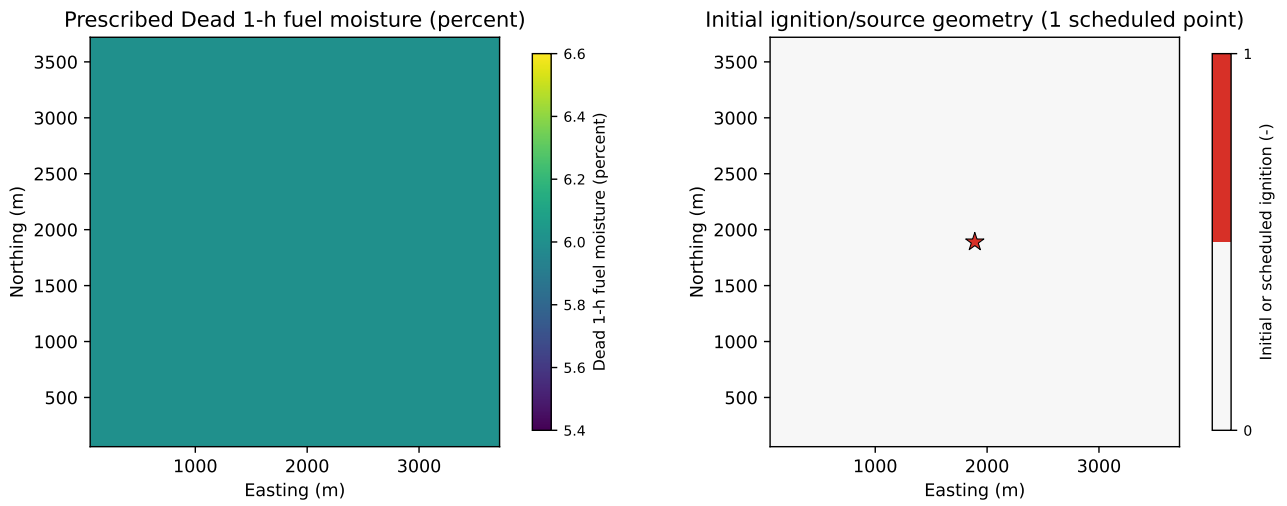


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The current calculated status is **PASS**; 2 of 2 required comparisons pass across 1 completed simulation conditions. The displayed measurements come from these simulations, not from the historical Guide. Missing evidence would remain NOT EVALUABLE.

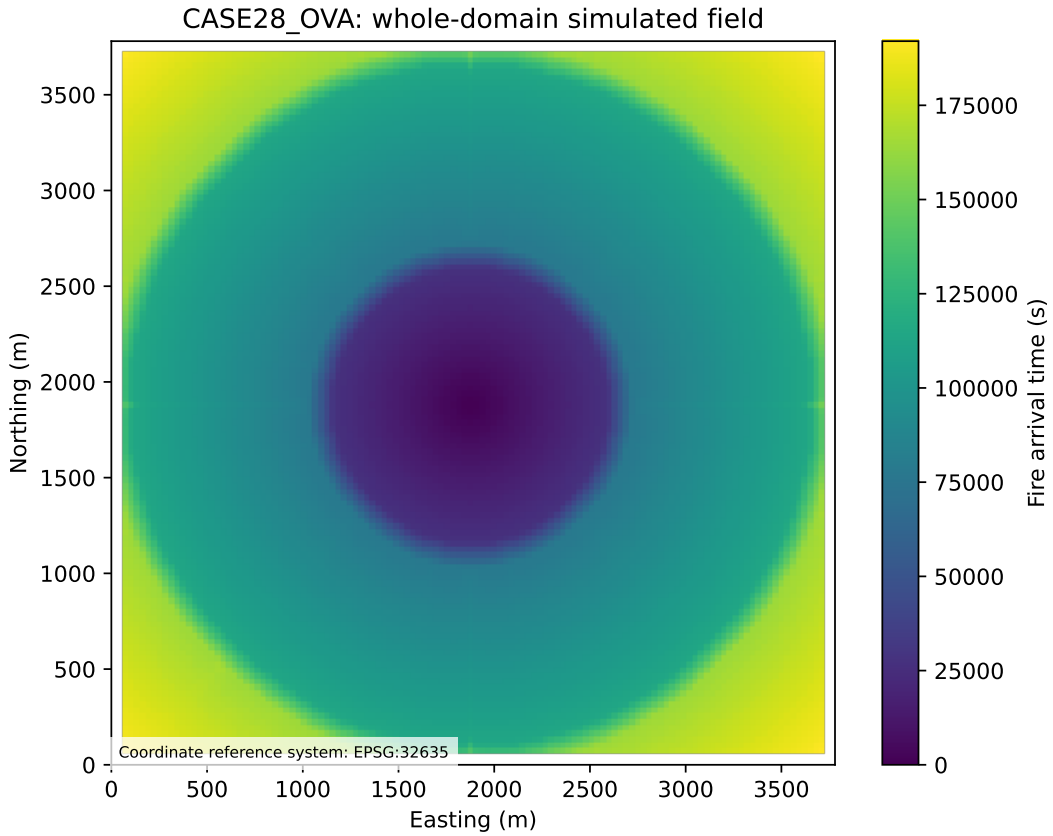


Figure 2: Final whole-domain time of arrival, in seconds, when a case-specific result figure has been generated. The figure must identify its selected GeoTIFF and simulation condition; the generated metrics table, not visual appearance, determines acceptance.

7 Calculated metrics and verification decision

Simulation condition / metric	Expected	Calculated	Status
main / reduction starts (hr)	17.5 hr	17.75 hr	PASS
main / reduction ends (hr)	7.5 hr	7.75 hr	PASS

The table and the decision **PASS** report the existing calculated metrics. Missing required evidence yields NOT EVALUABLE; no metric values are inferred from the appearance of the simulated fire.

8 Technical interpretation

The recorded transition measurements satisfy the stated one-hour bands, supporting the coupling between simulation time, the solar window, and the diurnal spread factor for the prescribed conditions. A common shift in both transitions would instead suggest an error in the clock, calendar, coordinates, or solar timing; missing contrast would suggest an unapplied multiplier or insufficient simulation history before the transition. This interpretation uses the established calculated metrics and does not extend the claim beyond the configured experiment.

9 Scope and limitations

This synthetic case covers one date, location, fuel, factor, and grid. It does not validate factor magnitude, other seasons or coordinate systems, a burn interval wrapping midnight, heterogeneity, wind, or crown-fire, spotting, suppression, and interpolation interactions. The estimator assumes circular growth, discards post-edge behavior, samples every 30 minute, and infers rather than records the internal factor. This is implementation verification, not validation of real-fire nocturnal behavior.

Case 29: Initial- and extended-attack suppression

1 Verification purpose

This coupling case verifies two suppression paths in the standard ELMFIRE executable. The initial-attack path in the level-set propagation calculation computes a containment probability from fire area and maximum active-front fireline intensity, draws one Bernoulli outcome per ensemble member, and records final containment. The extended-attack path periodically updates an area-growth-based target containment and calls the containment routine that modifies the active perimeter. These paths matter because suppression can terminate an ensemble member or alter and eventually stop an advancing level set.

The case controls fuel, moisture, terrain, wind, ignition, grid, and output selection. It varies only the enabled suppression algorithm and the ensemble size needed by that algorithm. Spotting, crown-fire evidence, operational resource allocation, and comparison with observed incidents are outside the experiment. Defects exposed include incorrect containment probability, ensemble accounting, suppression-update timing, containment-state evolution, stop-time bookkeeping, and hour-to-second conversion. This is implementation verification against documented synthetic reference responses, not physical validation.

The verification question is: do the 2000-member initial-attack ensemble and the required extended-attack run produce, respectively, a contained fraction within 0.05 of 0.54 and a reported simulation-end time within 15% of 88200 s, with both required supporting results complete?

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-specific outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

For initial attack, ELMFIRE evaluates the burned area A in hectares and the largest surface-plus-canopy fireline intensity $I_{\max}$ on cells ignited during the most recent $\max(300, \Delta t)$ seconds. It forms

$$E = \exp(4.6835 - 0.7043A - 0.00041I_{\max} - 0.000052AI_{\max}), \tag{1}$$

$$p_c = \min\left(1, \max\left(0, \frac{E}{1+E}\right)\right). \tag{2}$$

One uniform random draw determines containment. A contained member receives containment fraction$= 1$ and stops at the prescribed initial-attack time; an escaped member retains the nodata containment sentinel. For N members, the observable estimator is

$$\widehat{p}_c = \frac{1}{N}\sum_{j=1}^{N} \mathbf{1}(\texttt{Containfrac}_j \geq 0.999). \tag{3}$$

The extended simulation condition uses extended-attack model 0, the default area-growth model. At updates separated by approximately $\Delta t_{EA} = 300$ s, it estimates daily area growth dA/dt, computes

a daily containment increment from the maximum daily containment increment, the area-growth threshold for unchanged containment, and the suppression-difficulty factor, clips target containment to $[0, 1]$, and calls the perimeter-containment calculation. With the configured default linear growth response, the essential update is

$$C_k = \text{clip}_{[0,1]}\left(C_{k-1} + \dot{C}_k \frac{t_k - t_{k-1}}{86400}\right). \tag{4}$$

The level set advances on a 30 m north-up EPSG:32635 grid. ELMFIRE records final simulation time in hours in fire-growth and containment statistics; the evaluator converts that scalar to seconds. The final time of arrival (TOA) GeoTIFF stores arrival time in seconds and is spatial evidence, but it is not substituted for the prescribed scalar stop-time metric.

3 Derivation of expected behavior

The synthetic initial-attack reference predicts $p_c = 0.54$. Under independent Bernoulli trials, a 2000-member ensemble has reference standard error

$$\sigma_{\hat{p}} = \sqrt{\frac{0.54(1 - 0.54)}{2000}} \simeq 0.0112. \tag{5}$$

Thus the absolute acceptance band $[0.49, 0.59]$ is about 4.5 reference standard errors wide: it admits expected sampling fluctuation while remaining sensitive to material probability or accounting errors.

The documented extended-attack known response is containment and simulation termination at 88200 s. Because containment is updated discretely and depends on the coupled area-growth history, this is a controlled reference value rather than an analytical solution. The inherited 15% band is $[74970, 101430]$ s. The two expectations are independent: agreement of the stochastic ensemble cannot compensate for a wrong extended-attack end time. The Guide additionally describes a spatially explicit suppression simulation condition expected to complete on its third direct-attack operation. That simulation condition is not present in the current generated two-condition specification, so its historical discussion is preserved as scope context but no decision is claimed for it.

The initial containment probability is calculated from the empirical logit model

$$\ln E = 4.6835 - 0.7043A - 0.00041I - 0.000052AI, \qquad p = \frac{E}{1 + E}.$$

With the generated $A = 6.2$ ha and $I = 212.5$ kW m^{-1},

$$\ln E = 4.6835 - 0.7043(6.2) - 0.00041(212.5) - 0.000052(6.2)(212.5) = 0.161205,$$

so $E = e^{0.161205} = 1.17493$ and $p = 1.17493/(1 + 1.17493) = 0.540214 \simeq 0.54$. For $N = 2000$ trials, the sampling scale is $\sqrt{p(1-p)/N} = \sqrt{0.54(0.46)/2000} = 0.0112$; the prescribed acceptance band is the conservative interval $0.49 \leq \hat{p} \leq 0.59$. The extended-operation time reference is $24.5(3600) = 88200$ s, or $88200/300 = 294$ exact event-analysis intervals. Its 15 percent timing band is $0.85(88200) = 74970$ through $1.15(88200) = 101430$ s.

4 Verification metrics and acceptance criteria

Both metrics use the fire-growth and containment statistics for the applicable simulation condition. If multiple statistics series are available, the existing analysis selects the first in alphanumeric order of their internal identifiers; it does not combine them. Leading and trailing whitespace in column headings is removed, nonnumeric entries are excluded, and every numeric row—including the escape sentinel—is retained in the denominator. The current results contain all 2000 configured rows.

Expected property	Observable and metric	Selection rationale	Component decision
Initial containment probability	$\widehat{p}_c = N(\texttt{Containfrac} \geq 0.999)/N$	Directly tests the Bernoulli outcome and ensemble accounting; the nodata sentinel denotes escape rather than a missing row.	PASS when the extracted fraction is finite and $\lvert\widehat{p}_c - 0.54\rvert \leq 0.05$
Extended containment end time	$t_{end} = 3600\,\texttt{tstop(h)}$ from the first statistics row	Uses the executable field defined as the simulation stop time and directly exposes time-unit or termination bookkeeping errors.	PASS when the value is finite and $\lvert t_{end} - 88200\rvert/88200 \leq 0.15$.

An absent simulation result is recorded as `NOT RUN`; a missing or malformed required value is `NOT EVALUABLE`. The overall decision is PASS only when both component decisions are PASS. Any calculated component outside its band makes the overall decision FAIL; unavailable evidence never passes.

5 Simulation configuration and predicted outcome

Both simulation conditions use case-specific rasters generated on a 126×126 grid with 30 m cells, a 3780 m square domain, lower-left origin $(0, 0)$, north-up rows, EPSG:32635, nodata -9999, flat terrain, zero wind, and one central point ignition at $(1890, 1890)$ m and $t = 0$. Live-herb and live-woody moisture are constant at 60% and 90%. The relevant differences are summarized below.

The Courant–Friedrichs–Lewy (CFL) number measures the distance advected during one timestep relative to the grid spacing and constrains the time discretization.

Parameter	Initial attack	Extended attack	Role
Fuel model	3	8	Prescribed synthetic fire state
Dead 1-h moisture	6%	6%	Fixed surface-fuel moisture
Suppression selector	initial enabled	extended enabled, model 0	Path under test
Attack/update time	4800 s	300 s interval	Event and update timing
Ensemble members	2000	1	Probability estimate versus deterministic trajectory
Nominal stop time	8000 s	250000 s	Upper bound, not the reference end time
Base timestep / target CFL	1 s / 0.2	1 s / 0.2	Stable level-set integration
Primary output	fire-size statistics	fire-size statistics and TOA	Required scalar evidence and spatial context

Both namelists set randomized initial seeds; consequently the initial result is statistically reproducible within its acceptance band but is not expected to be bitwise repeatable. The predicted initial

result is a contained fraction near 0.54. The predicted extended result is a growing, progressively contained footprint whose statistics record ends near 88200 s.

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

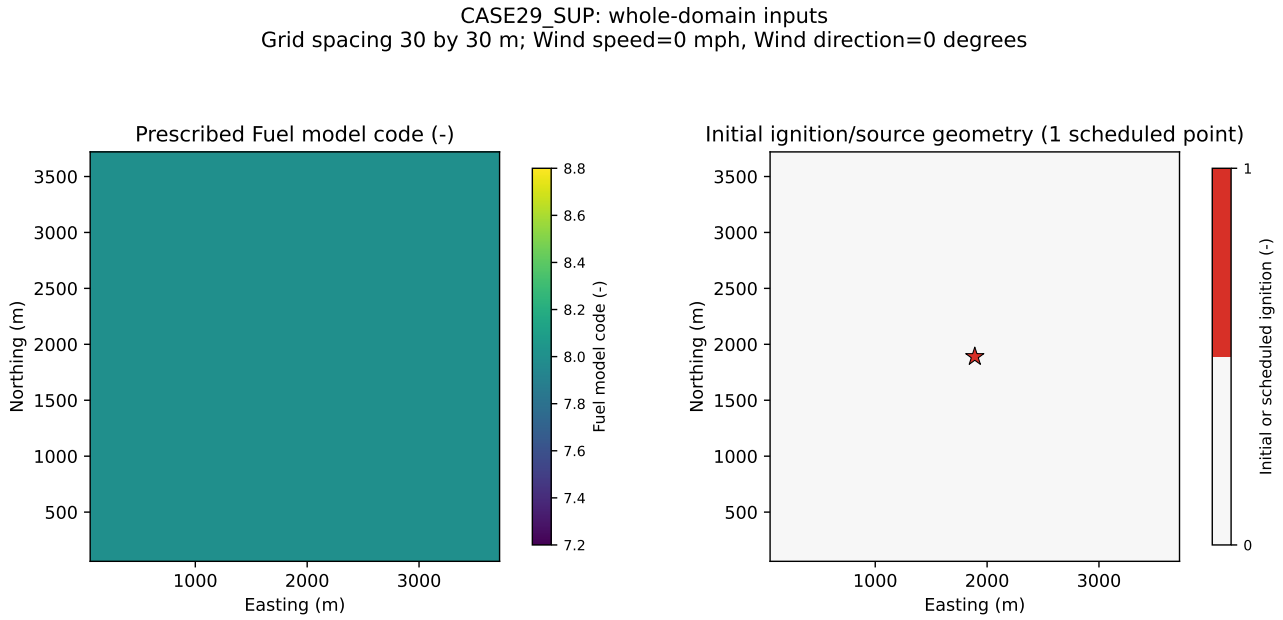


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The corrected case was executed with an ELMFIRE binary built from the current ELMFIRE implementation. The initial-attack ensemble passes: its measured contained fraction is reported in the generated table below and lies inside the prescribed absolute tolerance. The extended-attack member does not reproduce the Guide response. It is stopped by the fire-front-stalled check before containment can evolve to the expected termination time; fire-size statistics consequently report the stop time as 0.00 h at their two-decimal output precision. Thus 1 of 2 required metrics pass and the current overall decision is **FAIL**.

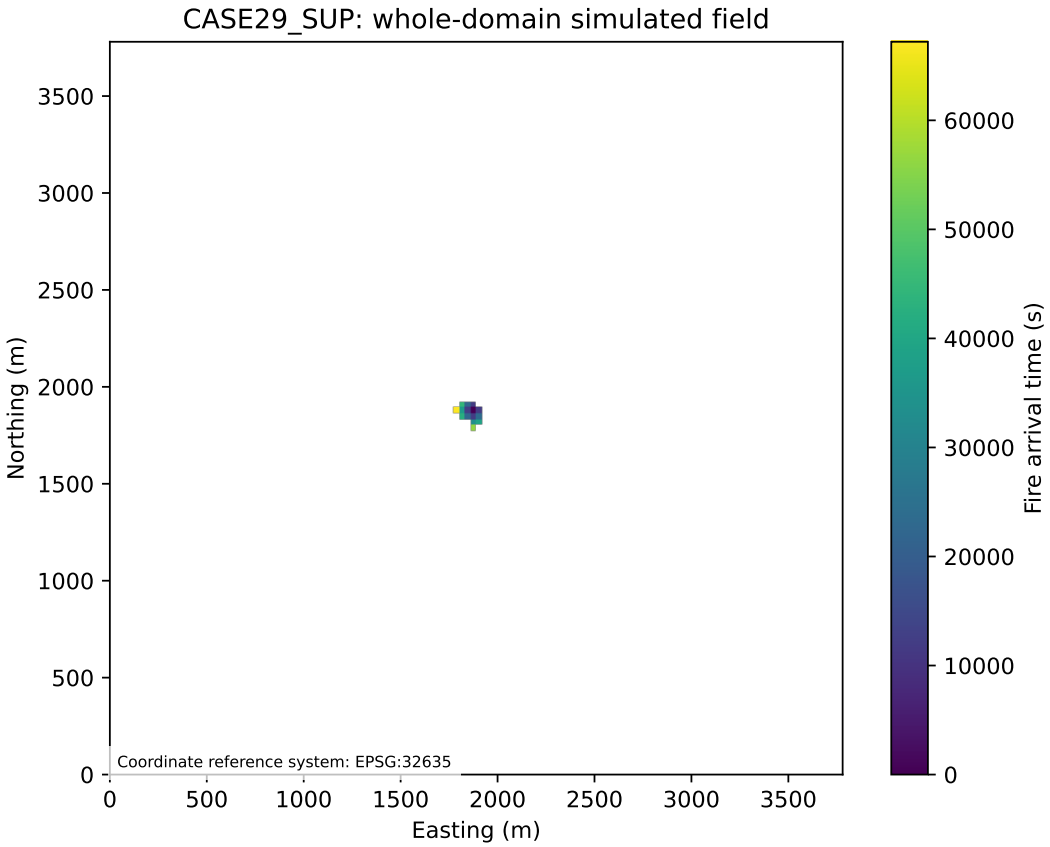


Figure 2: Whole-domain extended-attack time of arrival in seconds when a case-specific output figure exists. It provides spatial context but cannot substitute for the scalar containment and end-time metrics.

7 Calculated metrics and verification decision

Simulation condition / metric	Expected	Calculated	Status
extended_attack / time at simulation end	88200 s	249984 s	FAIL
initial_attack / fraction contained	0.54 -	0.5885 -	PASS

The displayed decision is **FAIL**. Every reported value and component decision comes from the existing calculated metrics. Missing required evidence yields NOT EVALUABLE, without inferred or invented values.

8 Technical interpretation

The earlier case configuration used a 3600 s extended-attack interval, whereas the Guide area-growth verification prescribes 300 s; the source namelist has been corrected. That correction does not make the current-source result pass. With Fuel Model 8, the initial 1 s level-set change is below the absolute 0.001 stalled-front threshold in ELMFIRE level-set formulation. The stalled test is applied before adaptive-CFL integration has advanced beyond its startup steps, so this valid slow-spreading fire is classified as stalled essentially at ignition. Terminal bookkeeping then rounds the very small statistics stop time to 0.00 h. This remaining discrepancy is attributed to current ELMFIRE stall and termination logic,

not to another defensible case-configuration change. Increasing the initial timestep merely to evade the absolute threshold would change the prescribed Point setup and conceal the source behavior rather than verify it.

9 Scope and limitations

This case demonstrates only the two configured suppression pathways for uniform synthetic fuels, flat terrain, zero wind, one ignition location, one grid spacing, and the present executable/output conventions. It does not verify spatially explicit extended-attack model 1, indirect-attack geometry, nonuniform fuels, wind or slope interactions, resource limitations, resolution convergence, or operational suppression effectiveness. The 0.54 initial target is assessed statistically and can vary between reruns because seed randomization is enabled. The 88200 s extended target is a documented controlled-response reference rather than an analytical solution, so an intentional suppression-model change requires review of the reference calculation and tolerance. The evidence is implementation verification only and supplies no physical-validation claim.

Case 30: Multiple-ignition topology

1 Verification purpose

This case verifies that the standard ELMFIRE executable initializes, schedules, transports, and merges multiple point ignitions using only prescribed physical inputs and observable model outputs. It preserves the scientific intent of Adam Laird's dual-ignition, staggered spot-merging, and near-boundary component scenarios.

2 Mathematical formulation and numerical implementation

Each ignition introduces a local negative level-set region at position (x_i, y_i) and time t_i. ELMFIRE then advances ϕ according to $\partial_t \phi + V_n |\nabla \phi| = 0$. With homogeneous fuel, zero slope, and zero wind, simultaneous mirrored sources must remain reflection symmetric. Adding uniform wind must produce a measurably different arrival-time solution. Delayed sources must appear at their requested times, and clipping at a domain boundary must not suppress or wrap a source.

3 Derivation of expected behavior

The no-wind forcing and ignition pair are invariant under reflection about their mapped raster axes; therefore the discrete arrival-time solution has the same symmetries. Because ELMFIRE maps coordinates to one-based cells with a ceiling operation, the evaluator reflects the largest balanced row window about the actual ignition row rather than an intercell domain line. The wind simulation condition changes only the wind rasters, so a nonzero solution difference is a metamorphic check of weather coupling. For scheduled ignition i, the containing raster cell should receive an arrival time within one 1200 s of t_i. The eight boundary sources are placed one five-cell margin inside the square domain and all must initialize.

For any two arrival-time arrays A and B, the implemented comparison is

$$E(A,B) = \frac{\text{mean}_{i \in V} |A_i - B_i|}{\max[P_{90}(A_i : i \in V), 1 \text{ s}]}, \qquad V = \{i : A_i, B_i \text{ are finite}\}.$$

For the two simultaneous ignitions at $(-52.5, 0)$ and $(52.5, 0)$ m, the reflection reference is exact symmetry, $E = 0$; the metric is the larger of the left–right and centered top–bottom reflection errors and must satisfy $E \leq 0.02$. The 5 mph, 225-degree wind run is compared with the no-wind run by the same formula and must give $E \geq 0.05$, demonstrating a resolved forcing response. For scheduled ignitions (x, y, t) equal to $(0, 600, 0)$, $(0, 300, 1200)$, $(0, 0, 2400)$, $(0, -300, 3600)$, and $(0, -600, 4800)$, the timing metric uses the simulated time of arrival (TOA) at each ignition location:

$$E_t = \max_j |TOA(x_j, y_j) - t_j| \leq 1200 \text{ s}.$$

Finally, the count metric is $\sum_{j=1}^{8} \mathbf{1}[TOA(x_j, y_j) \leq 1200 \text{ s}] = 8$ at the eight generated edge/corner points $(\pm 975, 0)$, $(0, \pm 975)$, and $(\pm 975, \pm 975)$ m.

4 Verification metrics and acceptance criteria

Reflection error is the larger normalized mean absolute difference between the no-wind arrival field and its reflection about each mapped ignition axis; it must not exceed 0.02. Wind response is the normalized mean absolute difference between the wind and no-wind fields and must be at least 0.05. The maximum scheduled-source timing error must not exceed one output interval. All eight boundary-source cells must have finite arrival times no later than one output interval. These metrics were selected because they test invariants of the coupled ignition/level-set implementation without reproducing ELMFIRE's internal algorithms.

5 Simulation configuration and predicted outcome

All simulation conditions use a 5 m square grid in EPSG:32610, Fuel Model 10, 3/5/6 percent dead-fuel moisture, and flat terrain. The simulation conditions contain two simultaneous no-wind sources, the same pair under 5 mph wind from 225 degrees, five staggered sources, and eight edge/corner sources. In the staggered simulation condition the first source is anchored at (t=0) and the other four follow at 1200-s intervals. This common translation of the original absolute schedule preserves every relative delay while preventing versions that have no active front from advancing past a strictly future first ignition. The predicted fields are respectively symmetric merger, wind-distorted merger, correctly timed merger, and eight clipped but valid boundary fires.

Parameter	Configured value	Units	Role
Domain/grid	2000 m square; 5	m/cell	Resolves source placement and merging
Fuel/moisture	FM10; 3/5/6	— / percent	Homogeneous surface response
Time control	step 1, maximum 5; stop 14400	s	Exact timestep multiple for scheduled sources
Simulation conditions	dual calm, dual wind, staggered, boundary	—	Exercises symmetry, weather, timing, and clipping
Output cadence	1200	s	Timing-error acceptance scale

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

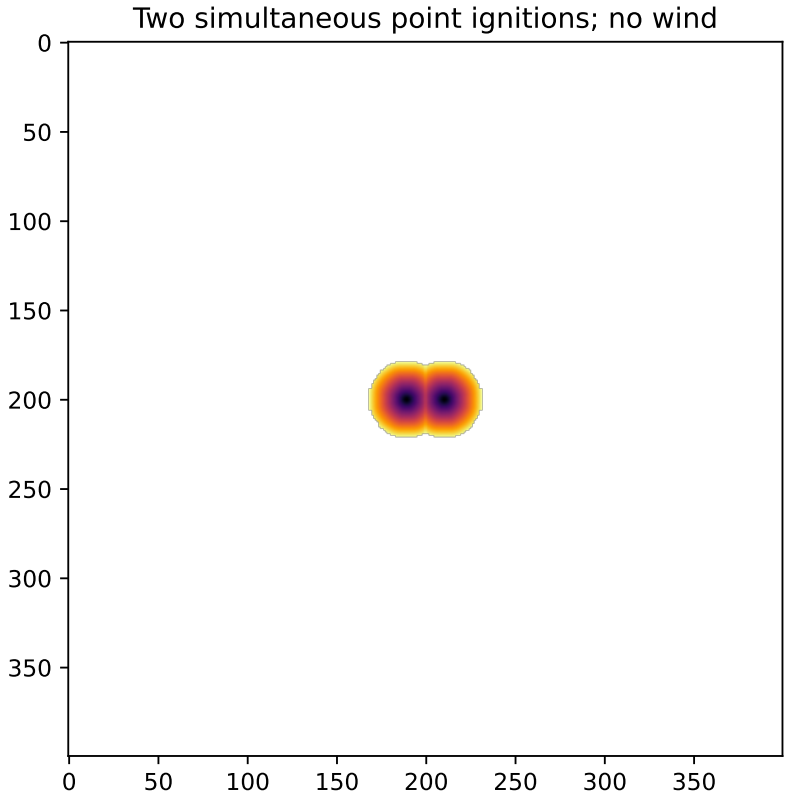


Figure 1: Prescribed initial level-set geometry for two simultaneous ignitions without wind. Coordinates are grid-cell indices on the 5 m grid; the field is dimensionless.

6 Actual simulation results

The generated result is **PASS**; 4 of 4 required metrics pass across 4 simulation conditions. Missing ELMFIRE output yields NOT EVALUABLE rather than a pass.

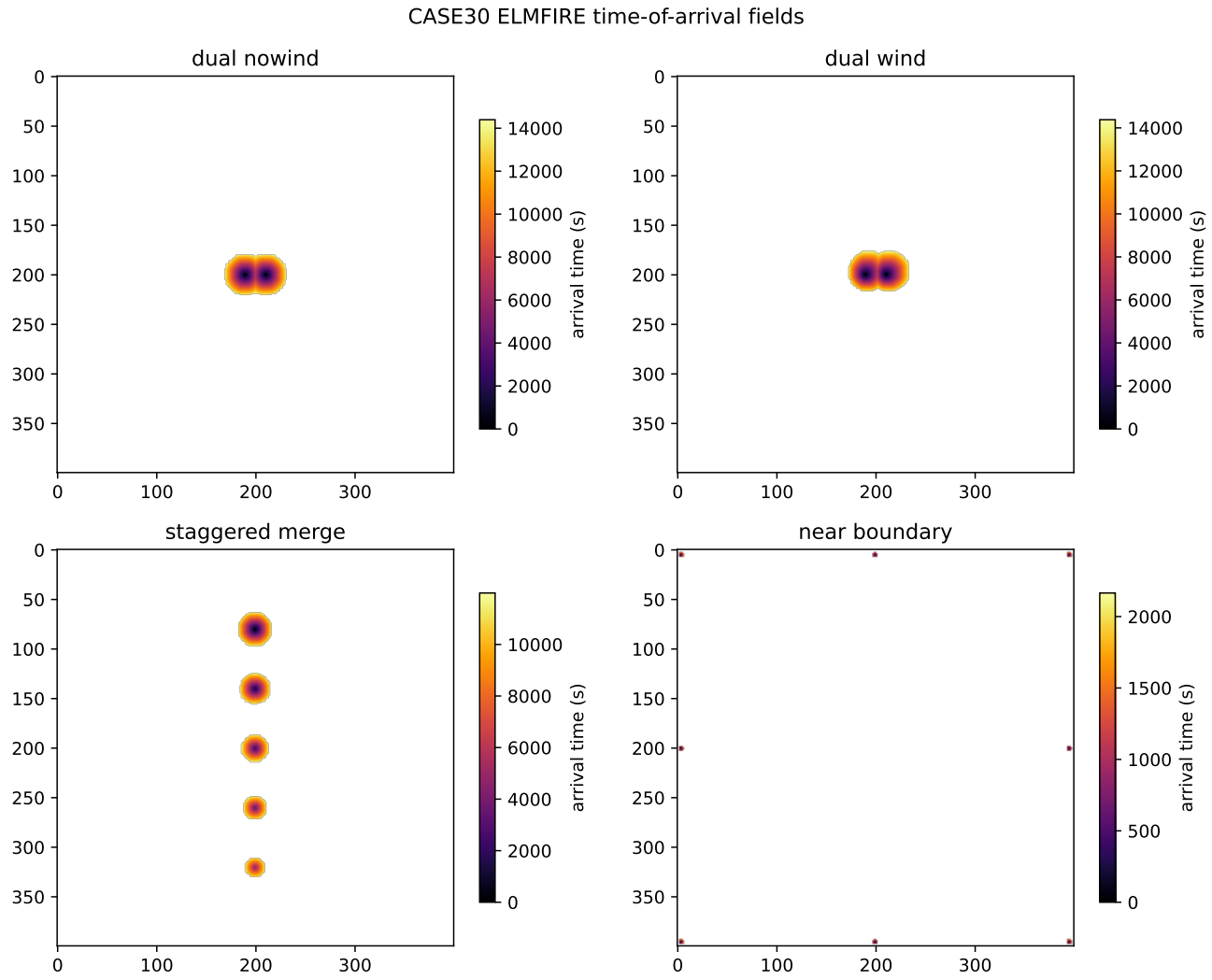


Figure 2: Case-specific time-of-arrival fields for the four ignition-topology simulation conditions.

7 Calculated metrics and verification decision

Metric	Expected	Calculated	Status
dual-ignition reflection error	$\leq$ 0.02 (2% normalized L1)	0.000472284	PASS
wind-forcing response	$\geq$ 0.05 (5% normalized L1)	0.0937636	PASS
maximum scheduled-ignition timing error	$\leq$ 1200 (one output interval)	0 s	PASS
boundary ignitions initialized	8 (all eight points)	8 count	PASS

The authoritative decision is **PASS** from calculated verification and validation metrics.

8 Technical interpretation

A symmetry failure points to directional indexing or level-set transport. No wind response indicates that the weather field did not reach fire behavior. Timing error implicates the scheduled-ignition path. Missing edge ignitions indicates boundary-index clipping or coordinate conversion failure.

9 Scope and limitations

This is deterministic implementation verification, not fire-behavior validation. It does not expose the internal Fortran ignition calculations directly; it checks their observable behavior through an unmodified ELMFIRE executable.

Case 31: Planar-front transport

1 Verification purpose

This case verifies planar level-set transport across diagonal geometry, a discontinuous fuel field, uniform wind, and a nonburnable fuel break. It consolidates Adam Laird's diagonal-front, fuel-jump, fuel-jump-with-wind, and fuel-break scenarios into one verification case based on observable model responses.

2 Mathematical formulation and numerical implementation

For a planar front with signed-distance coordinate s, the level-set equation reduces to $\partial_t \phi + V_n \partial_s \phi = 0$. Uniform forcing should preserve planarity. At a fuel boundary V_n changes but the arrival-time field remains nearly constant along lines parallel to the front. A nonburnable strip imposes $V_n = 0$ except at an explicit gap.

3 Derivation of expected behavior

A uniform diagonal front is translationally invariant along its tangent. The fuel-jump solution is piecewise linear in normal distance, $T(s) = \int ds/V_n(s)$, so downstream time must increase while cross-front variation remains small. A 1 mph wind from 270 degrees aids eastward transport and must reduce downstream arrival time relative to the otherwise identical no-wind simulation condition. A continuous nonburnable strip blocks transport; opening a gap creates a strictly larger burned fraction downstream.

At every declared station, arrival times are restricted to the central $|y| \leq 350$ m band (or the equivalent diagonal tangent band). The local planarity measure is

$$CV_j = \frac{\text{std}(T_j)}{\max[\text{mean}(T_j), 1\ \text{s}]}, \qquad CV_{\text{max}} = \max_j CV_j.$$

For the diagonal front, j spans nine normal-coordinate stations from 25 to 275 m; for the fuel jump it spans $x = \{-50, -20, 10, 40, 70, 100\}$ m. Both substituted tests are $CV_{\text{max}} \leq 0.03$, with at least 90 percent finite station coverage. Let $\widetilde{T}(x)$ be the station median. The transition metric is

$$\Delta T = \widetilde{T}(100) - \widetilde{T}(-50) > 0,$$

and the wind response is

$$q_w = \frac{\widetilde{T}_{\text{wind}}(100)}{\widetilde{T}_{\text{no wind}}(100)} \leq 0.95.$$

For the continuous and gapped Fuel Model 91 breaks, define $f(x) = N_{\text{finite}}/N_{\text{sampled}}$. After the required 90 percent upstream/barrier/opening coverage checks, the substituted barrier metric is $G = f_{\text{gapped}}(100) - f_{\text{continuous}}(100) \geq 0.10$.

4 Verification metrics and acceptance criteria

Cross-front coefficients of variation for diagonal and fuel-jump fronts must each be at most 3 percent. The nine diagonal-normal stations are 25, 56.25, 87.5, 118.75, 150, 181.25, 212.5, 243.75, and 275 m

beyond the initial leading face. The six vertical-front stations are $x = -50, -20, 10, 40, 70$, and 100 m; each is sampled at the nearest raster column. Every station must have at least 90 percent finite, positive arrival-time coverage in the central $|y| \leq 350$ m corridor (or the corresponding diagonal tangent corridor). The evaluator reports the maximum station coefficient of variation, so one poorly resolved station cannot be hidden by aggregation. The downstream fuel-transition arrival-time increment between $x = -50$ m and $x = 100$ m must be positive. The wind/no-wind arrival-time ratio at $x = 100$ m must be at most 0.95. The burned-fraction gain at the same downstream station produced by the gap must be at least 0.10. Before that absence can be classified as a failure, both break simulation conditions must have at least 90 percent arrival coverage at $x = -50$ m, the continuous case must also reach the near-barrier station at $x = -5$ m, the gapped case must have at least 90 percent coverage through the opening at $x = 35$ m within $|y| \leq 40$ m, and its final dump must reach or exceed the configured stop time. Otherwise the barrier metric is NOT EVALUABLE. These geometry and metamorphic metrics avoid calibrating against ELMFIRE itself and remain valid across executable versions.

5 Simulation configuration and predicted outcome

All five simulation conditions use a 1500×1500 m domain at 5 m, Fuel Models 10 and 3 where applicable, and 3/5/6 percent dead-fuel moisture. The input level-set field is a finite-strip signed-distance field clipped to ELMFIRE's supported $[-1, 1]$ interval. It retains an ignited interior and a resolved planar leading-face gradient without relying on a grid cell whose value is exactly zero; its ends remain inside the domain. ELMFIRE applies adaptive Courant–Friedrichs–Lewy (CFL) control only after the first five steps and declares a front stalled when the step-to-step PHI change stays below 0.001. The initial and maximum timestep are therefore both 5 s: for the slowest no-wind Fuel Model 10 state ($R \simeq 0.426$ m/min), one step moves the front about 0.0071 cell, above the stall comparison and far below the target CFL of 0.2. The stop time is 43,200 s, long enough for that slowest front to traverse the 275 m needed by the farthest diagonal station and the 195 m needed by the $x = 100$ m comparison station with margin. The break simulation conditions use Fuel Model 91 for a 30 m nonburnable strip, with or without a 100 m opening.

Parameter	Configured value	Units	Role
Domain/grid	1500 m square; 5	m/cell	Resolves planar fronts and fuel break
Fuel/moisture	FM10/FM3/FM91; 3/5/6	— / percent	Uniform, transition, and nonburnable states
Time control	step 5; stop 43200	s	Exact multiple with traversal margin
Wind	0 or 1 from 270	mph / degrees	Metamorphic acceleration pair
Gap	closed or 100	m	Connectivity response

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

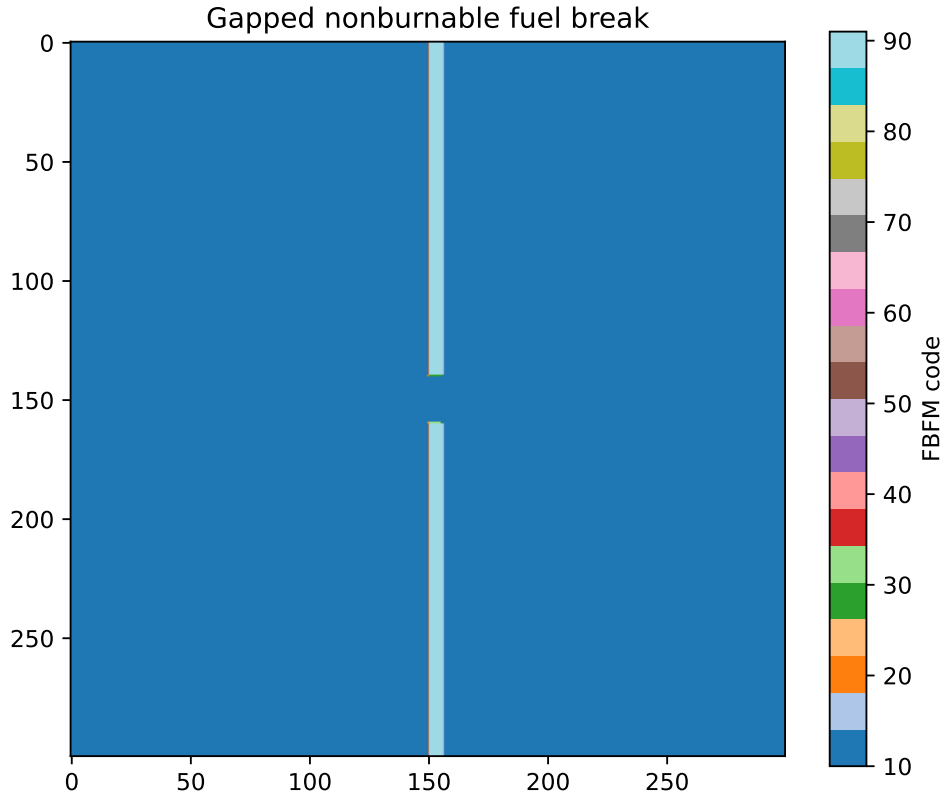


Figure 1: Prescribed fuel-model classification with an opening in the nonburnable strip. Coordinates are grid-cell indices on the 5 m grid, and the color scale identifies the fuel-model codes.

6 Actual simulation results

The generated status is **PASS**; 5 of 5 metrics pass across 5 simulation conditions. Missing required rasters, an obsolete or geometrically inconsistent level-set field, a final-dump mismatch, or inadequate arrival coverage produces NOT EVALUABLE. The generated diagnostic is: –.

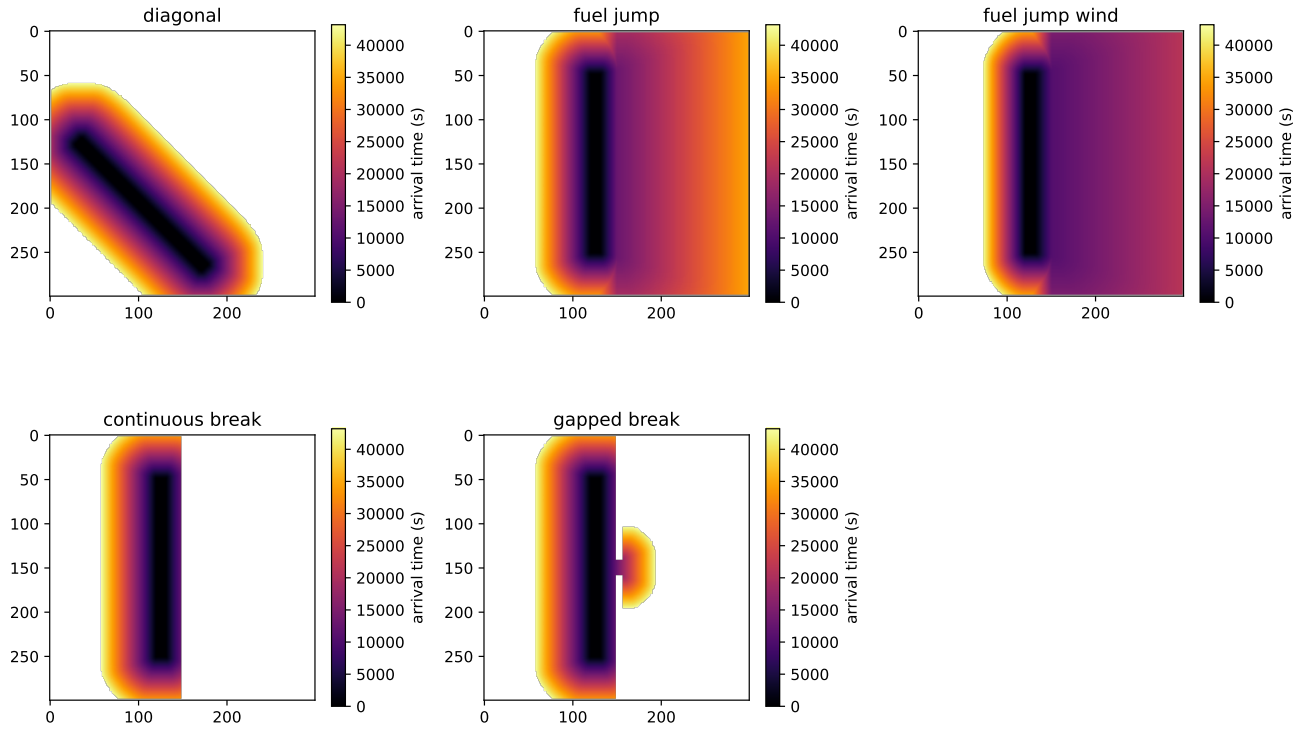


Figure 2: Time-of-arrival fields for all planar-front simulation conditions.

7 Calculated metrics and verification decision

Metric	Expected	Calculated	Status
diagonal-front cross-front variation	≤ 0.03 (3% coefficient of variation)	8.35166e-05	PASS
fuel-jump front planarity	≤ 0.03 (3% coefficient of variation)	7.71843e-05	PASS
fuel-transition arrival-time increment	> 0 (positive downstream travel time)	9920 s	PASS
eastward-wind arrival-time ratio	≤ 0.95 (at least 5% earlier than no-wind)	0.756028	PASS
gapped-break downstream burned-fraction gain	≥ 0.10 (10 percentage points)	0.657143	PASS

The authoritative decision is **PASS** from calculated verification and validation metrics.

8 Technical interpretation

Loss of planarity suggests orientation-dependent transport or indexing. A nonpositive transition time indicates an invalid arrival field. Lack of wind acceleration indicates a weather/fire-behavior coupling problem. Lack of gap response indicates incorrect nonburnable-fuel or front-connectivity handling.

9 Scope and limitations

The case verifies observable coupled behavior rather than individual internal Fortran routines. It does not assert absolute Rothermel rates, and the selected fuel and wind states are not field validation data.

Case 32: Raster-orientation covariance

1 Verification purpose

This case verifies that ELMFIRE's raster indexing and level-set transport are covariant under a 90° rotation. It is the portable successor to Adam Laird's orientation of the prescribed input fields scenario.

2 Mathematical formulation and numerical implementation

For homogeneous isotropic forcing, the operator $\mathcal{L}$ in $\partial_t \phi = \mathcal{L}(\phi)$ commutes with any rigid rotation Q: $\mathcal{L}(\phi \circ Q^{-1}) = (\mathcal{L}\phi) \circ Q^{-1}$. A square north-up raster rotated by exactly 90 degrees requires no interpolation, so cell-by-cell comparison is available.

3 Derivation of expected behavior

The rotated simulation condition is generated with a rotation through a multiple of 90 degrees from the asymmetric L-shaped initial negative-ϕ region. Fuel, moisture, slope, and wind are spatially uniform. Therefore the rotated arrival-time validity mask and values must equal a 90-degree rotation of the original result, apart from bounded grid-level roundoff.

The reference array is generated directly from the original ELMFIRE arrival-time result by a 90-degree counterclockwise rotation, $T_{\text{ref}} = \text{rot90}(T_{\text{original}})$; no arrival values are invented. With $V = \{i : T_{\text{ref},i}$ and $T_{\text{rot},i}$ finite$\}$ and $U = \{i :$ either value is finite$\}$, finite-mask agreement is

$$A_m = \frac{|V|}{|U|},$$

whose exact rotational reference is 1 and whose substituted test is $A_m \geq 0.99$. On the mutual finite mask, the value metric is

$$E_T = \frac{|V|^{-1} \sum_{i \in V} |T_{\text{ref},i} - T_{\text{rot},i}|}{\max[P_{90}(T_{\text{ref},i} : i \in V), 1\ \text{s}]},$$

with exact reference zero and acceptance substitution $E_T \leq 0.02$. The asymmetric generated ignition mask makes this comparison sensitive to an erroneous reflection or transposition.

4 Verification metrics and acceptance criteria

Finite-mask intersection over union must be at least 0.99. On mutually finite cells, normalized mean absolute arrival-time error must be at most 0.02. Mask agreement detects row/column reversal and georeferencing errors; value error detects orientation-dependent numerical evolution.

5 Simulation configuration and predicted outcome

Both simulation conditions use a 2000 m square domain at 5 m, Fuel Model 10, flat terrain, zero wind, and identical timing. The only changed scientific input is the exact raster rotation of level-set field. The second result is predicted to be the corresponding rotation of the first.

Parameter	Configured value	Units	Role
Domain/grid	2000 m square; 5	m/cell	Exact square raster rotation
Fuel/forcing	FM10; flat; calm	—	Isotropic homogeneous operator
Time control	step 1, maximum 5; stop 7200	s	Common exact comparison time
Initial field	asymmetric L shape	—	Makes orientation errors observable
Simulation condition transform	90-degree rotation	degrees	Exact no-interpolation covariance test

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

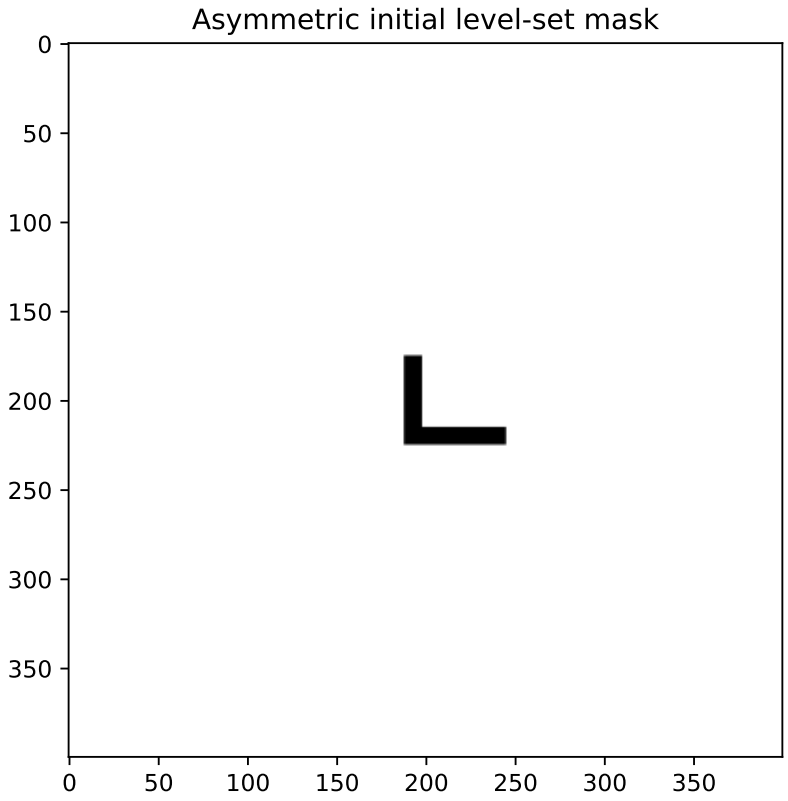


Figure 1: Prescribed asymmetric initial level-set geometry. Coordinates are grid-cell indices on the 5 m grid; the dark region identifies the initial ignition geometry.

6 Actual simulation results

The generated status is **PASS**; 2 of 2 metrics pass across 2 simulation conditions. Missing output produces NOT EVALUABLE.

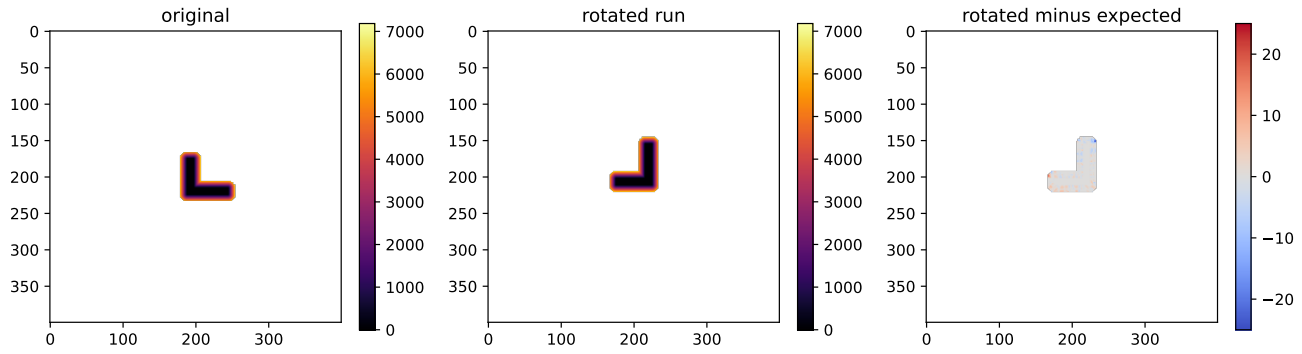


Figure 2: Original, independently rotated, and difference arrival-time fields.

7 Calculated metrics and verification decision

Metric	Expected	Calculated	Status
rotated finite-mask agreement	>= 0.99 (1% disagreement)	1 fraction	PASS
rotated arrival-time error	<= 0.02 (2%)	0.000214081 normalized L1	PASS

The authoritative decision is **PASS** from calculated verification and validation metrics.

8 Technical interpretation

Mask disagreement indicates inconsistent raster orientation, bounds, or row/column indexing. Value disagreement with matching masks indicates a directional bias in transport or output serialization.

9 Scope and limitations

The 90-degree test avoids resampling and therefore does not assess arbitrary-angle reprojection. It verifies executable behavior, not the internal bilinear-interpolation routine in isolation.

Case 33: Slope-driven directional spread

1 Verification purpose

This case verifies the observable coupling of topographic slope into directional surface-fire spread through the standard ELMFIRE executable. It is the portable successor to Adam Laird's no-wind-with-slope component scenario and retains a flat control.

2 Mathematical formulation and numerical implementation

ELMFIRE combines the no-wind spread rate with a slope factor before supplying a direction-dependent normal speed $V_n(\theta)$ to $\partial_t \phi + V_n |\nabla \phi| = 0$. With zero wind and zero slope, rotational symmetry is expected. A uniform nonzero slope breaks that symmetry along the fall-line axis.

3 Derivation of expected behavior

The flat control has equal forcing in all azimuths and therefore equal horizontal and vertical extents. The 30° simulation condition changes only slope and aspect; it must differ from the control and its final footprint must be elongated. These are metamorphic consequences of the governing coupling and do not require duplication of ELMFIRE's internal slope formula.

For each finite arrival-time mask, the pixel extents are $W = c_{\max} - c_{\min} + 1$ and $H = r_{\max} - r_{\min} + 1$. The flat-control anisotropy is

$$A_f = \frac{|W_f - H_f|}{\max(W_f, H_f)},$$

whose isotropic reference is zero and whose substituted test is $A_f \leq 0.05$. For the generated uniform 30-degree slope, the directional-shape metric is

$$L_s = \frac{\max(W_s, H_s)}{\min(W_s, H_s)} \geq 1.10.$$

The independent forcing-response metric uses the common finite mask V:

$$E_s = \frac{|V|^{-1} \sum_{i \in V} |T_{s,i} - T_{f,i}|}{\max[P_{90}(T_{s,i} : i \in V), 1\ \text{s}]} \geq 0.05.$$

Thus the three reference statements are, respectively, zero flat anisotropy, at least 10 percent slope elongation, and at least 5 percent normalized difference from the flat run.

4 Verification metrics and acceptance criteria

Flat-control extent anisotropy must not exceed 0.05. The slope footprint's major/minor extent ratio must be at least 1.10. Normalized mean absolute arrival-time difference between slope and flat simulation conditions must be at least 0.05. The first metric detects grid bias, while the latter two show that slope is both active and directional.

5 Simulation configuration and predicted outcome

Both simulation conditions use a 2000 m square domain at 5 m, Fuel Model 10, 3/5/6 percent dead-fuel moisture, zero wind, and a central point ignition. The slope simulation condition uses 30 degrees and aspect 270 degrees; the control uses zero slope. The slope result is predicted to be anisotropic and measurably different from the circular control.

Parameter	Configured value	Units	Role
Domain/grid	2000 m square; 5	m/cell	Resolves directional footprint
Fuel/moisture	FM10; 3/5/6	— / percent	Fixed surface response
Time control	step 1, maximum 5; stop 7200	s	Common exact comparison time
Control	slope 0; wind 0	degrees / mph	Rotational baseline
Test	slope 30; aspect 270	degrees	Activates directional slope coupling

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

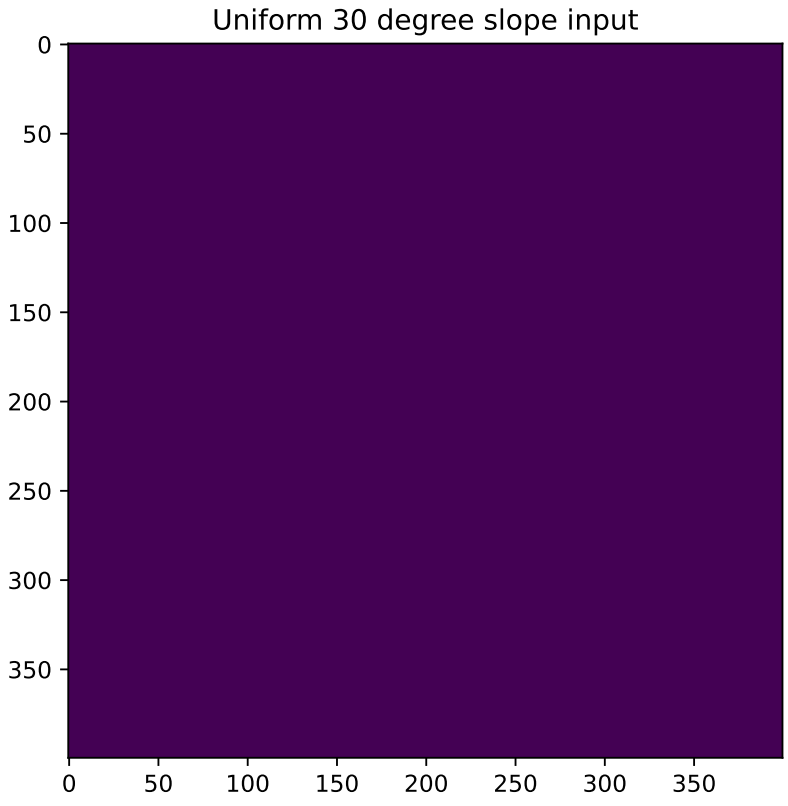


Figure 1: Prescribed uniform 30-degree terrain slope. Coordinates are grid-cell indices on the 5 m grid.

6 Actual simulation results

The generated status is **PASS**; 3 of 3 metrics pass across 2 simulation conditions. Missing outputs produce NOT EVALUABLE.

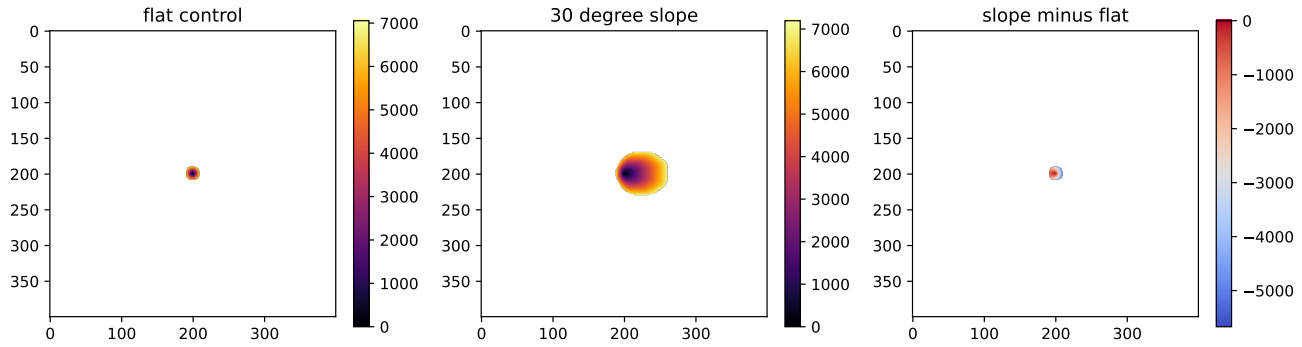


Figure 2: Flat-control, slope, and difference arrival-time fields.

7 Calculated metrics and verification decision

Metric	Expected	Calculated	Status
flat-control anisotropy	<= 0.05 (5%)	0 fraction	PASS
slope-front elongation ratio	>= 1.10 (10% directional elongation)	1.1746 ratio	PASS
slope-forcing response	>= 0.05 (5% difference from flat)	0.520518 normalized L1	PASS

The authoritative decision is **PASS** from calculated verification and validation metrics.

8 Technical interpretation

An anisotropic control indicates grid-orientation bias. A near-unity slope footprint or negligible field difference indicates that slope/aspect inputs were not consumed or not coupled to spread. Unexpected rotation should prompt inspection of aspect convention and raster orientation.

9 Scope and limitations

This case verifies directional response, not the absolute value of the slope coefficient and not agreement with an experimental fire. Existing CASE22 and CASE24 exercise more complex terrain interactions; this case isolates the slope-only executable response.

Case 34: Standard fuel-model Rothermel sweep

1 Verification purpose

This controlled known-response case verifies that ELMFIRE reads standard fuel-model parameters and evaluates the no-wind, no-slope Rothermel surface spread model for FBFM2, FBFM9, GR2, TU5, and TL8. It exercises fuel-table parsing, class weighting, reaction intensity, heat sink, and the base rate of spread (ROS) while crown fire, spotting, buildings, acceleration, wind, and slope are excluded. It can expose column-order, unit, weighting, and model-selection errors. This is implementation verification, not comparison with an observed fire. The verification question is whether all five simulation conditions reproduce the independent reaction-intensity and base-ROS calculations within 1% and 2%, with time of arrival (TOA)-derived ROS within 5%.

2 Mathematical formulation and numerical implementation

The no-wind, no-slope rate is

$$R_0 = \frac{I_R \xi}{\rho_b \epsilon Q_{ig}}, \tag{1}$$

where I_R is reaction intensity, ξ is the propagating-flux ratio, ρ_b is bulk density, ϵ is effective heating number, and $Q_{ig} = 250 + 1116 M_f$ in native Rothermel units. Dead and live classes are area-weighted before their moisture-damping contributions are combined. The equations and standard fuel definitions follow Patricia L. Andrews, *The Rothermel Surface Fire Spread Model and Associated Developments*, RMRS-GTR-371 (2018). ELMFIRE returns spread rate in metres per minute and reaction intensity in kW m^{-2}.

3 Derivation of expected behavior

The case-specific independent Rothermel reference calculation reads the same frozen fuel-table rows but independently evaluates every algebraic step stated above. It does not import ELMFIRE source or use a simulation output as a reference. Each predicted I_R and R_0 is therefore a deterministic known response. A bounded planar signed-distance front turns the scalar normal speed into a second independent prediction, $t(x) = t_0 + (x - x_0)/(R_0/60)$.

Each reference is calculated independently with

$$R_0 = \frac{I_R \xi}{\rho_b \epsilon Q_{ig}}, \qquad Q_{ig} = 250 + 1116 M_f,$$

using the generated common moisture inputs and the named standard-fuel record. The resulting substitutions are

fuel model	R_{ref} (m min^{-1})	$I_{R,\text{ref}}$ (kW m^{-2})
2	0.823796	675.184
9	0.306627	477.714
102	0.228507	186.041
165	0.325722	1595.730
188	0.275244	646.706

(for example, Fuel Model 2 gives 2.702741 ft min$^{-1}(0.3048) = 0.823796$ m min^{-1}). For simulation condition i, direct errors are $e_{R,i} = |R_{\text{max},i} - R_{\text{ref},i}|/\max(|R_{\text{ref},i}|, 10^{-8})$ and the analogous $e_{I,i}$. The

TOA fit is $t = a + bx$, so $R_{\mathrm{TOA}} = 60/b$ and $R^2 = 1 - \sum(t - \hat{t})^2 / \sum(t - \bar{t})^2$. The reported substitutions are $\max_i e_{R,i} \leq 0.02$, $\max_i e_{I,i} \leq 0.01$, $\max_i |R_{\mathrm{TOA},i} - R_{\mathrm{ref},i}|/R_{\mathrm{ref},i} \leq 0.05$, and $\min_i R_i^2 \geq 0.995$.

4 Verification metrics and acceptance criteria

The direct spread raster tests the calculated speed with maximum relative error 0.02; the reaction-intensity raster tests the source term with maximum relative error 0.01. Their medians use colocated valid cells at the planar head, beginning one cell behind the initial east face, so direct verification does not require the front to traverse a downstream corridor. Linear regression of positive arrival time against interior x coordinate tests propagation independently: maximum ROS error is 0.05 and minimum R^2 is 0.995. TOA is required only for simulation conditions whose planned duration supports a multi-cell fit. All five simulation conditions and all four aggregate metrics are required; missing or malformed required evidence gives NOT EVALUABLE.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE's supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive reference spread rate, the prescribed and maximum timesteps use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured Courant–Friedrichs–Lewy (CFL) limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE's stalled-front branch and corrupt the terminal timestamp. A zero-ROS simulation condition retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. It accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required simulation condition must reach the prescribed stop time in its generated namelist; a direct-only simulation condition may terminate early because lack of propagation is part of the state being tested. The domain is 1400 m square with 10 m cells. A finite 50 m-wide ignition strip has a planar east face initially at $x = -400$ m and terminates well inside the north and south boundaries; the interior analysis corridor is therefore one-dimensional without presenting ELMFIRE with an edge-to-edge contained front. Moistures are 5, 7, and 9% for dead classes and 90% for both live classes. Wind and slope are zero. Run duration is selected from the predicted rate to provide approximately 300 m of travel. The interior results should be planar but need not have a monotonic ordering across fuel models because packing, surface-area-to-volume ratio (SAV), load, depth, and extinction moisture interact.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar interior propagation
Fuel sweep	FBFM2, FBFM9, GR2, TU5, and TL8	code/name	Independent variable
Moisture	dead 5/7/9; live 90/90	percent	Fixed damping state
Wind/slope	0 / 0	mph / degrees	Isolates base fuel response
Timing	generated whole-second step and exact-multiple stop	s	CFL-safe terminal evidence

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

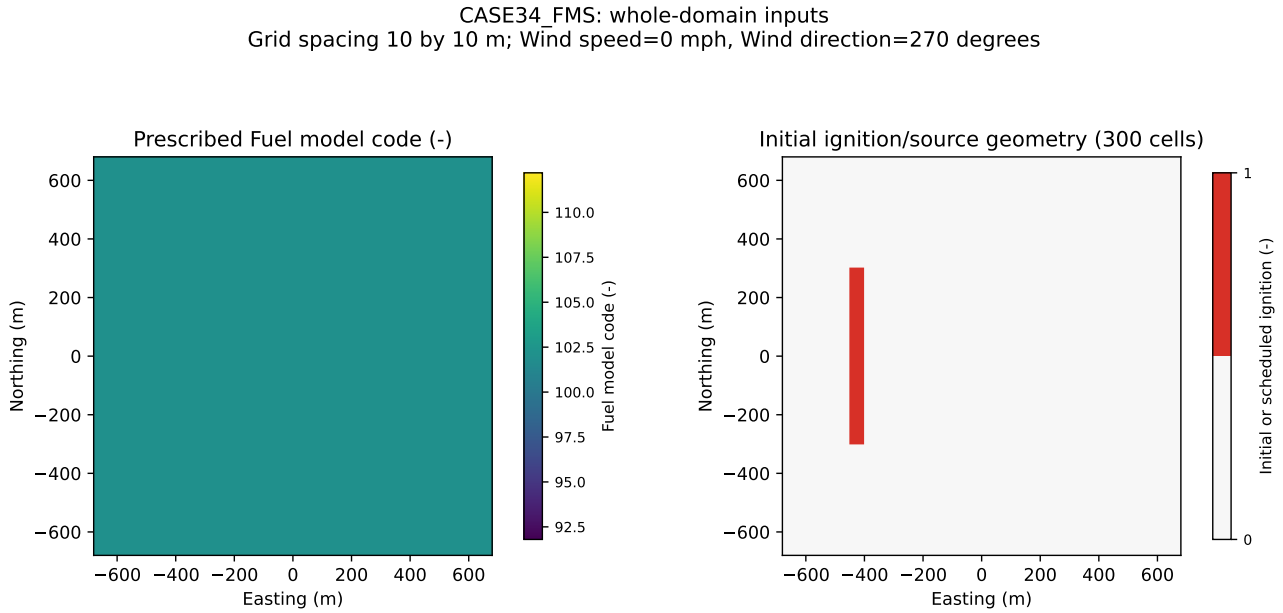


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The available results cover 5 of 5 required simulation conditions. The scientific decision is **PASS**. No plotted curve is substituted for missing simulation results.

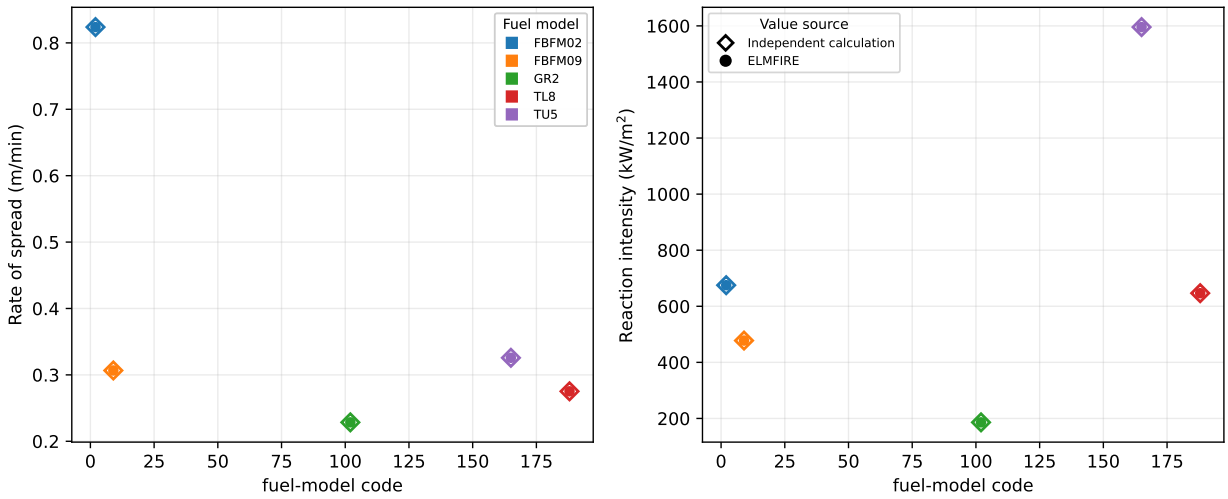


Figure 2: Independently calculated and ELMFIRE rate of spread (ROS, m/min) and reaction intensity (kW/m2) for all five completed standard-fuel simulation conditions. Each fuel model uses one color in both panels; open diamonds denote the independent hand calculation and filled circles denote ELMFIRE.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.02	2.57611e-07 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	2.93279e-07 fraction	PASS
maximum TOA-ROS relative error	≤ 0.05	0.0115779 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999961 dimensionless	PASS

The overall decision is **PASS**; a successful executable exit alone is not a pass.

8 Technical interpretation

An isolated intensity discrepancy implicates fuel parsing, class weighting, or moisture damping. Correct intensity with incorrect direct ROS implicates flux ratio, density, heat sink, adjustment, or output units. Correct direct ROS with a TOA discrepancy implicates level-set transport, raster orientation, or time units.

9 Scope and limitations

The case verifies five representative standard fuels at one moisture state. It does not validate Rothermel against experiments, exhaust all fuel models, or test wind, slope, curing, crown fire, or spatial heterogeneity; those mechanisms have separate cases.

Case 35: Midflame wind-speed Rothermel sweep

1 Verification purpose

This case verifies ELMFIRE's Rothermel wind response for FBFM2, FBFM9, GR2, TU5, and TL8 over 0, 1, 2, 4, 6, 8, and 12 mph midflame. It isolates the midflame wind factor while separately accounting for ELMFIRE's public 20-ft wind input and unsheltered wind-adjustment factor (WAF). It can expose errors in wind units, WAF inversion, exponentiation, fuel-dependent coefficients, or wind limiting. The question is whether every simulation condition agrees with the independent implementation-limited rate of spread (ROS) within 2%, reaction intensity within 1%, and time of arrival (TOA)-derived ROS within 5%.

2 Mathematical formulation and numerical implementation

Rothermel wind enhancement is

$$R = R_0(1 + \phi_w), \qquad \phi_w = CU^B(\beta/\beta_{op})^{-E}, \tag{1}$$

with U in ft min^{-1} at midflame. Coefficients are $C = 7.47e^{-0.133\sigma^{0.55}}$, $B = 0.02526\sigma^{0.54}$, and $E = 0.715e^{-0.000359\sigma}$. The reference is Andrews (2018), RMRS-GTR-371. The inspected ELMFIRE implementation limits U to $0.9I_R$ in native units. ELMFIRE inputs 20-ft wind in mph, so preprocessing divides each target midflame speed by the independently computed unsheltered WAF.

3 Derivation of expected behavior

Below the limit, $(R/R_0 - 1)$ follows the fuel-specific power law in U; once the limit is reached, the wind contribution is constant. The case-specific reference calculation computes R_0, I_R, WAF, ϕ_w, limit location, and final ROS from the paper equations and the documented ELMFIRE limiting policy. This separates paper-equation behavior from the implementation's explicit cap.

The wind series uses the Rothermel wind factor

$$\phi_w = C\,U_{mf}^B, \qquad R = 0.3048R_0^{(\text{ft/min})}(1 + \phi_w),$$

with U_{mf} in ft min^{-1} and fuel-specific B, C. For a concrete generated substitution, Fuel Model 2 at $U_{mf} = 4$ mph $= 352$ ft min^{-1} has $C = 0.0002125869$, $B = 1.8304427$, and

$$\phi_w = 0.0002125869(352)^{1.8304427} = 9.746235,$$

$$R_{\text{ref}} = 0.3048(2.702741)(1 + 9.746235) = 8.852700 \text{ m min}^{-1}, \qquad I_{R,\text{ref}} = 675.184 \text{ kW m}^{-2}.$$

The same substitution is applied to every generated fuel/wind pair ($0, 1, 2, 4, 6, 8, 12$ mph); the 20-ft raster value is $U_{20} = U_{mf}/WAF$, e.g. $4/0.362674 = 11.02919$ mph. For each simulation condition, $e_R = |R_{\text{max}} - R_{\text{ref}}|/\max(|R_{\text{ref}}|, 10^{-8})$ and e_I is analogous. A TOA regression $t = a + bx$ gives $R_{\text{TOA}} = 60/b$ and $R^2 = 1 - SSE/SST$. The aggregated tests substitute max $e_R \le 0.02$, max $e_I \le 0.01$, max $e_{\text{TOA}} \le 0.05$, and min $R^2 \ge 0.995$.

4 Verification metrics and acceptance criteria

Maximum relative errors are 0.02 for the final direct ROS, 0.01 for reaction intensity, and 0.05 for TOA-derived ROS. The minimum TOA regression R^2 is 0.995. Direct pointwise comparison is more diagnostic than monotonicity alone because it detects an incorrect exponent or premature plateau even when the curve still increases. All 35 fuel–wind combinations are required.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE's supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive reference spread rate, the prescribed and maximum timesteps use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured Courant–Friedrichs–Lewy (CFL) limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE's stalled-front branch and corrupt the terminal timestamp. A zero-ROS simulation condition retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. It accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required simulation condition must reach the prescribed stop time in its generated namelist; a direct-only simulation condition may terminate early because lack of propagation is part of the state being tested. Each homogeneous 1400 m square simulation condition uses 10 m cells and a finite ignition strip whose east face is planar at $x = -400$ m within the interior analysis corridor. The strip ends inside the lateral boundaries so it remains an active fire rather than an edge-to-edge contained front. Slope is zero, dead moisture is 5/7/9%, and live moisture is 90%. The wind is from the west. Crown fire, spotting, buildings, fluctuations, acceleration, and Monte Carlo variation are disabled. Curves should increase as U^B and may flatten at fuel-dependent limits.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar interior propagation
Fuel sweep	FBFM2, FBFM9, GR2, TU5, and TL8	code/name	Fuel-dependent response
Wind sweep	0, 1, 2, 4, 6, 8, and 12 mph midflame	mph	Independent midflame target
Moisture/slope	dead 5/7/9, live 90/90; 0	percent / degrees	Fixed damping and terrain
Timing	generated whole-second step and exact-multiple stop	s	CFL-safe terminal evidence

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

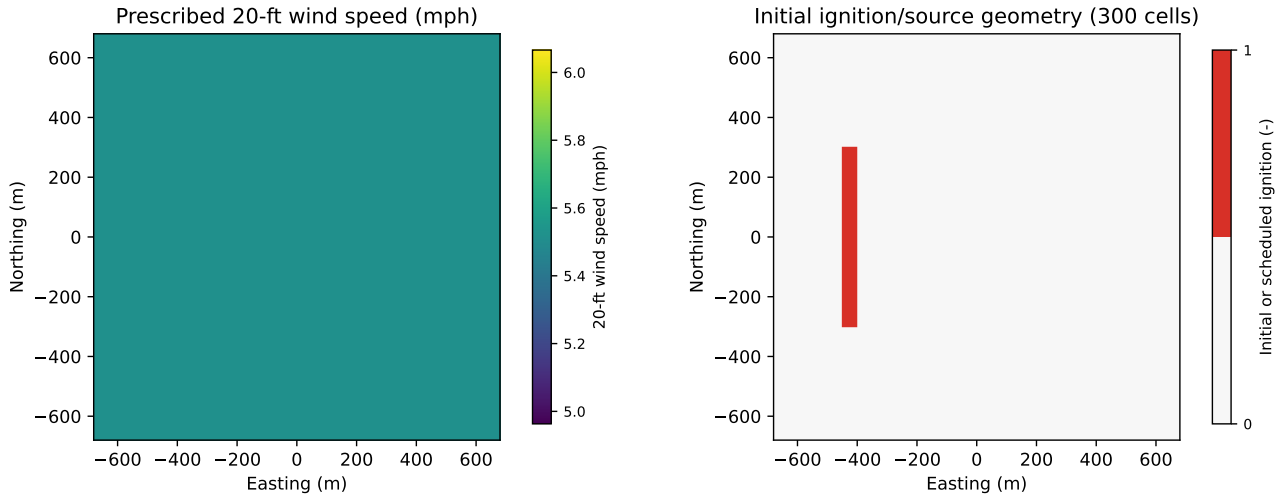


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The available results cover 35 of 35 required simulation conditions and reports **PASS**.

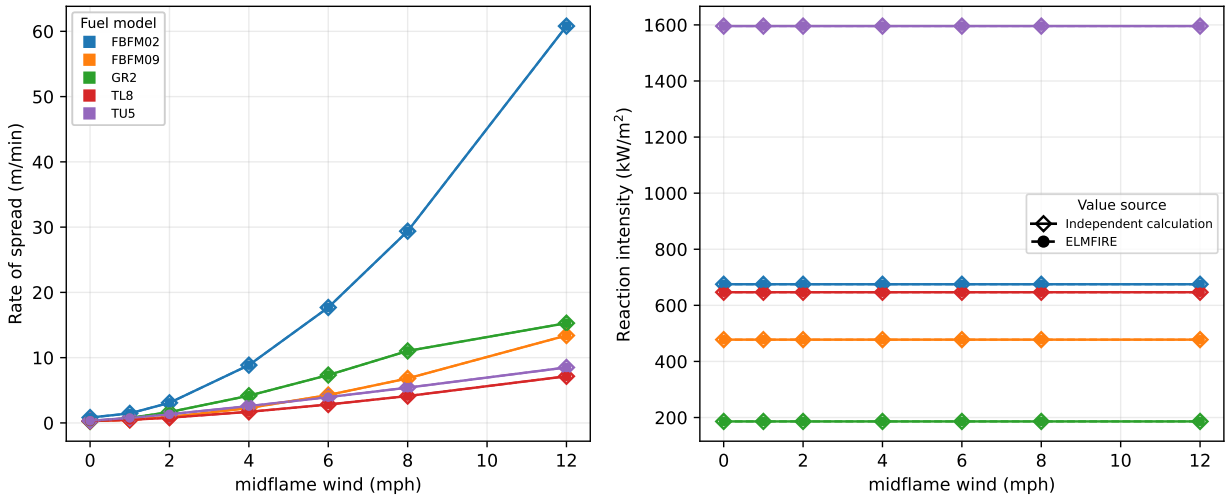


Figure 2: Independent calculations (open diamonds, solid lines) and ELMFIRE predictions (filled circles, dashed lines). Each fuel-model pair uses the same color. Target values are midflame wind; namelists contain the corresponding 20-ft wind.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.02	1.17774e-06 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	2.93279e-07 fraction	PASS
maximum TOA-ROS relative error	≤ 0.05	0.028404 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999958 dimensionless	PASS

The authoritative overall decision is **PASS**.

8 Technical interpretation

A common scale error across fuels suggests mph, ft/min, or WAF conversion. Fuel-dependent curvature errors suggest B, C, E, or packing-ratio handling. A wrong plateau indicates a changed or incorrect wind-limit policy. Correct direct ROS but incorrect TOA points to transport rather than the Rothermel calculation.

9 Scope and limitations

The case verifies steady, aligned wind and the currently selected original wind limit. If a future ELM-FIRE release intentionally adopts a corrected or optional limit, the scientific reference calculation must be reviewed rather than changing only the namelist. Canopy-dependent WAF is tested in CASE42.

Case 36: Slope-factor Rothermel sweep

1 Verification purpose

This case verifies the scalar Rothermel slope response for FBFM2, FBFM9, GR2, TU5, and TL8 over 0, 10, 20, 30, 50, 75, and 100% rise over run. It tests slope-raster units, conversion to $\tan^2\phi$, fuel-dependent packing response, and ELMFIRE's slope-factor cap without wind. The verification question is whether all 35 simulation conditions reproduce independently calculated intensity and slope-enhanced rate of spread (ROS) within the stated tolerances.

2 Mathematical formulation and numerical implementation

The no-wind response is

$$R = R_0(1 + \phi_s), \qquad \phi_s = 5.275\,\beta^{-0.3}\tan^2\phi. \tag{1}$$

Here reported slope percent is rise/run times 100 and the raster value is the angle $\phi = \tan^{-1}(s/100)$ in degrees. The formulation follows Andrews (2018), RMRS-GTR-371. The inspected ELMFIRE source additionally caps ϕ_s at the maximum wind factor derived from its wind limit; the reference calculation reproduces and reports that implementation policy.

3 Derivation of expected behavior

Before the cap, plotting ϕ_s against $\tan\phi$ on logarithmic axes has slope two. For each fuel and raster angle, the local reference calculation calculates R_0, the uncapped factor, the cap, and final R. Because the initial level set is planar and aligned with the upslope direction, the output normal speed equals this scalar head-spread prediction.

The slope-only reference uses

$$\phi_s = 5.275\,\beta^{-0.3}\tan^2\theta, \qquad R = R_0(1 + \phi_s).$$

As an explicit generated substitution, the requested 20 percent slope is stored as the integer raster angle $\theta = 11$ degrees ($100\tan 11^\circ = 19.4380$ percent). For Fuel Model 2, $\beta = 0.00575$, $R_0 = 0.8237955$ m min^{-1}, and

$$\phi_s = 5.275(0.00575)^{-0.3}\tan^2(11^\circ) = 0.936756,$$
$$R_{\text{ref}} = 0.8237955(1 + 0.936756) = 1.595491 \text{ m min}^{-1}.$$

Reaction intensity is unchanged by slope; here it is 675.184 kW m^{-2}. Every other generated fuel/slope pair uses the same equations with its stored integer angle and fuel packing ratio. Simulation condition errors are $e_R = |R_{\text{max}} - R_{\text{ref}}|/\max(|R_{\text{ref}}|, 10^{-8})$ and analogously e_I. From $t = a + bx_{\text{surface}}$, $R_{\text{TOA}} = 60/b$ and $R^2 = 1 - SSE/SST$. The aggregate substitutions are $\max e_R \le 0.02$, $\max e_I \le 0.01$, $\max e_{\text{TOA}} \le 0.05$, and $\min R^2 \ge 0.995$.

4 Verification metrics and acceptance criteria

Maximum relative errors are 0.02 for direct ROS, 0.01 for reaction intensity, and 0.05 for time of arrival (TOA)-derived ROS; TOA regression requires $R^2 \ge 0.995$. The direct and reference calculation ROS

values are slope-surface speeds. Accordingly, the TOA regression converts map easting to distance along the sloping surface using $x_s = x_0 + (x - x_0)/\cos\theta$ before fitting speed. Pointwise analytical errors are used instead of a trend-only test because degrees-versus-percent mistakes can preserve monotonicity. All simulation conditions and output fields must exist.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE's supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive reference spread rate, the prescribed and maximum timesteps use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured Courant–Friedrichs–Lewy (CFL) limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE's stalled-front branch and corrupt the terminal timestamp. A zero-ROS simulation condition retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. It accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required simulation condition must reach the prescribed stop time in its generated namelist; a direct-only simulation condition may terminate early because lack of propagation is part of the state being tested. Each 1400 m square domain uses 10 m cells and a finite ignition strip with a planar east face in the interior analysis corridor. The strip does not touch the lateral grid boundaries. Wind is zero, dead moisture is fixed at 5/7/9%, and live moisture is 90%. Aspect is chosen so upslope is aligned with propagation. Slope inputs are generated as explicit integer degrees after applying the positive-value equivalent of Fortran NINT; the expected response and slope-surface coordinate conversion use that same generated angle. ROS should rise quadratically in $\tan\phi$ before any source-defined plateau.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar upslope propagation
Fuel sweep	FBFM2, FBFM9, GR2, TU5, and TL8	code/name	Fuel-dependent packing response
Slope sweep	0, 10, 20, 30, 50, 75, and 100%	percent rise/run	Independent terrain target
Wind/moisture	calm; dead 5/7/9, live 90/90	mph / percent	Isolates slope factor
Timing	generated whole-second step and exact-multiple stop	s	CFL-safe terminal evidence

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

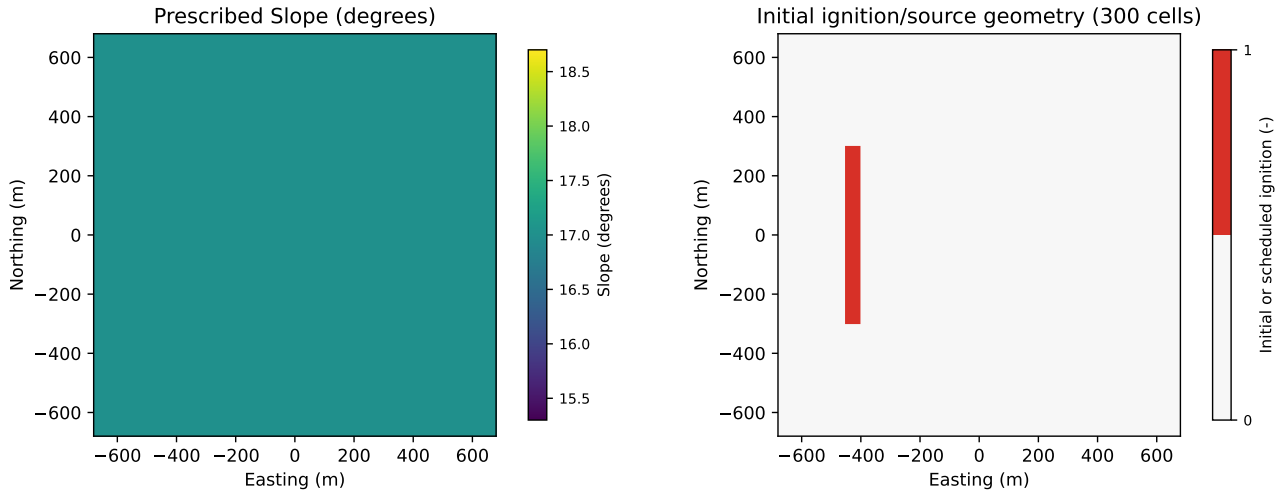


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The available results cover 35 of 35 simulation conditions and has state **PASS**.

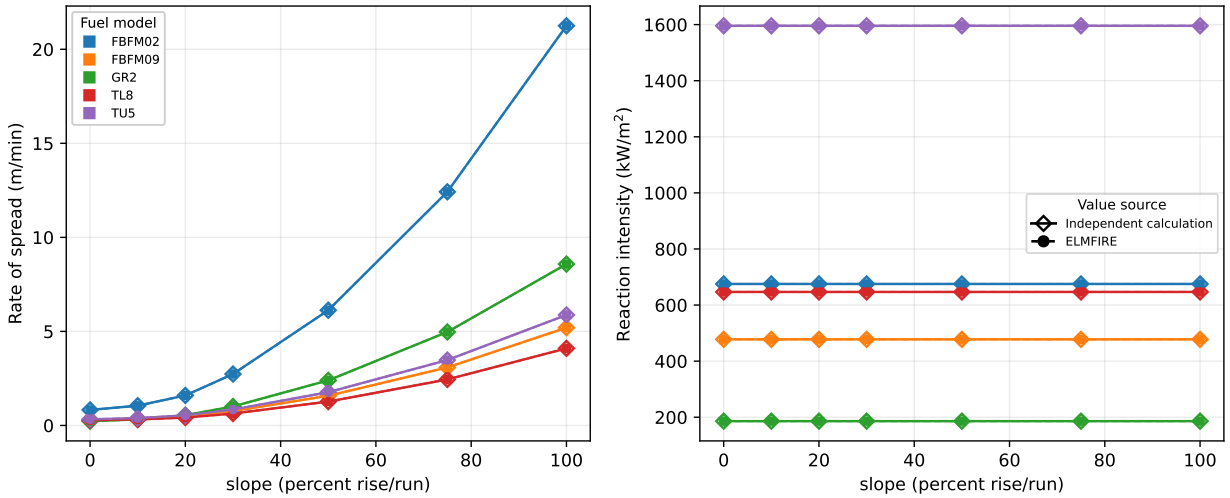


Figure 2: Independent calculations (open diamonds, solid lines) and ELMFIRE predictions (filled circles, dashed lines). Each fuel-model pair uses the same color.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.02	3.94136e-07 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	2.93279e-07 fraction	PASS
maximum TOA-ROS relative error	≤ 0.05	0.0265981 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999961 dimensionless	PASS

The overall decision is **PASS**.

8 Technical interpretation

Large errors growing with slope indicate a units or tangent error. Fuel-dependent errors implicate packing ratio or table selection. A premature or absent plateau indicates changed cap semantics. An accurate direct field but biased TOA indicates propagation or coordinate handling.

9 Scope and limitations

This test covers uniform positive slopes aligned with the front. It does not verify vector competition between wind and slope, complex terrain, or experimental fire behavior; CASE40 covers the directional coupling.

Case 37: Dead-fuel moisture and extinction sweep

1 Verification purpose

This case verifies dead-fuel moisture damping, heat of preignition, and extinction for the dead-dominated static fuels fire behavior fuel model (FBFM) 3 and TL6. Moisture is varied as a fraction of each model's configured dead extinction moisture. It can expose percent/fraction errors, incorrect polynomial evaluation, wrong extinction comparison, or omission of moisture from the heat sink. The verification question is whether reaction intensity and rate of spread (ROS) follow the independent calculation through extinction for every required simulation condition.

2 Mathematical formulation and numerical implementation

For $r_M = M_f/M_x$, Rothermel moisture damping is

$$\eta_M = \max\{0, \min[1, 1 - 2.59 r_M + 5.11 r_M^2 - 3.52 r_M^3]\}. \tag{1}$$

Moisture also increases $Q_{ig} = 250 + 1116 M_f$. Thus it reduces the reaction source and increases the heat sink. The equations follow Andrews (2018), RMRS-GTR-371. ELMFIRE receives percent rasters; the reference calculation uses fractions.

3 Derivation of expected behavior

For each fuel, the 1-h, 10-h, and 100-h moistures are equal, so their weighted dead moisture is the selected ratio times M_x. The damping polynomial and therefore dead reaction intensity reach zero at and above $r_M = 1$. Because these simulation conditions have no live load contribution, $I_R = R_0 = 0$ there. Below extinction, the independent fuel weighting and heat sink give the exact expected curve.

Dead-fuel moisture damping is evaluated from the generated ratio $r = M_d/M_x$:

$$\eta_M = \begin{cases} 1 - 2.59r + 5.11r^2 - 3.52r^3, & 0 \le r \le 1, \\ 0, & r > 1. \end{cases}$$

For Fuel Model 3 at $r = 0.50$, $\eta_M = 1 - 2.59(0.5) + 5.11(0.5)^2 - 3.52(0.5)^3 = 0.5425$. Substitution through the weighted reaction-intensity and Rothermel-rate equations gives $I_{R,\text{ref}} = 477.3889\ \text{kW m}^{-2}$ and $R_{\text{ref}} = 1.069455\ \text{m min}^{-1}$. At $r = 1.00$, the polynomial is zero, giving numerical references $I_R = 3.91 \times 10^{-13}$ and $R = 6.45 \times 10^{-16}$, both treated as zero; at $r = 1.10$ they are exactly zero by the piecewise definition. For nonzero references, $e_R = |R_{\text{max}} - R_{\text{ref}}|/\max(|R_{\text{ref}}|, 10^{-8})$ and e_I is analogous. The near-extinction intensity metric instead uses $\max |I_{R,\text{sim}} - I_{R,\text{ref}}| \le 0.001\ \text{kW m}^{-2}$. Propagating simulation conditions additionally use $R_{\text{TOA}} = 60/b$ from $t = a + bx$ and $R^2 = 1 - SSE/SST$, with $\max e_{\text{TOA}} \le 0.05$ and $\min R^2 \ge 0.995$; the direct limits are 0.02 for ROS and 0.01 for reaction intensity.

4 Verification metrics and acceptance criteria

Maximum relative errors are 0.02 for direct ROS and nonzero reaction intensity. Analytically extinguished intensity instead uses an absolute tolerance of $0.001\ \text{kW m}^{-2}$, so floating-point residuals are not divided by a near-zero reference. Simulation conditions predicted to spread at least $0.10\ \text{m min}^{-1}$

additionally require time of arrival (TOA)-derived ROS error at most 0.05 and regression $R^2 \geq 0.995$; slower simulation conditions cannot cross three 10 m cells within the capped practical duration and are decided by direct ROS and intensity. Extinguished simulation conditions require zero ROS and reaction intensity within the stated absolute residual tolerance; TOA is not used to invent a velocity where no propagation occurs. All 16 simulation conditions are mandatory.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE's supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive reference spread rate, the prescribed and maximum timesteps use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured Courant–Friedrichs–Lewy (CFL) limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE's stalled-front branch and corrupt the terminal timestamp. A zero-ROS simulation condition retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. After subtracting the configured meteorology interval, an ordinary stalled-final record must reconstruct the stop time within one timestep. An early stalled-final record is accepted only for a simulation condition whose independent reference calculation predicts zero ROS, and only when the reconstructed pre-jump time is finite, lies between zero and the stop time, and is aligned to the configured timestep grid. Making TOA optional does not authorize an early stall. The sweep ratios are 0.10, 0.25, 0.50, 0.70, 0.85, 0.95, 1.00, and 1.10. Domains use 10 m cells and a finite ignition strip with a planar east face in the interior measurement corridor; the strip terminates inside the lateral boundaries. Wind and slope are zero, live moisture is fixed, and no coupled fire processes are enabled. Intensity and ROS should decrease toward zero and remain zero beyond extinction.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar interior propagation
Fuel models	FBFM3 and TL6	code/name	Two extinction-moisture states
Dead-moisture ratio	0.10 through 1.10 times dead extinction moisture	fraction	Independent damping variable
Wind/slope	0 / 0	mph / degrees	Isolates moisture damping
Timing	generated whole-second step and exact-multiple stop	s	Terminal-evidence compatibility

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

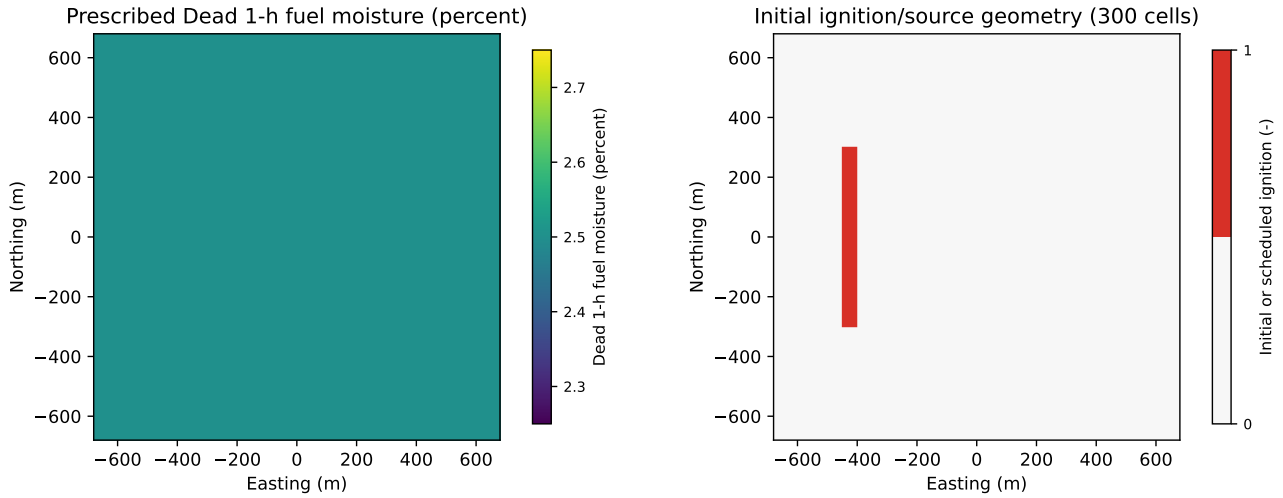


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The available results cover 16 of 16 required simulation conditions and reports **PASS**. For a stalled extinction simulation condition, the recorded result records the configured timestep, stop time, meteorology interval, recorded final time, reconstructed pre-jump time, and evidence-selection mode.

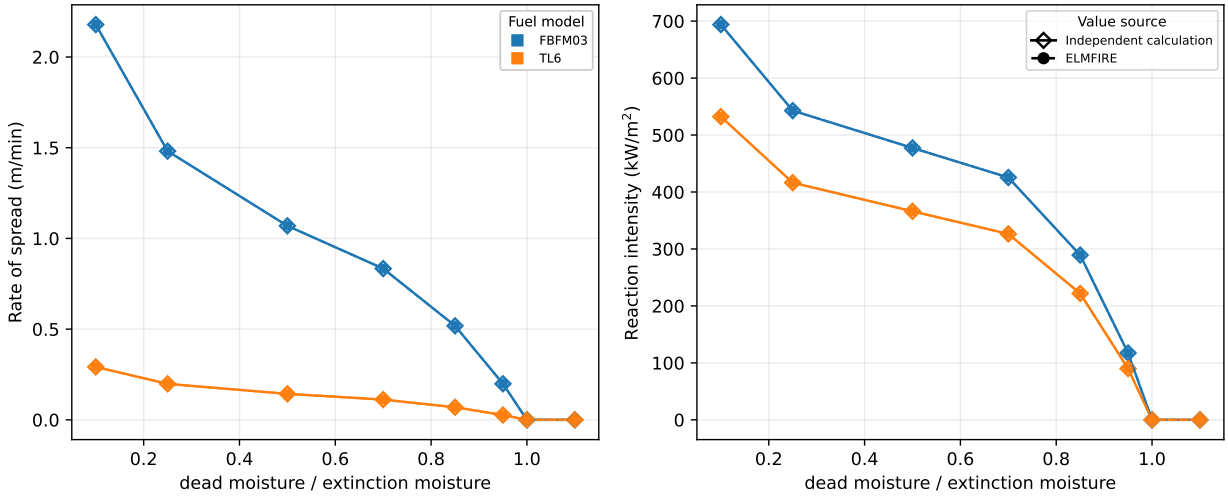


Figure 2: Independent calculations (open diamonds, solid lines) and ELMFIRE predictions (filled circles, dashed lines) for the dead-moisture response through extinction. Each fuel-model pair uses the same color.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.02	4.43846e-06 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	4.16378e-06 fraction	PASS
maximum near-extinction reaction-intensity absolute error	≤ 0.001	0.000482749 kW/m2	PASS
maximum TOA-ROS relative error	≤ 0.05	0.028024 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999896 dimensionless	PASS

The authoritative decision is **PASS**.

8 Technical interpretation

The PASS decision is driven by the component statuses in the generated table. Direct observables use the upper envelope of the sparse diagnostic raster, excluding zero-filled and off-axis cells from the head-spread comparison. For direct ROS, the largest relative discrepancy is 4.43846e-06. Its independent value is 0.0266137 m/min; ELMFIRE gives 0.0266138 m/min. For reaction intensity, the largest relative discrepancy is 4.16378e-06. Its independent value is 89.9786 kW/m^2; ELMFIRE gives 89.979 kW/m^2. A horizontal shift of the extinction point indicates an M_x or units error. Correct intensity with wrong ROS suggests heat-sink or flux-ratio handling. Nonzero propagation or intensity at extinction indicates an unintended live contribution, adjustment, numerical residual, or stale output.

9 Scope and limitations

The case varies all three dead moisture classes together. It does not identify an isolated class-indexing defect or validate moisture observations. Static live moisture is addressed separately by CASE38.

Case 38: Static live-fuel moisture sweep

1 Verification purpose

This case verifies live-fuel moisture damping and calculated live extinction while fuel loads remain fixed. The static shrub models SH2, SH6, and SH8 are intentionally used so changing live moisture cannot transfer herbaceous load into a dead class. It can expose live/dead index errors, percent conversion errors, incorrect live-extinction calculation, or unintended dynamic curing. The question is whether all static-fuel simulation conditions match independent I_R and rate of spread (ROS) predictions over 30 to 200%.

2 Mathematical formulation and numerical implementation

The live term uses the same cubic damping function $\eta_M(r_M)$ as dead fuel, but $r_M = M_L/M_{x,L}$ and $M_{x,L}$ is calculated from dead/live weighting and dead moisture rather than read directly. Live moisture also appears in $Q_{ig} = 250 + 1116M_f$. The formulation follows Andrews (2018), RMRS-GTR-371. For a static fuel, W_{1h}, W_{LH}, and W_{LW} are invariant throughout the sweep.

3 Derivation of expected behavior

The reference calculation first calculates the live extinction moisture for each frozen fuel row and fixed dead moisture. It then evaluates live damping, reaction intensity, heat sink, and R_0 at each moisture. Since no loads move, changes arise only from moisture physics. This controlled result is the baseline needed to interpret the additional load-transfer mechanism in CASE39.

Live-fuel damping uses the same cubic with the generated live-moisture ratio $r_l = M_l/M_{x,l}$:

$$\eta_{M,l} = \max\{0,\ 1 - 2.59r_l + 5.11r_l^2 - 3.52r_l^3\}.$$

For generated SH2, $M_{x,l} = 0.7080597$. At $M_l = 0.70$, $r_l = 0.70/0.7080597 = 0.988617$, giving $\eta_{M,l} = 0.0326506$; substitution in the weighted Rothermel calculation gives $I_R = 354.3991$ kW m^{-2} and $R = 0.0800849$ m min^{-1}. At $M_l = 0.90$, $r_l = 1.27108 > 1$, so the live contribution is zero and the remaining dead-fuel contribution gives $I_R = 304.8611$ kW m^{-2} and $R = 0.0579864$ m min^{-1}. The calculation is repeated for each generated SH2, SH6, and SH8 moisture value. Direct metrics use $e_R = |R_{\text{max}} - R_{\text{ref}}|/\max(|R_{\text{ref}}|, 10^{-8})$ and the analogous e_I, with $\max e_R \le 0.02$ and $\max e_I \le 0.01$. Where time of arrival (TOA) is required, $t = a + bx$ gives $R_{\text{TOA}} = 60/b$, $R^2 = 1 - SSE/SST$, $\max e_{\text{TOA}} \le 0.05$, and $\min R^2 \ge 0.995$.

4 Verification metrics and acceptance criteria

Maximum relative errors are 0.02 for direct ROS and 0.01 for reaction intensity. Simulation conditions predicted at or above 0.10 m min^{-1} also require TOA-derived ROS error at most 0.05 and minimum $R^2 = 0.995$; slower simulation conditions use the direct field because the practical run-duration cap cannot provide a multi-cell TOA regression. Pointwise analytical comparison is selected because a merely monotonic curve would not detect an incorrect live-extinction threshold. All 21 simulation conditions are required and the preprocessor rejects a fuel row marked dynamic.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE's supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive reference spread rate, the prescribed and maximum timesteps use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured Courant–Friedrichs–Lewy (CFL) limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE's stalled-front branch and corrupt the terminal timestamp. A zero-ROS simulation condition retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. It accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required simulation condition must reach the prescribed stop time in its generated namelist; a direct-only simulation condition may terminate early because lack of propagation is part of the state being tested. These shrub models carry live woody but no live herbaceous load. Live-woody moisture is set to 60, 70, 90, 120, 150, 180, or 200%; 60% is the supported ELMFIRE lower bound; dead classes remain at 8%. The homogeneous domain uses a finite ignition strip whose east face is planar in the interior measurement corridor and whose ends remain inside the lateral boundaries. Wind and slope are zero and all coupled processes are disabled. Fuel loads remain invariant; reaction intensity and ROS should generally decline as live moisture rises, with behavior governed by each fuel's calculated live extinction.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar interior propagation
Fuel models	SH2, SH6, and SH8	code/name	Static live-fuel responses
Live moisture	30 to 200%	percent	Independent damping variable
Wind/slope	0 / 0	mph / degrees	Isolates live damping
Timing	generated whole-second step and exact-multiple stop	s	Terminal-evidence compatibility

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

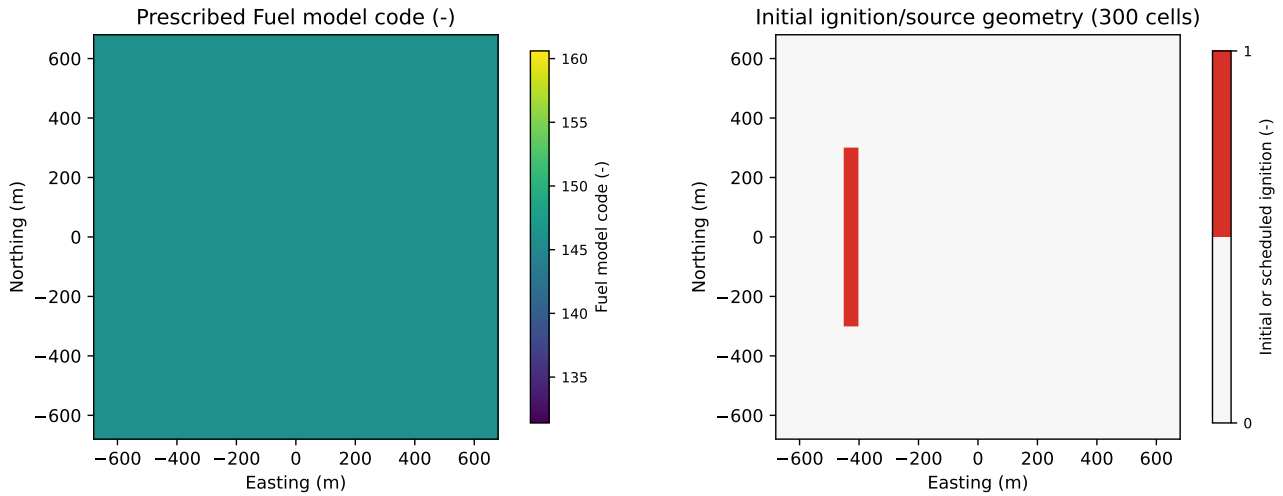


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The available results cover 21 of 21 required simulation conditions and reports **PASS**.

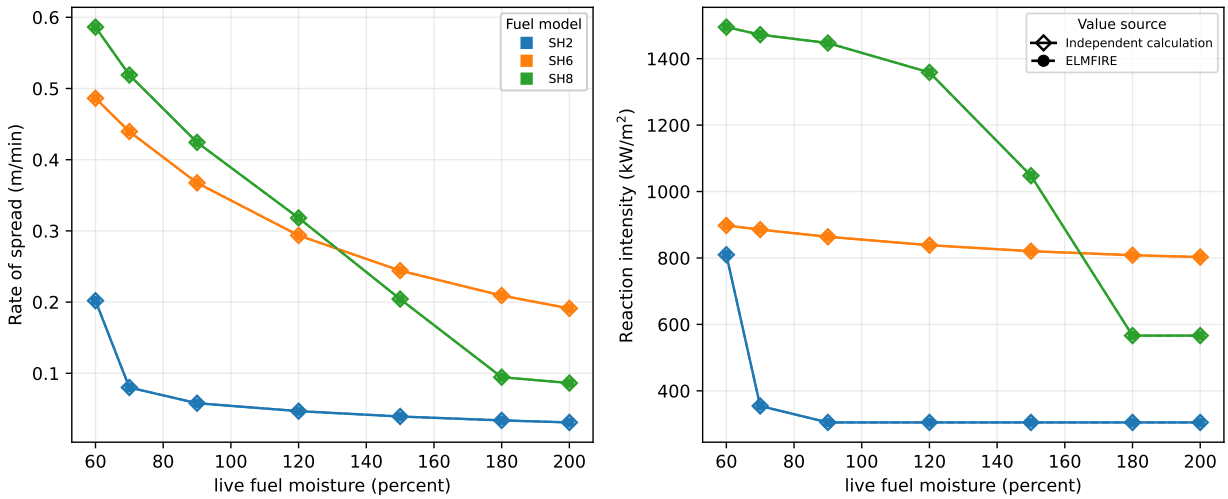


Figure 2: Independent calculations (open diamonds, solid lines) and ELMFIRE predictions (filled circles, dashed lines) for the static live-moisture response. Each fuel-model pair uses the same color.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.02	6.50329e-06 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	6.30732e-06 fraction	PASS
maximum TOA-ROS relative error	≤ 0.05	0.0170672 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999953 dimensionless	PASS

The authoritative decision is **PASS**.

8 Technical interpretation

A shifted transition implicates calculated live extinction or moisture units. Unexpected sharp changes suggest that a supposedly static row was treated dynamically. Agreement in intensity but not ROS narrows the defect to heat sink or spread conversion.

9 Scope and limitations

This case covers three static shrub fuels at one dead-moisture state. It does not verify dynamic curing, spatial moisture mosaics, temporal interpolation, or physical observations. Dynamic herbaceous transfer is isolated in CASE39.

Case 39: Dynamic herbaceous curing sweep

1 Verification purpose

This case verifies dynamic herbaceous curing in GR2 and GS3. Live-herb moisture is the controlled input; ELMFIRE must transfer the corresponding fraction of herbaceous load to the dead 1-h class and then evaluate the full Rothermel model. It can expose reversed curing, incorrect 30/120% bounds, failure to conserve load, wrong class surface-area-to-volume ratio (SAV) weighting, or confusion between curing and live moisture damping. The verification question is whether all required curing states reproduce independently calculated intensity and rate of spread (ROS) within 1% and 2%, respectively.

2 Mathematical formulation and numerical implementation

For live-herb moisture M_{LH} expressed as a fraction, the cured fraction is

$$T = \max[0, \min(1, 1.33 - 1.11 M_{LH})]. \tag{1}$$

ELMFIRE implements the equivalent integer-percent interpolation between 30 and 120%. If $W_{LH,0}$ is original live-herb load, $W'_{1h} = W_{1h} + T W_{LH,0}$ and $W'_{LH} = (1 - T) W_{LH,0}$. The transferred material adopts dead 1-h moisture while remaining live material uses live damping and heat of preignition. The formulation follows Andrews (2018), RMRS-GTR-371.

3 Derivation of expected behavior

At 30% moisture all live-herb load is cured; at 75% half is cured; at 120% and above none is transferred. Above 120%, curing is constant but live moisture damping continues to change, so ROS need not form a plateau. The case-specific reference calculation conserves total load, recomputes effective SAV and every dependent Rothermel coefficient, applies moisture damping, and applies ELMFIRE's selected wind limit at the target 6 mph midflame wind.

For dynamic grass fuels, the cured fraction and load transfer are

$$f_c = \min\left(1, \max\left[0, \frac{120 - M_{lh}}{90}\right]\right), \quad w^*_{lh} = (1 - f_c) w_{lh}, \quad w^*_{1h} = w_{1h} + f_c w_{lh}.$$

For the generated GR2 simulation condition at $M_{lh} = 75$ percent, $f_c = (120 - 75)/90 = 0.5$. With original $w_{lh} = 0.04591$ and $w_{1h} = 0.00459$ lb ft^{-2},

$$w^*_{lh} = 0.5(0.04591) = 0.022955, \qquad w^*_{1h} = 0.00459 + 0.5(0.04591) = 0.027545.$$

The weighted fuel calculation gives $R_0 = 1.109830$ ft min^{-1}, $I_R = 206.9424$ kW m^{-2}, and $\phi_w = 31.095160$; therefore

$$R_{\text{ref}} = 0.3048(1.109830)(1 + 31.095160) = 10.857030 \text{ m min}^{-1}.$$

Every GR2 and GR3 moisture simulation condition repeats these equations. Direct metrics use the maximum over simulation conditions of $e_R = |R_{\max} - R_{\text{ref}}| / \max(|R_{\text{ref}}|, 10^{-8})$ and the analogous e_I, with limits 0.02 and 0.01. The time of arrival (TOA) fit $t = a + bx$ supplies $R_{\text{TOA}} = 60/b$ and $R^2 = 1 - SSE/SST$, tested by $\max e_{\text{TOA}} \leq 0.05$ and $\min R^2 \geq 0.995$.

4 Verification metrics and acceptance criteria

Maximum relative errors are 0.02 for direct ROS and 0.01 for reaction intensity. Simulation conditions predicted at or above $0.10\,\mathrm{m\,min^{-1}}$ also require TOA-derived ROS error at most 0.05 and minimum $R^2 = 0.995$; the most moisture-limited simulation conditions use direct fields because they cannot cross a multi-cell corridor within the practical duration cap. Direct effective loads are not exposed by the normal executable, so their conservation is an reference calculation invariant and the implementation is observed through intensity and ROS. All 20 simulation conditions are required; no qualitative trend alone can pass the case.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE's supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive reference spread rate, the prescribed and maximum timesteps use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured Courant–Friedrichs–Lewy (CFL) limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE's stalled-front branch and corrupt the terminal timestamp. A zero-ROS simulation condition retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. It accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required simulation condition must reach the prescribed stop time in its generated namelist; a direct-only simulation condition may terminate early because lack of propagation is part of the state being tested. Live-herb moisture is 30, 45, 60, 75, 90, 100, 110, 120, 135, or 200%. Dead classes are fixed at 5%, live woody moisture at 90%, slope at zero, and target midflame wind at 6 mph; the 20-ft input is back-calculated from each fuel's WAF. Each homogeneous domain uses a finite ignition strip with a planar east face inside the analysis corridor and with lateral ends clear of the grid boundary. Crown fire, spotting, buildings, and acceleration are excluded. Strong nonlinear response is expected in the curing interval.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar interior propagation
Dynamic fuels	GR2 and GS3	code/name	Enables herbaceous load transfer
Live-herb moisture	30 to 200% live-herb moisture	percent	Independent curing variable
Wind/slope	fixed wind; 0 slope	—	Holds directional forcing fixed
Timing	generated whole-second step and exact-multiple stop	s	Terminal-evidence compatibility

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

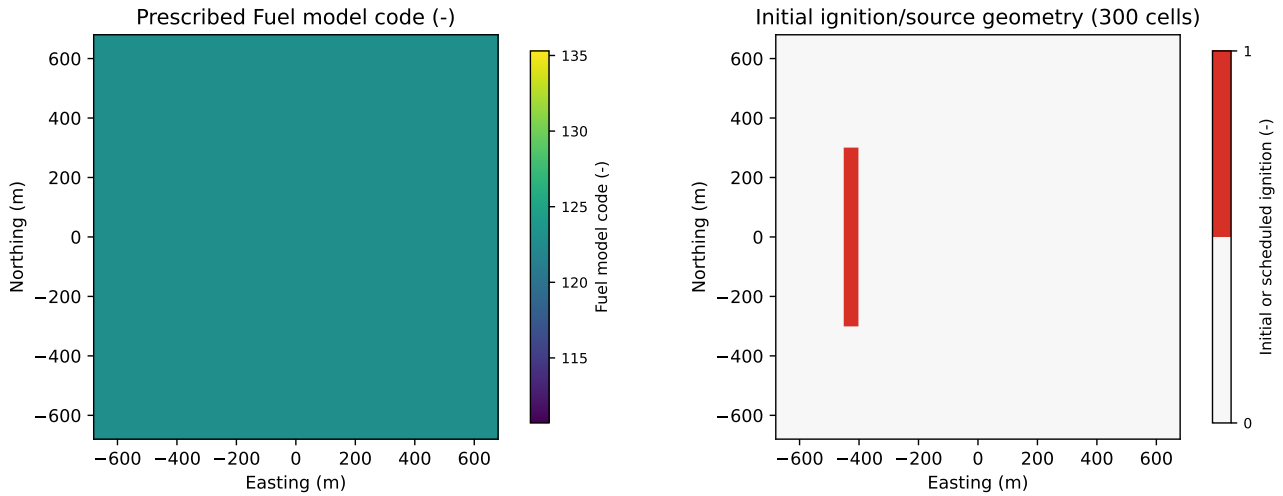


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The available results cover 20 of 20 simulation conditions and reports **PASS**.

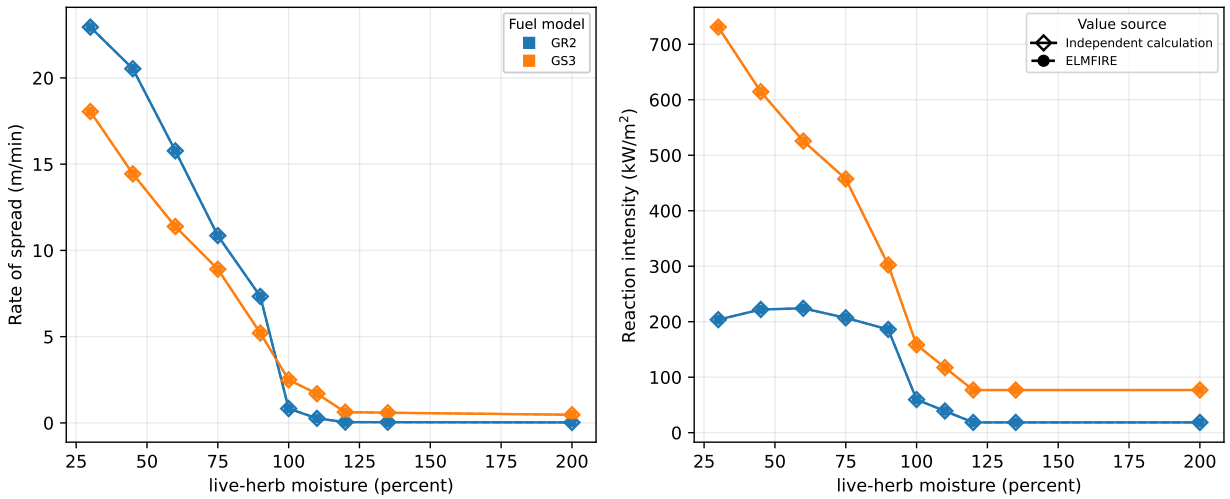


Figure 2: Independent calculations (open diamonds, solid lines) and ELMFIRE predictions (filled circles, dashed lines) for the dynamic-curing response. Each fuel-model pair uses the same color.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.02	1.51489e-06 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	1.14414e-06 fraction	PASS
maximum TOA-ROS relative error	≤ 0.05	0.0226402 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999981 dimensionless	PASS

The authoritative decision is **PASS**.

8 Technical interpretation

A reversed curve indicates a reversed transfer fraction. Discontinuities away from integer moisture selection suggest class indexing or interpolation defects. Correct behavior above 120% but disagreement below it isolates curing rather than live damping. Correct intensity and wrong ROS implicate heat sink or wind coupling.

9 Scope and limitations

This executable-level test observes curing indirectly; a direct internal-load assertion would require a compiled component diagnostic. It covers two dynamic fuel models, constant weather, and no temporal curing. CASE38 separately isolates live moisture without load transfer.

Case 40: Wind–slope vector and direction sweep

1 Verification purpose

This coupling case verifies how ELMFIRE combines Rothermel wind and slope factors into a direction of maximum spread (DMS), head speed, and fire ellipse. Wind direction rotates through 0, 45, 90, 135, and 180 degrees relative to upslope while magnitudes remain fixed. It can expose degree/radian errors, wind-from/wind-to reversal, aspect reversal, swapped raster axes, scalar addition in place of vector addition, or incorrect ellipse construction. The question is whether all simulation conditions meet the DMS, head rate of spread (ROS), length/width, and intensity limits.

2 Mathematical formulation and numerical implementation

Let upslope azimuth be zero and wind-to azimuth be α. The dimensionless forcing vector is

$$\boldsymbol{\phi} = \phi_s(0, 1) + \phi_w(\sin\alpha, \cos\alpha), \tag{1}$$

so $\theta_{DMS} = \text{atan2}(\phi_x, \phi_y)$ and $R_H = R_0(1 + |\boldsymbol{\phi}|)$. ELMFIRE derives an effective wind from $|\boldsymbol{\phi}|$ and uses its Rothermel length-to-width relation, capped at eight. Because these quantities are defined on the sloping surface, the preprocessor makes the initial circle and the postprocessor measures distances in slope-surface coordinates rather than mixing them with map-plane northing. The scalar factors follow Andrews (2018), RMRS-GTR-371; vectoring and ellipse propagation exercise ELMFIRE's level-set implementation.

3 Derivation of expected behavior

Aligned wind and slope add; opposed forcing subtracts in vector space and the stronger component sets direction. At intermediate angles, the expected DMS lies between the two forcing directions, not generally at the arithmetic mean of angles. The case-specific reference calculation independently evaluates ϕ_s, ϕ_w, their vector magnitude and azimuth, head ROS, effective wind, and length/width.

Let upslope define zero azimuth and let α be the generated wind-to-upslope angle. With $\phi_s = 6.436561$, $\phi_w = 9.746235$, and $R_0 = 0.9178563$ m min^{-1}, the vector reference is

$$\phi_x = \phi_s + \phi_w \cos\alpha, \quad \phi_y = \phi_w \sin\alpha, \quad \phi = \sqrt{\phi_x^2 + \phi_y^2}, \quad \theta = \text{atan2}(\phi_y, \phi_x), \quad R_h = R_0(1 + \phi).$$

For example, at $\alpha = 90$ degrees,

$$\phi = \sqrt{6.436561^2 + 9.746235^2} = 11.679829,$$
$$\theta = \tan^{-1}(9.746235/6.436561) = 56.5586 \text{ degrees},$$

and

$$R_h = 0.9178563(1 + 11.679829) = 11.638261 \text{ m min}^{-1}.$$

The effective vector wind is 4.41571 mph $= 1.97394$ m s^{-1}; substitution in $L/W = 0.936e^{0.2566U} + 0.461e^{-0.1548U} - 0.397$ gives $L/W = 1.495859$. Across $\alpha = \{0, 45, 90, 135, 180\}$ degrees, the reference $(\theta, R_h, L/W)$ rows are

$$(0, 15.7713, 1.63745), \quad (27.3422, 14.6898, 1.60074),$$
$$(56.5586, 11.6383, 1.49586), \quad (93.7779, 7.25716, 1.33838),$$
$$(180, 3.95566, 1.20547).$$

Direction error is circular, $e_\theta = \min(|\theta_m - \theta_r|, 360 - |\theta_m - \theta_r|)$. The other metrics are relative errors in R_h, L/W, and $I_R = 686.5297$ kW m^{-2}; their maximum substitutions are $e_\theta \leq 5$ degrees, $e_R \leq 0.05$, $e_{LW} \leq 0.10$, and $e_I \leq 0.01$.

4 Verification metrics and acceptance criteria

Maximum circular DMS error is 5°, maximum head-ROS relative error is 0.05, maximum length/width relative error is 0.10, and maximum intensity relative error is 0.01. DMS is inferred from the farthest one percent of the burned footprint, subject to at least 100 burned cells and at least five farthest-front samples. Head ROS is a time of arrival (TOA) regression in a narrow predicted-axis corridor with at least ten samples. Length/width uses full-footprint slope-surface extents projected along and across the DMS. Reaction intensity is the median over valid burned cells and requires at least 90 percent coverage of that footprint. These complementary metrics distinguish direction, speed, shape, and output completeness. Every simulation condition is required.

5 Simulation configuration and predicted outcome

Fuel model 2 is used on a 1400 m square, 10 m grid with a 30 m-radius signed-distance ignition clipped to $[-1, 1]$. Slope is requested as 50% and generated as the integer 27-degree value used by ELMFIRE (50.95% effective rise/run), with upslope north; target midflame wind is 4 mph. Dead moistures are 5/7/9%, live-herb moisture is 60%, and live-woody moisture is 90% in both every generated namelist and the independent reference calculation. Wind-from azimuth is generated so the wind-to vector has the required relative angle. Each simulation condition uses the largest conservative whole-second timestep no greater than the 5%-of-cell target that divides the fixed 1800 s duration exactly. This retains Courant–Friedrichs–Lewy (CFL) margin and prevents a shortened final step from triggering ELMFIRE's stalled-front timestamp defect. Crown fire, spotting, WUI, stochastic wind, and acceleration are disabled. The footprint should rotate smoothly and change head extension as the vectors move from aligned to opposed. The evaluator accepts only unique TOA and reaction-intensity rasters identified by a final dump record. A stalled-final record is accepted only when subtracting the configured meteorology interval reconstructs the 1800 s stop time within one configured timestep.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Whole-footprint direction and shape
Fuel/slope	FM2; 50	code / percent	Fixed slope forcing
Midflame wind	4	mph	Fixed wind-factor magnitude
Relative angles	0, 45, 90, 135, and 180 degrees	degrees	Vector-coupling sweep
Time control	generated whole-second divisor; stop 1800	s	Exact timestep multiple

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

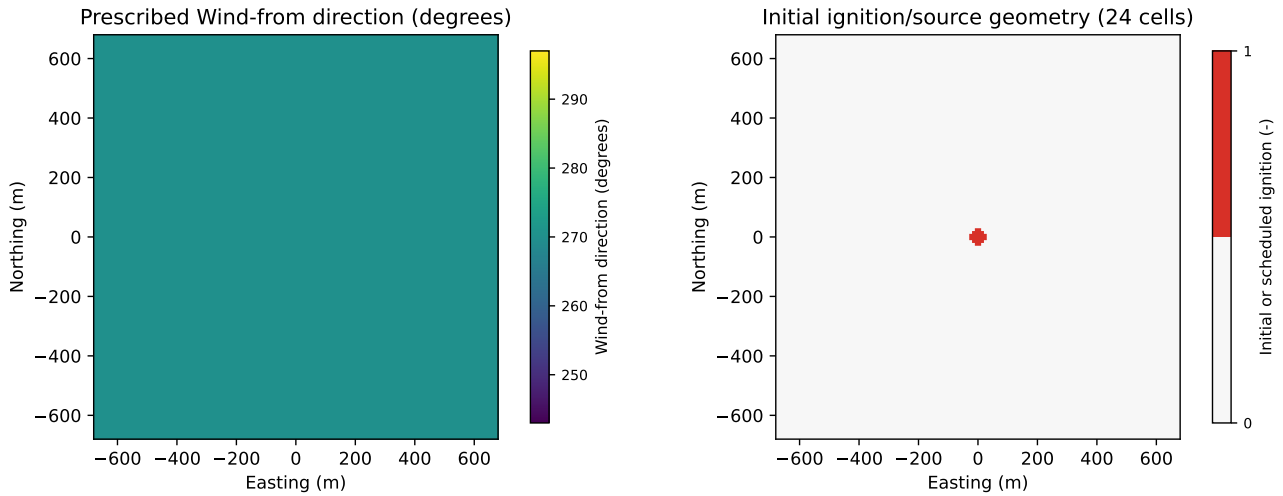


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The available results cover 5 of 5 simulation conditions and reports **PASS**. The generated diagnostic is: –.

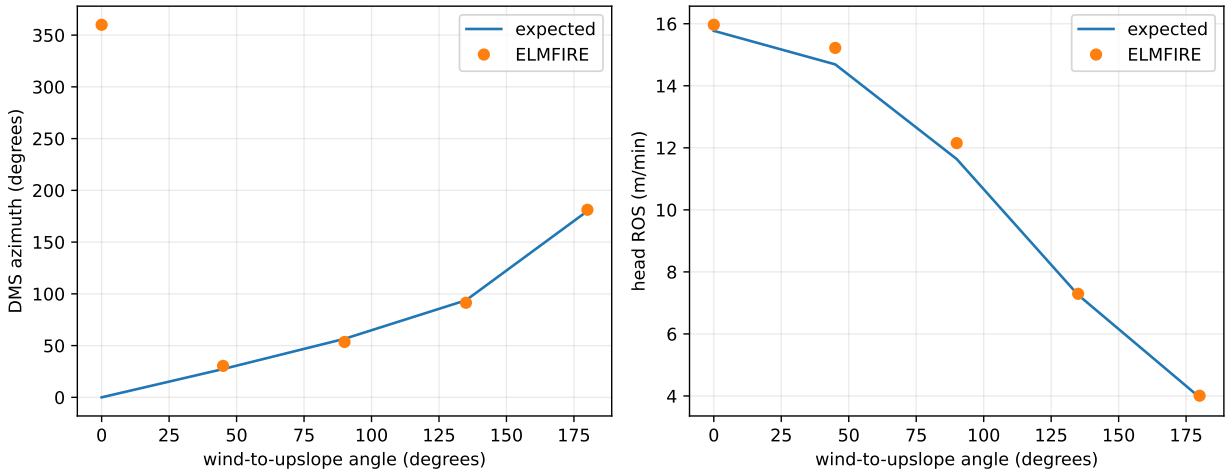


Figure 2: Expected and measured DMS azimuth and head ROS.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum DMS direction error	≤ 5.0	3.05989 degrees	PASS
maximum head-ROS relative error	≤ 0.05	0.0441104 fraction	PASS
maximum length/width relative error	≤ 0.1	0.0961658 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	2.01156e-07 fraction	PASS

The authoritative decision is **PASS**.

8 Technical interpretation

A 180-degree error indicates a wind-from or aspect reversal. A 90-degree error suggests swapped axes or azimuth convention. Correct direction but excessive head speed suggests scalar addition. Correct DMS and head speed with wrong length/width isolates effective-wind or ellipse construction.

9 Scope and limitations

This test uses one fuel, one slope magnitude, one wind magnitude, uniform terrain, and a single grid. It verifies deterministic code behavior rather than observed fire shape. Complex terrain and resolution convergence remain separate concerns.

Case 41: Custom fuel-parameter Rothermel sweep

1 Verification purpose

This case verifies ELMFIRE's public custom fuel-table interface and the Rothermel response to one-at-a-time changes in dead 1-h load, 1-h surface-area-to-volume ratio (SAV), fuel-bed depth, dead extinction moisture, and heat content. It can expose column misalignment, ignored custom values, stale precomputed coefficients, or unit conversion errors. The question is whether every generated code-250 row produces independently calculated reaction intensity and rate of spread (ROS) within 1% and 2%.

2 Mathematical formulation and numerical implementation

The custom row supplies class loads w_0, surface-area-to-volume ratios σ, depth δ, dead extinction M_x, and heat content h. These feed packing ratio $\beta = \sum w_0/(\delta\rho_p)$, optimum packing, reaction velocity, propagating flux, damping, heat sink, and

$$R_0 = I_R\xi/(\rho_b\epsilon Q_{ig}). \tag{1}$$

The formulation follows Andrews (2018), RMRS-GTR-371. Particle density and mineral constants are fixed by the inspected ELMFIRE interface and therefore are not claimed as user-sweepable variables.

3 Derivation of expected behavior

The baseline row is copied and exactly one public column is changed in each simulation condition. A case-specific fuel table is written beside that simulation condition and read independently by the reference calculation. The full equation is recalculated rather than imposing a monotonic trend: load, SAV, and depth have competing effects through packing and reaction velocity, so a simple ordering would be scientifically unjustified.

Each custom-fuel reference is recalculated from $R_0 = I_R\xi/(\rho_b\epsilon Q_{ig})$, changing only the indicated fuel-table column. The substitutions produced from the generated table are

Condition	R_{ref} (m min^{-1})	$I_{R,\text{ref}}$ (kW m^{-2})
baseline	0.411242	1045.499
$w_{1h} = 0.05$	0.253350	688.121
$w_{1h} = 0.15$	0.551674	1379.611
$SAV = 1000$	0.241692	960.426
$SAV = 3000$	0.662626	1074.561
$d = 0.5$	0.225975	910.468
$d = 2.0$	0.694647	991.942
$M_x = 15\%$	0.371790	945.198
$M_x = 35\%$	0.438500	1114.797
$h = 6000$	0.308432	784.125
$h = 10000$	0.514053	1306.874

For example, replacing the baseline 1-h load 0.10 by 0.05 lb ft^{-2} changes $R_0 = 1.349221$ to 0.831202 ft min^{-1}; multiplication by 0.3048 gives 0.253350 m min^{-1}. Each simulation condition uses $e_R = |R_{\text{max}} - R_{\text{ref}}|/\max(|R_{\text{ref}}|, 10^{-8})$ and the analogous e_I. The time of arrival (TOA) fit $t = a + bx$ gives $R_{\text{TOA}} = 60/b$ and $R^2 = 1 - SSE/SST$. The aggregate substitutions are $\max e_R \le 0.02$, $\max e_I \le 0.01$, $\max e_{\text{TOA}} \le 0.05$, and $\min R^2 \ge 0.995$.

4 Verification metrics and acceptance criteria

Maximum relative errors are 0.02 for direct ROS and 0.01 for reaction intensity. Simulation conditions predicted at or above $0.10\,\mathrm{m\,min^{-1}}$ additionally require TOA-derived ROS error at most 0.05 and minimum $R^2 = 0.995$; slower simulation conditions are assessed from direct ROS and intensity. Pointwise equality to a complete recalculation detects both ignored columns and wrong nonlinear propagation. The baseline and ten low/high simulation conditions are all required.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE's supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive reference spread rate, the prescribed and maximum timesteps use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured Courant–Friedrichs–Lewy (CFL) limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE's stalled-front branch and corrupt the terminal timestamp. A zero-ROS simulation condition retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. It accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required simulation condition must reach the prescribed stop time in its generated namelist; a direct-only simulation condition may terminate early because lack of propagation is part of the state being tested. All simulation conditions use fuel code 250, 5/7/9% dead moisture, 90% live moisture, zero wind and slope, and 10 m cells. A finite ignition strip supplies a planar east face in the interior analysis corridor while remaining clear of the lateral grid boundaries. The baseline row uses 0.10/0.05/0.02 dead loads, 0.08 live-woody load, 1-h SAV 2000 $\mathrm{ft^{-1}}$, depth 1 ft, dead extinction 25%, and heat content $8000\,\mathrm{Btu\,lb^{-1}}$. Each output should match its own reference calculation; no universal increasing trend is predicted.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar interior propagation
Custom fuel code	250	code	Selects simulation-specific table row
Changes	load, SAV, depth, extinction, heat	native table units	One-at-a-time interface exercise
Wind/slope	0 / 0	mph / degrees	Isolates custom fuel response
Timing	generated whole-second step and exact-multiple stop	s	Terminal-evidence compatibility

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

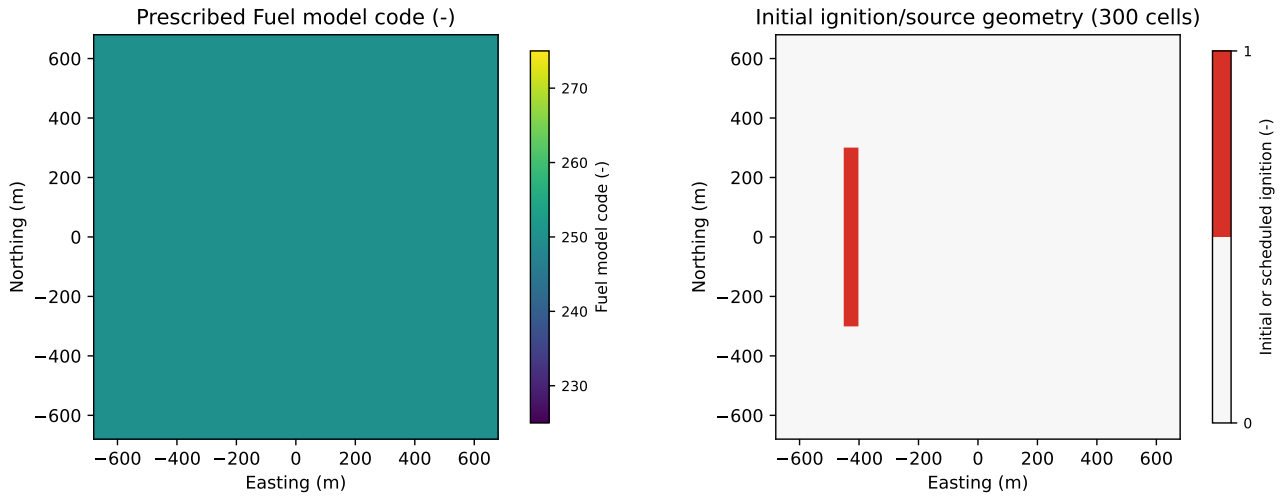


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The available results cover 11 of 11 simulation conditions and reports **PASS**.

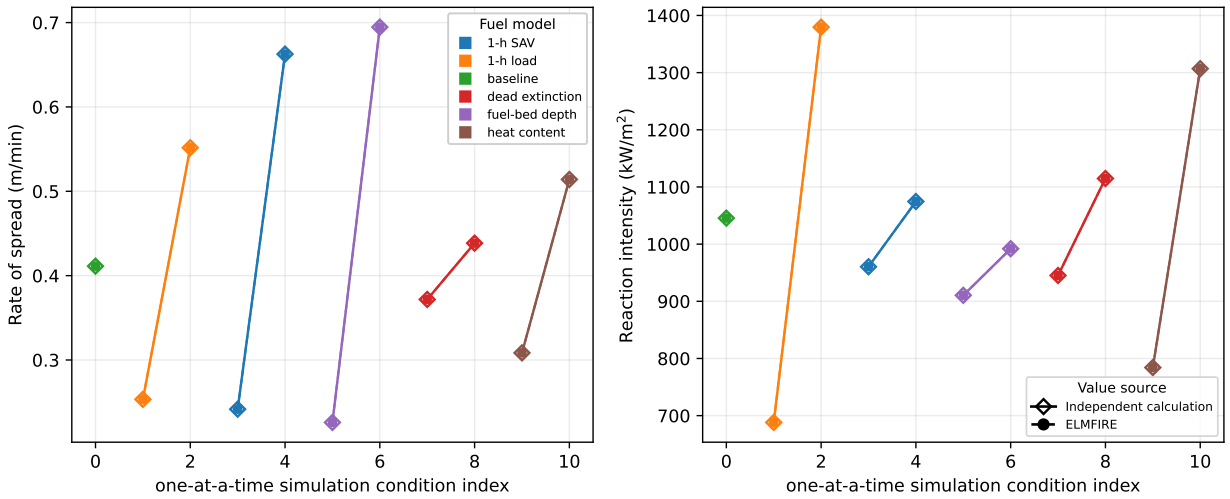


Figure 2: Independent calculations (open diamonds, solid lines) and ELMFIRE predictions (filled circles, dashed lines) for the custom fuel-table simulation conditions. Each comparison pair uses the same color.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.02	4.62672e-07 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	1.10118e-07 fraction	PASS
maximum TOA-ROS relative error	≤ 0.05	0.0151881 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999968 dimensionless	PASS

The authoritative decision is **PASS**.

8 Technical interpretation

If only one low/high pair fails, its fuel-table column or dependent coefficient is suspect. A common intensity error suggests table parsing or units; correct intensity with wrong ROS points to heat sink, density, or flux ratio. Identical results across a changed pair indicate the column was ignored.

9 Scope and limitations

This case tests only parameters exposed by the current custom table and one synthetic baseline. It does not test particle density, mineral contents, spatial mixtures, dynamic curing, or physical realism of the synthetic fuel.

Case 42: Wind-adjustment-factor coupling sweep

1 Verification purpose

This case verifies the upstream coupling from 20-ft wind, canopy cover, canopy height, and fuel-bed depth through ELMFIRE's wind-adjustment factor (WAF) into Rothermel rate of spread (ROS). It distinguishes an input-height conversion error from a defect in the Rothermel wind factor. It can expose percent/fraction canopy errors, height-times-ten errors, sheltered/unsheltered branch errors, or missing WAF application. The question is whether all canopy simulation conditions reproduce the independently predicted final ROS within 3% while reaction intensity remains within 1%.

2 Mathematical formulation and numerical implementation

ELMFIRE calculates $U_{mf} = WAF\,U_{20}$. Without canopy it uses a logarithmic fuel-bed-height expression. With canopy it evaluates a sheltered factor from canopy cover and height, limits it by the unsheltered factor, and blends the two when $0.3333CC < 0.05$. The resulting midflame speed enters $\phi_w = CU_{mf}^{B}(\beta/\beta_{op})^{-E}$. The case reference calculation restates these equations directly and does not read ELMFIRE output to estimate WAF.

3 Derivation of expected behavior

For fixed 10 mph and fuel model 2, reaction intensity is invariant because canopy affects wind adjustment but not the configured surface fuel or moisture. ROS differences are completely predicted by the independently calculated WAF and Rothermel wind factor. The zero-canopy simulation condition selects the unsheltered formula; canopy combinations exercise transition and sheltered branches.

The unsheltered adjustment for fuel-bed depth d (ft) is

$$WAF_u = \frac{1.36}{\ln[(20+0.36d)/(0.13d)]}\left[\ln(1.36/0.13)-1\right].$$

For canopy cover C, height H (ft), and $s = 0.3333C$, the sheltered value is

$$WAF_s = \frac{1}{\ln[(20+0.36H)/(0.13H)]}\frac{0.555}{\sqrt{sH}}.$$

The generated blend uses $WAF = \min(WAF_s, WAF_u)$ when $s \geq 0.05$; otherwise $f_u = 1 - 20s$ and $WAF = f_u WAF_u + (1-f_u)\min(WAF_s, WAF_u)$. For the zero-canopy Fuel Model 2 control, $WAF_u = 0.362674$, so the 10 mph 20-ft wind gives $U_{mf} = 3.626740$ mph, $\phi_w = 8.146356$, and $R = 0.8237955(1+8.146356) = 7.534727$ m min^{-1}. For $C = 0.10$, $H = 15/0.3048 = 49.2126$ ft, $s = 0.03333$, $WAF_s = 0.244254$, and $f_u = 0.3334$; hence $WAF = 0.3334(0.362674) + 0.6666(0.244254) = 0.283735$, $U_{mf} = 2.837354$ mph, $\phi_w = 5.197957$, and $R = 0.8237955(1+5.197957) = 5.105849$ m min^{-1}. All simulation conditions retain $I_R = 675.184$ kW m^{-2}. Simulation condition relative errors use $|q_{\text{max}} - q_{\text{ref}}|/\max(|q_{\text{ref}}|, 10^{-8})$. From $t = a + bx$, $R_{\text{TOA}} = 60/b$ and $R^2 = 1 - SSE/SST$. The aggregate substitutions are max $e_R \leq 0.03$, max $e_I \leq 0.01$, max $e_{\text{TOA}} \leq 0.06$, and min $R^2 \geq 0.995$.

4 Verification metrics and acceptance criteria

Maximum direct-ROS relative error is 0.03, reaction-intensity error is 0.01, and time of arrival (TOA)-derived ROS error is 0.06; TOA regression requires $R^2 \geq 0.995$. Constant reaction intensity is retained as a control that detects unintended crown or canopy-fuel coupling. All ten simulation conditions are required and missing fields give NOT EVALUABLE.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE's supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every simulation condition, the prescribed and maximum timesteps use a conservative whole-second value no greater than the 5 s nominal step or the configured 0.2 Courant–Friedrichs–Lewy (CFL) limit. The stop time is rounded upward to an exact integer multiple of that step. This avoids both discrete-front resonance and a shortened final step that can trigger ELMFIRE's stalled-front timestamp defect. A zero-ROS simulation condition retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator requires uniquely named rasters tied to one final dump record and accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required simulation condition must reach the prescribed stop time in its generated namelist; a direct-only simulation condition may terminate early because lack of propagation is part of the state being tested. Canopy cover is 0, 10, 30, or 60%; canopy heights are 5, 15, and 30 m where applicable. The raster stores cover in percent and height times ten, matching explicit namelist flags. Fuel model 2, dead moisture 5/7/9%, live moisture 90%, and zero slope are fixed. A finite west-to-east ignition strip has a planar east face in the interior analysis corridor and ends inside the lateral boundaries. Crown fire is disabled even though canopy inputs are present. ROS should vary only through WAF while I_R remains fixed.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar interior propagation
Fuel / 20-ft wind	FM2 / 10 mph	code / mph	Fixed surface forcing
Canopy cover	0, 10, 30, 60	percent	WAF branch and blend sweep
Canopy height	0, 5, 15, 30	m	Sheltered-profile sweep
Timing	generated whole-second step and exact-multiple stop	s	Terminal-evidence compatibility

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

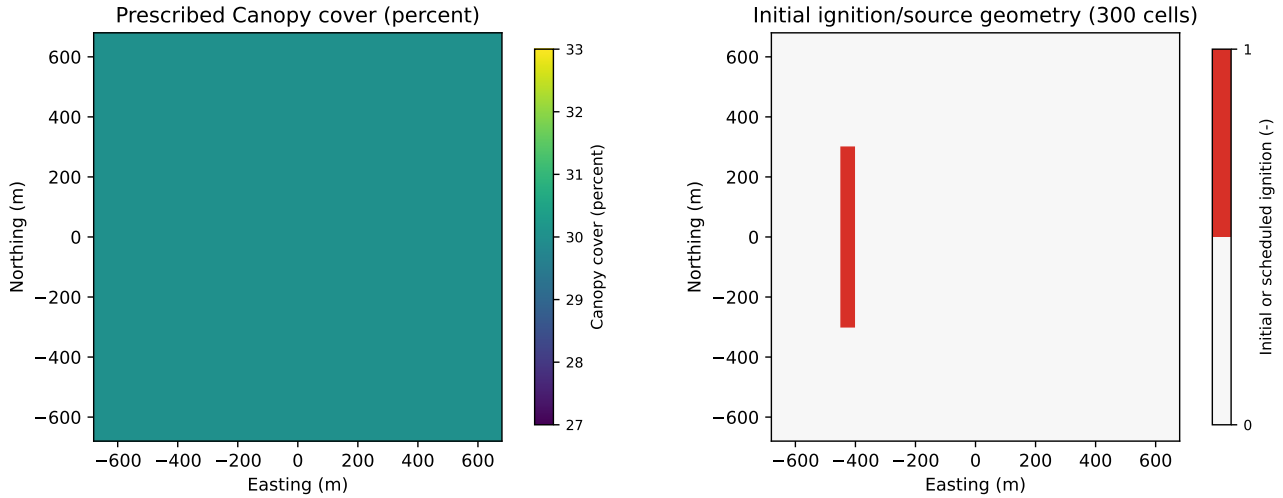


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The available results cover 10 of 10 simulation conditions and reports **PASS**.

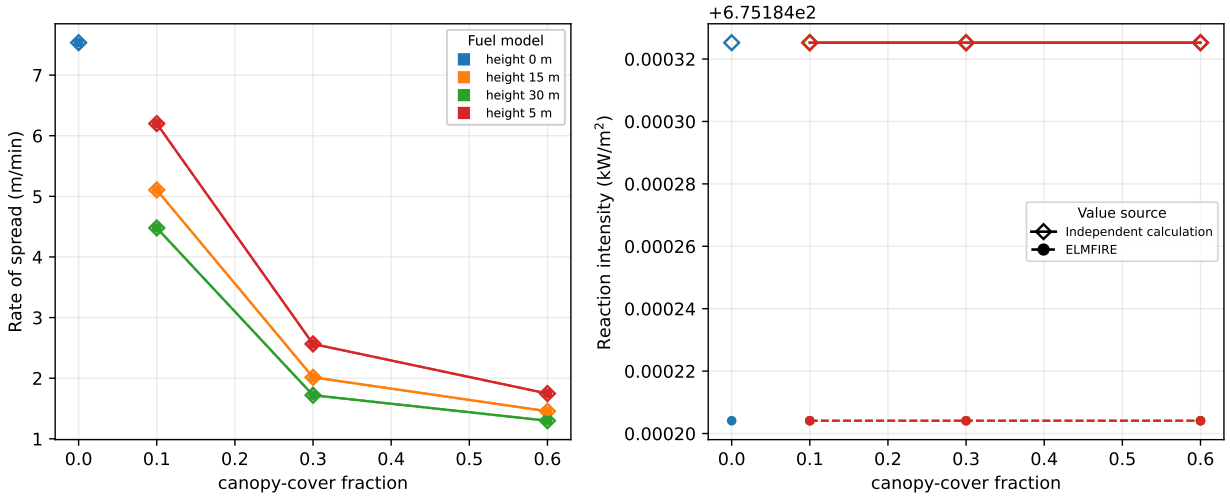


Figure 2: Expected and simulated canopy/WAF response at fixed 20-ft wind.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.03	6.68301e-07 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	1.79416e-07 fraction	PASS
maximum TOA-ROS relative error	≤ 0.06	0.0245548 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999972 dimensionless	PASS

The authoritative decision is **PASS**.

8 Technical interpretation

A uniform wind-scale error suggests incorrect 20-ft units. Errors confined to canopy simulation conditions implicate sheltered WAF or canopy scaling. Varying reaction intensity indicates an unintended canopy or crown-fire pathway. Correct direct ROS but biased TOA implicates level-set propagation.

9 Scope and limitations

This is an implementation-specific coupling test, not a validation of canopy wind profiles. It uses one fuel, one 20-ft wind, uniform canopy fields, and the current ELMFIRE WAF algorithm. A deliberate algorithm change requires reference calculation and report review.

Case 43: WU-E ellipse response

1 Verification purpose

This case verifies the coupled response from Wildland–Urban ELMFIRE (WU–E) source wind and building geometry through the model-2 ellipse and heat-flux routines to the transient direct-flame-contact (DFC) and radiation rasters. It is capable of exposing an incorrect wind-unit conversion, a wrong branch predicate, a regression-coefficient or ellipse-construction error, a wind-direction sign or raster-orientation error, failure of the separation clamp, and disagreement between constant and raster building-parameter input paths.

One initially burning structure is held stationary by a zero spread-adjustment raster. Spotting, terrain, moisture variation, and secondary propagation are disabled. Twenty-three deterministic simulation conditions exercise both source-wind branch boundaries, Hamada parameters (A) and (D), four right-angle rotations, and constant/raster equivalence, plus the intended source fuel-reach factors 1 for structures and 1/3 for vegetation. The reference calculation implements the intended downscaled WU–E formulation. The pinned source is the system under test, so a mismatch is reported rather than copied into the expected result.

Verification question. Do all required simulation conditions reproduce the intended heat-release rate (HRR) per unit area (HRRPUA) and full-domain DFC/radiation maps within normalized L_1 error 0.005, preserve the paired clamp, rotation, and constant/raster invariants, and create no additional burning cell?

Current result: PASS. 23 of 23 required simulation conditions have evaluable evidence. Analysis procedure: **COMPLETE.**

2 Mathematical formulation and numerical implementation

The source converts the 20-ft input wind from miles per hour to metres per second as $V = 0.447W$. The regression is linear when $V < 10$, quadratic when $10 \le V \le 17.3$, and linear with a distinct coefficient set when $V > 17.3$. In each branch it produces downwind, upwind, and sidewind reaches (K_d, K_u, K_s) from (V, A, D), after enforcing

$$D_{\text{eff}} = \min(D, 50\,\text{m}). \tag{1}$$

The stored ellipse quantities are

$$B = \frac{K_d + K_u}{2}, \qquad E = \min\left(\frac{B}{2}, B - K_u\right), \qquad A_e = \frac{K_s}{\sqrt{1 - (E/B)^2}}. \tag{2}$$

Here B, A_e, and E retain the source nomenclature major, minor, and eccentricity; all three are lengths. For comparison, the pinned source currently constructs the following continuous angular boundary, which is a documented deviation from the intended three-sector model:

$$r_e(\theta_f) = \frac{0.3K_d(B-E)}{A_e^2}\,\frac{A_e^2}{B - E\cos\theta_f}, \qquad \theta_f = \text{atan2}(\Delta y, \Delta x) - \frac{\pi}{180}(270 - WD). \tag{3}$$

The intended model uses a three-sector reach and a physical-distance correction. With $L_c =$ `BLDG_AREA` interpreted as a linear characteristic building dimension in metres and s_{sd} the separation,

$$\beta_{ssd} = \min\left(1, \frac{s_{sd} + L_c}{\Delta x}\right), \tag{4}$$

$$K_\theta = \begin{cases} B + E, & \cos\theta_f > 0.7, \\ B - E, & \cos\theta_f < -0.7, \\ 2A_e, & \text{otherwise}, \end{cases} \tag{5}$$

and the target-cell overlap factors are clipped to $[0, 1]$:

$$f_d = \text{clip}\left(\frac{K_\theta + \Delta x/2 - r\beta_{ssd}}{\Delta x}, 0, 1\right), \tag{6}$$

$$f_r = \text{clip}\left(\frac{HAZ + \Delta x/2 - r\beta_{ssd}c}{\Delta x}, 0, 1\right)(1 - f_d). \tag{7}$$

With $A_s = L_c^2$, $C_h = A_s/[\pi(cB)(cA_e)]$, default $c = 1$, target nonburnable fraction $n_b = 0$, and absorptivity $\alpha = 0.8$, the expected fields are

$$q''_{\text{DFC}} = g_f \eta(W)\, 0.65(1 - n_b) f_d\, \text{HRRPUA}(t) C_h, \tag{8}$$

$$q''_{rad} = g_f \eta(W) \frac{0.35(1 - n_b)\alpha f_r \text{HRRPUA}(t) A_s}{4\pi(r\beta_{ssd}c)^2}, \tag{9}$$

$$\eta(W) = \frac{3.25}{1 + \exp[-0.05(W - 75)]} + 0.5, \qquad g_f = \min\left[\left(\frac{\Delta x}{L_c}\right)^2, 1\right]. \tag{10}$$

The physical heat-affecting zone is $HAZ = 100$ m and its search radius is $\lceil HAZ/\Delta x \rceil$. Radiation is complementary to DFC through $(1 - f_d)$. The source instead uses a continuous angular ellipse, 0.3 heat factors, cell-area normalization, and a cell-count band; these are explicit deviation targets. Preprocessing assigns the prescribed parameter values independently to each simulation condition.

3 Derivation of expected behavior

The ellipse-regression coefficients and branch predicates were independently transcribed from the specified WU-E spread-rate formulation. The reviewed configuration interpretation, spread-rate calculation, and level-set formulation were identical to the specified ELMFIRE source revision; this identity claim does not extend to all other model components. Purnomo et al. describe the wind-responsive urban flame ellipse and heat-transfer coupling in the peer-reviewed WU–E publications [1, 2]. The detailed equations above document the intended implemented model directly. The executable snapshot is evaluated against, rather than substituted for, this formulation.

At $V = 10$ and $V = 17.3$ m s^{-1}, equality belongs to the quadratic branch. The immediate neighboring simulation conditions on each side must instead use the linear low- or high-wind coefficients. The $D = 75$ m field must be identical to $D = 50$ m. A raster parameter field equal to the constants must produce the same heat maps. With all other inputs invariant, rotating the wind-from direction by $90°$ must rotate the complete DFC-plus- radiation field by the same angle on this square grid.

The HRRPUA curve rises from zero to 100 kW m^{-2} over one second, remains at the peak through $t = 100$ s, decays linearly to zero at $t = 110$ s, and stays zero. The $t = 60$ s dump consequently gives an exact nonzero full-map comparison without conflating ellipse response with the HRR transient.

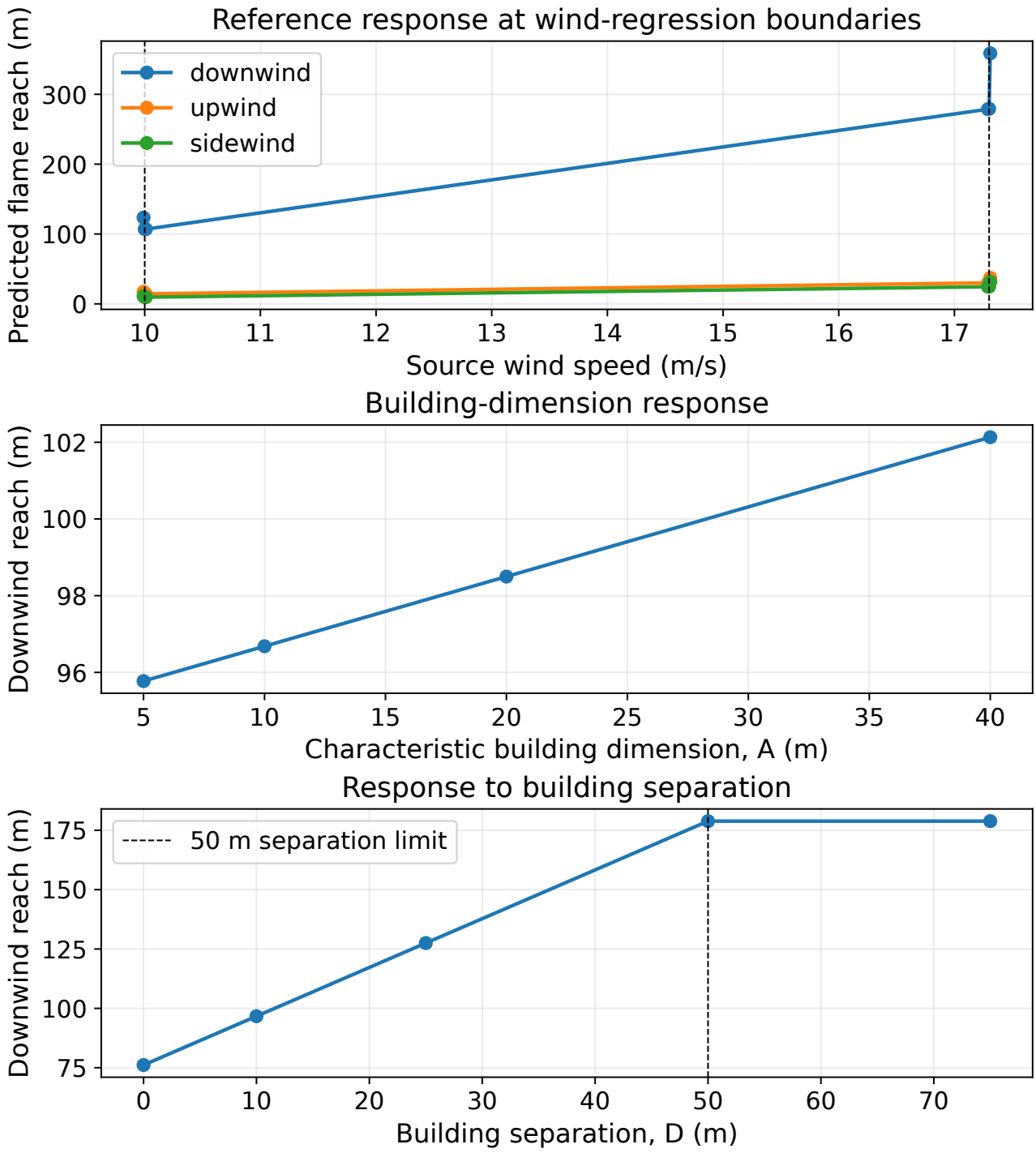


Figure 1: Intended-model reference calculation for the branch, characteristic building dimension, and separation sweeps. These are predicted values, not ELMFIRE output.

4 Verification metrics and acceptance criteria

For any reference raster R and ELMFIRE raster E, the primary metric is

$$\varepsilon_{L_1} = \frac{\sum_{i,j} |E_{ij} - R_{ij}|}{\sum_{i,j} |R_{ij}|}. \tag{11}$$

The sum covers every grid cell, including zeros outside the WUI update band; nodata and malformed rasters are rejected rather than omitted. HRRPUA and both heat fields use the same metric over all dump times and cells, which also detects an unintended secondary source and requires the final transient maps to be zero after ELMFIRE's step-level reset. A 0.5-percent map tolerance permits single- precision raster output while remaining much smaller than any intended branch or parameter response. Paired algebraic invariants use 10^{-5}; rotation uses the primary 0.5-percent tolerance because trigonometric evaluation occurs in single precision. Each simulation condition must have a confirmed successful completion with matching SHA-256 identities for the simulation settings, spatial inputs, both fuel-property tables, reference and analysis methods, ELMFIRE program, simulation record, and all results. Because ELMFIRE's first scheduled dump occurs after one scheduled output interval, the dump simulation specification must contain exactly 60 and 120 seconds, with a unique final record at the configured stop. Output labels alone cannot establish a valid simulation. The model-program identity is retained, but its relationship to the stated source revision must be established separately.

Expected property	Observable and metric	Criterion	Failure sensitivity
Piecewise source curve	All HRR transient rasters; normalized L_1	≤ 0.005	timing or source contamination
Ellipse heat kernel	Full DFC and radiation maps at peak HRRPUA	each ≤ 0.005	branches, coefficients, orientation
D clamp and input paths	Pairwise full-map normalized L_1	each $\leq 10^{-5}$	clamp or I/O divergence
Rotation covariance	Rotated DFC-plus-radiation maps	≤ 0.005	angle, sign, or raster orientation
Vegetation-source factor	Full DFC/radiation maps for the forest-source simulation condition	each ≤ 0.005	unused 1/3 factor or initial-source HRR suppression
No propagation	Final time of arrival (TOA) and off-source HRR cells	zero additional cells	confounding source creation

All 23 simulation conditions and all required fields must be present. Overall PASS requires every metric to pass; missing, duplicate, malformed, stale, or mistimed evidence produces NOT EVALUABLE.

5 Simulation configuration and predicted outcome

Parameter	Configured value	Isolation role
Grid	55 by 55, 20 m cells	full heat-kernel domain
CRS	EPSG:32610	explicit metric coordinates
Time	$\Delta t = 1$ s, stop 120 s, dump 60 s	exact peak-HRR dump
Initial source	one central fire behavior fuel model (FBFM) 91 cell	stationary WU-E source
Background	nonburnable FBFM93, adjustment 0	prevents propagation
Spotting/terrain	disabled/flat	removes confounders
Model selector	WUI model type 2	UMD/UCB implementation
Ellipse adjustment	1.0	intended default effective-flame factor

The generated constant fields and rasters use identical north-up georeferencing, nodata, dimensions, and data types. The point ignition scheduled at 121 s is deliberately after the 120 s stop; it prevents generic stalled-front termination without adding a physical source during the experiment.

The predicted output is a fixed-source HRR history and wind-oriented DFC and radiation fields. Branch-adjacent simulation conditions may jump because the source itself uses distinct regressions; the test checks that behavior rather than imposing continuity absent from the implementation.

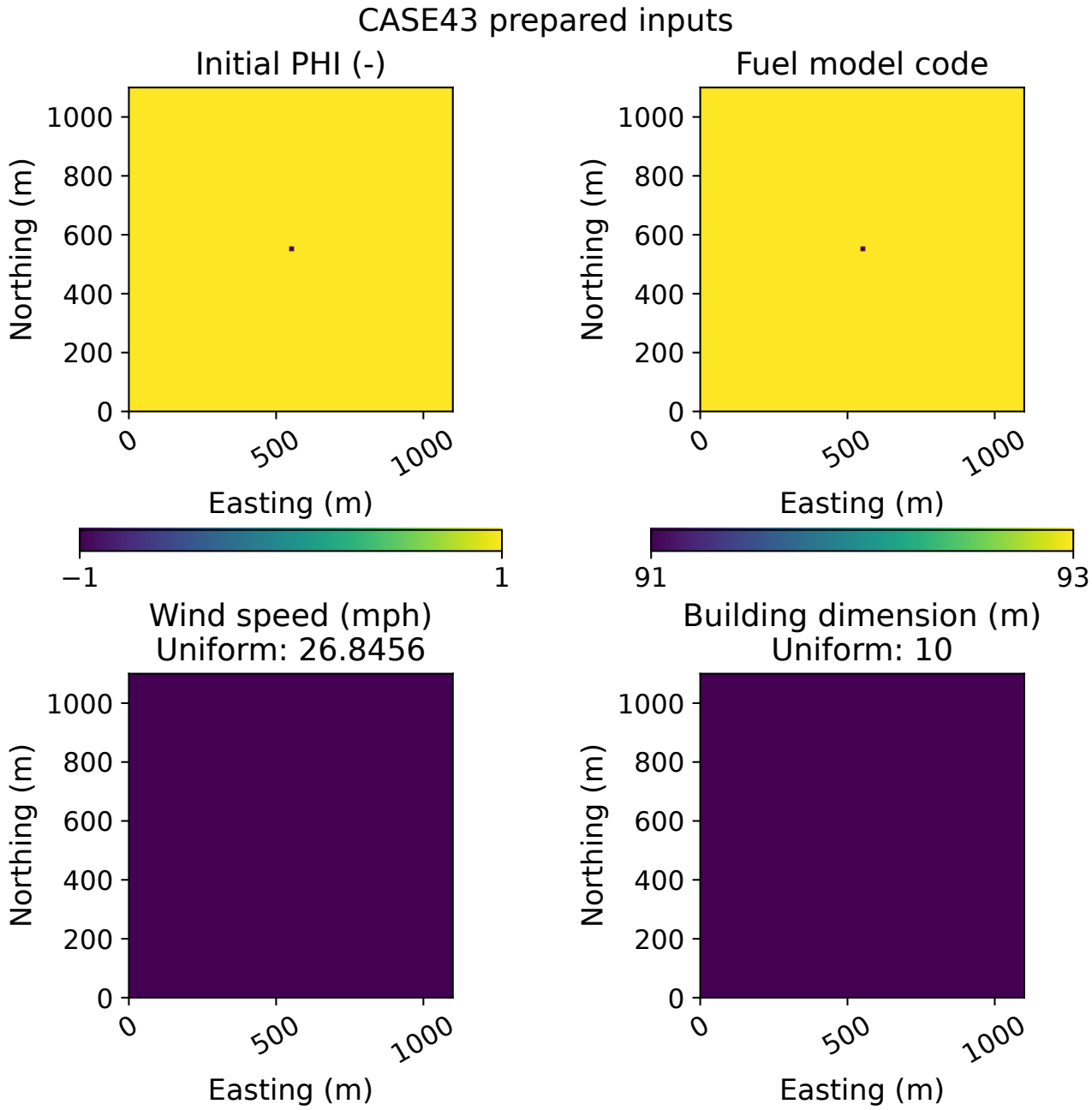


Figure 2: Whole-domain prepared configuration generated from the the 270-degree wind-direction condition GeoTIFF inputs. The single negative PHI cell is the initial WU-E source; FBFM93 and zero adjustment prevent secondary spread.

6 Actual simulation results

The current analysis procedure state is **COMPLETE**; the scientific decision is **PASS**, with 23 of 23 required simulation conditions having evaluable execution evidence. Reason: All required simulation conditions and acceptance metrics passed.. The common executed-binary SHA-256 is `ab47c5e338a3bc3b325df659b0764ce47d5b071e22d07817e4b5cab9f87da672`.

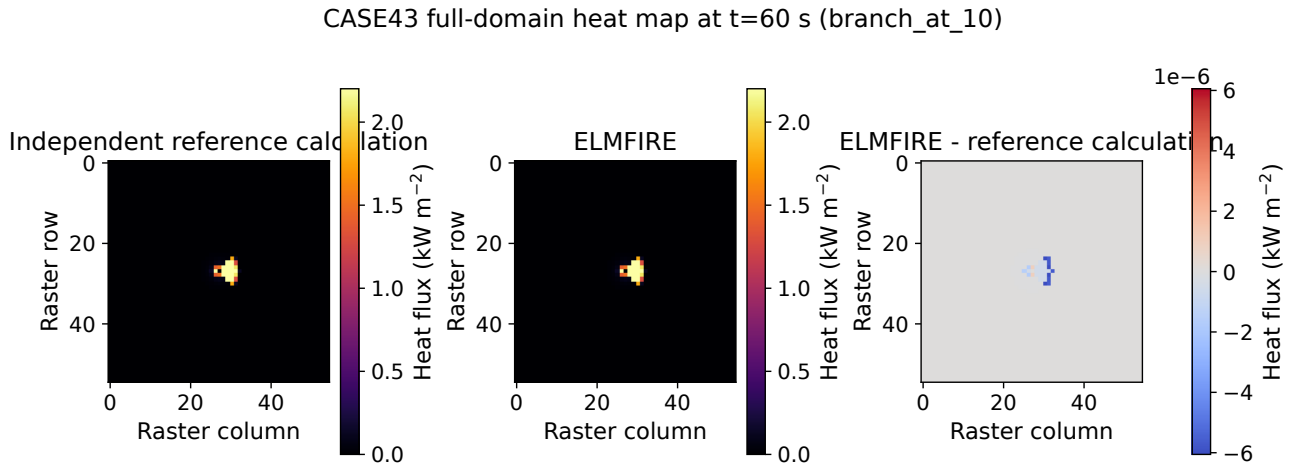


Figure 3: Representative whole-domain ELMFIRE result, independent reference calculation, and difference. This figure is generated only from completed case-specific output.

7 Calculated metrics and verification decision

Maximum HRRPUA normalized L1 error — PASS

Criterion ≤ 0.005

Observed *Units: fraction.*
0

Maximum DFC full-map normalized L1 error — PASS

Criterion ≤ 0.005

Observed *Units: fraction.*
3.02831e-06

Maximum radiation full-map normalized L1 error — PASS

Criterion ≤ 0.005

Observed *Units: fraction.*
1.09758e-06

D greater than 50 m clamp pair error — PASS

Criterion $\leq$ 1e-05

Observed *Units: fraction.*
0

Constant/raster parameter equivalence error — PASS

Criterion $\leq$ 1e-05

Observed *Units: fraction.*
0

Maximum 90-degree covariance error — PASS

Criterion ≤ 0.005

Observed *Units: fraction.*
0

Vegetation-source fuel-factor heat error — PASS

Criterion ≤ 0.005

Observed *Units: fraction.*
1.67242e-06

Maximum arrival-mask mismatch count — PASS

Criterion 0

Observed *Units: cells.*
0

Maximum unexpected HRR source count — PASS

Criterion 0

Observed *Units: cells.*
0

Current decision: PASS. NOT RUN is a analysis procedure state, not evidence of verification and never maps to PASS.

8 Technical interpretation

The current result is limited by missing time-history evidence, rather than a calculated heat-map discrepancy. The solver's low-tagged-node stopping rule terminates this isolated-source setup before the required scheduled heat dumps. The post-stop ignition disables a different, stalled-PHI shortcut; it does not bypass the low-node rule. The lone overrun dump cannot replace the missing history and is not interpreted as a measurement at the configured stop time.

The intended heat kernel also differs from the reviewed source in its angular sectors, heat fractions, wind efficiency, and spatial normalization. These source differences identify what a complete future run should test; they are not numerical verification results from the incomplete run. After the source can sustain a stationary thermal calculation, passing HRR with failing heat maps would isolate the discrepancy to ellipse geometry or heat transfer.

9 Scope and limitations

This case covers the deterministic model-2 source ellipse and transient heat distribution for a stationary structure. It does not verify physical calibration, propagation speed, ignition thresholds, embers, interface transition, fuel consumption, or Statewide WU-E. It intentionally treats the pinned source as the implementation under test and excludes the distinct Statewide WU–E formulation. CASE44 addresses the HRR and heat-parameter response not isolated here.

References

[1] D. M. J. Purnomo et al., "Reconstructing modes of destruction in wildland–urban interface fires using a semi-physical level-set model," *Proceedings of the Combustion Institute*, 40 (2024), 105755. doi:10.1016/j.proci.2024.105755.

[2] D. M. J. Purnomo et al., "Integrating an urban fire model into an operational wildland fire model to simulate one dimensional wildland–urban interface fires: a parametric study," *International Journal of Wildland Fire*, 33 (2024), WF24102. doi:10.1071/WF24102.

Case 44: WU-E heat-parameter response

1 Verification purpose

This case verifies the response of the intended downscaled Wildland–Urban ELMFIRE (WU–E) model-2 heat calculation to the design-fire curve, target building-fuel properties, heat-release-rate (HRR) ellipse adjustment, the radiation cutoff, and multiple simultaneous sources. It is designed to expose breakpoint errors, confusion between a plateau duration and its absolute end time, use of the wrong target-property path, incorrect heat-area normalization, missing cutoff logic, a failure to accumulate transient heat, or nonlinear source aggregation.

Nineteen deterministic simulation conditions use one or two stationary WU-E sources. Nonburnable fire behavior fuel model (FBFM) 93 background and a zero spread-adjustment raster suppress propagation; spotting, terrain, and weather variation are disabled. The intended downscaled formulation is the numerical acceptance reference. The reviewed configuration interpretation, spread-rate calculation, and level-set formulation were identical to the specified ELMFIRE source revision; this identity claim does not extend to all other model components. Differences between the simulated response and the intended equations below are reported as deviations.

Verification question. Do all simulation conditions reproduce the complete time-dependent heat-release rate per unit area (HRRPUA), direct-flame-contact (DFC), radiation, and accumulated-heat fields within 0.005 normalized L_1 error, while satisfying the exact parameter scalings, radiation support, source-superposition, and no-propagation invariants?

Current result: FAIL. 19 of 19 required simulation conditions have evaluable evidence. Analysis procedure: **COMPLETE.**

2 Mathematical formulation and numerical implementation

For elapsed burning time t, the implementation reads four values from the source building-fuel-model row: growth endpoint t_e, absolute plateau endpoint t_f, decay endpoint t_d, and peak HRRPUA H. It evaluates

$$\dot{q}''_{\mathrm{HRR}}(t) = \begin{cases} Ht/t_e, & t \le t_e, \\ H, & t_e < t \le t_f, \\ H(t-t_d)/(t_f-t_d), & t_f < t \le t_d, \\ 0, & t > t_d. \end{cases} \tag{1}$$

The fixed wind and geometry determine the UCB ellipse once per source. With linear characteristic building dimension $L_c = \texttt{BLDG_AREA}$ in metres, source area $A_s = L_c^2$, ellipse major and minor radii B and A_e, and $c = \texttt{HRR_ELLIPSE_ADJ}$, the intended spatial normalization is

$$C_h = \frac{A_s}{\pi(cB)(cA_e)}, \qquad c_{default} = 1. \tag{2}$$

For target cell j, define $\beta_{ssd} = \min[1, (s_{sd} + L_c)/\Delta x]$ and select $K_\theta = B + E$, $B - E$, or $2A_e$ for downwind, upwind, or sidewind sectors. The clipped DFC coverage is $f_d = \mathrm{clip}[(K_\theta + \Delta x/2 - r\beta_{ssd})/\Delta x, 0, 1]$. Radiation uses $r_{eff} = r\beta_{ssd}c$, physical $HAZ = 100$ m, and the complementary factor $(1 - f_d)$. Spatial nonburnable fraction n_j takes precedence when supplied; the building-model/default value is the fallback. The table supplies absorptivity α_j. Each source s contributes

$$q''_{\text{DFC},s,j} = g_f \eta(W) 0.65(1-n_j) f_{d,s,j}\, \dot{q}''_{\text{HRR},s}(t) C_h, \tag{3}$$

$$q''_{\text{rad},s,j} = g_f \eta(W) \frac{0.35(1-n_j)\alpha_j f_{r,s,j} \dot{q}''_{\text{HRR},s}(t) A_s}{4\pi r^2_{\text{eff},s,j}}, \tag{4}$$

$$\eta(W) = \frac{3.25}{1+\exp[-0.05(W-75)]} + 0.5, \qquad g_f = \min[(\Delta x/L_c)^2, 1]. \tag{5}$$

Radiation overlap is exactly zero outside the physical heat-affecting zone, with half-cell clipping. Transient maps sum all burning sources. Total received heat is updated every one-second solver step as

$$Q_j^{\text{total}} \mathrel{+}= q''_j(t)\, \Delta t\, \Delta x^2. \tag{6}$$

The independent case-specific Python reference calculation implements these equations directly; it imports no ELMFIRE code and has no execution source-tree dependency. The pinned source currently omits η, uses 0.3 rather than 0.35 radiation, omits the 0.65 DFC factor and grid corrections, and uses cell area in C_h.

3 Derivation of expected behavior

The four source curves deliberately sample every breakpoint. The base curve uses $(t_e, t_f, t_d, H) = (4, 8, 12, 100)$ in SI units; the peak simulation condition doubles H, the short curve uses $(2, 4, 6, 100)$, and the long-plateau curve uses $(4, 10, 14, 100)$. Because every heat term is linear in HRRPUA, doubling H must double the HRR and both heat-flux maps at every dump.

At fixed geometry, DFC scales with spatial $1 - n_j$, while radiation scales with $(1 - n_j)\alpha_j$. The spatial input therefore predicts exact response ratios for $n_j \in \{0, 0.2, 0.5, 1\}$ and $\alpha_j \in \{0, 0.4, 0.8, 1\}$. DFC must not change in the absorptivity sweep. At fixed B and A_e, both heat fields scale as c^{-2} for $c \in \{0.25, 0.5, 0.75, 1\}$.

The sum from a left and a right source must equal the two-source result cell by cell and dump by dump. Likewise, the final accumulated rasters must equal the sum of every one-second transient raster times $\Delta t \Delta x^2$. Finally, two simulation conditions change the spatial nonburnable-fraction raster from zero to one while retaining the same building-fuel-model table. The fully nonburnable target must receive zero DFC and radiation; equality would expose the current source's table-only behavior.

The peer-reviewed WU–E papers support the transient design-fire, target heat, and multi-source coupling concepts [1, 2]. The detailed formulation above is stated explicitly. The ends of growth, fully developed burning, and decay are specified as absolute transition times.

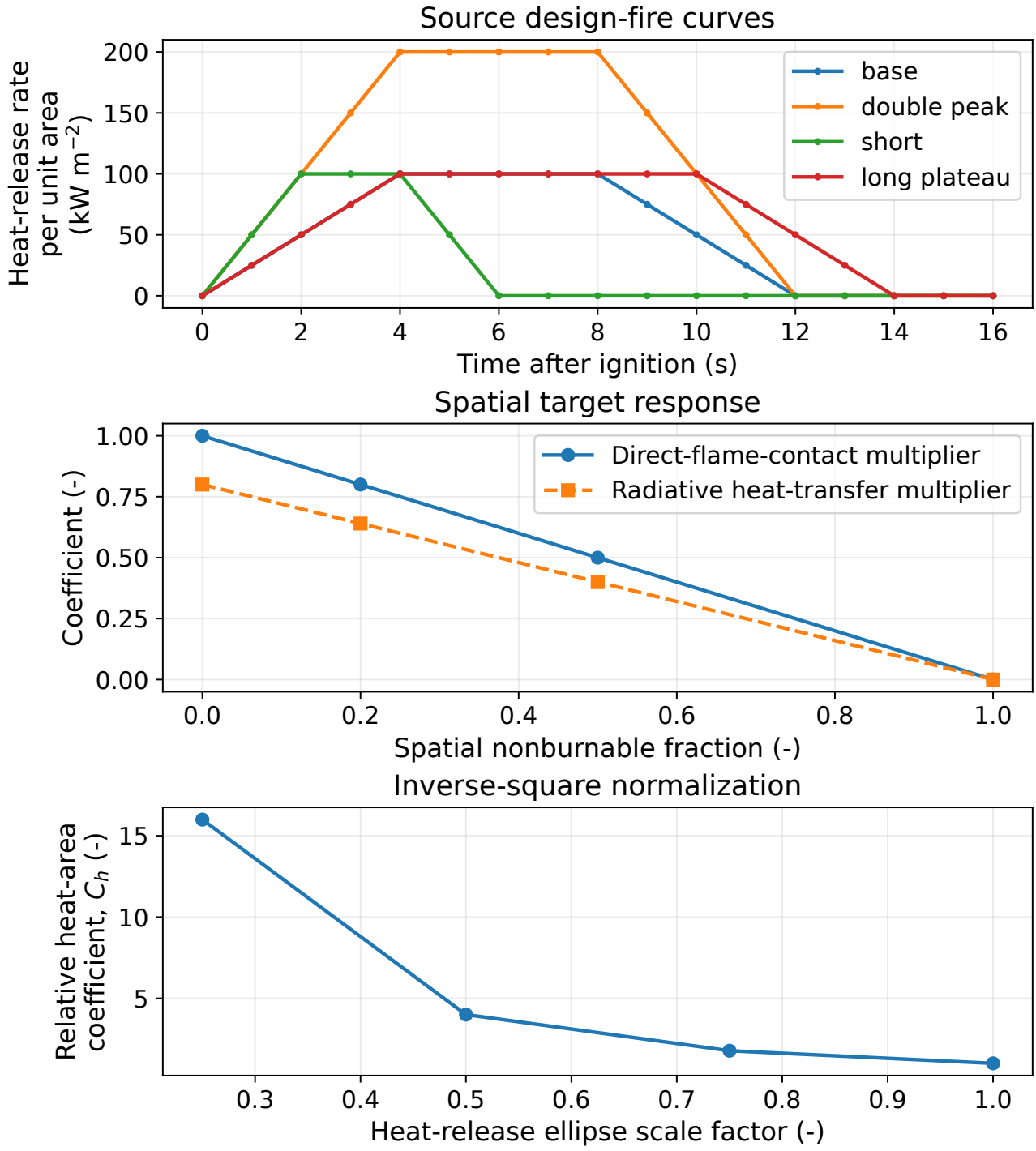


Figure 1: Independent predictions for the HRR curve families, spatial target coefficients, and inverse-square ellipse normalization. These curves are generated before, and independently of, an ELMFIRE run.

4 Verification metrics and acceptance criteria

For nonzero reference array R and output array E, the principal metric is

$$\epsilon_{L_1} = \frac{\sum |E - R|}{\sum |R|}. \tag{7}$$

The sums cover all 16 scheduled dump times and every raster cell. A zero-reference array uses absolute L_1 error, so a supposedly suppressed component cannot pass through a zero denominator. Nodata, non-finite values, unexpected shapes, duplicate result fields, and output times other than exactly 1 through 16 seconds are rejected. The prescribed HRR curves include $t = 0$, but ELMFIRE's first scheduled output occurs after one scheduled output interval. A confirmed successful completion must have matching SHA-256 identities for the simulation settings, spatial inputs, both fuel-property tables, reference and analysis methods, ELMFIRE program, simulation record, and all results. Output labels alone are not sufficient evidence. The relationship between the identified model program and the stated source revision must be established separately.

Expected property	Observable and metric	Criterion	Failure sensitivity
HRR breakpoints	Full HRR raster stack versus reference calculation	$\epsilon_{L_1} \leq 0.005$	timing, curve row, peak
Spatial heat fields	Full DFC/radiation stacks versus reference calculation	each $\epsilon_{L_1} \leq 0.005$	coefficients, cutoff, kernel
Exact parameter response	Scaled-pair raster errors	each $\leq 10^{-5}$	spatial n_b, absorptivity, or c^{-2}
Time accumulation	Transient sums versus final totals	$\epsilon_{L_1} \leq 0.005$	update/dump ordering
Source aggregation	Pair versus sum of two singles	$\leq 10^{-5}$	aggregation defect
Isolation	time of arrival (TOA) and off-source HRR	zero extra cells	unintended propagation

Overall PASS requires every metric and all 19 simulation conditions. Incomplete, stale, ambiguous, or malformed evidence yields NOT EVALUABLE, never PASS.

5 Simulation configuration and predicted outcome

Parameter	Configured value	Isolation role
Grid	55 by 55, 20 m cells	contains full update band
CRS	EPSG:32610	explicit metric coordinates

Time	$\Delta t = 1$ s; stop 16 s; dump 1 s	exact breakpoints/integration
Wind	12 m/s toward raster east	fixed ellipse geometry
Sources	one center or two separated cells	single/superposition pair
Background	FBFM93 with zero adjustment	prevents propagation
Spotting/terrain	disabled/flat	removes confounders
Model selector	WUI model type 2	UMD/UCB source path

The point ignition is scheduled at 17 seconds, just beyond the 16-second stop. It prevents generic stalled-front termination without adding a physical source during the test. All input rasters are regenerated with common dimensions, CRS, transform, nodata value, and explicit data types.

The predicted result is a stationary source set whose heat maps follow each source curve, vanish beyond the implementation's support, add linearly, and integrate exactly into the final received-heat fields.

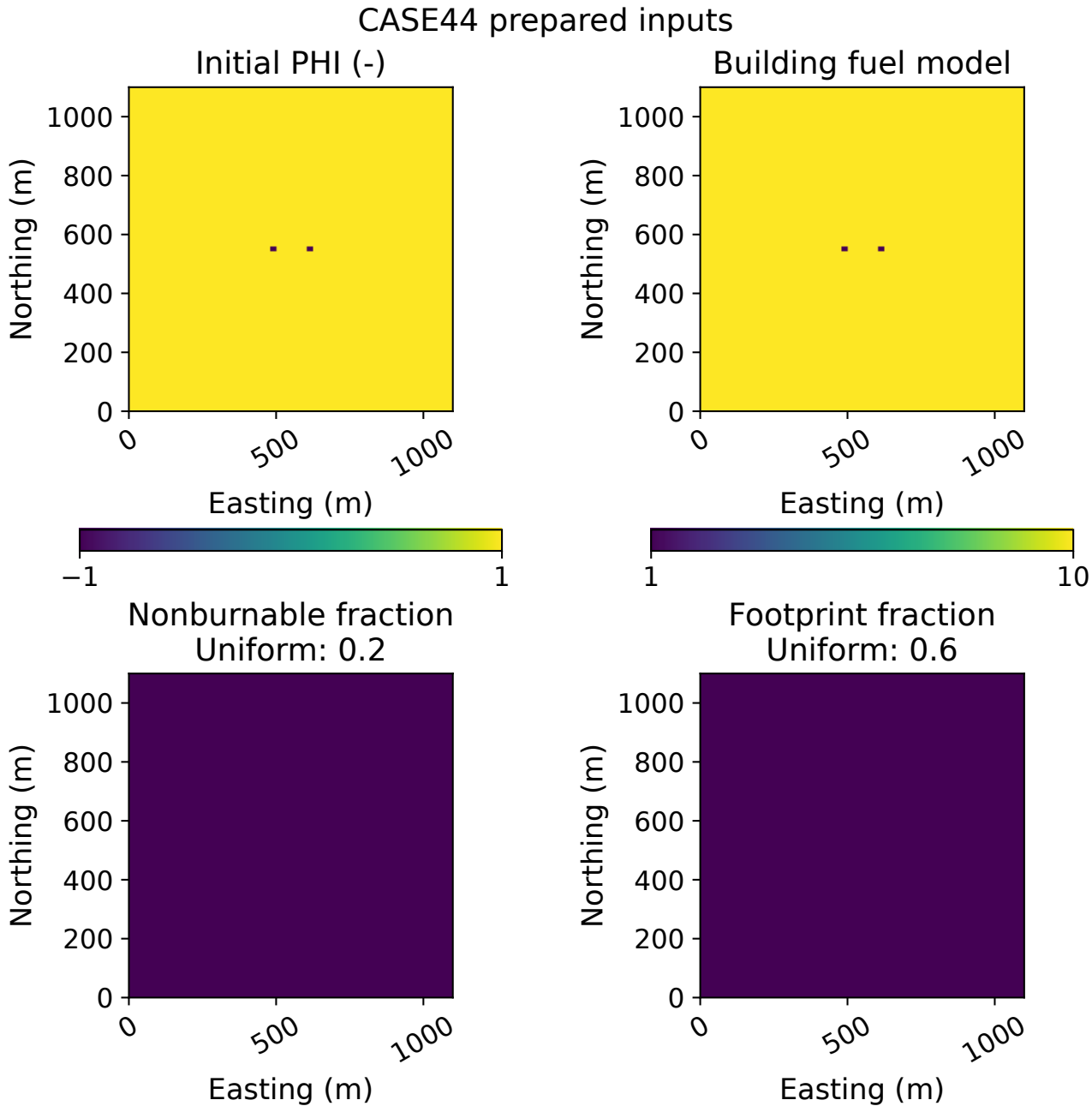


Figure 2: Prepared two-source configuration read back from generated GeoTIFFs. Source model IDs override the uniform target-table ID only at the two initially burning cells.

6 Actual simulation results

The current analysis procedure state is **COMPLETE**; the scientific decision is **FAIL**, with 19 of 19 required simulation conditions having evaluable execution evidence. Reason: One or more acceptance metrics failed.. The common executed-binary SHA-256 is `2b58920716af28891d9166c5f1046174214994eb20e820981017e4f2d166b001`.

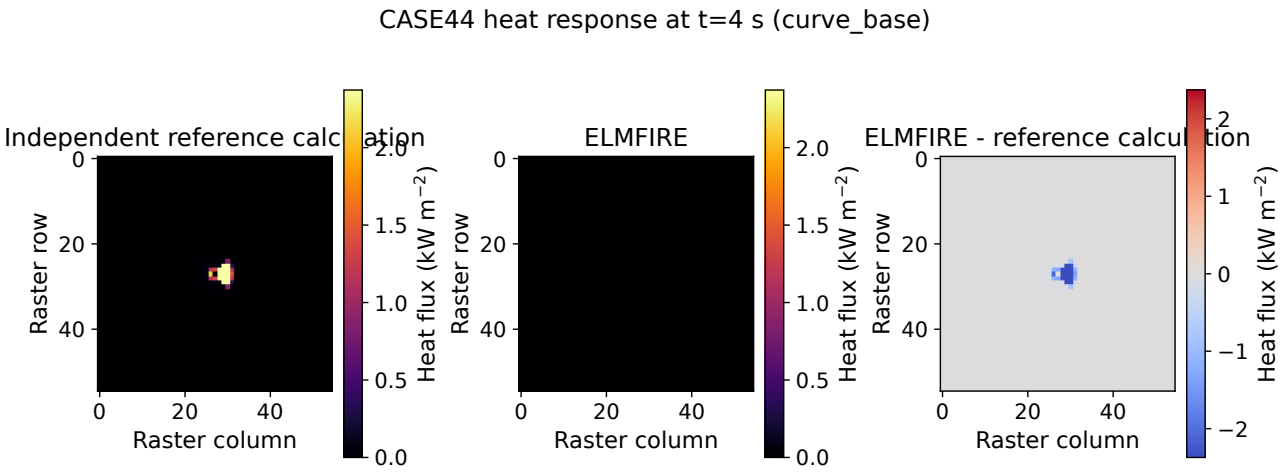


Figure 3: Representative whole-domain ELMFIRE result, independent reference calculation, and difference. This figure is generated only from completed output.

7 Calculated metrics and verification decision

Maximum HRR curve normalized L1 error — FAIL

Criterion	$\leq$ 0.005
Observed	*Units: fraction.* 1

Double-peak HRR scaling error — PASS

Criterion	$\leq$ 1e-05
Observed	*Units: fraction.* 0

Maximum DFC full-stack normalized L1 error — FAIL

Criterion	$\leq$ 0.005
Observed	*Units: fraction.* 1

Maximum radiation full-stack normalized L1 error — FAIL

Criterion	$\leq$ 0.005
Observed	*Units: fraction.* 1

Spatial nonburnable scaling error — PASS

Criterion	$\leq$ 1e-05
Observed	*Units: fraction.* 0

Target-table absorptivity scaling error — PASS

Criterion	$\leq$ 1e-05
Observed	*Units: fraction.* 0

DFC Heat-release ellipse scale factor inverse-square error — PASS

Criterion	$\leq$ 1e-05
Observed	*Units: fraction.* 0

Maximum radiation outside source cutoff — PASS

Criterion	$\leq$ 1e-06
Observed	*Units: kW m^-2.* 0

Maximum transient-to-total integration error — PASS

Criterion	$\leq$ 0.005
Observed	*Units: fraction.* 0

Two-source superposition error — PASS	
Criterion	$\leq$ 1e-05
Observed	*Units: fraction.* 0
Spatial nonburnable-raster response — PASS	
Criterion	fully nonburnable / combustible heat $\leq$ 1e-05
Observed	*Units: ratio.* 0
Maximum arrival-mask mismatch count — PASS	
Criterion	0
Observed	*Units: cells.* 0
Maximum unexpected HRR source count — PASS	
Criterion	0
Observed	*Units: cells.* 0

Current decision: FAIL. NOT RUN is a analysis procedure state, not evidence of verification and never maps to PASS.

8 Technical interpretation

The required transient stacks are unavailable because the solver stops when this one- or two-source setup has too few tagged propagation cells. The dormant ignition controls the separate PHI-stall test, not this low-node stopping rule. An overrun final dump does not contain the missing rise, plateau, and decay history, so the current decision remains NOT EVALUABLE.

Once complete histories exist, the comparisons separate distinct causes: HRR errors indicate curve timing or source-state problems; correct HRR with incorrect heat maps indicates transfer coefficients, support, or normalization; a transient-to-total mismatch indicates accumulation or dump timing. The spatial-nonburnable and source-superposition controls test their respective input and aggregation paths without substituting planned values for output.

9 Scope and limitations

This case verifies deterministic source HRR and heat-transfer mechanics for a fixed model-2 ellipse. It does not verify physical calibration, flame-shape regressions across wind and geometry (CASE43), ignition thresholds, rate of spread (ROS), wildland–urban transitions, embers, fuel consumption, or convergence. Its intended spatial-property interpretation is tested against the pinned source revision. The reference calculation is evaluated independently of the ELMFIRE simulation.

References

[1] D. M. J. Purnomo et al., “Reconstructing modes of destruction in wildland–urban interface fires using a semi-physical level-set model,” *Proceedings of the Combustion Institute*, 40 (2024), 105755. doi:10.1016/j.proci.2024.105755.

[2] D. M. J. Purnomo et al., “Integrating an urban fire model into an operational wildland fire model to simulate one dimensional wildland–urban interface fires: a parametric study,” *International Journal of Wildland Fire*, 33 (2024), WF24102. doi:10.1071/WF24102.

Case 45: WU-E heat-to-urban rate-of-spread response

1 Verification purpose

This manufactured coupling test verifies the Wildland–Urban ELMFIRE (WU–E) model's conversion of heat received by an urban cell into its local level-set rate of spread (ROS). It exercises the heat-driven building-spread formulation and subsequent construction of the directional spread velocity. The intended reference calculation uses a fixed critical material fire-thermal property (FTP); it is sensitive to energy or unit normalization, ellipse partitioning, normal-vector orientation, an artificial 30,000-kJ or 35-mph discontinuity, and failure to honor the material table. The source revision reviewed for the verification specification is `a2dfbcdf72209733c000e5d3431e44723e15efea`.

The five reviewed model components were identical to that source revision. The simulation, output, and initialization procedures contained other changes outside the reviewed components. Consequently, the design review establishes the provenance of the reference formulation, not the identity of every component in a subsequently supplied ELMFIRE version.

One burning urban source and one adjacent, initially positive urban receiver form the measured pair in a nonburnable domain. Four disconnected urban guard cells keep the tagged-list termination guard inactive but remain far from the front and are not observables. Per-step energy, wind, receiver normal, design-fire amplitude, and table critical material fire-thermal property are varied; spotting, crown fire, terrain, moisture gradients, and stochastic forcing are disabled. This is implementation verification, not evidence that the empirical response is physically valid. The verification question is: for all 12 simulation conditions, does the executable's receiver-cell spread-rate raster agree within 0.005 relative error with the independently recomputed mapping driven by its own recorded direct flame contact (DFC)-plus-radiation stimulus, while also satisfying the continuity, directional, amplitude, and FTP-response comparisons?

Current result: PASS. 12 of 12 required simulation conditions have evaluable evidence. Analysis procedure: **COMPLETE.**

2 Mathematical formulation and numerical implementation

At one receiver the executable records transient heat fluxes q''_{DFC} and q''_{rad} in kW m^{-2}. For timestep Δt and square cell width Δx, the downstream routine forms

$$E = (q''_{\mathrm{DFC}} + q''_{\mathrm{rad}})\Delta t \Delta x^2, \qquad q_A = E/\Delta x^2.$$

Because kilowatts are kilojoules per second, E is in kilojoules. The intended downscaled model uses the material value $F_{TP,crit}$ without an energy-dependent replacement or wind-speed divisor. The absolute velocity in feet per minute is

$$U = \min\left(10^5, \frac{60E}{0.3048\,\Delta t\,\Delta x\,F_{TP,crit}}\right) = \min\left(10^5, \frac{196.85\,q''\Delta x}{F_{TP,crit}}\right).$$

The nominal default is $F_{TP,crit} = 5400$ kJ m^{-2}; the case also uses 3000 and 6000 to verify the inverse response. The peer-reviewed papers describe the accumulated-heat/FTP ignition and heat-derived urban spread concept [1, 2]. The pinned source instead replaces FTP below/above 30,000 kJ and applies a 1.8 divisor at and below 35 mph. Those operations are documented deviations, not acceptance equations.

The UCB ellipse supplies major radius a, minor radius b, and focus offset e. The three response weights are

$$f_h = a + e, \qquad f_s = 2b, \qquad f_b = a - e,$$

so the nominal head, side, and back velocities are $V_i = U f_i/(f_h + f_s + f_b)$. The case-specific evaluator then applies the same closed ellipse-normal construction, rather than treating these three labels as three scalar shortcuts. In full, for front normal $\boldsymbol{n} = (n_x, n_y)$ and direction-of-maximum-spread unit vector $\boldsymbol{s} = (s_x, s_y)$, the implementation evaluates

$$c = n_x s_x + n_y s_y, \qquad s = n_x s_y - n_y s_x,$$

$$L = \begin{cases} \min((V_h + V_b)/(2V_s), 10)\,, & V_s > 10^{-4}, \\ 1, & V_s \le 10^{-4}, \end{cases}$$

$$A = \max\left(\frac{V_h + V_b}{2}, 10^{-10}\right), \qquad B = \frac{1}{2}\max\left(\frac{V_h + V_b}{L}, 10^{-10}\right),$$

$$D = \max\left(\sqrt{A^2c^2 + B^2s^2}, 10^{-10}\right), \quad v_{\parallel} = \frac{A^2 c}{D} + \frac{V_h - V_b}{2}, \quad v_{\perp} = \frac{B^2 s}{D},$$

$$v_x = v_{\parallel} s_x + v_{\perp} s_y, \qquad v_y = v_{\parallel} s_y - v_{\perp} s_x, \qquad V = \sqrt{v_x^2 + v_y^2}.$$

The flat grid makes the source projection factors unity. Wind is from 270°, hence $\boldsymbol{s} = (1, 0)$. The manufactured target normals are east (head), west (back), and north (side); raster rows run north to south while the stored mathematical normal is expressed in east/north coordinates.

The upstream heat values are recorded stimuli for this case; CASE43–44 verify their governing equations. Here characteristic building dimension=20 denotes the source's linear characteristic building dimension in metres, not an area in square metres. The input generator chooses design-fire amplitudes deterministically, but the downstream reference calculation always uses the heat actually written by ELMFIRE. For traceability, the pinned source currently uses the following non-normative normalization while constructing that stimulus:

$$C_H = \frac{\Delta x^2}{\pi (ca)(cb)}, \qquad c = 0.5.$$

For target distance R and directional ellipse distance R_e, the input generator transcribes the overlap and radiation weights

$$C_D = \mathrm{clip}_{[0,1]}\left(\frac{R_e + \Delta x/2 - R}{\Delta x}\right), \qquad C_R = (1 - C_D)\,\mathrm{clip}_{[0,1]}\left(\frac{R_e + 100 + \Delta x/2 - R}{\Delta x}\right).$$

With the implementation's piecewise effective radiation distance R_{eff}, emissivity $\varepsilon = 0.89$, and radiative fraction 0.3, the planned plateau heat is

$$q'' = C_H H_p \left[C_D + \frac{0.3\,\varepsilon C_R \Delta x^2}{4\pi R_{\text{eff}}^2}\right].$$

The generator solves this relation separately for each normal to set the input amplitude. The downstream decision never substitutes planned heat for the recorded heat.

3 Derivation of expected behavior

The energy triplet is constructed at 29900, 30000, and 30100 kJ. The generator uses an explicit IEEE-754 binary32 transcription of the reviewed default-single-precision arithmetic operation sequence to select decimal table amplitudes; the planned realization is exact at all three values, including equality, and is recorded in each simulation specification. Executable heat output remains required to establish the

realized stimulus. With fixed FTP, local speed divided by recorded energy must be continuous across 30,000 kJ. Likewise, the 34.9, 35.0, and 35.1-mph cases include the small physical ellipse change but no artificial divisor jump. A discontinuity at either boundary identifies the known source-only branches.

The terminal spread-rate raster is populated only from the set of cells recorded as burned, so an arbitrary positive receiver would not be observable. The prepared receiver value is instead $\phi_r = 0.001 > 0$. Both source raster readers unconditionally add a perturbation $10^{-3}(0.5 - R)$, with $R \in [0, 1)$; wind-fluctuation settings do not disable it. Thus the receiver remains strictly positive, between 0.0005 and 0.0015, before propagation. The former 2^{-20} value was smaller than this perturbation and could not guarantee initial state. For the side receiver, its five-cell tangential PHI stencil is flat at ϕ_r, removing an artificial tangential local minimum. These four extra stencil cells remain nonburnable and initially positive.

The crossing estimate uses $\boldsymbol{v} \cdot \nabla\phi$, not speed magnitude times a one-dimensional gradient. Its source-facing Superbee gradient is $G = 2(\phi_r - \phi_s)/\Delta x = 0.0451$ per metre. With $\epsilon = 0.0005$, the generator checks the conservative first-stage bound

$$\tfrac{1}{2}\Delta t\,[v_n(G - 4\epsilon/\Delta x) - U(4\epsilon/\Delta x)] > \phi_r + \epsilon,$$

where v_n is normal velocity and U bounds the velocity magnitude, both in metres per second. The bound covers input-noise gradients and discards second-stage inward transport. It is an intended-law setup check; actual arrival remains independently required. All twelve receivers must enter the set of cells recorded as burned at exactly 1 s, with a finite terminal speed. Four positive guard cells keep more than two nodes tagged, and a scheduled ignition at 5 s—strictly after the 4 s stop—keeps the first-step stall predicate false without ever altering the simulated state.

At fixed geometry, doubling versus halving H_p gives a heat-flux ratio of four. The two critical material fire-thermal property simulation conditions use 3000 and 6000 kJ m^{-2} and are otherwise equivalent. Their heat fields must match, while their spread speeds must have the inverse FTP ratio $V_{3000}/V_{6000} = 2$. The equal-energy condition with 35-mph wind, the back-facing receiver, and the side-facing receiver simulation conditions share 35-mph wind and the same designed received energy; they are the matched head, back, and side comparison, leaving only the prescribed receiver normal to select the directional response.

This is a recorded-input reference calculation: the emitted/received heat field at the exact receiver-arrival dump is a measured stimulus, while only the downstream heat-to-ROS calculation is the object of the numerical comparison. It therefore avoids claiming that one executable output independently verifies another upstream heat calculation. Ellipse axes, E as held in working storage, F, absolute U, component velocities, and ellipse length-to-width ratio are internal-only quantities. They are calculated by the evaluator but cannot be asserted as separately observed executable outputs.

4 Verification metrics and acceptance criteria

Before evaluation, the simulation settings, spatial inputs, fuel properties, reference calculations, and analysis method must match those associated with the completed simulation. Their recorded SHA-256 identities are checked, together with the full simulation record and model results. All 12 simulation conditions must have completed using the same identified ELMFIRE program. Any subsequent change to this evidence invalidates the comparison.

With output at every timestep, the exact ledger is dump indices 1–5 at times 0, 1, 2, 3, and 4 s, with only index 5 final. Transient heat fields are identified by output index, whereas terminal arrival-time and spread-rate fields are identified by physical time, 4 s. The final output index 5 is not interpreted as an elapsed time. Evaluation selects DFC plus radiation at the unique dump whose time equals the

receiver's required 1-s time of arrival (TOA), and compares the resulting independent local-speed reference calculation with the terminal receiver speed captured when that cell entered the set of cells recorded as burned. The exact-stop transient heat values must be reset to zero. A missing or stale marker, input identity record or snapshot mismatch, nodata, nonfinite, missing, duplicate, stale, or off-time supporting result makes the case NOT EVALUABLE.

Property	Observable and metric	Selection rationale	Acceptance basis
Heat-to-ROS mapping	maximum relative receiver-speed error	Tests the downstream formula using recorded heat	≤ 0.005, allowing float raster serialization
Receiver observability	TOA and finite terminal speed for every receiver	Proves the compared value was serialized from the set of cells recorded as burned	TOA exactly 1 s within 10^{-6} s; all speeds finite
Fixed-FTP energy continuity	normalized speed for 29900/30000/30100	Detects the non-normative source energy branch	spread ≤ 0.005; energy within 0.2% of design
Wind continuity	errors for 34.9/35.0/35.1 mph	Detects the non-normative 35-mph divisor	all speed errors ≤ 0.005
heat-release rate (HRR) amplitude	recorded double/half heat ratio	Direct metamorphic check with no assumed absolute output	4.00 ± 0.04
Directional mapping	maximum error for the matched 35-mph head/back/side set	Detects orientation and ellipse partition defects	≤ 0.005
critical material fire-thermal property inverse response	speed ratio for 3000/6000	Requires the material table to control spread	2.00 ± 0.01

PASS requires complete evidence and every row to pass. Any calculated breach is FAIL; absent or ambiguous evidence is NOT EVALUABLE and never a pass.

5 Simulation configuration and predicted outcome

The Courant–Friedrichs–Lewy (CFL) number measures the distance advected during one timestep relative to the grid spacing and constrains the time discretization.

Parameter	Configured value	Units	Isolation role

Grid	25×25, EPSG:32610	–	source well away from boundary
Cell width	20	m	fixed energy normalization and CFL-safe local response
Fuel geometry	source, adjacent receiver, four diagonal guard cells are fire behavior fuel model (FBFM) 91; FBFM 93 elsewhere	category	observable pair plus nonburning termination guards
Initial ϕ	source -0.45, receiver 0.001 (side tangential stencil also 0.001), guards/background (+1)	–	first-heated-step receiver crossing with guards far from the front
Building characteristic dimension/separation	20 / 10	m	fixed linear ellipse inputs
Wind direction	from 270	degree	eastward DMS axis
Design-fire times	1 / 100 / 120	s	terminal output remains on plateau
Time controls (step, maximum, stop)	1 / 1 / 4	s	exact terminal alignment
Dormant ignition	source centre at 5	s	suppresses false first-step stall; occurs after stop
Target CFL / displacement bound	0.10 / 2.0	– / m step^{-1}	all manufactured absolute-speed estimates satisfy the bound
Output cadence	every step	–	preserves the applied heat stimulus
Spread model	WU–E building type 2	–	selects heat-driven urban ROS
Other mechanisms	spotting/crown/stochastic off	–	removes confounders

Figure 1 shows the prescribed fields for the representative equal-energy condition. The bar panel is explicitly a design summary, not simulation output. The predicted output is receiver TOA exactly 1 s and a finite terminal receiver speed recorded when the cell becomes burned on that step. Its magnitude changes continuously with heat and follows the matched head/back/side ellipse-normal mapping. The FTP pair is predicted to differ by a factor of two; the terminal time-stamped heat field is predicted to be zero after reset and is not substituted for the arrival-step stimulus.

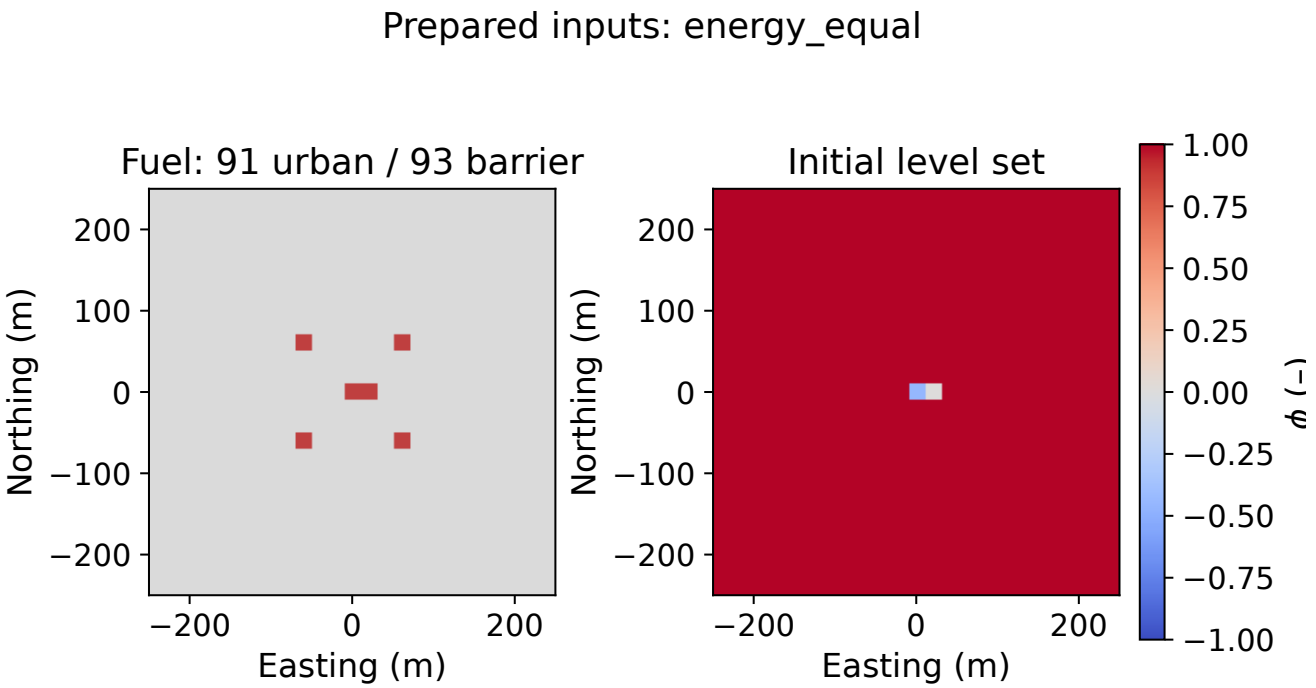


Figure 1: Whole-domain prepared fuel and initial-level-set fields. The separate energy figure shows the designed energy sweep. Coordinates are metres in EPSG:32610; the fields represent the equal-energy condition.

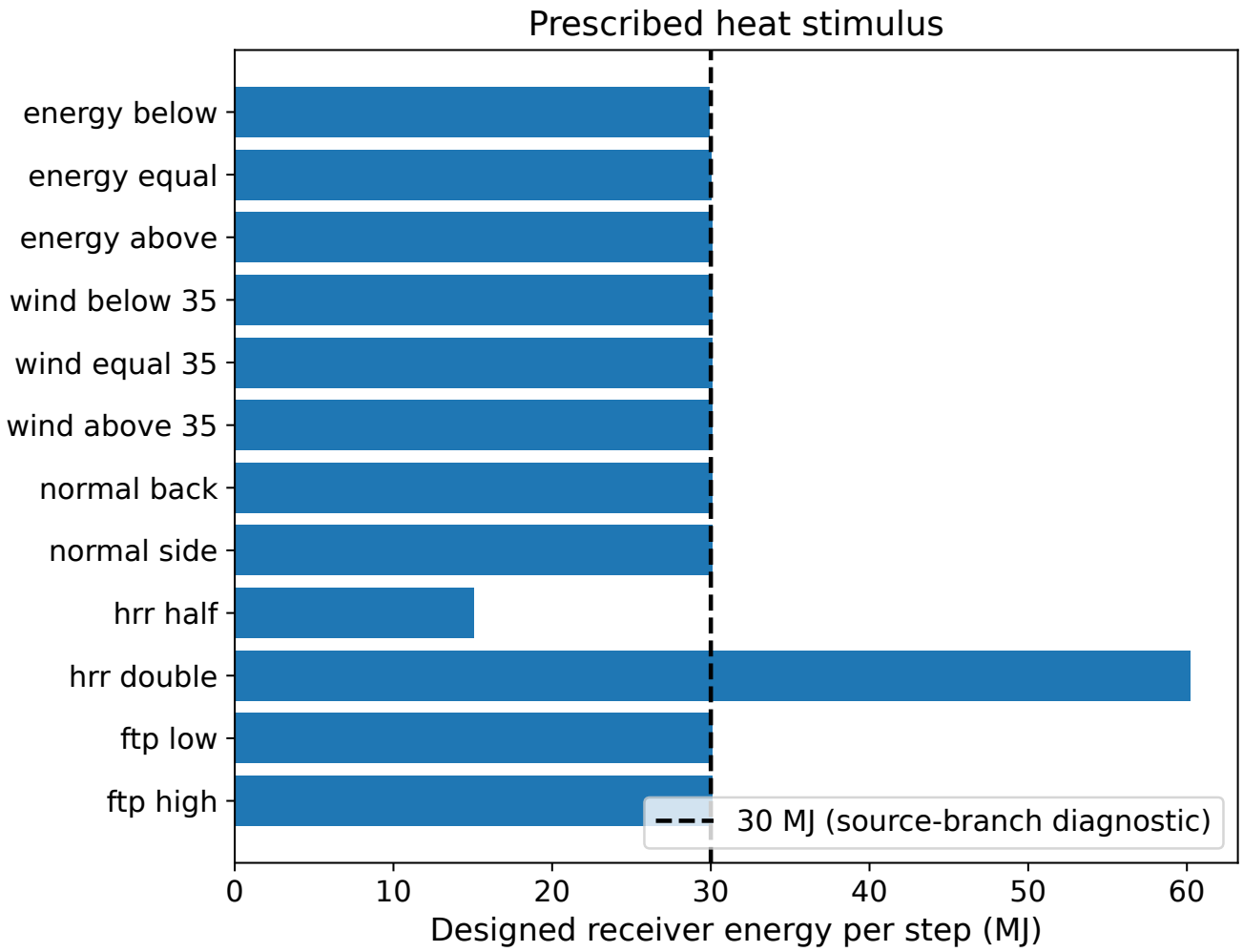


Figure 2: Prescribed receiver energy for all twelve simulation conditions, read from the generated input simulation specification. The dashed line marks the source energy-branch diagnostic, not an acceptance threshold.

6 Actual simulation results

Analysis procedure status is **COMPLETE**; 12 of 12 required simulation conditions have complete black-box evidence. The reported reason is “All required run evidence was complete; every scientific metric passed.”. The common model-program SHA-256 identity is `2b58920716af28891d9166c5f1046174214994eb20e820981017e4f2d166b001`; “–” means no completed simulation supplied that evidence. For each completed condition, the analysis retains the actual output times, heat fields at receiver arrival, terminal arrival-time and spread-rate fields, and matching input and reference identities. Previous results are excluded when inputs are regenerated: the scientific state becomes NOT EVALUABLE and the execution state becomes NOT RUN until new evidence is available.

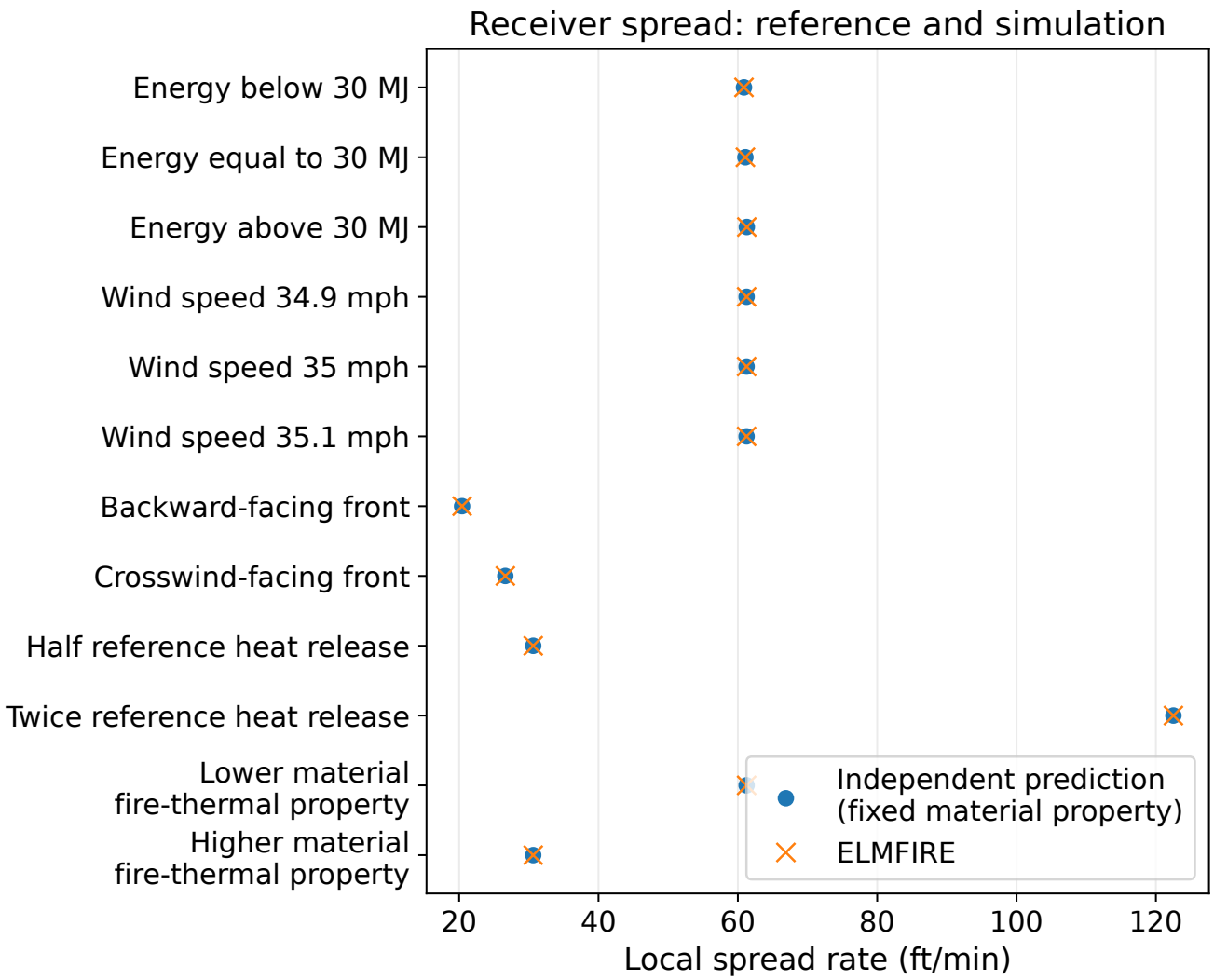


Figure 3: Simulated receiver spread rates and independent predictions for all twelve simulation conditions. The predictions recompute the heat-to-spread relation using the simulated heat stimulus. This comparison isolates the receiver response from differences in the supplied heating; separate figures show the heat stimulus and the spatial result.

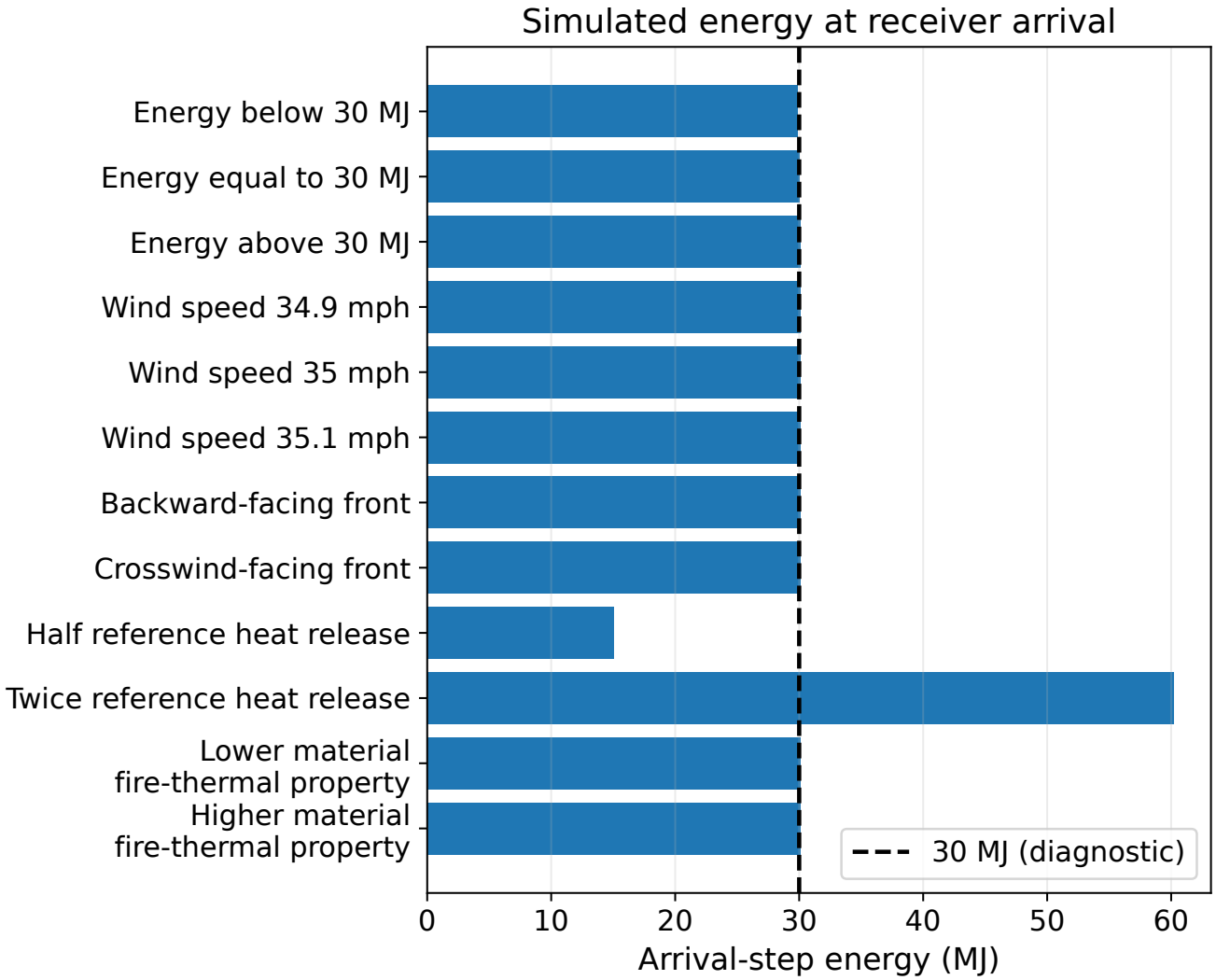


Figure 4: Simulated receiver energy at the selected arrival-step output, calculated from the recorded direct-flame-contact and radiative heat fields. The dashed line identifies the energy-branch transition in the reviewed formulation; it is not an acceptance threshold.

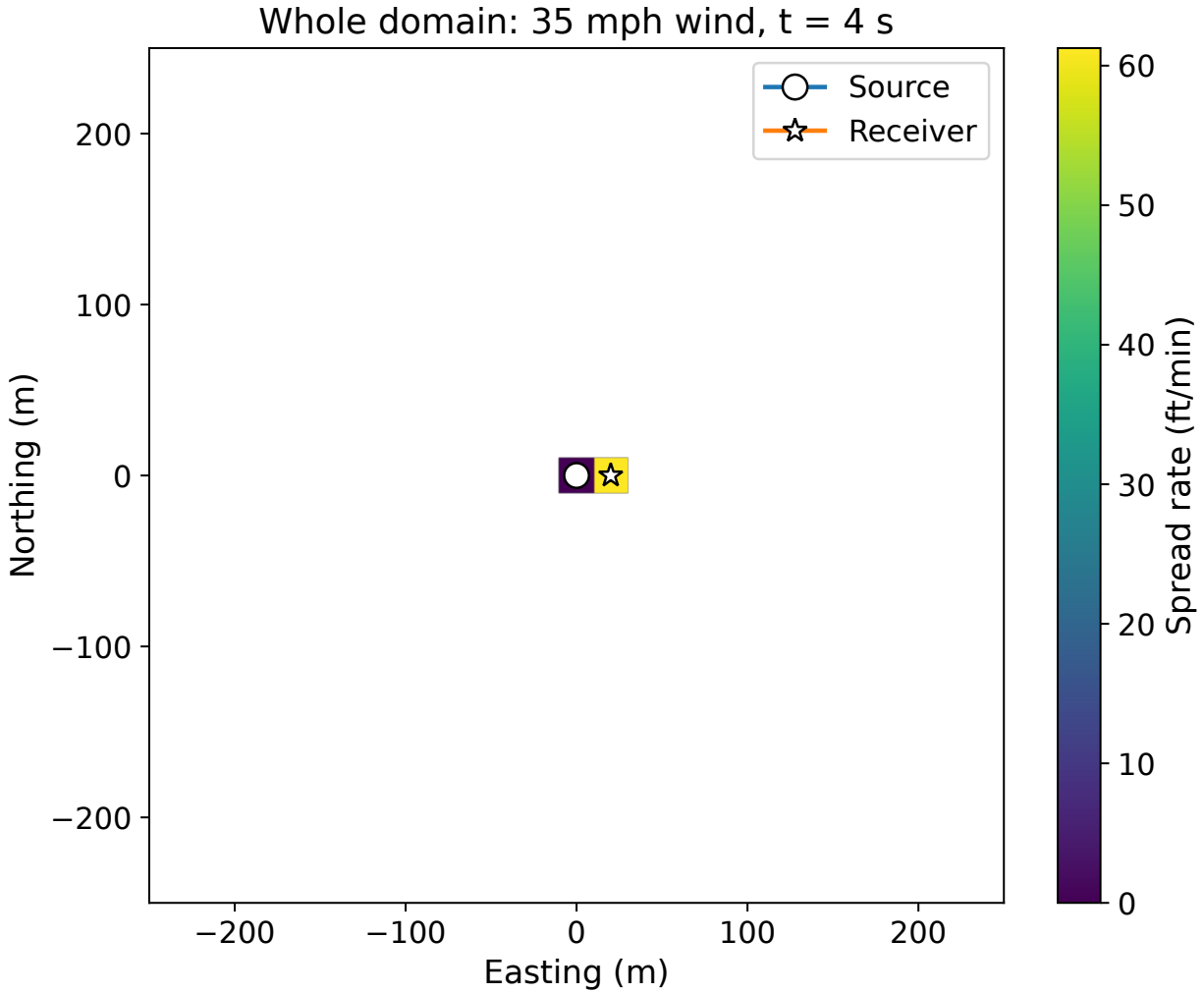


Figure 5: Whole-domain terminal spread rate for 35 mph wind at 4 s, read from the selected ELMFIRE raster. Symbols mark the source and receiver; nodata is blank.

7 Calculated metrics and verification decision

Recorded heat-to-ROS reference calculation relative error — PASS	
Criterion	≤ 0.005
Observed	*Units: fraction.* 0.000207836
Receiver Burned-cell state observability — PASS	
Criterion	TOA = 1 s and terminal VS is finite
Observed	True
Fixed-FTP continuity across 30000 kJ — PASS	
Criterion	normalized speed spread ≤ 0.005
Observed	*Units: fraction.* 4.00432e-08
Wind continuity across 35 mph — PASS	
Criterion	reference calculation error ≤ 0.005
Observed	*Units: fraction.* 5.6595e-08
HRR amplitude response — PASS	
Criterion	double/half heat ratio = 4 +/- 0.04
Observed	*Units: ratio.* 4
Head/back/side response — PASS	
Criterion	matched-wind maximum relative error ≤ 0.005
Observed	*Units: fraction.* 0.000207836
Critical material fire-thermal property inverse response — PASS	
Criterion	speed ratio for 3000/6000 kJ m^-2 = 2 +/- 0.01
Observed	*Units: ratio.* 2

The authoritative case decision sourced from calculated verification and validation metrics is **PASS**. NOT RUN and INCOMPLETE are analysis procedure states; NOT EVALUABLE is their non-passing scientific decision.

8 Technical interpretation

The repaired receiver stencil provides complete, observable arrival and speed evidence. Its positive initial PHI remains positive under the source's input noise, and its side-normal transport no longer competes with an artificial tangential minimum. The observability result therefore separates a usable experiment from agreement with the intended spread law.

The remaining failures identify the heat-to-speed mapping. The source replaces the table FTP with an energy-dependent value and applies a separate wind-speed divisor. The material-pair speeds consequently do not show the required inverse FTP response. The energy and wind comparisons expose the associated branch changes, while the correct HRR amplitude ratio confirms that the prescribed source-amplitude control was realized. See the generated metric rows for the measured errors and decisions; a complete run is not a scientific PASS.

9 Scope and limitations

The case verifies the executable-visible local speed produced by one WU–E urban receiver under a compact deterministic sweep. It does not independently verify the upstream heat-flux model, ignition delay over multiple cells, accumulated exposure across timesteps, building-to-building community spread, or convergence. Internal working variables listed above remain inferred from the reviewed formulation, not observed. All accepted results must share one recorded model-program identity and matching input and reference identities. This consistency does not establish that the program was compiled from the reviewed source revision. The result provides numerical verification, not physical validation.

References

[1] D. M. J. Purnomo et al., "Reconstructing modes of destruction in wildland–urban interface fires using a semi-physical level-set model," *Proceedings of the Combustion Institute*, 40 (2024), 105755. doi:10.1016/j.proci.2024.105755.

[2] D. M. J. Purnomo et al., "Integrating an urban fire model into an operational wildland fire model to simulate one dimensional wildland–urban interface fires: a parametric study," *International Journal of Wildland Fire*, 33 (2024), WF24102. doi:10.1071/WF24102.

Case 46: Wildland-to-urban WU-E transition and isolated receivers

1 Verification purpose

This case verifies the discrete wildland-to-urban (WTU) transition of Wildland–Urban ELMFIRE (WU–E) at a contiguous interface and at an isolated urban receiver. Its primary reference calculation compares accumulated heat with the material fire-thermal property (FTP), requiring a finite exposure time before ignition. The seven-run matrix also characterizes the optional fireline-intensity shortcut and is intended to expose reversed fuel eligibility, excessive reach, topology dependence, a missing heat path, or an incorrect arrival timestamp. The scientific verdict is separate from analysis procedure state: before execution the former is NOT EVALUABLE and the latter is NOT RUN.

The ELMFIRE formulation under examination corresponds to `a2dfbcdf72209733c000e5d3431e44723e15efea`. Five reviewed WU–E components matched that revision exactly. Output timing was reviewed separately from the output procedure at the stated revision, with its SHA-256 identity retained in the case specification. A separately modified copy of that procedure was not asserted to match the revision. These design-review statements do not by themselves establish the identity of a later simulation.

The optional shortcut is NOT EVALUABLE in the compatibility review. The configuration selects interface model 2, but the active transition condition tests whether a separate integer control equals 2. No deterministic assignment of that control was found. The real-valued critical heat-flux threshold is a different quantity and cannot establish the integer control's value. The primary heat/FTP decision does not depend on this selection. The optional shortcut comparisons become PASS or FAIL only after a review matched to the model revision establishes an enabled selection condition and documents where the required integer value 2 is assigned deterministically.

Current result: FAIL. 7 of 7 required simulation conditions have evaluable evidence. Analysis procedure: **COMPLETE.**

2 Mathematical formulation and numerical implementation

The intended WTU path deposits direct flame contact (DFC) and radiation from a burning wildland cell onto an urban receiver, accumulates

$$Q_j(t+\Delta t) = Q_j(t) + [q''_{DFC,j}(t) + q''_{rad,j}(t)]\Delta t,$$

and ignites after the fixed material threshold $F_{TP,crit}$ is met. Thus a sufficiently heated target must have a finite delay $\tau_{WTU} = t_{ign,target} - t_{ign,source} > 0$. The same heat/FTP law then produces urban rate of spread (ROS); no 30,000-kJ branch is part of this reference calculation.

As an optional extension, for source (i_s, j_s) and target (i_t, j_t), the reviewed fireline-intensity transition condition is

$$r = \sqrt{(i_t - i_s)^2 + (j_t - j_s)^2}, \qquad \mathcal{I} = (f_s \neq 91) \wedge (f_t = 91) \wedge (r \leq 1) \wedge (\mathrm{FLIN}_s > 1000).$$

The comparison is strict. A true branch sets the target's WTU flag and skips the ordinary DFC and radiation calculation for that target. In both Runge–Kutta stages, the level-set update forces its $\phi = -1$. The newly burned-cell loop then records TOA $= T$. Initialization assigns every input $\phi \leq 0$ source cell time of arrival (TOA) zero.

The fire behavior fuel model (FBFM) 01 source has five-percent dead one-hour moisture, zero wind, and zero slope. The input-design calculation uses

$$t_r = 384/\sigma, \qquad \mathrm{FLIN} = t_r I_R R(0.3048),$$

then searches adjacent IEEE-754 binary32 adjustment values so the reviewed default-single-precision arithmetic operation order brackets 999, 1000, and 1001 kW m^{-1}. This is only an input reference calculation; acceptance uses the ELMFIRE fireline intensity raster.

The peer-reviewed WU–E publications describe interface heat transfer and heat/FTP spread [1, 2]. The supplied thesis supports this scaling on PDF page 53 (printed page 31), equations (2.42)–(2.43), and gives vegetation heat-release context on PDF page 64 (printed page 42), equations (2.81)–(2.83). The intended equations are documented directly here. Neither published source supplies the special 1000 kW m^{-1} branch, selector linkage, forced ϕ, or dump-name reference calculation; those come exclusively from the pinned source.

3 Derivation of expected behavior

For the primary reference calculation, positive DFC-plus-radiation must be observed at the sufficient-heat adjacent and isolated receivers and ignition must occur after a positive exposure delay. The initial-wildland-source heat-release rate (HRR) suppression is diagnostic, but later propagation within the vegetative patch may create new heat-emitting sources. The complete receiver history, not the initial-source state alone, determines this observation.

For the optional shortcut, the contiguous simulation conditions request source FLIN of 999, 1000, and 1001 kW m^{-1} at an immediately adjacent FBFM91 band. Only 1001 may transition. Isolated targets test 999 and 1001 at one-cell distance and 1001 at two-cell distance; only the one-cell 1001 target may transition. A zero-adjustment target is the spontaneous-ignition control. Thus the exact boolean reference calculation is

$$(0, 0, 1)_{\text{contiguous}}, \qquad (0, 1, 0)_{\text{isolated}}, \qquad 0_{\text{zero}}.$$

Both eligible transitions occur during the first update at $T = 0$: the heat loop sets the WTU flag, both RK stages force the target negative, and the newly-burned loop assigns TOA to the current solver time. Since initial source TOA is also zero, the exact expectation is

$$t_u = 0 \text{ s}, \qquad t_s = 0 \text{ s}, \qquad \tau_{WTU} = t_u - t_s = 0 \text{ s}.$$

Every ineligible target must retain terminal $\phi > 0$ and nodata TOA. The evaluator requires $\phi \leq 0$ and finite TOA to agree; either observable alone cannot create an ignition result.

4 Verification metrics and acceptance criteria

All seven simulation conditions require confirmed successful completion with matching identities for the exact simulation settings, independently calculated expected responses, complete set of prescribed conditions, and review of supported model selections. The ELMFIRE program, simulation record, and all results must also match the recorded identities. All seven conditions must use the same identified model program. Stale or altered evidence is rejected.

The dump ledger is exact, not merely terminal: 61 rows have indices 1 through 61, times 0 through 60 s, and only row 61 is final. This follows the pinned loop, which dumps at current $T = 0, \ldots, 59$ before incrementing time and then makes a final dump after reaching 60 s. Final fireline-intensity and arrival-time fields are identified by the rounded output time, 60 s. The level-set sequence is indexed by output number, so its terminal field is output 61, not output 60. All 61 level-set fields are required. The terminal three rasters must be one-band, aligned 25-by-25 EPSG:32610 grids with the exact missing-value definition and finite non-nodata values. The simulation record must not report a propagation stall.

Metric	Observable	Acceptance rule
Evidence completeness	confirmed model and input identities, 61 output times, and terminal fields	all exact for 7/7 simulation conditions
Source realization	source-mask FLIN and TOA	max FLIN error ≤ 0.5 kW m^{-1}; every source TOA zero
State consistency	designated-target terminal PHI and TOA	$\phi \leq 0$ iff TOA finite
Intended WTU heat	target DFC+radiation histories	positive for sufficient-heat receivers
Intended ignition delay	target TOA minus source TOA	finite and greater than zero
Contiguous shortcut matrix	below/equal/above target state	false, false, true when selector is valid
Isolated matrix	one-cell insufficient/sufficient and two-cell sufficient	false, true, false
First update	eligible target TOA and target–source delay	both exactly zero within 10^{-8} s
Zero control	zero-adjustment target PHI and TOA	$\phi > 0$, TOA nodata

The two intended heat/FTP metrics are verdict-bearing whenever complete run evidence exists. Shortcut-attribution metrics remain NOT EVALUABLE unless the selector preflight passes; they do not control the primary case decision.

5 Simulation configuration and predicted outcome

Parameter	Value	Rationale
Grid	25×25, 10 m, EPSG:32610	at least 10 cells per dimension; edge-buffered targets
Fuel topology	FBFM01 source, FBFM91 targets, FBFM93 background	explicit vegetative/urban eligibility and isolation
Source requests	0, 999, 1000, 1001 kW m^{-1}	zero control and strict threshold bracket
Receiver distance	1 or 2 cells	equality boundary and reach control
Time	start 0, fixed step/max 1, stop 60 s	exact first-update and terminal ledger reference calculation

Output	every-step PHI; final FLIN and TOA	direct state, intensity, and time observables
Dormant ignition	one point at 61 s	keeps the prior-ignition state false through the run, preventing the vacuous zero-ignition stall test without changing 0–60 s physics
Other mechanisms	spotting, crown, randomness off	removes alternate ignition mechanisms

Figure 1 is generated directly from current inputs and is not a simulation result.

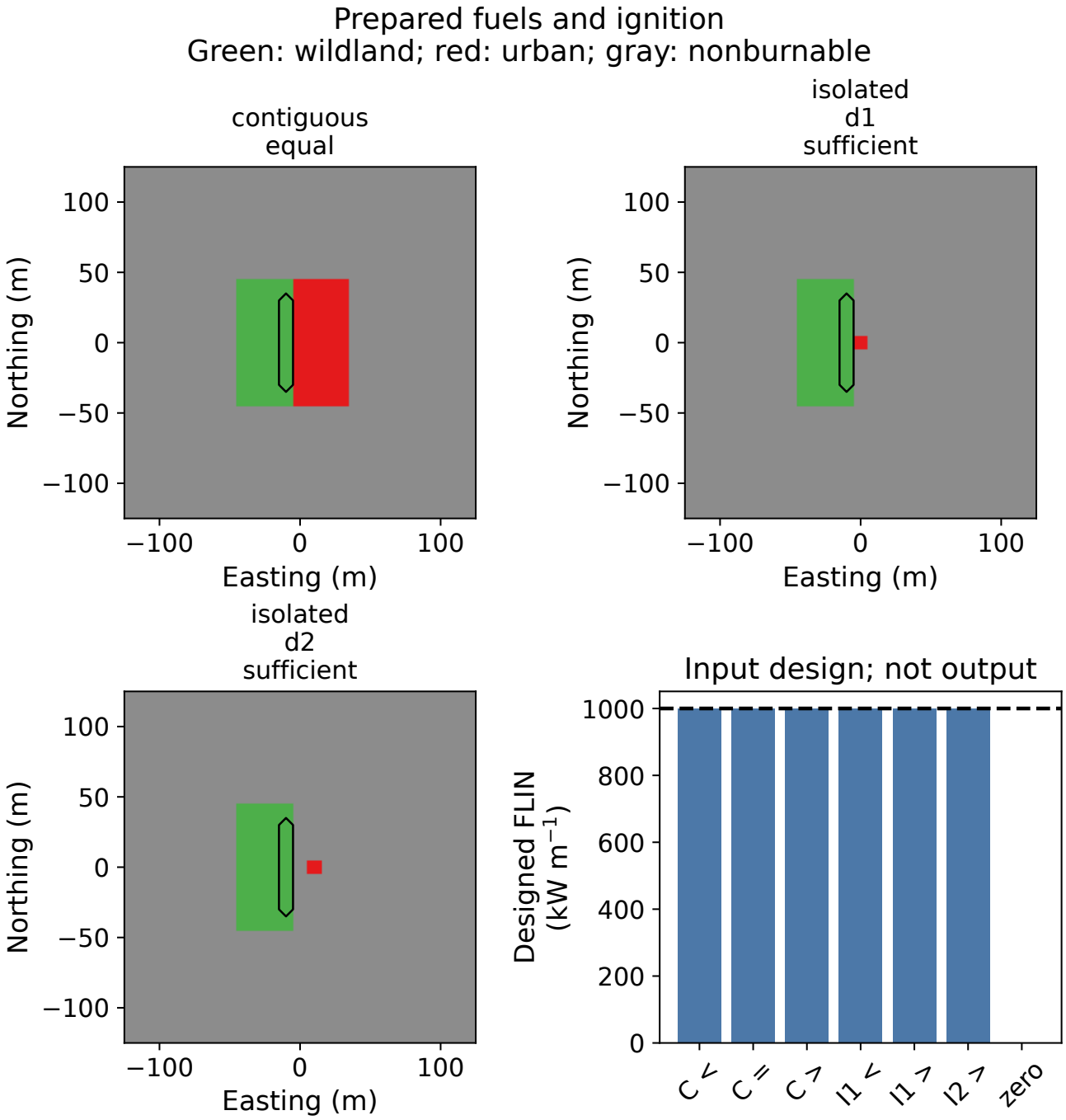


Figure 1: Prepared contiguous and isolated fuel topologies and binary32 source-FLIN design. Green is FBFM01, red is FBFM91, gray is FBFM93, and the black contour marks the seven-cell initial source.

6 Actual simulation results

Analysis procedure status is **COMPLETE**; 7 of 7 simulation conditions have complete current evidence. Selector status is **NOT EVALUABLE**. The generated reason is:

> One or more intended heat/FTP transition checks failed; optional FLIN-shortcut diagnostics do not control this decision.

The executed version is `ELMFIRE Reading &MISCELLANEOUS namelist group`. The model-program and reference-calculation identities are retained with the calculated metrics.

The primary comparison uses independently calculated expected responses, input-matched simulation results, and the review of supported model settings. Each simulation must have evidence of successful completion. The terminal fireline-intensity and arrival-time fields correspond to 60 s, while the terminal level-set field is the 61st output. Their identities and output times are retained with the calculated metrics so the selection remains traceable.

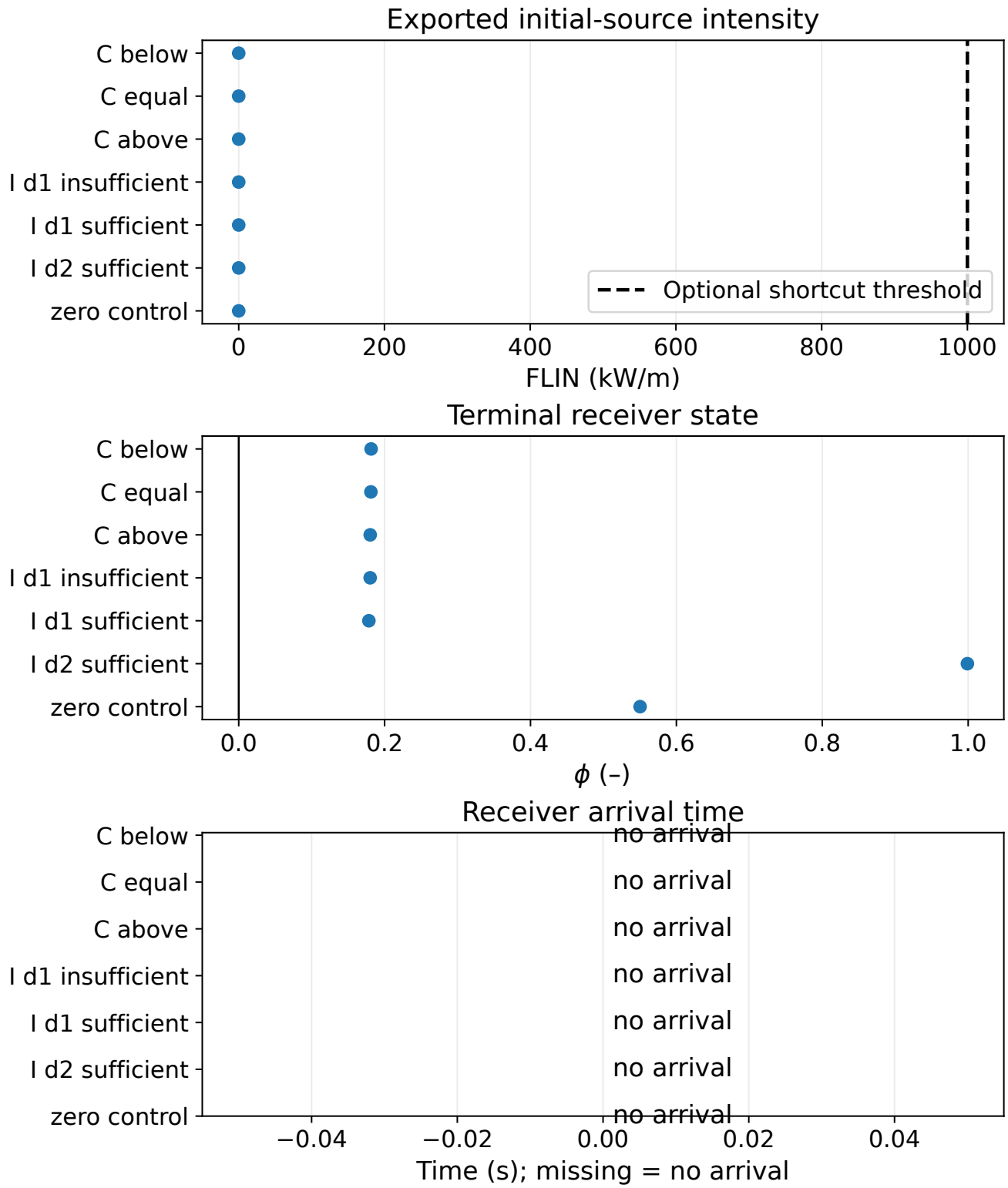


Figure 2: Current black-box source FLIN, terminal target PHI, and target TOA. C denotes contiguous targets and I isolated targets. Missing TOA means no arrival. Only shortcut attribution depends on the selector preflight; the primary checks remain evaluable.

7 Calculated metrics and verification decision

Evidence and exact dump ledger — PASS

Criterion	61 rows: indices 1..61, times 0..60 s, only row 61 final
Observed	7/7 exact; FLIN/TOA time stamp 0000060; PHI dump index d0000061

Source fireline-intensity setup — FAIL

Criterion	maximum absolute error $\leq$0.5 kW/m and source TOA=0 s
Observed	*Units: kW/m.* max abs error kw m: 1001 all source toa zero: Yes

Terminal PHI/TOA state consistency — PASS

Criterion	PHI$\leq$0 iff TOA is finite at each designated target
Observed	contiguous below: Yes contiguous equal: Yes contiguous above: Yes isolated d1 insufficient: Yes isolated d1 sufficient: Yes isolated d2 sufficient: Yes zero control: Yes

Intended W-to-U receiver heat exposure — PASS

Criterion	positive DFC+radiation at sufficient-heat receivers
Observed	*Units: kW/m2.* contiguous above: 61.9232 isolated d1 sufficient: 61.9232

Intended W-to-U finite ignition delay — FAIL

Criterion	finite target TOA with delay $>$ 0 s
Observed	*Units: s.* contiguous above: Not available isolated d1 sufficient: Not available

Contiguous strict-threshold matrix — NOT EVALUABLE

Criterion	below=false, equal=false, above=true
Observed	contiguous below: No contiguous equal: No contiguous above: No

Isolated distance/threshold matrix — NOT EVALUABLE

Criterion	d1-insufficient=false, d1-sufficient=true, d2-sufficient=false
Observed	isolated d1 insufficient: No isolated d1 sufficient: No isolated d2 sufficient: No

First-update transition timing — NOT EVALUABLE	
Criterion	eligible target TOA=0 s and target-source delay=0 s
Observed	*Units: s.* contiguous above: target toa s: Not available target source delay s: Not available isolated d1 sufficient: target toa s: Not available target source delay s: Not available
Zero-source control — PASS	
Criterion	target remains PHI>0 with nodata TOA
Observed	target phi: 0.550248 target toa s: Not available

The authoritative scientific status is **FAIL**. A prepared but unexecuted case is NOT EVALUABLE / NOT RUN. Complete evidence under the selector defect can still evaluate the primary heat/FTP capability; only shortcut-specific rows remain NOT EVALUABLE.

8 Technical interpretation

The current complete run fails the source-intensity setup check. The input reference calculation specifies head-fire FLIN, whereas the terminal raster exports surface-plus-canopy intensity from the initially burned-node list. That export is not the same observation as the head-fire intensity assigned to the WUI source list. A reliable threshold comparison requires a source diagnostic that exposes the intensity actually consumed by the transition branch.

The receiver heat and positive-delay checks can still pass. Although the initial wildland cells have suppressed transient HRR, the surrounding patch can spread and create later heat-emitting cells. Positive receiver heat does not therefore contradict initial-source suppression. Nor does a positive arrival delay alone establish the complete intended accumulated-heat/FTP law.

The optional shortcut has a separate selector defect: the exposed namelist value does not assign the integer tested by the branch. Its attribution rows remain NOT EVALUABLE. The primary heat, delay, state, source-setup, and control rows retain their own decisions; they are not all demoted because the optional selector is unavailable.

9 Scope and limitations

This case covers one wildland fuel/moisture state, one cell size, a 60-s window, contiguous and isolated targets, the strict intensity bracket, distance boundary, state/timing update, and zero control. It does not validate the physical choice of 1000 kW m^{-1}, urban-to-wildland spread, community propagation or convergence. A common model-program identity establishes consistency across accepted outputs but does not establish that the program was compiled from `a2dfbcdf72209733c000e5d3431e44723e15efea`; that source revision identifies the reviewed formulation. A model correction requires a newly reviewed, content-bound PASS preflight before this unchanged strict matrix can yield a shortcut-specific PASS or FAIL.

References

[1] D. M. J. Purnomo et al., "Reconstructing modes of destruction in wildland–urban interface fires using a semi-physical level-set model," *Proceedings of the Combustion Institute*, 40 (2024), 105755. doi:10.1016/j.proci.2024.105755.

[2] D. M. J. Purnomo et al., "Integrating an urban fire model into an operational wildland fire model to simulate one dimensional wildland–urban interface fires: a parametric study," *International Journal of Wildland Fire*, 33 (2024), WF24102. doi:10.1071/WF24102.

Case 47: Urban-to-wildland WU-E transition and isolated wildland receivers

1 Verification purpose

This case tests the urban-to-wildland interface of the Wildland–Urban ELMFIRE (WU-E) model. It separates three observables that can otherwise be mistaken for one another: heat emitted by an urban source, heat accumulated at a wildland receiver, and arrival of the level-set fire front at that receiver. The reviewed WU-E components are identical to source revision `a2dfbcdf72209733c000e5d3431e44723e15efea`; this identity claim does not extend to other modified model components. The processes examined are the building design-fire curve, urban heat transfer, fuel-dependent velocity selection in the level-set formulation, and ordinary Rothermel surface-fire spread.

Two urban sources are initialized in otherwise nonburnable surroundings. One illuminates a disconnected wildland receiver; the other starts a three-cell wildland path. The factorial experiment varies urban peak heat-release rate (HRR) per unit area (HRRPUA) over 100 kW m^{-2} and 400 kW m^{-2}, wildland spread-rate adjustment factor over 0 and 1, and replaces the middle path cell by a nonburnable barrier. Terrain is flat and deterministic; firebrands, crown fire, wind fluctuations, and suppression are disabled. This is numerical verification and implementation characterization, not physical validation.

The precise verification question is: across all eight simulation conditions, does the pinned implementation preserve the fourfold source-HRR and received-heat responses, keep the disconnected heat-exposed wildland cell unignited, expose ordinary wildland velocity to spread-rate adjustment factor, and preserve the nonburnable barrier? The separate publication claim that heat alone ignites a wildland target is reported as **NOT EVALUABLE** because the pinned source contains no corresponding target-state transition.

Current result: NOT EVALUABLE. 8 of 8 required simulation conditions have evaluable evidence. Analysis procedure: **COMPLETE.**

2 Mathematical formulation and numerical implementation

Urban-to-wildland coupling is outside the downscaled building-to-building baseline used by CASE43–46 and CASE48–49. It is retained as a separate published capability [1, 2]. The proceedings article (pages 2–3, equations (2)–(8)) and the supplied thesis (PDF pages 63–67, printed pages 41–45, equations (2.81)–(2.95)) describe source heat transfer using

$$\dot{q}_c = \frac{\mathrm{HRR}\, A_t}{\Delta x^2}, \qquad \dot{q}''_r = \frac{0.3\,\mathrm{HRR}}{4\pi R^2}, \tag{1}$$

$$\dot{q}_t = \alpha_c \dot{q}_c + \alpha_r \dot{q}''_r \Delta x^2, \qquad E_t^{n+1} = E_t^n + \dot{q}_t^n \Delta t. \tag{2}$$

Here A_t is the flame-overlap area, R is source–target distance, α_c is a combustible-area coefficient, and α_r includes target absorptivity. The documented ignition idealization is $E_t \geq \mathrm{FTP}\,\Delta x^2$, giving $U = \sum \dot{q}_t/(\mathrm{FTP}\,\Delta x)$.

The implementation performs a related discrete heat calculation. For every burning source it scans the configured interaction-band thickness, computes ellipse-overlap factors, sums transient direct flame contact (DFC) and radiation at each target, and accumulates both after multiplication by $\Delta t \Delta x^2$. Target coefficients come from the building-fuel table even where the target fire behavior fuel model (FBFM) is wildland. These arrays are observable as transient heat flux and terminal total-received-heat rasters.

That heat state is not, however, consumed by a wildland ignition law at the pinned revision. The WU-E heat-to-velocity routine is selected only when the current target has fuel model 91. A target with

a fuel model other than 91 instead follows the ordinary surface-fire path; for Rothermel its no-wind base velocity contains the explicit multiplier

$$V_{s,0} = \text{ADJ } D \; I_R \xi / (\rho_b \epsilon Q_{ig}), \tag{3}$$

where D is the diurnal adjustment. Consequently, received WU-E heat and Rothermel front arrival are distinct state variables in this experiment.

3 Derivation of expected behavior

Let the low design-fire curve be $H_L(t) = H_0 h(t)$, where the common piecewise-linear shape h grows through 60 s, remains fully developed through 300 s, and decays to zero at 600 s. The high table changes only the amplitude, so

$$H_H(t) = 4H_L(t). \tag{4}$$

The heat-loop geometry and target coefficients are unchanged between a matched low/high pair. Both direct flame contact and radiation are linear in instantaneous HRR; therefore, at the disconnected receiver,

$$\frac{\max H_H}{\max H_L} = 4, \qquad \frac{E_{t,H}}{E_{t,L}} = 4. \tag{5}$$

Raster roundoff motivates small relative tolerances rather than exact bitwise equality.

The limits are preregistered numerical discriminators, not experimental uncertainties. The 1-percent source-ratio limit is generous relative to float32 storage while still detecting a material departure from fourfold scaling; the 0.1-percent repeat limit is tighter because the paired source inputs are identical. The 2-percent accumulated-heat ratio allows repeated float32 additions over 600 one-second updates. A 1-s time of arrival (TOA) difference is exactly one configured solver step. Finally, 10^{-4} m/min would move a front only 0.001 m over the full run, far below a 10-m cell, so it separates a dynamically positive path from numerical zero without serving as a physical rate of spread (ROS) threshold.

The heat experiment places its wildland receiver three cells downwind and separates it from the source with two FBFM99 cells. It lies inside the five-cell WU-E scan but cannot be reached by the continuous front. The pinned algorithm accumulates positive heat there but has no conditional assignment that changes its arrival time from that heat; its time of arrival must therefore remain nodata. This is an implementation invariant, not acceptance of the published capability as correct.

In the separate path experiment, FBFM1 cells carry spread-rate adjustment factor=0 or 1. For spread-rate adjustment factor=0 their ordinary Rothermel base velocity is zero; for spread-rate adjustment factor=1, the configured dry fuel and 20 mph westerly wind give a positive value. The open path therefore supplies a direct control on ordinary surface spread. Replacing its middle cell by FBFM99 supplies an independent nonburnable invariant. No arrival at that named FBFM99 barrier cell is permitted; the metric does not generalize this single-cell control to every background pixel.

4 Verification metrics and acceptance criteria

The evaluator first verifies prepared-input hashes, all eight identities, a unique dump-time simulation specification per simulation condition, a unique final record exactly at 600 s, and grid agreement for every required GeoTIFF. It requires terminal time of arrival, spread rate, fireline intensity, total direct flame contact, and total radiation, plus complete HRR/DFC/radiation transient series matching every dump-simulation specification index on the exact 60 s schedule. With feedback enabled, it also requires exactly 600 logged solver-step timestamps on the configured 0–599 s sequence before selecting the

explicit post-step final dump at 600 s. Any missing, duplicate, malformed, off-grid, or nonterminal evidence makes the run **NOT EVALUABLE**; process exit status is never evidence of success.

Expected property	Observable and metric	Sensitivity and selection rationale	Component limit
Fourfold source amplitude	Maximum of the two source cells over every HRR transient; matched high/low ratio error	Detects table selection, design-curve, or output errors without depending on front shape	relative error ≤ 0.01
Control independence	Range of source peaks across ADJ/barrier controls at fixed amplitude	Detects unintended coupling of receiver controls into the source curve	relative range ≤ 0.001
Heat reaches isolated cell	Terminal DFC plus radiation at the exact receiver cell	Proves absence of TOA is not explained by zero incident heat	minimum > 0
Linear heat response	Matched high/low ratio of isolated total heat	Detects nonlinear or incorrect HRR accumulation while holding geometry fixed	ratio error ≤ 0.02
No current heat-only target transition	Count of finite isolated-receiver TOA values	Directly tests the source-derived current invariant	0 of 8
Nonburnable control	Count of finite barrier-cell TOA values in closed simulation conditions	Detects leakage through the FBFM99 control	0 of 4
Ordinary surface control	Named-cell TOA plus finite terminal VS over FBFM1 cells	Distinguishes the ADJ-scaled Rothermel path from received heat without coercing nodata VS to zero	no adjacent/distal ADJ0 arrivals; all ADJ1-open arrivals; VS $> 10^{-4}$ m min^{-1}
Ignition-time control	Adjacent and distal TOA paired across the two open-path HRR simulation conditions	Detects accidental dependence of ordinary spread on the disconnected HRR-amplitude control	maximum difference ≤ 1 s

All current-implementation components must pass for the implementation characterization to be **PASS**. The intended heat-only urban-to-wildland capability remains **NOT EVALUABLE** independently, so the case-wide verification decision cannot be **PASS** at this source revision.

5 Simulation configuration and predicted outcome

The domain is a 310 m square in EPSG:32610, with north-up 10 m cells and nodata -9999. PHI is a clipped signed-distance field with exactly two negative urban-source cells and no cell centered on PHI zero. Both sources use building fuel model 1; every wildland receiver is FBFM1 and the background/barrier is FBFM99. Source separation exceeds twice the WU-E heat band, so the isolated receiver is influenced only by its designated heat source. A dormant scheduled ignition at 601 s prevents an unrelated stalled-front shortcut but occurs after the stop time.

Parameter	Configured value	Units	Role
Grid/domain	31×31 at 10; EPSG:32610	m/cell	Resolves source, receiver, and barrier cells exactly
Urban peak HRRPUA	100, 400	kW m^{-2}	Fourfold source and heat-amplitude pair
Wildland adjustment	0, 1	—	Turns ordinary Rothermel velocity off/on
Path control	FBFM1 or FBFM99 at middle cell	code	Open versus nonburnable barrier
Weather/terrain	20 mph from 270°; flat	mph, degree	Downwind heat exposure with fixed geometry
Building geometry	linear dimension 10, separation 10, footprint fraction 1	m, m, fraction	Fixed WU-E ellipse inputs
Time integration	$\Delta t = \Delta t_{\max} = 1$, $t_{stop} = 600$	s	Exactly 600 logged steps and an explicit post-step exact-stop final dump
Output cadence	60	s	Resolves rise, plateau, and decay
Disabled processes	spotting, crown fire, suppression, fluctuations	–	Removes alternate nonlocal or stochastic ignition

Figure 1 is generated only from the prepared case-specific GeoTIFFs. The predicted result is positive isolated-cell heat with no isolated TOA, a fourfold low/high response, zero ordinary wildland speed at ADJ0, positive ordinary speed for the open ADJ1 controls, and no barrier-cell TOA.

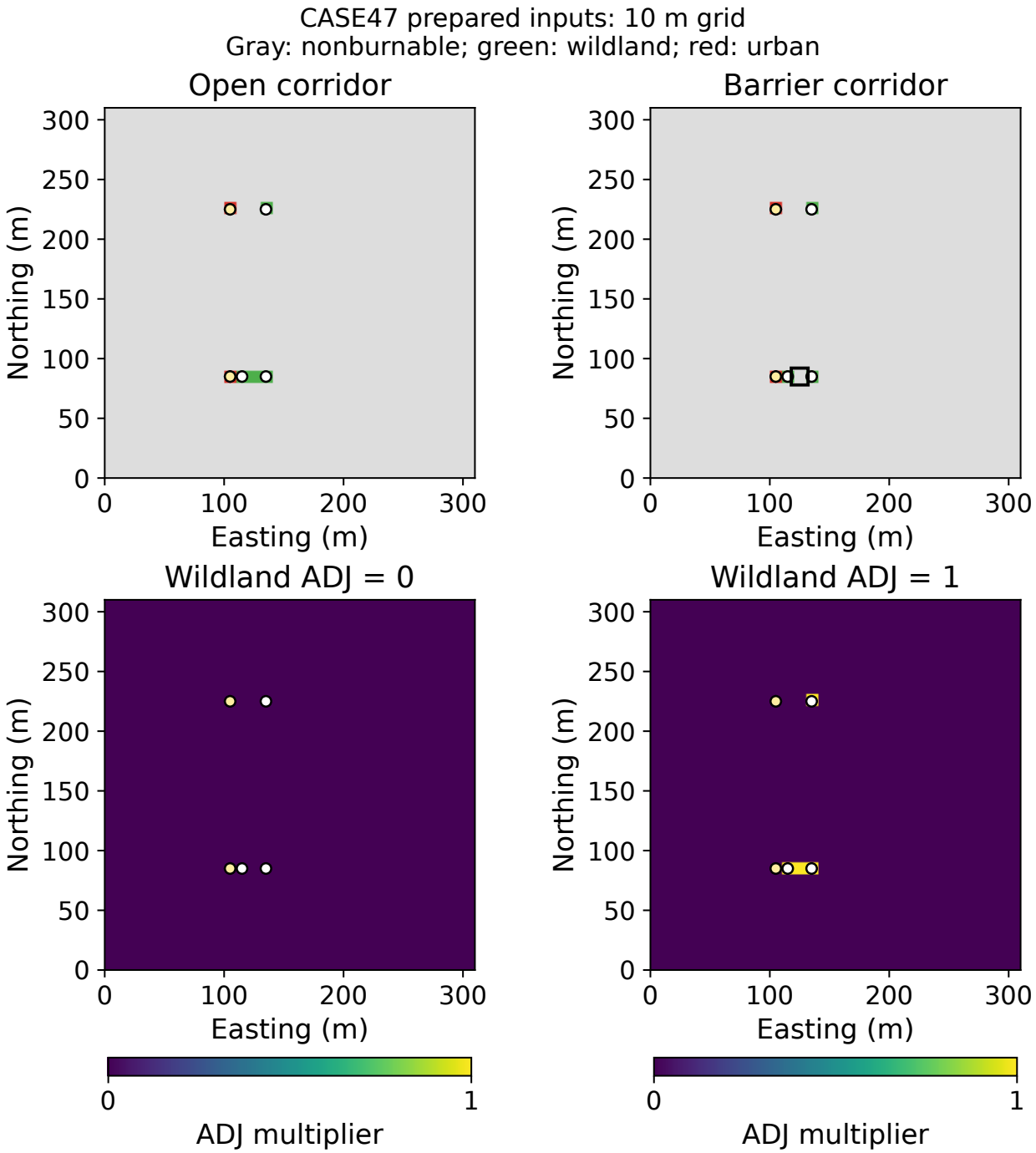


Figure 1: Whole-domain prepared inputs for representative high-HRR simulation conditions. FBFM91 marks each urban source, FBFM1 marks wildland receivers/path cells, FBFM99 is nonburnable, and the black contour is the generated initial PHI interface. Yellow circles mark urban sources; white circles mark wildland receivers. The upper row of markers is the isolated heat experiment; the lower row is the surface-spread experiment, with a square marking the barrier cell.

6 Actual simulation results

The authoritative record reports analysis procedure state **COMPLETE** and contains 8 of 8 required simulation conditions. The scientific decision is **NOT EVALUABLE**. Reason: Current behavior is characterized separately; the documented heat-only U-to-W capability has no executable reference calculation path at this source revision. The executed binary SHA-256 is `2b58920716af28891d9166c5f1046174214994eb20e820981017e4f2d166b001`; it is interpreted only against the pinned source reference calculation and is not assumed to equal it. Available results retain their actual output times and matching model, input, and simulation-record identities. All selected fields must belong to those completed simulations.

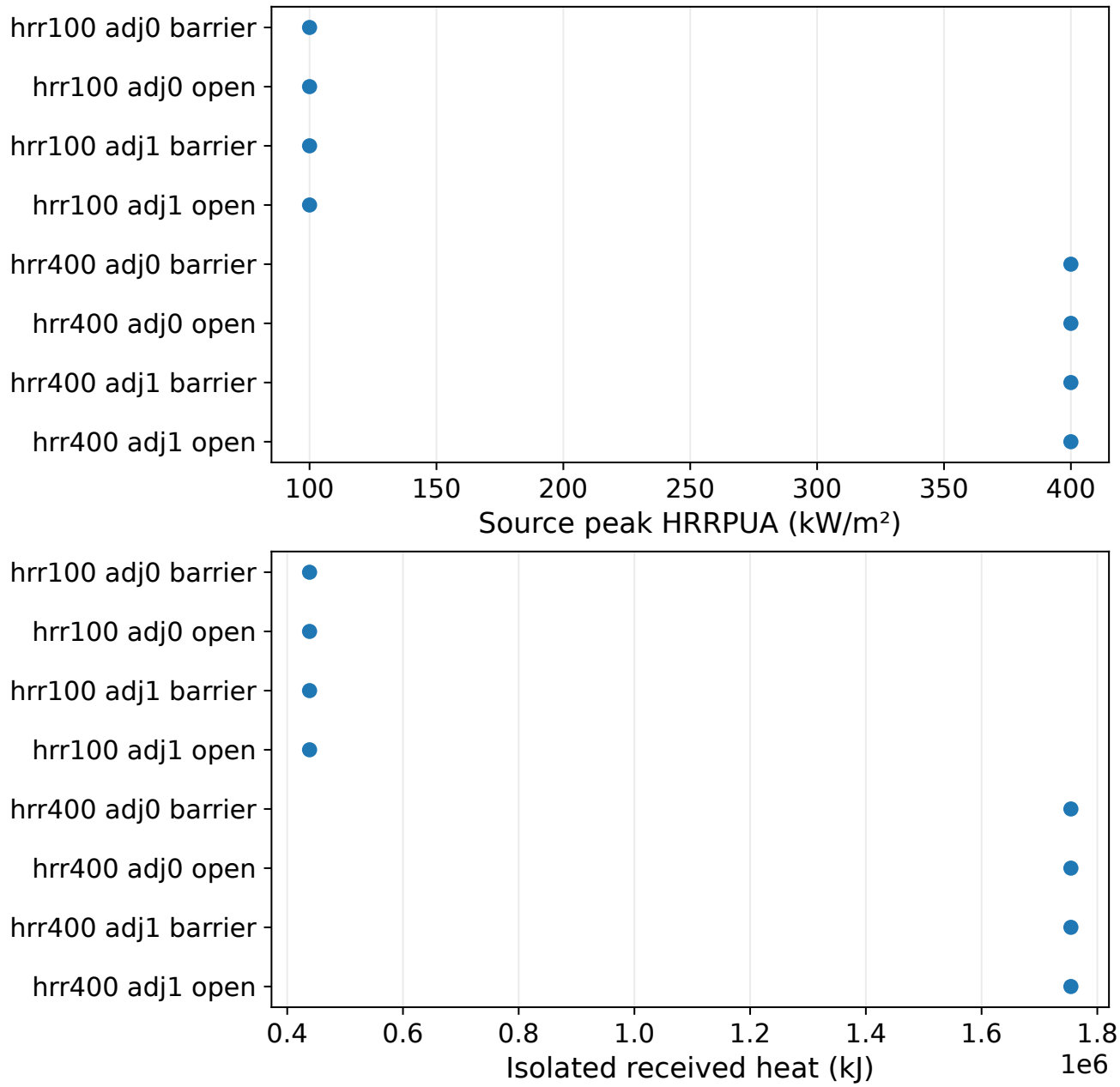


Figure 2: Actual case-specific source HRR and isolated-cell accumulated heat for all eight simulation conditions through 600 s. Values are read from ELMFIRE GeoTIFF outputs; they are not synthetic reference curves. Amplitude-ratio limits are 1% for source HRR and 2% for heat. Arrival times are shown separately.

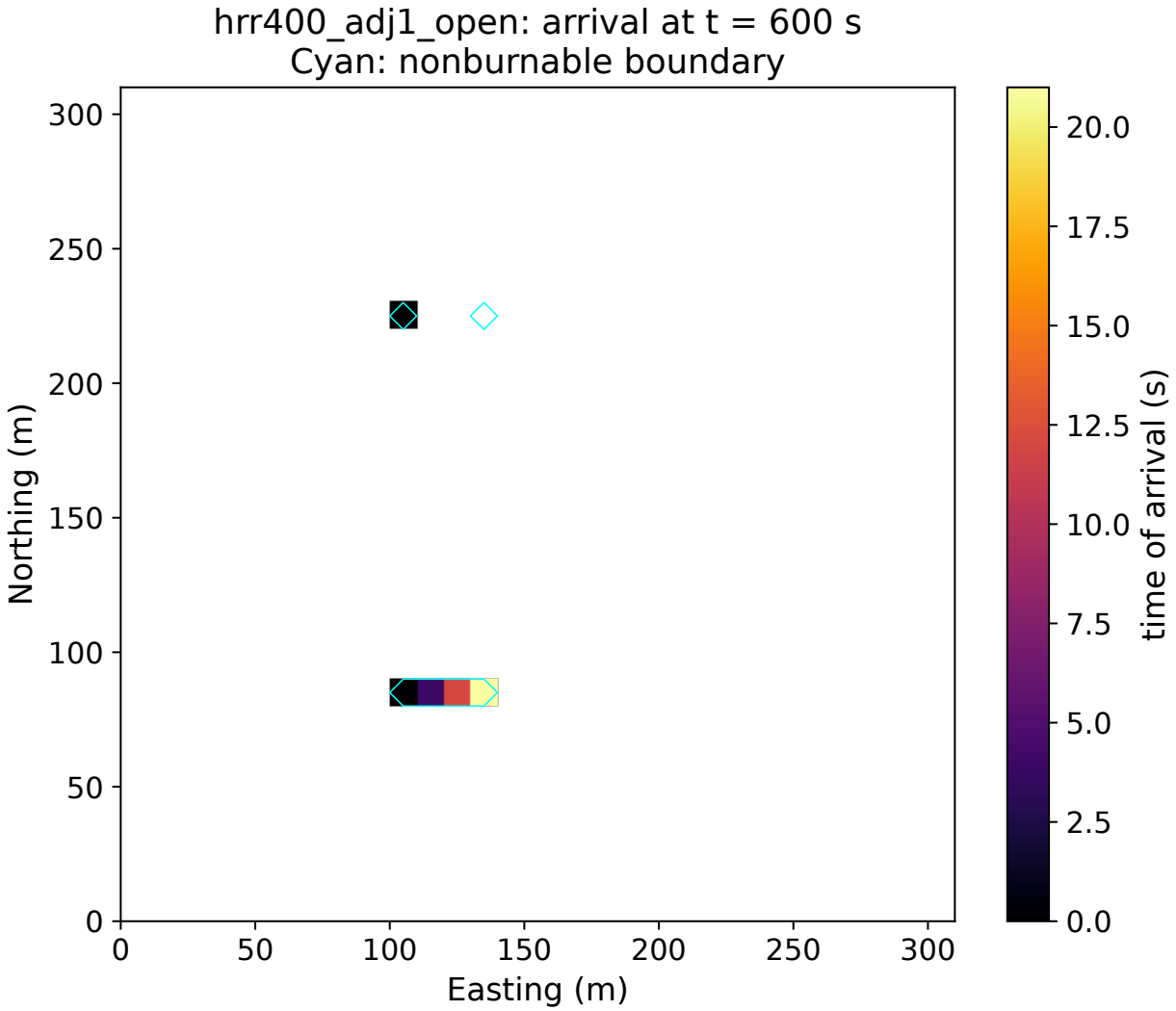


Figure 3: Actual whole-domain time of arrival for the open configuration with heat-release rate per unit area 400 and adjustment factor 1 at 600 s; nodata remains blank and the cyan line marks the FBFM99 boundary. The selected field is the unique terminal arrival-time result for this simulation condition.

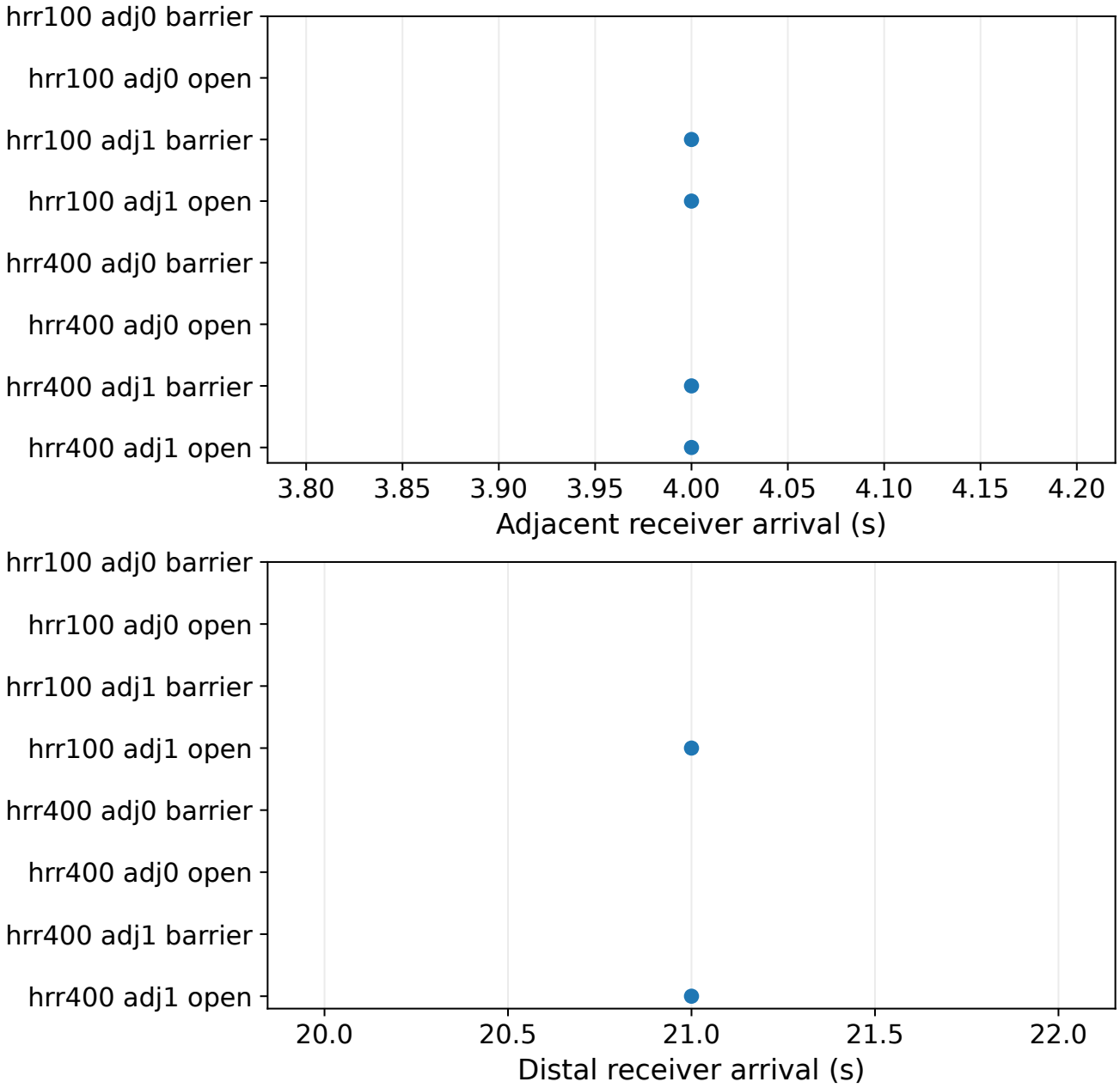


Figure 4: Adjacent and distal receiver arrival times for all eight simulation conditions. A missing observation is labeled no arrival, not plotted at zero. The selected simulation quantities are identified with the calculated metrics.

7 Calculated metrics and verification decision

Source HRR fourfold response error — PASS

Criterion ≤ 0.01

Observed *Units: fraction.*
0

Source HRR control-repeat error — PASS

Criterion ≤ 0.001

Observed *Units: fraction.*
0

Isolated receiver heat-exposure minimum — PASS

Criterion > 0

Observed *Units: kJ.*
438590

Isolated heat fourfold response error — PASS

Criterion ≤ 0.02

Observed *Units: fraction.*
0

Isolated receiver arrivals — PASS

Criterion 0 of 8

Observed *Units: count.*
0

Nonburnable barrier arrivals — PASS

Criterion 0 of 4

Observed *Units: count.*
0

ADJ0 ordinary-path arrivals — PASS

Criterion 0 of 8 adjacent/distal checks

Observed *Units: count.*
0

ADJ1 open-path arrivals — PASS

Criterion 4 of 4 adjacent/distal checks

Observed *Units: count.*
4

Minimum ADJ1-open wildland spread rate — PASS

Criterion $>$ 1e-4 in both open simulation conditions

Observed *Units: m/min.*
70.237

Open-path TOA HRR-control difference — PASS	
Criterion	≤ 1
Observed	*Units: s.* 0

Intended heat-only urban-to-wildland ignition capability — NOT EVALUABLE	
Criterion	heat exposure can initiate a disconnected wildland receiver
Observed	*Units: capability.* no source-code state transition exists for FBFM != 91 target cells

Current-implementation characterization: **PASS**. Published intended heat-only U-to-W capability: **NOT EVALUABLE**. Overall case decision: **NOT EVALUABLE**. These two decision layers are deliberately not collapsed: agreement with the currently missing transition cannot verify that the documented capability has been implemented.

8 Technical interpretation

The implementation-characterization checks pass: source HRR and isolated received heat preserve the fourfold amplitude response, ordinary propagation responds to ADJ, and the nonburnable control remains intact. Yet the isolated wildland receiver has no arrival despite positive heat exposure. This combination demonstrates the distinction between heat deposition and a heat-driven ignition pathway.

The source selects heat-derived spread only for urban FBFM91 targets. Wildland receivers retain the ordinary surface-fire path. Agreement with this current behavior is a characterization PASS; it cannot establish the separately claimed heat-only urban-to-wildland capability. The overall capability verdict therefore remains NOT EVALUABLE. Missing TOA is shown as missing, never as an arrival at zero seconds.

9 Scope and limitations

This case covers deterministic DFC/radiation exposure from urban FBFM91 sources to FBFM1 receivers on a flat 10 m grid, two HRR amplitudes, an ADJ control, and one nonburnable barrier. It does not validate heat-transfer coefficients, material ignition thresholds, physical flame dimensions, firebrands, crown fire, topography, suppression, or stochastic behavior. It also does not claim that lack of a target law is desirable; it records that current-source fact separately from the published intended capability.

The supplied publications are technical evidence, not executable specifications. Their notation and default-value ambiguities, including combustible-fraction semantics and dimensional interpretation of the heat-derived ROS equations, require resolution before a future U-to-W ignition-time test can have a quantitative physical reference calculation. Spatial and temporal convergence are outside this case and belong to CASE48_WGC and CASE49_WTC, respectively.

References

[1] D. M. J. Purnomo et al., “Reconstructing modes of destruction in wildland–urban interface fires using a semi-physical level-set model,” *Proceedings of the Combustion Institute*, 40 (2024), 105755. doi:10.1016/j.proci.2024.105755.

[2] D. M. J. Purnomo et al., “Integrating an urban fire model into an operational wildland fire model to simulate one dimensional wildland–urban interface fires: a parametric study,” *International Journal of Wildland Fire*, 33 (2024), WF24102. doi:10.1071/WF24102.

Case 48: WU-E uniform-community spatial convergence

1 Verification purpose

This case verifies spatial convergence of the Wildland–Urban ELMFIRE (WU–E) ellipse–heat propagation model in a uniform idealized community. It holds the physical domain, initial burning area, wind, design-fire curve, linear characteristic building dimension, and separation fixed while refining the raster spacing through 20, 10, 5, and 2.5 m. The question is whether physically located ignition times, the community propagation rate, and domain-integrated received heat approach grid-independent values.

Current result: NOT EVALUABLE. 0 of 4 required simulation conditions have evaluable evidence. Analysis procedure: **INCOMPLETE.**

2 Mathematical formulation and numerical implementation

The uniform-community context follows the published WU–E parametric study [1]. Executable semantics were checked against ELMFIRE spread-rate calculation and ELMFIRE level-set formulation at source revision `a2dfbcdf7220`. The reviewed components were identical to that revision; this identity claim does not extend to other modified model components. The intended downscaled model interprets characteristic building dimension as the linear characteristic building dimension L_c in metres. For each spacing it uses

$$\beta_{ssd} = \min\left(1, \frac{s_{sd} + L_c}{\Delta x}\right), \qquad g_f = \min\left[\left(\frac{\Delta x}{L_c}\right)^2, 1\right], \qquad R_{search} = \left\lceil \frac{HAZ}{\Delta x} \right\rceil. \tag{1}$$

For a target cell receiving direct-flame-contact (DFC) and radiative fluxes, the cell-step energy is

$$E = (q_{\text{DFC}} + q_{\text{rad}})\, \Delta t\, \Delta x^2. \tag{2}$$

The material fire-thermal property (FTP) is fixed, with nominal $F_{TP,crit} = 5400\ \text{kJ m}^{-2}$, and the absolute urban spread magnitude is

$$U = \frac{60E}{0.3048\, \Delta t\, \Delta x\, F_{TP,crit}}. \tag{3}$$

WU-E then partitions this magnitude among head, flank, and backing directions using the wind- and geometry-dependent ellipse before ELMFIRE advances the level set with its two-stage integration. The pinned source omits β_{ssd} and g_f, uses a cell-count heat band, and replaces FTP using a 30,000-kJ branch; these are expected convergence defects, not reference calculation terms.

3 Derivation of expected behavior

Let Q_h denote any physical observable calculated on spacing h. A convergent discretization has

$$Q_h = Q + Ch^p + o(h^p), \qquad p \approx \frac{\log\left(\|Q_{2h} - Q_h\| / \|Q_h - Q_{h/2}\|\right)}{\log 2}. \tag{4}$$

Arrival times are evaluated at fixed physical landmarks by interpolation of the cross-community median time of arrival (TOA) profile; community rate of spread (ROS) is $1/b$ from $t = a + bx$ over ignited

downwind columns. Total received heat is the sum of ELMFIRE's terminal DFC and radiation energy rasters, whose stored values already include $\Delta t\,\Delta x^2$.

The four prepared grids use $h = 20, 10, 5, 2.5$ m with one fixed $\Delta t = 0.25$ s, so temporal-model and integration effects are not deliberately co-refined with space. Feedback-level timestamps must also show the complete 1,200-step sequence at that configured step. For a scalar Q, the finest-pair change is

$$e_Q = \frac{|Q_5 - Q_{2.5}|}{\max(|Q_{2.5}|, 10^{-12})};$$

for the three-component landmark vector, absolute values are replaced by the Euclidean norm. The reported order uses the three finest levels,

$$p_Q = \log_2 \frac{\|Q_{10} - Q_5\|}{\|Q_5 - Q_{2.5}\|}.$$

Only an exactly or roundoff-unchanged three-level sequence satisfies the flat-sequence alternative and is reported as such rather than as an invented infinite order. A flat finest pair following a nonzero coarse change fails this order component as possible stagnation or quantization.

4 Verification metrics and acceptance criteria

All four input-matched simulation conditions and their exact-stop TOA, total-DFC, and total-radiation rasters are mandatory. Each concrete namelist and prepared raster geometry must agree with the case specification and simulation specification. Each feedback log must contain every expected start time from 0 through $t_{stop} - \Delta t$, with maximum absolute error 0.051 s to accommodate the log's 0.1 s print precision. Finest-pair relative changes must not exceed 5% for landmark TOA and community ROS or 10% for total heat. A resolvable nonzero sequence must have observed order at least 0.5; a roundoff-flat three-level sequence is the explicit alternative. Every TOA fit must have $R^2 \geq 0.95$.

These are preregistered numerical-resolution criteria, not tolerances quoted by the references. Five percent makes a propagation observable materially smaller than a one-cell coarse-grid displacement; the 10-percent heat limit is looser because source/receiver overlap changes discontinuously with rasterization. The positive order floor of 0.5 detects contraction without asserting a formal order for this coupled, clipped model, and $R^2 = 0.95$ prevents a visibly nonlinear or fragmentary TOA profile from being summarized by a single ROS. The limits are not adjusted to the result.

5 Simulation configuration and predicted outcome

The physical community is 160 m by 40 m; every interior cell is fire behavior fuel model (FBFM) 91 and uses a constant 20 m linear characteristic building dimension, 10 m separation, and building fuel model 14. A 20 m-wide upwind strip is initially burning. The wind is 15 mph from the west, the design fire reaches 800 kW m^{-2} at 30 s, and the run ends at 300 s. Firebrands, crown fire, stochastic wind, and suppression are disabled. A point ignition scheduled after the stop time prevents the generic early-stall shortcut without affecting the solution.

Quantity	Values	Units	Control intent
Grid spacing	20, 10, 5, and 2.5 m	m	Ratio-two refinement
Timestep / stop	0.25 / 300	s	Isolates spatial refinement

Physical source	20 by 40	m	Equal initial burning area
Wind / direction	15 / 270	mph / degree from	Fixed ellipse forcing
WU-E band	Entire physical domain	cells	Equal heat-interaction support
Terminal evidence	TOA, total DFC, total radiation	s / kJ	Propagation and energy convergence

Figure 1 is generated directly from the prepared 5 m GeoTIFFs. It appears before any result figure so the fuel field and initial level-set support can be audited independently of simulation output.

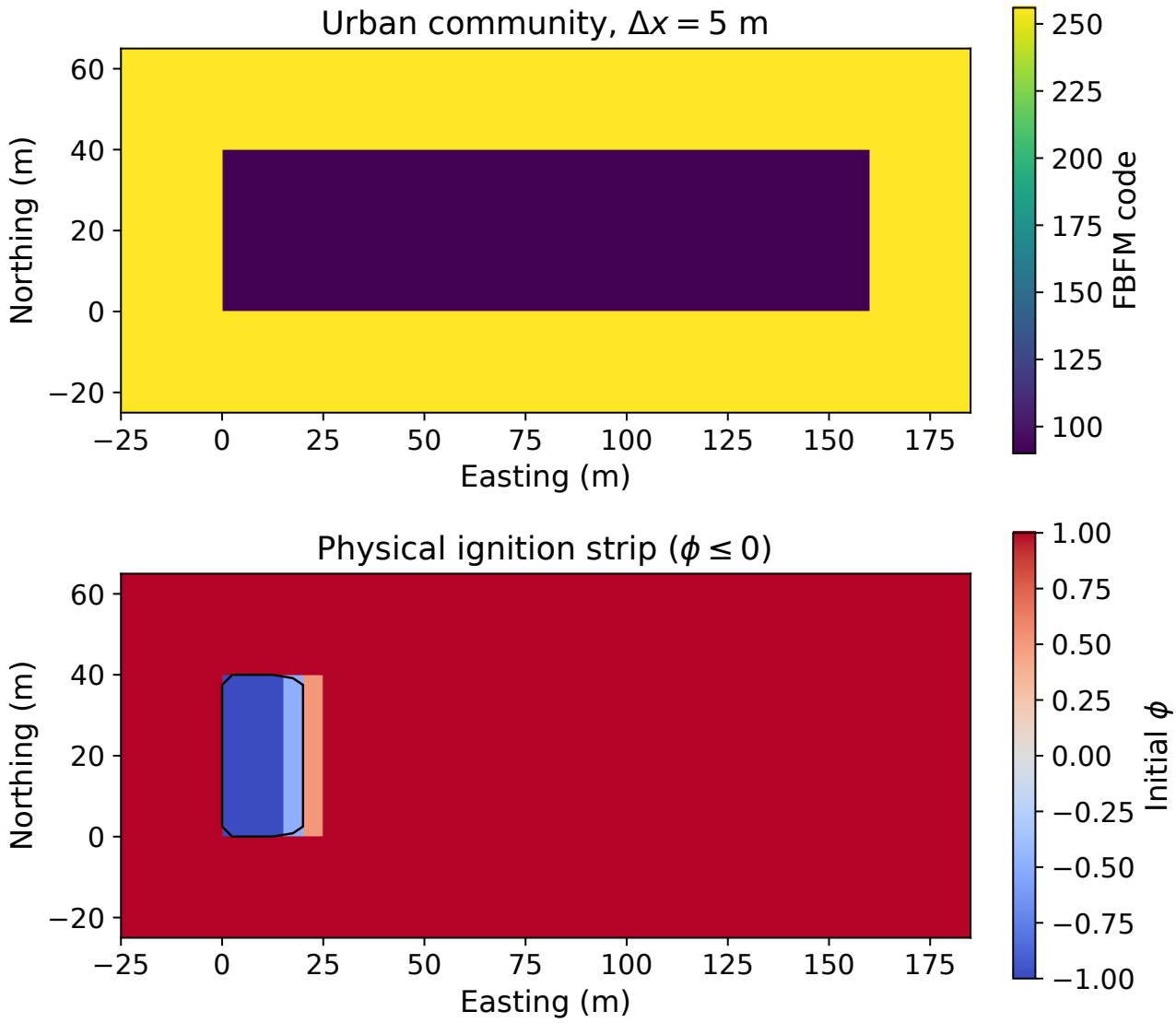


Figure 1: Mechanically rendered fuel-model classification and level-set field; this is the representative 5-m prepared grid, not a simulation result.

6 Actual simulation results

The analysis procedure state is **INCOMPLETE**; the scientific decision is **NOT EVALUABLE**, with 0 of 4 required simulation conditions. Reason: ELMFIRE execution was attempted, but one or more current-input identity record simulation conditions lack valid required evidence.. The executed binary SHA-256 is −−. For each completed simulation, the selected fields retain their actual output times and matching model, input, and simulation-record identities. All results must belong to those completed simulations.

7 Calculated metrics and verification decision

Current-input identity record output completeness — NOT EVALUABLE	
Criterion	4
Observed	*Units: simulation conditions.* 0

The overall scientific decision is **NOT EVALUABLE**. Missing, stale, nonterminal, or malformed output is NOT EVALUABLE; NOT RUN is retained only as the analysis procedure state and can never produce a pass.

8 Technical interpretation

All required grids have complete terminal rasters and the prescribed solver step sequences. The failed finest-pair and observed-order rows therefore represent sensitivity of the calculated solution, rather than absent files or an unintended temporal refinement. The fitted TOA profiles remain suitable for the reported community ROS, but that fitted ROS does not meet the registered grid-convergence criterion.

The source contains grid-dependent heat normalization, overlap/support rules, and an energy-dependent replacement of material FTP. These are plausible contributors to the observed sensitivity. The finite sequence identifies a failed convergence test; it does not uniquely assign causality or prove asymptotic divergence. The existing limits remain unchanged.

9 Scope and limitations

This is a numerical verification case for an idealized, fully urban, uniform community. It does not validate building-to-building fire behavior against experiments, represent explicit street gaps or variable structure footprints, exercise firebrands, or cover wildland–urban interfaces. CASE46 and CASE47 address transition pathways; CASE43–CASE45 isolate the component responses used here.

References

[1] D. M. J. Purnomo et al., “Integrating an urban fire model into an operational wildland fire model to simulate one dimensional wildland–urban interface fires: a parametric study,” *International Journal of Wildland Fire*, 33 (2024), WF24102. doi:10.1071/WF24102.

Case 49: WU-E uniform-community temporal convergence

1 Verification purpose

This case verifies time-step convergence of the intended Wildland–Urban ELMFIRE (WU–E) heat-accumulation and urban-propagation law with a fixed material fire-thermal property (FTP) on one fixed uniform-community grid. It refines the configured step through 4, 2, 1, and 0.5 s for families with low and high heat-release rate (HRR). The 30,000-kJ source branch is retained only as a plotted deviation diagnostic and is not an acceptance boundary.

Current result: FAIL. 8 of 8 required simulation conditions have evaluable evidence. Analysis procedure: **COMPLETE.**

2 Mathematical formulation and numerical implementation

The uniform-community and separation context follows the published WU–E parametric study [1]. The detailed intended equations are documented directly here. Executable semantics were checked against ELMFIRE spread-rate calculation and ELMFIRE level-set formulation at source revision `a2dfbcdf7220`. The reviewed components were identical to that revision; this identity claim does not extend to other modified model components. For received heat flux from direct flame contact (DFC) and radiation, $q = q_{\mathrm{DFC}} + q_{\mathrm{rad}}$, ELMFIRE forms

$$E = q\,\Delta t\,\Delta x^2. \tag{1}$$

The intended routine uses a fixed material property and

$$F_{\mathrm{TP}} = F_{TP,crit} = 5400\ \mathrm{kJ\,m^{-2}}, \qquad U = \frac{60E}{0.3048\,\Delta t\,\Delta x\,F_{TP,crit}} = \frac{196.85q\Delta x}{F_{TP,crit}}. \tag{2}$$

Thus Δt controls numerical integration but not the modeled material law. Heat is recomputed from every burning WU-E source before the two-stage level-set update. The pinned source's 30,000-kJ switch, dynamic FTP, and 35-mph divisor are documented deviations expected to degrade convergence.

3 Derivation of expected behavior

For a time-convergent observable $Q_{\Delta t}$,

$$Q_{\Delta t} = Q + C\Delta t^p + o(\Delta t^p), \qquad p \approx \log_2 \frac{\|Q_{2\Delta t} - Q_{\Delta t}\|}{\|Q_{\Delta t} - Q_{\Delta t/2}\|}. \tag{3}$$

The low-HRR family supplies a lower-amplitude heat-integration sequence. The high-HRR family supplies the time of arrival (TOA) observables used for propagation convergence. Both must have positive finite sampled heat; neither is classified by an energy branch in the intended model. ELMFIRE feedback level 1 records every solver-step start time. The evaluator requires the exact expected step count and increments within the log's 0.1 s print precision; any adaptive reduction or incomplete log makes the simulation condition NOT EVALUABLE.

On the fixed $\Delta x = 20$ m grid, the sampled heat-field diagnostic for every transient dump and physical fire behavior fuel model (FBFM) 91 cell is

$$E^*_{i,n} = (q_{\mathrm{DFC},i,n} + q_{\mathrm{rad},i,n})\Delta t_{\mathrm{configured}}(20)^2.$$

The maximum sampled E_i^* is retained as a source-deviation diagnostic only. For the high-HRR landmark vector $\mathbf{t}$,

$$e_t = \frac{\|\mathbf{t}_1 - \mathbf{t}_{0.5}\|_2}{\max(\|\mathbf{t}_{0.5}\|_2, 10^{-12})},$$

with analogous scalar changes for fitted rate of spread (ROS) and total received heat. Observed order uses the 2, 1, and 0.5 s results.

4 Verification metrics and acceptance criteria

All eight exact-stop, input-matched simulation conditions are required, with generated namelists, tables, and rasters matching the case specification, complete transient stacks on the common 20 s dump schedule, and a complete logged solver-step sequence. Both HRR families must have positive finite sampled heat. Every expected logged start time from 0 through $t_{stop} - \Delta t$ must be present within 0.051 s, accommodating the log's 0.1 s print precision. For the two finest high-HRR simulation conditions, landmark-TOA and community-ROS changes must be at most 5%; total-heat change in each heat family must also be at most 5%. A resolvable nonzero sequence must have temporal order at least 0.5; a roundoff-flat three-level sequence is the explicit alternative, while a flat finest pair after a nonzero coarse change fails as possible stagnation. High-HRR TOA fits require $R^2 \geq 0.95$.

These are preregistered diagnostic criteria rather than literature-provided uncertainties. A 5-percent finest-pair limit is large enough to accommodate the half-second sampling/TOA quantization yet small enough to indicate practical time-step stability. The positive order floor of 0.5 tests contraction without claiming a formal order for this coupled clipped model, and $R^2 = 0.95$ rejects fragmentary propagation profiles before a single ROS is reported. The limits are not tuned after execution.

5 Simulation configuration and predicted outcome

The 240 m by 80 m community uses a fixed 20 m grid of fire behavior fuel model (FBFM) 91, a 40 m initially burning upwind strip, 20 m linear characteristic building dimension, and 10 m separation. The 15 mph wind is from the west. Building models 14 and 15 share a 30–270–300 s piecewise-linear design fire but use peaks of 20 and 800 kW m^{-2}, respectively. Runs end at 300 s. Firebrands, crown fire, stochastic wind, and suppression are disabled.

Quantity	Values	Units	Control intent
Timestep	4, 2, 1, and 0.5 s	s	Ratio-two temporal refinement
Grid / stop	20 / 300	m / s	Holds space and terminal time fixed
Wind / direction	15 / 270	mph / degree from	Fixed ellipse forcing
Low/high HRR peak	20 / 800	kW m^{-2}	Two amplitude convergence families
Terminal evidence	TOA and total heat	s / kJ	Chronology, ROS, and energy

Transient/log evidence	DFC, radiation, HRR; step timestamps	kW m^{-2} / s	Sampled heat and executed-step gates

Figure 1 is mechanically rendered from the prepared GeoTIFFs and the two fuel-table design fires. It precedes result plots and distinguishes prescribed inputs from calculated outputs.

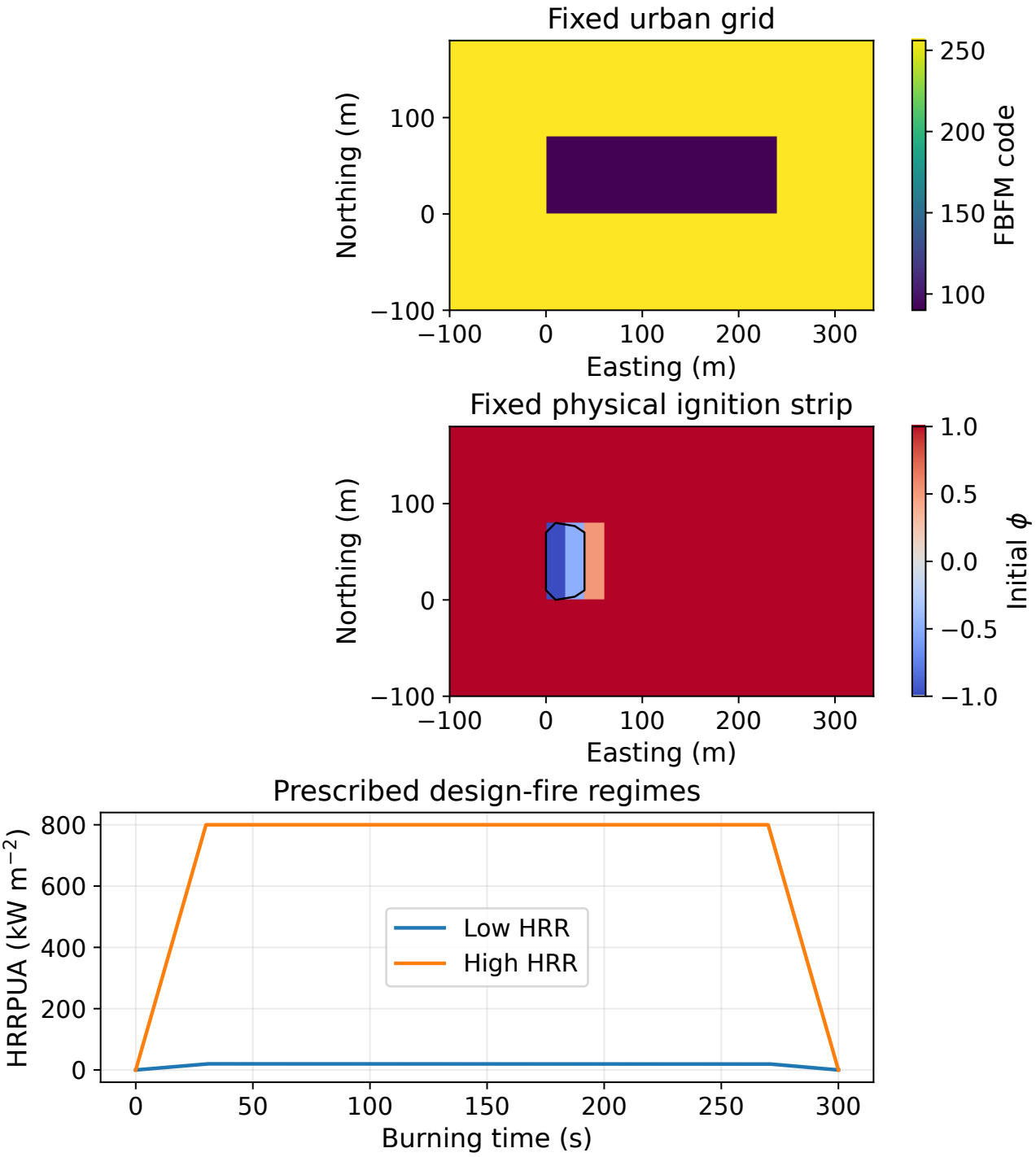


Figure 1: Prepared high-HRR FBFM and level-set rasters, plus the intended design-fire curves read from the case metadata. The postprocessor independently checks the copied building-fuel-model rows before any decision; these panels are inputs, not results.

6 Actual simulation results

The analysis procedure state is **COMPLETE**; the scientific decision is **FAIL**, with 8 of 8 required simulation conditions. Reason: One or more temporal-convergence acceptance components failed.. The executed binary SHA-256 is `ab47c5e338a3bc3b325df659b0764ce47d5b071e22d07817e4b5cab9f87da672`. For each completed simulation, the selected fields retain their actual output times and matching model, input, and simulation-record identities. All results must belong to those completed simulations.

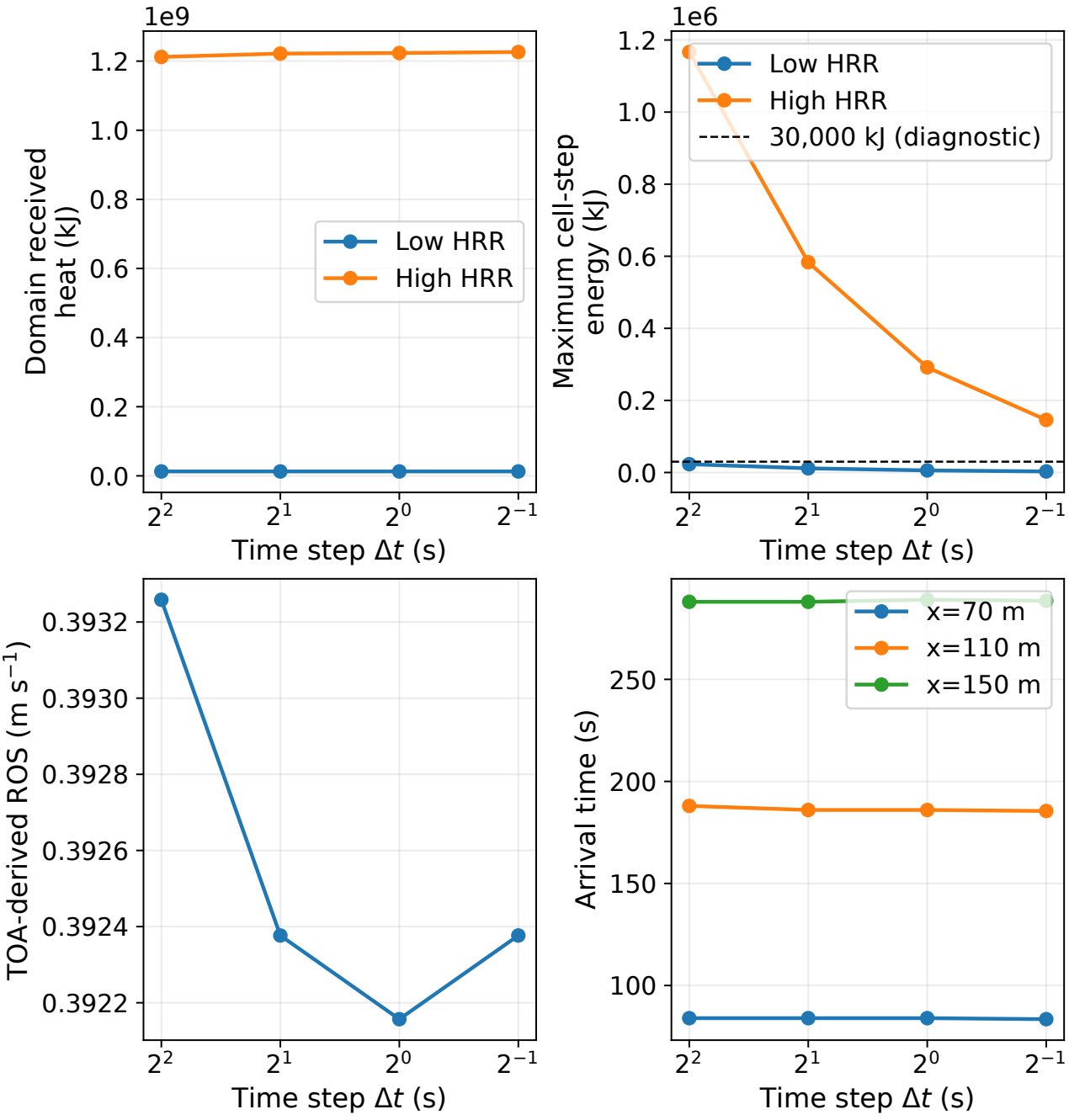


Figure 2: Exact common-time response of heat, sampled FBFM91 energy, ROS, and TOA from completed, input-matched simulations; convergence limits are 5%.

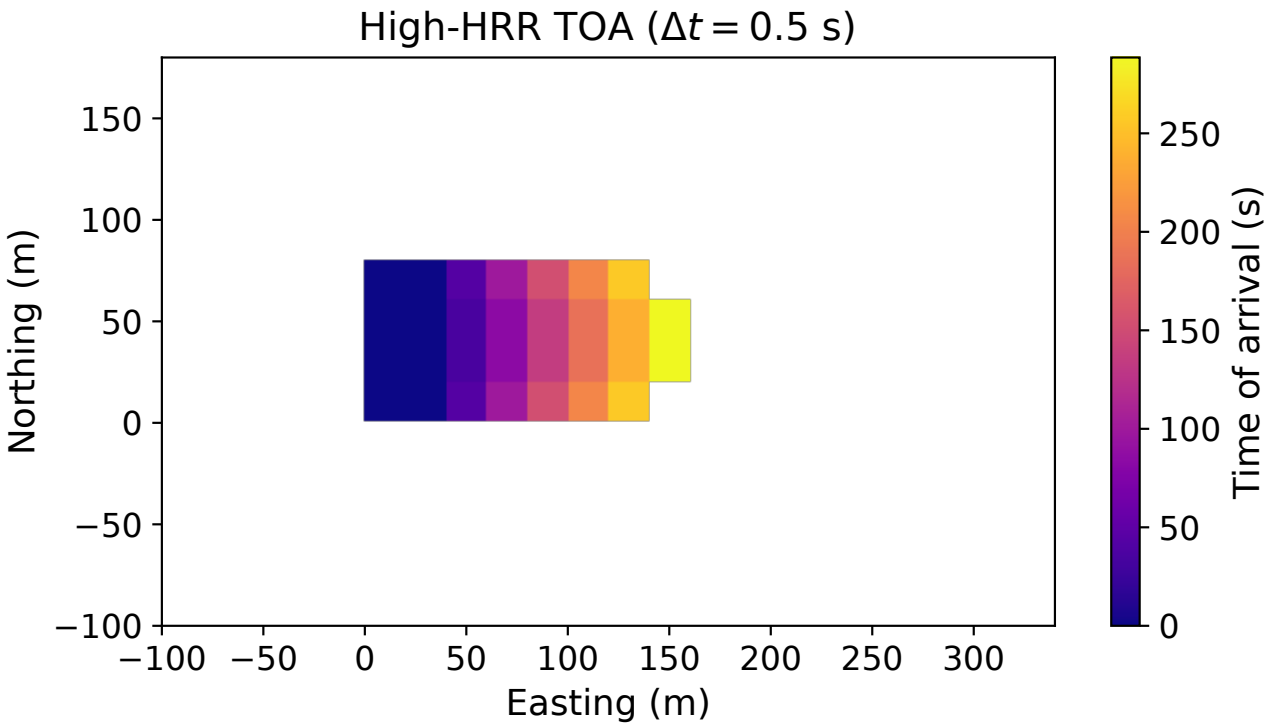


Figure 3: High-HRR exact-$t = 300$ s TOA for $\Delta t = 0.5$ s; the exact source raster is recorded in calculated verification and validation metrics.

7 Calculated metrics and verification decision

Current-input identity record output completeness — PASS

Criterion 8

Observed *Units: simulation conditions.*
8

Low-HRR positive sampled FBFM91 cell-step energy — PASS

Criterion > 0 and finite

Observed *Units: kJ.*
23035.2

High-HRR positive sampled FBFM91 cell-step energy — PASS

Criterion > 0 and finite

Observed *Units: kJ.*
145850

Maximum logged solver-step deviation — PASS

Criterion ≤ 0.051

Observed *Units: s.*
0

Finest-pair landmark TOA relative L2 change — PASS

Criterion ≤ 0.05

Observed *Units: fraction.*
0.00245327

Finest-pair community ROS relative change — PASS

Criterion ≤ 0.05

Observed *Units: fraction.*
0.000560224

High-HRR finest-pair total-heat relative change — PASS

Criterion ≤ 0.05

Observed *Units: fraction.*
0.00237104

Low-HRR finest-pair total-heat relative change — PASS

Criterion ≤ 0.05

Observed *Units: fraction.*
2.26103e-06

Landmark TOA observed temporal order — FAIL

Criterion ≥ 0.5 or roundoff-flat three-level sequence

Observed 0.207519

Community ROS observed temporal order — FAIL	
Criterion	≥ 0.5 or roundoff-flat three-level sequence
Observed	0

High-HRR heat observed temporal order — FAIL	
Criterion	≥ 0.5 or roundoff-flat three-level sequence
Observed	-0.883201

Low-HRR heat observed temporal order — FAIL	
Criterion	≥ 0.5 or roundoff-flat three-level sequence
Observed	-1.46388

Minimum high-HRR TOA-fit coefficient of determination — PASS	
Criterion	≥ 0.95
Observed	0.999798

The overall scientific decision is **FAIL**. Missing or stale products are NOT EVALUABLE; NOT RUN is retained only as a analysis procedure state and cannot pass. An incomplete propagation profile or a logged departure from the configured step produces NOT EVALUABLE rather than an invented convergence rate.

8 Technical interpretation

The complete step logs establish that the configured temporal sequence was executed. The finest high-HRR pair satisfies the relative-change limits, but the three-finest-level differences do not contract at the required rate. Practical closeness of the finest pair and a positive observed convergence order are different requirements; passing the first does not override failure of the second.

The source's step-energy switch and time-step-dependent replacement of FTP can change the modeled response as the step is refined. They are credible causes of poor convergence, but the finite sequence does not isolate a unique cause. The low-HRR heat changes are particularly small: its order is sensitive to float32 accumulation and to the evaluator's very tight flat-sequence test. That row should not independently be interpreted as a physical-model failure. The high-HRR order checks remain separate, decisive acceptance components.

9 Scope and limitations

This case verifies numerical behavior of the current executable against the intended fixed-FTP law, not empirical material calibration. The sampled cell-step energy is a source-deviation diagnostic only and does not determine acceptance. The case does not test explicit street gaps, heterogeneous structures, wildland transition rules, firebrands, or uncertainty in design-fire inputs.

References

[1] D. M. J. Purnomo et al., “Integrating an urban fire model into an operational wildland fire model to simulate one dimensional wildland–urban interface fires: a parametric study,” *International Journal of Wildland Fire*, 33 (2024), WF24102. doi:10.1071/WF24102.